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# **Computation of Multi-Particle Phase-Space Integrals and Their Applications in High-Energy Physics**

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# Resumo

A crescente precisão experimental alcançada pelas experiências realizadas em colisores de partículas exige previsões teóricas com um grau de rigor comparável. Uma parte substancial destes processos é governada pela interação forte, descrita pela Cromodinâmica Quântica (QCD). A energias elevadas, esta teoria pode ser tratada perturbativamente, organizando as previsões físicas como uma expansão em potências da constante de acoplamento forte,  $\alpha_s$ . Contudo, aumentar a precisão de um cálculo não corresponde apenas a acrescentar diagramas de Feynman: exige controlar uma estrutura complexa de contribuições de emissão real e virtual, bem como as divergências que surgem em cada uma delas.

As divergências infravermelhas constituem uma das principais dificuldades dos cálculos perturbativos em QCD. As contribuições de emissão real tornam-se singulares quando um partão é emitido com energia muito baixa, o limite *soft*, ou quando dois partões se tornam colineares. Estas singularidades estão associadas à emissão de radiação não resolvida, isto é, de partões cuja energia ou separação angular é demasiado pequena para serem distinguidos como estados finais distintos. Paralelamente, as amplitudes que incluem correções virtuais de loop apresentam polos infravermelhos associados às mesmas regiões cinemáticas. Embora estas divergências se cancelem em observáveis físicos suficientemente inclusivos, as componentes individuais do cálculo não são finitas. É, por isso, necessário reorganizar a previsão teórica de modo a isolar as singularidades e a efetuar o seu cancelamento de forma analítica e controlada.

O formalismo de subtração por antenas fornece um enquadramento adequado para esse objetivo. Neste contexto, uma antena é formada por dois partões ligados por cor, designados por radiadores, entre os quais ocorre a emissão de radiação. A radiação não resolvida é então descrita em termos destas antenas, permitindo tratar de forma sistemática as regiões singulares do espaço de fases. As funções de antena reproduzem os limites *soft* e colineares relevantes de forma local no espaço de fase. Quando são subtraídas às contribuições reais, eliminam as singularidades antes da integração numérica. Depois de integradas analiticamente sobre o espaço de fase da radiação não resolvida, originam polos em regularização dimensional que cancelam os polos das contribuições virtuais. O cancelamento destas singularidades permite que cada contribuição seja integrada numericamente em quatro dimensões.

Esta dissertação apresenta o desenvolvimento de *AntCalc*, um pacote simbólico para a construção e integração automatizadas de funções de antena em QCD. O projeto procura automatizar etapas que requerem implementações específicas para cada antena, reunindo num mesmo fluxo as convenções de normalização, construção e integração. O trabalho centra-se nas antenas quark-antiquark sem massa, considerando a configuração cinemática final-final, na qual ambos os radiadores da antena são partículas do estado final, até à ordem next-to-next-to-leading order (NNLO). Para além deste objetivo, é explorada uma extensão ao regime de quarks massivos, através da implementação da antena  $A_3^0(Q, g, \bar{Q})$ , que descreve a emissão de um glúon entre um quark e um antiquark massivos.

A base teórica da dissertação inclui uma introdução à QCD perturbativa, aos campos de quarks e glúons e à construção de amplitudes a partir de diagramas de Feynman. São discutidas as divergências ultravioleta, tratadas através da renormalização, e as divergências infravermelhas, que se cancelam

mediante a combinação das diferentes contribuições perturbativas reais e virtuais à mesma ordem. Utiliza-se regularização dimensional, na qual o número de dimensões do espaço-tempo é tomado como  $d = 4 - 2\epsilon$ . Neste enquadramento, as divergências manifestam-se como polos em  $1/\epsilon$ , e as quantidades integradas são expressas por séries de Laurent. O cancelamento dos polos pode, assim, ser acompanhado explicitamente ao combinar as diferentes contribuições para uma quantidade física.

É também apresentada a estrutura das funções de antena e a sua decomposição em cor. As antenas são obtidas a partir de amplitudes físicas normalizadas e organizadas de acordo com os radiadores e as partículas emitidas. A fatorização do espaço de fase é essencial neste processo, pois permite separar o espaço de fase reduzido, associado aos radiadores após a emissão não resolvida ser removida, do espaço de fase correspondente à própria emissão. A integração pode então ser efetuada apenas sobre o espaço de fase da radiação não resolvida, independentemente do processo duro em que a antena venha a ser inserida.

A integração analítica das funções de antena sobre o correspondente espaço de fases de antena constitui uma das etapas mais exigentes na obtenção das antenas integradas. Para realizar esta integração, as integrais de espaço de fases são reescritas como integrais de loop através da técnica de reverse unitarity. Estas são então reduzidas, através de identidades de integração por partes (IBP) e do algoritmo de Laporta, a um conjunto finito de integrais mestres. Em casos adequados, recorre-se também à redução de Passarino–Veltman. Uma vez identificados os integrais mestres e substituídas as respetivas expressões conhecidas, obtém-se a forma integrada da antena como uma expansão em  $\epsilon$ . O resultado da redução fica, deste modo, expresso num conjunto limitado de integrais mestres, cujas expansões em  $\epsilon$  determinam as partes singular e finita da antena integrada.

O pacote *AntCalc* foi implementado no ambiente *Wolfram Mathematica* e integra ferramentas especializadas para as diferentes fases do cálculo. *FeynArts* é utilizado para gerar os diagramas de Feynman relevantes, enquanto *FeynCalc* e *FeynHelpers* apoiam a construção e a manipulação das amplitudes. A redução IBP é realizada com *LiteRed2*. A arquitetura do pacote é orientada por perfis, isto é, por descrições estruturadas que especificam o processo, as amplitudes, as famílias de integrais, os fatores de normalização e os integrais mestres necessários. A informação específica de cada antena fica, assim, concentrada no respetivo perfil, enquanto as rotinas de construção e integração permanecem comuns aos diferentes casos.

O fluxo de trabalho divide-se em duas fases principais: construção e integração. Na fase de construção, o utilizador fornece a descrição do processo e o pacote gera os diagramas e amplitudes correspondentes. São depois efetuadas as manipulações algébricas necessárias para extrair a função de antena não integrada, ou as suas componentes de cor quando aplicável. Na fase de integração, essa expressão é convertida para uma forma apropriada à redução, associada à família de integrais correta e reduzida aos integrais mestres. Finalmente, a substituição dos resultados dos integrais mestres e a expansão em torno de  $\epsilon = 0$  produzem a antena integrada. Ao longo deste fluxo são efetuados testes de consistência e apresentados diagnósticos do cálculo.

Como exemplo detalhado, é apresentada a derivação da antena  $A_3^0$ , que descreve a emissão de um glúon entre um quark e um antiquark sem massa a *tree-level* (onde *tree-level* corresponde a contribuições de diagramas de Feynman sem loops). Este caso ilustra todo o percurso, desde a geração da amplitude até à expressão integrada. A validação local da antena é efetuada através dos seus limites infravermelhos. No limite *soft*, verifica-se que a expressão reproduz o fator eikonal esperado; nos limites colineares, confirma-se a recuperação da função de desdobramento (*splitting-function*) de Altarelli–Parisi apropriada. A concordância nestes limites constitui uma verificação local da expressão obtida, independente da comparação posterior da antena integrada com resultados da literatura.

Para além de  $A_3^0$ , *AntCalc* é aplicado a um conjunto de antenas sem massa até NNLO. São obtidas contribuições de dois, três e quatro partões, incluindo os conjuntos  $A_2^2$ ,  $A_3^1$  e  $A_4^0$ , bem como antenas com outras estruturas de cor, tais como  $B_4^0$  e  $C_4^0$ . Os resultados não integrados e integrados são comparados com resultados estabelecidos na literatura, verificando-se concordância. A concordância com os resultados publicados verifica, em simultâneo, as convenções de regularização dimensional e normalização, bem como o tratamento da estrutura de cor e do espaço de fase adotado na implementação.

A validação global é realizada através do cálculo do rácio hadrónico  $R$ , no limite de quarks sem massa, até NNLO em QCD, isto é, até  $\mathcal{O}(\alpha_s^2)$ . Este observável é definido como o rácio entre a secção eficaz inclusiva de produção de hádrões em aniquilação  $e^+e^-$  e a secção eficaz para a produção de um par de muões,

$$R(s) = \frac{\sigma(e^+e^- \rightarrow \text{hadrões})}{\sigma(e^+e^- \rightarrow \mu^+\mu^-)}.$$

Por ser inclusivo, constitui um teste rigoroso da consistência do formalismo: os polos infravermelhos provenientes das contribuições reais e virtuais devem cancelar exatamente. O cálculo do  $R$ -ratio a partir dos objetos integrados produzidos por *AntCalc* confirma esse cancelamento e reproduz o resultado finito conhecido até NNLO. A reprodução do  $R$ -ratio complementa, deste modo, os testes locais das antenas, verificando o cancelamento dos polos quando as diferentes contribuições são combinadas num observável físico.

A dissertação explora ainda a extensão ao caso massivo pela construção e integração da antena massiva  $A_3^0$ . A presença de quarks massivos nesta antena altera a cinemática, a estrutura do espaço de fase e os limites infravermelhos, eliminando, em particular, algumas singularidades colineares existentes no caso sem massa. Este cálculo fornece uma primeira implementação do fluxo massivo em *AntCalc* e identifica as alterações necessárias à construção e integração de futuras antenas massivas.

Em síntese, o trabalho desenvolvido mostra que as etapas de construção, redução e integração das funções de antena consideradas podem ser reunidas num único procedimento automatizado. O pacote *AntCalc* desenvolvido nesta tese reproduz resultados conhecidos para antenas sem massa até NNLO, valida os limites infravermelhos locais e recupera corretamente o  $R$ -ratio hadrónico. A implementação obtida pode servir de base à inclusão de novas classes de antenas sem exigir que todo o procedimento de construção e integração seja novamente implementado de raiz.

**Palavras chave:** QCD; Subtração por Antenas; NNLO; Singularidades Infravermelhas; *Mathematica*

# Abstract

The precision achieved by measurements at the Large Hadron Collider (LHC) places increasingly stringent demands on theoretical predictions. In perturbative Quantum Chromodynamics (QCD), the accuracy of these predictions is systematically improved by including higher-order corrections in the strong coupling constant,  $\alpha_s$ . Beyond leading order, however, real and virtual contributions develop infrared singularities associated with soft and collinear parton radiation. While these singularities cancel in infrared-safe observables when all relevant contributions are combined, the real and virtual terms are separately divergent and are defined over phase spaces of different particle multiplicities. Their numerical evaluation therefore requires a procedure that isolates the singular behaviour and implements its cancellation explicitly.

Antenna subtraction provides a systematic way to do this. It uses universal antenna functions to describe unresolved radiation between colour-connected hard partons. These functions are subtracted locally from the real-emission contribution and then integrated analytically in dimensional regularisation, making the cancellation of infrared poles against the virtual contributions explicit.

This thesis develops *AntCalc*, a symbolic framework in Wolfram Mathematica for constructing and integrating QCD antenna functions. The package combines *FeynArts*, *FeynCalc*, *FeynHelpers*, and *LiteRed2*, while keeping process-dependent information separate from the general calculation workflow. It generates the required amplitudes, constructs unintegrated antennae or colour components, maps the corresponding phase-space integrals onto integral families, and reduces them to master integrals through integration-by-parts identities.

The implementation reproduces the massless final-final quark-antiquark antennae through next-to-next-to-leading order. The tree-level three-parton antenna  $A_3^0$  is used as a detailed example and is checked in its soft and collinear limits. The integrated antenna functions are found to agree with existing results in the literature, while their combined consistency is further validated through the calculation of the massless hadronic  $R$ -ratio up to NNLO,  $O(\alpha_s^2)$ . The extension beyond the massless case is also explored through the implementation of the massive  $A_3^0$  antenna, demonstrating the applicability of the developed framework to antennae involving massive partons. Future work will extend *AntCalc* to further massive antennae, additional partonic configurations, and a more modular multi-language architecture.

**Keywords:** QCD; Antenna Subtraction; NNLO; Infrared Singularities; *Mathematica*

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# List of Abbreviations and Symbols

## Abbreviations and Acronyms

| Term            | Meaning                                                         |
|-----------------|-----------------------------------------------------------------|
| QCD             | Quantum Chromodynamics                                          |
| LO / NLO / NNLO | Leading / Next-to-Leading / Next-to-Next-to-Leading Order       |
| IR / UV         | Infrared / Ultraviolet                                          |
| MI              | Master Integral(s)                                              |
| RR / RV / VV    | double-Real, Real-Virtual and double-Virtual NNLO contributions |
| AP              | Altarelli–Parisi (splitting function)                           |
| CDR             | Conventional Dimensional Regularisation                         |
| KLN             | Kinoshita–Lee–Nauenberg (theorem)                               |
| SM              | Standard Model of Particle Physics                              |
| EFT             | Effective Field Theory                                          |
| SUSY            | Supersymmetry                                                   |
| MC              | Monte Carlo                                                     |
| IBP             | Integration by Parts                                            |

## Antenna and Regularisation Notation

| Symbol            | Meaning                                                                         |
|-------------------|---------------------------------------------------------------------------------|
| $X_n^l$           | Generic antenna symbol: $n$ is the final-state multiplicity, $l$ the loop order |
| $A, B, C$         | Quark-antiquark antenna families (implemented)                                  |
| $D, E$            | Quark-gluon antenna families (not implemented in the present work)              |
| $F, G, H$         | Gluon-gluon antenna families (not implemented in the present work)              |
| $X$ (undecorated) | Leading-colour component (from the raw “Lead” interference term)                |
| $\tilde{X}$       | Subleading-colour component (raw “Sublead” interference term)                   |
| $\hat{X}$         | Quark-loop component (raw “QL” interference term)                               |

| Symbol                     | Meaning                                                                                                            |
|----------------------------|--------------------------------------------------------------------------------------------------------------------|
| $\check{X}$                | Double-fermion-loop self-interference component (NNLO $A_2^2$ only)                                                |
| $S_\epsilon$               | $(4\pi)^{-\epsilon} e^{\epsilon\gamma_E}$ , the $\overline{\text{MS}}$ normalisation factor                        |
| $\Lambda_l$                | $(8\pi^2)^l$ , loop-order normalisation                                                                            |
| $G_k$                      | $(8\pi^2)^k$ , perturbative-order scale normalisation                                                              |
| $C(\epsilon, k)$           | $G_k S_\epsilon$ , combined normalisation factor                                                                   |
| $\mathcal{N}_X$            | Build-stage colour and coupling normalisation                                                                      |
| $\mathcal{N}(\epsilon, k)$ | Full integration normalisation, $C(\epsilon, k)/\Phi_2$                                                            |
| $\mathcal{T}$ -objects     | Process-wide colour- and kinematics-conscious objects assembled from antennae, used in the $R$ -ratio construction |
| $\Phi_n$                   | $d$ -dimensional $n$ -body phase-space volume                                                                      |

## General Kinematics

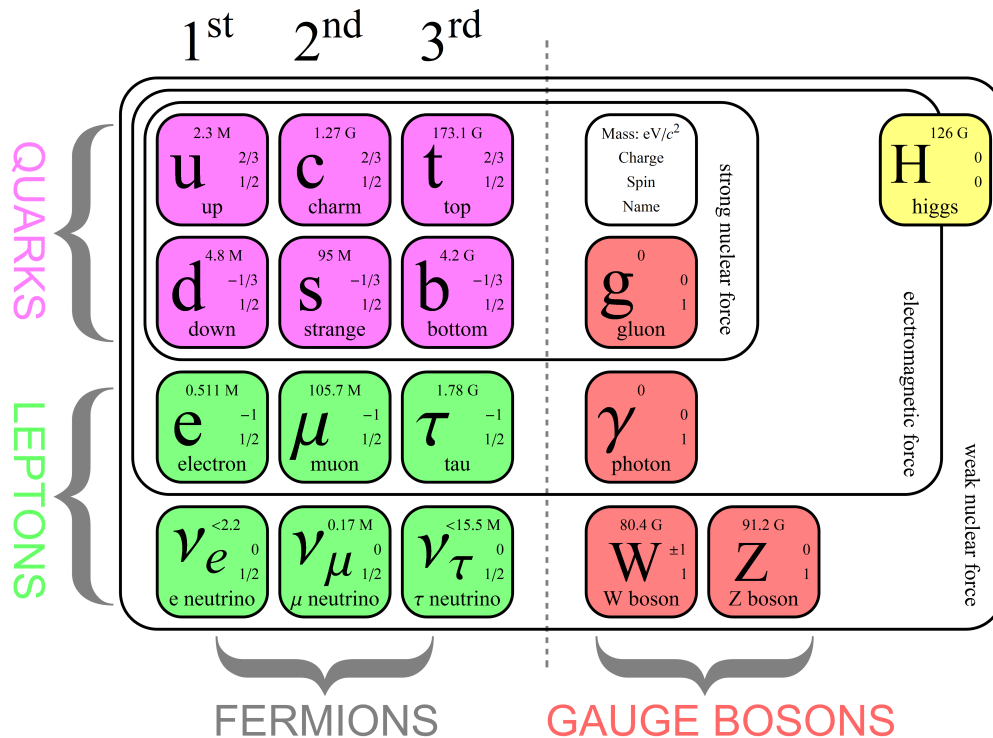
| Symbol                        | Meaning                                                             |
|-------------------------------|---------------------------------------------------------------------|
| $\alpha_s$                    | Strong coupling constant                                            |
| $\epsilon, d = 4 - 2\epsilon$ | Dimensional-regularisation parameter; spacetime dimension           |
| $s_{ij}$                      | Mandelstam invariant, $2p_i \cdot p_j$                              |
| $\mu$                         | Renormalisation/regularisation scale                                |
| $q^2$                         | Hard-process invariant mass squared                                 |
| $N$                           | Number of colours (= 3)                                             |
| $N_f$                         | Number of active quark flavours                                     |
| $C_F, C_A$                    | Quadratic Casimir invariants (fundamental / adjoint representation) |
| $g_s$                         | Strong coupling, $g_s = \sqrt{4\pi\alpha_s}$                        |

# Chapter 1

## Introduction

High-energy collider experiments, in particular proton-proton collisions at the Large Hadron Collider (LHC) at CERN, provide increasingly precise tests of the strong interaction and of the Standard Model of particle physics. Matching this experimental precision requires theoretical predictions with correspondingly high accuracy, making higher-order perturbative calculations in Quantum Chromodynamics (QCD) essential.

QCD is the quantum field theory describing the strong interaction. It is a non-Abelian gauge theory based on the  $SU(3)_c$  colour symmetry, with quarks as fermionic matter fields and gluons as the corresponding gauge bosons. Quarks occur in six flavours and transform in the fundamental representation of  $SU(3)_c$ , while gluons transform in the adjoint representation and themselves carry colour charge. The quarks and gluons within the Standard Model particle content are shown in Fig. 1.1.



**Figure 1.1:** A diagram representing the Standard Model of Particle Physics. In it, the six quark flavours are visible on the top left in purple, among the fermions, and the gluon is represented close as the topmost gauge bosons, in red. Image by Matic Lubej [1].

At sufficiently large momentum-transfer scales, the strong coupling  $\alpha_s$  becomes small as a conse-

quence of asymptotic freedom. This allows QCD observables involving a hard scale  $Q$  to be calculated perturbatively as an expansion in powers of  $\alpha_s(Q)$ . Successive perturbative orders—leading order (LO), next-to-leading order (NLO), next-to-next-to-leading order (NNLO), and beyond—generally improve the accuracy of theoretical predictions and reduce their residual dependence on unphysical renormalisation and factorisation scales. This improvement, however, comes with a rapid increase in the complexity of the required calculations. Obtaining higher-order perturbative predictions is not simply a matter of calculating additional Feynman diagrams. Real-emission and virtual-loop contributions develop infrared singularities in kinematic configurations involving soft (low energy) or collinear partons. These individual contributions are divergent, and their infrared singularities must be extracted and cancelled before numerical integration can be performed. For infrared-safe observables, such as the hadronic  $R$ -ratio defined in Chapter 5, the singularities cancel between the real and virtual contributions, leaving a finite physical result.

The antenna subtraction formalism provides a framework for extracting and cancelling these infrared singularities, rendering the individual contributions suitable for numerical integration. The formalism describes unresolved radiation between pairs of colour-connected hard partons through universal antenna functions. These functions reproduce the relevant infrared limits locally and, after analytic integration over the unresolved phase space in dimensional regularisation,  $d = 4 - 2\epsilon$ , yield Laurent series whose  $\epsilon$ -pole structure cancels that of the virtual contributions. Their systematic construction and integration are therefore central to higher-order QCD calculations within the antenna subtraction framework.

This thesis develops and validates *AntCalc*, a symbolic framework for the automated construction and integration of QCD antenna functions. The work focuses on massless final-final quark-antiquark antennae through next-to-next-to-leading order (NNLO), combining symbolic amplitude generation with reduction methods for phase-space integrals. The resulting implementation is tested globally through the calculation of the massless hadronic  $R$ -ratio, a well-known inclusive observable in  $e^+e^-$  annihilation that provides a benchmark for the complete implementation, up to NNLO. Locally, individual antenna functions are tested, where appropriate, by verifying that they reproduce the expected eikonal factors and Altarelli-Parisi (AP) splitting functions in the relevant unresolved limits.

The remainder of the thesis is organised as follows. Chapter 2 introduces the theoretical foundations required throughout this work, including perturbative QCD, infrared singularities, dimensional regularisation, and the antenna subtraction formalism. It defines the massless antenna functions and discusses their colour structure, phase-space factorisation, and role in fixed-order calculations. The chapter also introduces the normalisation conventions used throughout. Chapter 3 describes *AntCalc*'s framework, including its architecture, antenna construction and integration workflows, reduction procedures, and the conventions and validation tools used to obtain the results. Chapter 4 presents the antenna functions obtained using *AntCalc*, organised by final-state multiplicity, with the master integrals required for each antenna set presented after the corresponding antennae. The  $A_3^0$  antenna is discussed in detail, from its unintegrated construction to its integrated form and the validation of its soft and collinear limits. Selected NLO and NNLO antenna results are then presented more concisely. The chapter concludes with an exploratory extension of the  $A_3^0$  antenna to the massive case. Chapter 5 provides a global validation of the framework through the calculation of the massless hadronic  $R$ -ratio through NNLO. The relevant antenna contributions are combined, demonstrating the cancellation of infrared poles and agreement with the known finite result. Chapter 6 discusses future directions for *AntCalc*, including more efficient symbolic and reduction backends, completion of the massive antenna sector, general model support, and extensions to higher perturbative orders, particularly  $N^3$ LO. Finally, Chapter 7 summarises the main conclusions of the thesis.

## Chapter 2

# Theoretical Foundations

### 2.1 QCD: Fields, Interactions and Perturbative Amplitudes

QCD is the quantum field theory of the strong interaction, whose elementary particles are the quarks and the gluons. Quarks are fermions that carry colour charge and interact through the exchange of gluons. Owing to the non-Abelian gauge structure of QCD, gluons themselves also carry colour charge and, therefore, undergo self-interaction.

At sufficiently high energy scales, the strong coupling,  $\alpha_s$ , becomes sufficiently small for physical observables to be calculated perturbatively, as an expansion in powers of  $\alpha_s$ . In this framework, the interaction structure of QCD determines the amplitudes for quark and gluon radiation that are studied throughout this thesis.

#### 2.1.1 Quark and Gluon Fields

Quarks are spin-1/2 fields carrying flavour and a three-valued colour charge. They are QCD's matter fields. Gluons are the eight gauge fields associated with local  $SU(3)_c$  colour symmetry. The QCD Lagrangian describes their propagation and interaction. It is written as follows,

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4}G_{\mu\nu}^a G^{a\mu\nu} + \sum_f \bar{\psi}_f (i\gamma^\mu D_\mu - m_f) \psi_f. \quad (2.1)$$

In this expression,  $G_{\mu\nu}^a$  is the non-Abelian field-strength tensor,  $D_\mu$  is the covariant derivative,  $\psi_f$  is the quark field and  $m_f$  is the quark mass. The flavour label  $f$  is summed over the quark species included in the theory, while the adjoint colour index  $a$  runs from one to eight. This Lagrangian is cleanly divided into two pieces, the quark and the gluon pieces.

Defining the covariant derivative as,

$$D_\mu = \partial_\mu - ig_s T^a A_\mu^a \quad (2.2)$$

where  $T^a$  represents the generators of the fundamental representation of  $SU(3)_c$  and  $A_\mu^a$  is the gluon gauge field, the quark Lagrangian term may be expanded to,

$$\mathcal{L}_q = \sum_f [\bar{\psi}_f (i\gamma^\mu \partial_\mu - m_f) \psi_f + g_s \bar{\psi}_f \gamma^\mu T^a \psi_f A_\mu^a]. \quad (2.3)$$

Here  $g_s$  is the strong coupling. In the expanded quark term, the first piece describes free quark propagation

and the second quark-gluon interaction. This second piece is also responsible for generating the  $q\bar{q}g$  vertex.

We now also define the totally antisymmetric structure constants of the  $SU(3)_c$  Lie algebra  $f^{abc}$  as,

$$[T^a, T^b] = if^{abc}T^c. \quad (2.4)$$

They may be used to define the non-Abelian field-strength tensor  $G_{\mu\nu}^a$  as,

$$G_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g_s f^{abc} A_\mu^b A_\nu^c. \quad (2.5)$$

The first two terms of the field-strength tensor are Abelian-like and, in fact, they can be rewritten using the Abelian-like linear part of the non-Abelian field-strength tensor  $F_{\mu\nu}^a$ , as,

$$G_{\mu\nu}^a = F_{\mu\nu}^a + g_s f^{abc} A_\mu^b A_\nu^c, \quad F_{\mu\nu}^a \equiv \partial_\mu A_\nu^a - \partial_\nu A_\mu^a. \quad (2.6)$$

The second term arises because, as shown in Eq. (2.4), the  $SU(3)_c$  generators do not commute. The non-Abelian field-strength tensor is used to write the gluon Lagrangian term. After expanding it becomes,

$$\mathcal{L}_g = -\frac{1}{4} \left( F_{\mu\nu}^a \right)^2 - \frac{g_s}{2} f^{abc} F^{a\mu\nu} A_\mu^b A_\nu^c - \frac{g_s^2}{4} f^{abc} f^{ade} A_\mu^b A_\nu^c A^{d\mu} A^{e\nu}. \quad (2.7)$$

Here, much like in  $\mathcal{L}_q$ , the gluon Lagrangian is also divided into three terms. The first term is quadratic in the gluon field and, after gauge-fixing, describes free gluon propagation as [2],

$$\Delta_{ab,\mu\nu}(k) = \delta_{ab} \frac{i}{k^2} \left( -g_{\mu\nu} + (1-\lambda) \frac{k_\mu k_\nu}{k^2} \right), \quad \mathcal{L}_{\text{gf}} = -\frac{1}{2\lambda} \left( \partial^\mu A_\mu^a \right)^2 \quad (2.8)$$

where  $\mathcal{L}_{\text{gf}}$  is the gauge-fixing Lagrangian,  $\delta_{ab}$  is the Kronecker delta, and  $\lambda$  is the gauge parameter. Throughout this text the 't Hooft-Feynman convention is used, which makes  $\lambda = 1$ , following the default convention from *FeynArts* [3]. It is important to note, however, that physical observables are independent of the gauge choice. In addition, the last two terms in Eq. (2.7) contain cubic and quartic terms in the gluon fields that generate, respectively, the three- and four-gluon self-interaction vertices,  $ggg$  and  $gggg$ .

Finally, we note that the Feynman gauge adopted throughout this work, corresponding to  $\lambda = 1$ , is a particular choice within the class of covariant gauges. In a non-Abelian gauge theory such as QCD, covariant gauge fixing introduces Faddeev–Popov ghost fields. Their dynamics and interactions with the gluon field are described by the ghost Lagrangian [2],

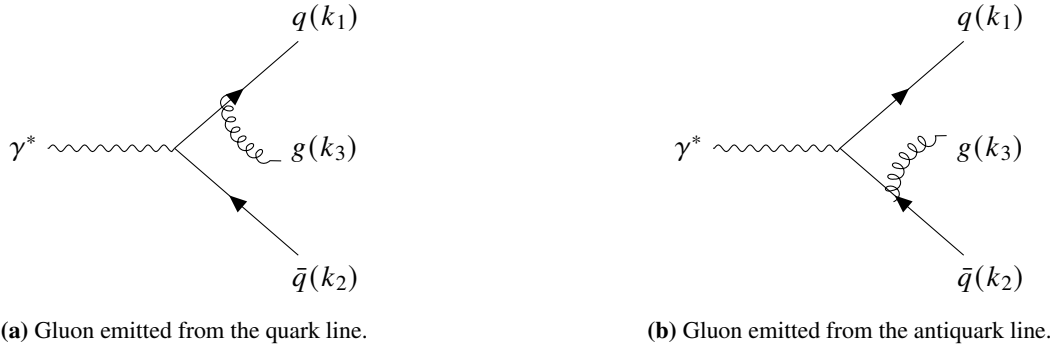
$$\mathcal{L}_{\text{ghost}} = \partial_\mu \eta^{a\dagger} \left( D_{ab}^\mu \eta^b \right). \quad (2.9)$$

This is the default ghost sector as used in *FeynArts* [3] and, as for the  $\lambda = 1$  choice above, is the one used for the duration of the present text and in *AntCalc*. These ghost fields cancel the unphysical timelike and longitudinal gluon polarisations that would, otherwise, propagate in covariant-gauge loop diagrams. This ensures that only the two physical transverse polarisation states contribute to physical amplitudes [2].

Together, the two pieces,  $\mathcal{L}_q$  and  $\mathcal{L}_g$ , constitute the gauge-invariant core of QCD. They generate the three elementary interaction vertices among the quark and gluon fields:  $q\bar{q}g$ ,  $ggg$  and  $gggg$  [2, 4].

### 2.1.2 From the QCD Lagrangian to Feynman Diagrams

The interaction terms in  $\mathcal{L}_q$  and  $\mathcal{L}_g$  determine the interaction vertices of QCD, which, together with the corresponding propagators, form the Feynman rules we use to compute scattering amplitudes perturbatively. Feynman diagrams are graphical representations of the individual contributions to the perturbative expansion of an amplitude. Each process may require more than one diagram to be fully represented; in that case, the complete amplitude is given as the sum of the amplitudes associated with all contributing diagrams. Fig. 2.1 depicts the two Feynman diagrams contributing to the process  $\gamma^* \rightarrow q(k_1)g(k_3)\bar{q}(k_2)$ , where a virtual photon splits into a quark-antiquark pair, with the gluon emitted from either the quark or antiquark line.



**Figure 2.1:** Tree-level diagrams for the process  $\gamma^* \rightarrow q(k_1)g(k_3)\bar{q}(k_2)$ .

The final amplitude in this case would then follow,

$$\mathcal{M}_{\gamma^* \rightarrow qg\bar{q}}^{(0)} = \mathcal{M}_{q\text{-emission}}^{(0)} + \mathcal{M}_{\bar{q}\text{-emission}}^{(0)}. \quad (2.10)$$

In the diagrams in Fig. 2.1, straight, curly, and wavy lines denote fermions (quarks), gluons and photons, respectively. These diagrams represent the reduced hadronic-current process with  $\gamma^*$  as an off-shell and colourless source. In the full process,  $e^+e^- \rightarrow \gamma^* \rightarrow qg\bar{q}$ , the photon is internal between the lepton (electron-positron) and hadron (quark-antiquark) currents. All other lines are external, meaning that they represent outgoing asymptotic partons, the quarks and gluons. Each of the diagrams features two vertices, where the lines meet. The vertices represent the interactions between the particles, which are determined by their respective Lagrangians. Each diagram contains an electromagnetic  $\gamma^* q\bar{q}$  current insertion and a QCD  $q\bar{q}g$  interaction vertex. The former follows from the QED sector, whereas the latter is generated by  $\mathcal{L}_{\text{QCD}}$ . A rate or cross section often depends on the squared amplitude  $|\mathcal{M}|^2$ , given as,

$$|\mathcal{M}|^2 = \sum_{D,D'} \mathcal{M}_D \mathcal{M}_{D'}^*, \quad (2.11)$$

where  $D$  and  $D'$  index the Feynman diagrams contributing to the amplitude, so that the sum includes both the diagonal  $D = D'$  terms and the interference between distinct diagrams.

An amplitude  $\mathcal{M}$  is computed from a Feynman diagram using Feynman rules. This ruleset is given in Section B.1.

## 2.2 Divergences in Perturbative QCD

### 2.2.1 Infrared Divergences

Beyond leading order in perturbation theory, QCD predictions receive contributions from both real and virtual corrections. Real corrections correspond to processes with additional parton emission in the final state, while virtual corrections arise from loop contributions to the amplitude. Both are required at a given perturbative order and can develop infrared singularities, which must ultimately cancel in infrared-safe physical observables.

The two QCD infrared limits, soft and collinear, can be derived explicitly as follows. We start by defining the physical system under study: consider two massless final-state partons  $i$  and  $j$  with four-momenta  $p_i$  and  $p_j$ . By momentum conservation, the four-momentum of the internal virtual propagator is  $P = p_i + p_j$ . The matrix element contains this propagator and in the square amplitude it appears as a  $1/P^2$  factor. Expanding it and applying the massless condition ( $p^2 = 0$ ), we get  $P^2 = 2p_i \cdot p_j = 2E_i E_j (1 - \cos \theta_{ij})$  [2]. Importantly, this quantity defines the Mandelstam invariants as  $s_{ij} = 2p_i \cdot p_j = P^2$ , in the massless regime<sup>1</sup>, which will serve as the natural integration variable throughout the work. The limits are then apparent: the soft limit is hit when one of the energies goes to zero and the collinear limit, when  $\theta_{ij}$  does. In both cases,

$$\lim_{E_i \rightarrow 0} s_{ij} = 0, \quad \lim_{\theta_{ij} \rightarrow 0} s_{ij} = 0. \quad (2.12)$$

At NLO, the process  $\gamma^* \rightarrow q(k_1)g(k_3)\bar{q}(k_2)$  illustrates the two types of single-unresolved infrared limits. The gluon becomes unresolved either when it is soft,  $k_3 \rightarrow 0$ , or when it becomes collinear with one of the neighbouring quarks,  $q \parallel g$  or  $g \parallel \bar{q}$ . The quarks can also participate in these collinear limits, but a single soft quark does not generate an independent infrared singularity. This distinction is not determined solely by the vanishing of an invariant  $s_{ij}$ , but by the soft and collinear factorisation properties of the matrix element; in particular, soft-gluon emission exhibits an eikonal singularity for which there is no analogue in the single-soft-quark limit.

At NNLO, two partons can become unresolved simultaneously, giving rise to additional infrared configurations. For a secondary  $q'\bar{q}'$  pair produced through gluon splitting, these include the double-soft limit in which  $q'$  and  $\bar{q}'$  become simultaneously soft, the collinear limit  $q' \parallel \bar{q}'$ , and triple-collinear configurations such as  $q \parallel q' \parallel \bar{q}'$ , in which the secondary pair becomes collinear with a hard parton. The double-soft  $q'\bar{q}'$  limit is singular despite the absence of a single-soft-quark singularity, as it is associated with the soft behaviour of the parent gluon. These and the other unresolved limits relevant to this thesis are summarised in Table 2.1.

| Unresolved limit                                     | Kinematic condition | Universal IR factor       |
|------------------------------------------------------|---------------------|---------------------------|
| $(1_q, 3_g, 2_{\bar{q}})$ — NLO real radiation       |                     |                           |
| Soft gluon                                           | $p_3 \rightarrow 0$ | $S_{132}$                 |
| Collinear                                            | $1_q \parallel 3_g$ | $P_{qg \rightarrow Q}(z)$ |
| $(1_q, 3_g, 4_g, 2_{\bar{q}})$ — NNLO real radiation |                     |                           |
| Soft gluon                                           | $p_3 \rightarrow 0$ | $S_{134}$                 |

<sup>1</sup>The general definition of the Mandelstam invariant is  $s_{ij} = (p_i + p_j)^2$ , but as  $p_k^2 = 0$  for any  $k$ , in this massless regime, then the massless Mandelstam invariant is defined as  $s_{ij} = 2p_i \cdot p_j$ .

| Unresolved limit                                                                | Kinematic condition                               | Universal IR factor                                                |
|---------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------|
| Collinear ( $qg$ )                                                              | $1_q \parallel 3_g$                               | $P_{qg \rightarrow Q}(z)$                                          |
| Collinear ( $gg$ )                                                              | $3_g \parallel 4_g$                               | $P_{gg \rightarrow G}(z)$                                          |
| Double soft                                                                     | $p_3, p_4 \rightarrow 0$                          | $S_{1342}$                                                         |
| Soft–collinear                                                                  | $p_3 \rightarrow 0, 4_g \parallel 2_{\bar{q}}$    | $S_{1;342}(z) P_{qg \rightarrow Q}(z)$                             |
| Triple collinear                                                                | $1_q \parallel 3_g \parallel 4_g$                 | $P_{134 \rightarrow Q}(w, z, y)$                                   |
| Double single collinear                                                         | $1_q \parallel 3_g, 2_{\bar{q}} \parallel 4_g$    | $P_{qg \rightarrow Q}(z) P_{\bar{q}g \rightarrow \bar{Q}}(y)$      |
| $(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}})$ — NNLO real radiation                |                                                   |                                                                    |
| Soft $q'\bar{q}'$ pair                                                          | $p_3, p_4 \rightarrow 0$                          | $S_{12}(3, 4)$ (double pole)                                       |
| Collinear ( $q'\bar{q}'$ )                                                      | $3_{q'} \parallel 4_{\bar{q}'}$                   | $P_{q\bar{q} \rightarrow G}(z)$                                    |
| Triple collinear                                                                | $1_q \parallel 3_{q'} \parallel 4_{\bar{q}'}$     | $P_{143 \rightarrow Q}^{\text{non-ident.}}(x, y)$                  |
| $(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}})$ — NNLO real radiation, identical flavour |                                                   |                                                                    |
| Triple collinear                                                                | $2_{\bar{q}} \parallel 3_q \parallel 4_{\bar{q}}$ | $\frac{1}{2} P_{234 \rightarrow \bar{Q}}^{\text{ident.}}(w, x, y)$ |

**Table 2.1:** Infrared limits of the real-radiation matrix elements for  $\gamma^* \rightarrow q\bar{q}$  through NNLO, listed by final-state parton configuration. Each configuration is colour-ordered, with the hard quark and antiquark first and last;  $p_i$  is the momentum of parton  $i$ ,  $\parallel$  denotes a collinear pair, and  $s_{i\dots k} = (p_i + \dots + p_k)^2$ . Limits related by  $q \leftrightarrow \bar{q}$  or by gluon relabelling are not listed separately. The subtraction counterterm of Eq. (2.13) must reproduce the matrix element in each of these limits [5]. Complete analytic expressions for the universal soft and collinear factors listed above are given in [5], Sections 8.1 and 8.2.2.

From an experimental point of view, a real emission that becomes soft or collinear may be unresolved, meaning that it cannot be distinguished from a configuration in which that emission is absent. For an infrared-safe observable, the two configurations yield the same value of the observable in the corresponding limit.

The virtual corrections contain infrared singularities related to those arising from unresolved real radiation. According to the Kinoshita–Lee–Nauenberg (KLN) theorem, these singularities cancel once all degenerate contributions to an infrared-safe physical observable are appropriately combined [6, 7]. The resulting observable is therefore finite, despite the presence of infrared divergences in the individual real and virtual contributions.

In practice, however, this cancellation cannot be performed directly at the integrand level, since the real and virtual contributions are separately divergent and are defined on phase spaces with different particle multiplicities. At NLO, the real contribution contains one additional final-state parton with respect to the LO  $n$ -parton process, and is integrated over the  $(n + 1)$ -particle phase space, whereas the virtual contribution is integrated over the  $n$ -particle phase space. A subtraction method resolves this mismatch by introducing a local counterterm  $d\sigma^S$  that reproduces the singular behaviour of the real contribution  $d\sigma^R$  in all unresolved limits. The counterterm can then be subtracted from the real contribution, rendering their difference finite and suitable for numerical integration. The same counterterm is analytically integrated over the unresolved one-particle phase space and added to the virtual contribution, where its explicit infrared poles cancel those of the virtual correction. Schematically, at NLO, this is expressed as,

$$\sigma = \int_{n+1} \left( d\sigma^R - d\sigma^S \right) + \int_n \left( d\sigma^V + \int_1 d\sigma^S \right) \quad (2.13)$$

where  $d\sigma^R$  represents the differential real radiation cross-section,  $d\sigma^V$  the differential virtual corrections cross-section and  $d\sigma^S$  the differential subtraction counterterm [8]. Importantly,  $d\sigma^S$  must reproduce  $d\sigma^R$ 's exact singular behaviour at every point in the phase space, such that the divergences cancel locally.

### 2.2.2 Ultraviolet Divergences

Infrared divergences are not the only kind of divergences we find in QCD. In loop diagrams, the loop momentum  $l$  is integrated over the whole of momentum space. Regions in which  $l$  becomes soft or collinear to an external massless momentum can give rise to infrared singularities, which contribute to the singular structure of  $d\sigma^V$ . In addition, loop integrals may become divergent when the magnitude of the loop momentum becomes arbitrarily large,  $|l| \rightarrow \infty$ . These are referred to as ultraviolet (UV) divergences.

The UV behaviour of a loop integral can be understood through simple power counting. In  $d$  dimensions, the loop integration measure can be written in spherical coordinates as,

$$d^d l = l^{d-1} dl d\Omega_{d-1}. \quad (2.14)$$

Whether the integral diverges or not depends on whether the integrand converges fast enough to counter the  $l^{d-1}$  measure term. Since each propagator contributes  $\sim 1/l^2$  to the integrand, schematically, a diagram with  $n_p$  propagators in the loop and no extra powers of  $l$  in the numerator becomes,

$$\int^\infty l^{d-1} \cdot \frac{1}{l^{2n_p}} = \int^\infty l^{d-1-2n_p} dl \quad (2.15)$$

which is UV divergent whenever  $d - 1 - 2n_p \geq -1 \Rightarrow d \geq 2n_p$  [4, 9].

Ultraviolet divergences arising in loop corrections require the parameters and fields appearing in the bare QCD Lagrangian to be renormalised. In particular, the bare strong coupling is related to the renormalised coupling  $\alpha_s(\mu)$  through a coupling renormalisation constant, which absorbs the corresponding ultraviolet divergences. The introduction of the renormalisation scale  $\mu$  leads to a scale-dependent renormalised coupling, whose evolution is governed by the QCD beta function. The coupling renormalisation relevant to the one-loop antenna functions considered in this work is discussed explicitly in Subsection 3.3.2.

## 2.3 Dimensional Regularisation

The subtraction scheme from Eq. (2.13) separates the NLO cross-section into contributions that can ultimately be evaluated numerically. The difference  $d\sigma^R - d\sigma^S$  is finite in the unresolved regions and can therefore be integrated numerically in four dimensions. The subtraction term must, however, still be integrated analytically over the unresolved one-parton phase space,

$$\int_1 d\sigma^S. \quad (2.16)$$

Since  $d\sigma^S$  is designed explicitly to carry over all of  $d\sigma^R$ 's divergent structures, this integral blows up once integrated over four dimensions, as does the virtual contribution  $d\sigma^V$ , whose infrared divergences originate from the loop integration, as explored in Subsection 2.2.2. To regulate these divergent contributions we continue the number of space-time dimensions away from four, via,

$$d = 4 - 2\epsilon \quad (2.17)$$

so that infrared divergences appear as explicit analytic poles in  $1/\epsilon$ . After analytic integration over the unresolved phase space, this leaves us with two  $\epsilon$ -dependent expressions, for the virtual corrections and for the now unresolved-parton-integrated subtraction cross-section,

$$\int_n \left( d\sigma^V + d\sigma_1^S \right). \quad (2.18)$$

From the KLN theorem, as described in Subsection 2.2.1, we know that these poles must cancel. This combination is therefore finite as  $\epsilon \rightarrow 0$ , allowing the remaining phase-space integration to be performed numerically in four dimensions.

Though the underlying principle is always the same, dimensional regularisation can be implemented using different schemes. The most common are 't Hooft-Veltman [10], Four-Dimensional Helicity [11] and Conventional Dimensional Regularisation (CDR) [2]. In CDR, all momenta and polarisation states are treated in  $d = 4 - 2\epsilon$  dimensions. This is the scheme used throughout the present work, since *AntCalc*'s primary objective is the simplification and democratisation of the antenna framework established in [5], which uses CDR throughout.

While using CDR, the integrated and unintegrated antennae considered in this work are normalised using the following convention. First, the antennae are expressed in the  $\overline{\text{MS}}$  (modified minimal subtraction)-convention using  $S_\epsilon = (4\pi)^\epsilon e^{-\epsilon\gamma_E}$  [5]. Second, they are also normalised in relation to the strong coupling constant expansion. For an antenna with  $n$  final-state partons and  $l$  loops, we define,

$$k = (n - 2) + l, \quad (2.19)$$

the perturbative-order index, which is zero for LO, one for NLO and two for NNLO, and introduce the normalisation factor

$$G_k = \left(8\pi^2\right)^{n-2} \Lambda_l = \left(8\pi^2\right)^{n-2} \left(8\pi^2\right)^l. \quad (2.20)$$

These two normalisations are then bundled into a general convention used throughout, which is written as,

$$\mathcal{N}(\epsilon, k) = \frac{C(\epsilon, k)}{\Phi_2}, \quad C(\epsilon, k) = \frac{G_k}{S_\epsilon^k} \quad (2.21)$$

using the  $d$ -dimensional two-particle phase-space measure  $\Phi_2$ . This is the general representation of the work's normalisation convention.

As discussed in Subsection 2.2.2, ultraviolet divergences are removed by expressing the bare strong coupling  $\alpha_0$  in terms of the renormalised coupling  $\alpha_s(\mu)$ . In the  $\overline{\text{MS}}$  scheme, the relation between the bare and renormalised couplings is given by [5],

$$\alpha_0 \mu_0^{2\epsilon} S_\epsilon = \alpha_s \mu^{2\epsilon} \left[ 1 - \frac{\beta_0}{\epsilon} \frac{\alpha_s}{2\pi} + \left( \frac{\beta_0^2}{\epsilon^2} - \frac{\beta_1}{2\epsilon} \right) \left( \frac{\alpha_s}{2\pi} \right)^2 + O(\alpha_s^3) \right], \quad (2.22)$$

$$\beta_0 = \frac{11N - 2N_f}{6}, \quad \beta_1 = \frac{34N^3 - 13N^2N_f + 3N_f}{12N}$$

where  $S_\epsilon$  denotes the conventional dimensional-regularisation factor, while  $\mu_0$  and  $\mu$  are the dimensional-regularisation and renormalisation scales associated with the bare and renormalised couplings, respectively. The coefficients  $\beta_0$  and  $\beta_1$  are the first two coefficients of the QCD beta function, with  $N = 3$  the

number of colours and  $N_f$  the number of active massless quark flavours [2, 5]. For the massless antenna functions considered here, the renormalisation scale is chosen as  $\mu^2 = q^2$ , with  $q^2$  the invariant mass squared of the antenna system.

## 2.4 The Antenna Subtraction Formalism

In perturbative QCD, computing the perturbative corrections to a certain jet observable  $J_m^{(n)}$ , requires summing over all partonic multiplicity channels up to the desired order.

At LO, the approximation to the  $m$ -jet cross section is computed as the integral over the appropriate phase space,

$$d\sigma_{\text{LO}} = \int_m d\sigma^B, \quad d\sigma^B = d\Phi_m(p_1, \dots, p_m; q) \frac{1}{S_m} |\mathcal{M}_m^0(p_1, \dots, p_m)|^2 J_m^m(p_1, \dots, p_m) \quad (2.23)$$

where  $d\sigma^B$  is the tree-level  $m$ -parton contribution to the  $m$ -jet cross section, built from the squared tree-level  $m$ -parton amplitude  $|\mathcal{M}_m^0(p_1, \dots, p_m)|^2$ . At LO every final-state particle forms its own jet, so the jet function carries  $m$  jets for  $m$  final-state particles,  $J_m^m$ .  $S_m$  is a symmetry factor for identical partons in the final state, and  $\Phi_m$  is the  $m$ -parton final-state phase space in  $d$  dimensions [5],

$$d\Phi_m(p_1, \dots, p_m; q) = \frac{d^{d-1}p_1}{2E_1(2\pi)^{d-1}} \dots \frac{d^{d-1}p_m}{2E_m(2\pi)^{d-1}} (2\pi)^d \delta^{(d)}(q - p_1 - \dots - p_m). \quad (2.24)$$

At NLO, the  $m$ -jet cross section receives a real-emission and a virtual contribution, built from an  $(m+1)$ -parton tree-level amplitude and from the interference between the  $m$ -parton tree-level and one-loop amplitudes, respectively, for a generic process,

$$\begin{aligned} d\sigma_{\text{NLO}}^R &= d\Phi_{m+1}(p_1, \dots, p_{m+1}; q) \frac{1}{S_{m+1}} |\mathcal{M}_{m+1}^0(p_1, \dots, p_{m+1})|^2 J_m^{(m+1)}(p_1, \dots, p_{m+1}), \\ d\sigma_{\text{NLO}}^V &= d\Phi_m(p_1, \dots, p_m; q) \frac{1}{S_m} 2 \operatorname{Re} \left( \mathcal{M}_m^1(p_1, \dots, p_m) \mathcal{M}_m^{0*}(p_1, \dots, p_m) \right) J_m^{(m)}(p_1, \dots, p_m). \end{aligned} \quad (2.25)$$

These contributions are summed to reach,

$$d\sigma_{\text{NLO}} = \int_{m+1} d\sigma_{\text{NLO}}^R + \int_m d\sigma_{\text{NLO}}^V. \quad (2.26)$$

At NNLO, the  $m$ -jet cross section receives three contributions: a double-real contribution, built from the  $(m+2)$ -parton tree-level amplitude; a real-virtual contribution, built from the interference between the  $(m+1)$ -parton tree-level and one-loop amplitudes; and a double-virtual contribution, built from the interference between the  $m$ -parton tree-level and two-loop amplitudes and from the squared  $m$ -parton one-loop amplitude, for a generic process,

$$\begin{aligned} d\sigma_{\text{NNLO}}^{RR} &= d\Phi_{m+2}(p_1, \dots, p_{m+2}; q) \frac{1}{S_{m+2}} |\mathcal{M}_{m+2}^0(p_1, \dots, p_{m+2})|^2 J_m^{(m+2)}(p_1, \dots, p_{m+2}), \\ d\sigma_{\text{NNLO}}^{RV} &= d\Phi_{m+1}(p_1, \dots, p_{m+1}; q) \frac{1}{S_{m+1}} 2 \operatorname{Re} \left( \mathcal{M}_{m+1}^1(p_1, \dots, p_{m+1}) \mathcal{M}_{m+1}^{0*}(p_1, \dots, p_{m+1}) \right) J_m^{(m+1)}(p_1, \dots, p_{m+1}), \\ d\sigma_{\text{NNLO}}^{VV} &= d\Phi_m(p_1, \dots, p_m; q) \frac{1}{S_m} \left[ 2 \operatorname{Re} \left( \mathcal{M}_m^2(p_1, \dots, p_m) \mathcal{M}_m^{0*}(p_1, \dots, p_m) \right) \right. \\ &\quad \left. + |\mathcal{M}_m^1(p_1, \dots, p_m)|^2 \right] J_m^{(m)}(p_1, \dots, p_m). \end{aligned} \quad (2.27)$$

These contributions are summed to reach,

$$d\sigma_{\text{NNLO}} = \int_{m+2} d\sigma_{\text{NNLO}}^{RR} + \int_{m+1} d\sigma_{\text{NNLO}}^{RV} + \int_m d\sigma_{\text{NNLO}}^{VV}. \quad (2.28)$$

### 2.4.1 Subtraction Terms

The contributions that form part of the NLO and NNLO  $m$ -jet cross sections, given in Eq. (2.26) and Eq. (2.28) respectively, are separately IR-divergent. Consequently, a direct numerical evaluation of the individual contributions is not possible, as the corresponding integrals are divergent. Our solution to this problem is to introduce subtraction terms that locally reproduce the infrared singular behaviour of the corresponding contributions throughout the relevant phase space.

#### 2.4.1.1 NLO

At NLO, a subtraction term,  $d\sigma_{\text{NLO}}^S$ , is introduced to locally reproduce the infrared singular behaviour of the real-emission contribution  $d\sigma_{\text{NLO}}^R$ . The  $m$ -jet cross section can then be written as,

$$d\sigma_{\text{NLO}} = \int_{m+1} \left( d\sigma_{\text{NLO}}^R - d\sigma_{\text{NLO}}^S \right) + \int_m \left( d\sigma_{\text{NLO}}^V + \int_1 d\sigma_{\text{NLO}}^S \right). \quad (2.29)$$

The subtraction term must reproduce the real-emission contribution in all single-unresolved limits, while remaining local in phase space and introducing no additional singularities. Consequently, the difference  $d\sigma_{\text{NLO}}^R - d\sigma_{\text{NLO}}^S$  is finite in these limits and can be integrated numerically in four dimensions. At the same time,  $d\sigma_{\text{NLO}}^S$  must be analytically integrable over the unresolved one-parton phase space. Its integrated form contains explicit poles in  $\epsilon$  that cancel those of the virtual contribution  $d\sigma_{\text{NLO}}^V$ .

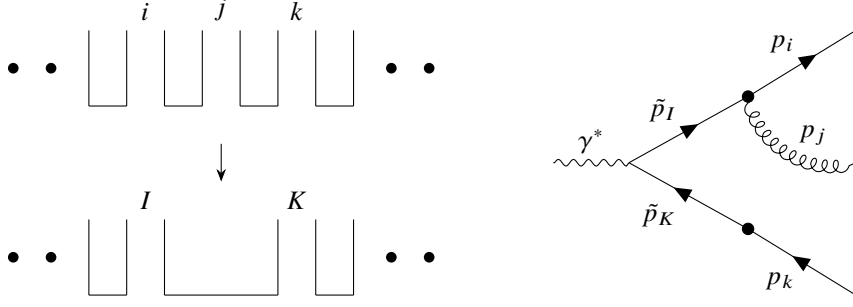
Within the antenna subtraction formalism,  $d\sigma_{\text{NLO}}^S$  is constructed from three-parton tree-level antenna functions  $X_{ijk}^0$ , which capture the unresolved radiation emitted between a pair of colour-connected hard radiators. It takes the schematic form,

$$d\sigma_{\text{NLO}}^S = \sum_{m+1} d\Phi_{m+1}(p_1, \dots, p_{m+1}; q) \frac{1}{S_{m+1}} \sum_j X_{ijk}^0 |\mathcal{M}_m(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1})|^2 J_m^{(m)}(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1}). \quad (2.30)$$

Here,  $i$  and  $k$  denote the hard radiators and  $j$  the unresolved parton. The momenta  $p_i, p_j, p_k$  are mapped onto the reduced on-shell momenta  $\tilde{p}_I$  and  $\tilde{p}_K$ , such that the reduced matrix element has the appropriate  $m$ -parton kinematics. For a  $q\bar{q}g$  antenna,  $i$  and  $k$  correspond to the quark and antiquark radiators, while  $j$  corresponds to the unresolved gluon. Parent partons  $I$  and  $K$  are colour connected. Daughter partons  $i$  and  $j$ , and  $j$  and  $k$  are also connected, but  $i$  and  $k$  are not. A diagram of these relationships is shown in Fig. 2.2.

The jet function  $J_m^m$  in the antenna subtraction term above does not depend on the individual momenta  $p_i, p_j$  and  $p_k$ , but only on  $\tilde{p}_I$  and  $\tilde{p}_K$ . One can therefore carry out the integration over the unresolved antenna phase space appropriate to  $p_i, p_j$  and  $p_k$  analytically, exploiting the factorisation of the phase space,

$$d\Phi_{m+1}(p_1, \dots, p_{m+1}; q) = d\Phi_m(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1}) d\Phi_{X_{ijk}}(p_i, p_j, p_k; q = \tilde{p}_I + \tilde{p}_K). \quad (2.31)$$



**Figure 2.2:** The colour connections in between parent and daughter partons before and after an emission, accompanied also by a schematic Feynman diagram using the momenta labels.

The integrated counterterm  $\int_1 d\sigma_{\text{NLO}}^S$  then takes the form,

$$\int_1 d\sigma_{\text{NLO}}^S = \sum_m d\Phi_m(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1}) \frac{1}{S_{m+1}} \sum_j \left( \int d\Phi_{X_{ijk}} X_{ijk}^0 \right) |\mathcal{M}_m(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1})|^2 J_m^{(m)}(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1}) \quad (2.32)$$

The integral  $\int d\Phi_{X_{ijk}} X_{ijk}^0$  defines the integrated antenna, which is then combined with the virtual contribution, rendering its contribution IR-finite. Setting  $m = 2$  in Eq. (2.31), we can see that the NLO antenna phase space  $d\Phi_{X_{ijk}}$  is proportional to the three-particle phase space,

$$d\Phi_3 = \Phi_2 d\Phi_{X_{ijk}}, \quad \Phi_2 = \int d\Phi_2. \quad (2.33)$$

#### 2.4.1.2 NNLO

Analogously, at NNLO, two subtraction counterterms,  $d\sigma_{\text{NNLO}}^S$  and  $d\sigma_{\text{NNLO}}^T$ , are introduced to reproduce the infrared singular behaviour of the double-real and real-virtual contributions,  $d\sigma_{\text{NNLO}}^{RR}$  and  $d\sigma_{\text{NNLO}}^{RV}$ , respectively. The  $m$ -jet cross section can then be written as,

$$\begin{aligned} d\sigma_{\text{NNLO}} &= \int_{m+2} \left( d\sigma_{\text{NNLO}}^{RR} - d\sigma_{\text{NNLO}}^S \right) + \int_{m+1} \left( d\sigma_{\text{NNLO}}^{RV} - d\sigma_{\text{NNLO}}^T \right) \\ &\quad + \int_m \left( d\sigma_{\text{NNLO}}^{VV} + \int_2 d\sigma_{\text{NNLO}}^S + \int_1 d\sigma_{\text{NNLO}}^T \right) \quad (2.34) \\ &= \int_m \left( d\sigma_{\text{NNLO}}^{VV} + \int_2 d\sigma_{\text{NNLO}}^S + \int_1 d\sigma_{\text{NNLO}}^T \right). \end{aligned}$$

At NNLO, the subtraction structure becomes more involved due to the presence of several single- and double-unresolved infrared configurations. The double-real and real-virtual subtraction counterterms are therefore decomposed into several contributions,

$$\begin{aligned} d\sigma_{\text{NNLO}}^S &= d\sigma^{S,a} + d\sigma^{S,b} + d\sigma^{S,c} + d\sigma^{S,d} \\ d\sigma_{\text{NNLO}}^T &= d\sigma^{T,a} + d\sigma^{T,b} + d\sigma^{T,c}. \end{aligned} \quad (2.35)$$

Each contribution accounts for a specific infrared singular configuration of the double-real or real-virtual matrix elements, as summarised in Table 2.2. For the double-real contribution, the different colour connections associated with the unresolved partons are illustrated schematically in Fig. 2.3.

The individual contributions to the subtraction terms are constructed from antenna functions. At NNLO, in addition to the tree-level three-parton antennae  $X_{ijk}^0$  already required at NLO, tree-level four-

| Configuration                                             | Subtraction Term |
|-----------------------------------------------------------|------------------|
| RR subtraction counterterm $-d\sigma_{\text{NNLO}}^S$     |                  |
| Single unresolved parton                                  | $d\sigma^{S,a}$  |
| Two colour-connected unresolved partons                   | $d\sigma^{S,b}$  |
| Two almost colour-unconnected unresolved partons          | $d\sigma^{S,c}$  |
| Two fully colour-unconnected unresolved partons           | $d\sigma^{S,d}$  |
| RV subtraction counterterm $-d\sigma_{\text{NNLO}}^T$     |                  |
| Explicit infrared poles from the one-loop matrix elements | $d\sigma^{T,a}$  |
| One of the partons becomes soft or collinear              | $d\sigma^{T,b}$  |
| Spurious poles introduced by the other two subterms       | $d\sigma^{T,c}$  |

**Table 2.2:** NNLO unresolved configurations and their matching subterm, part of the  $d\sigma_{\text{NNLO}}^S$  and  $d\sigma_{\text{NNLO}}^T$  subtraction counterterms.

parton antennae  $X_{ijkl}^0$  and one-loop three-parton antennae  $X_{ijk}^1$  are also required. The contributions to the double-real subtraction counterterm are given by,

$$\begin{aligned}
d\sigma^{S,a} &= \sum_{m+2} d\Phi_{m+2}(p_1, \dots, p_{m+2}; q) \frac{1}{S_{m+2}} \\
&\quad \sum_j X_{ijk}^0 |\mathcal{M}_{m+1}(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+2})|^2 J_m^{(m+1)}(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+2}), \\
d\sigma^{S,b} &= \sum_{m+2} d\Phi_{m+2}(p_1, \dots, p_{m+2}; q) \frac{1}{S_{m+2}} \\
&\quad \sum_{j,k} \left( X_{ijkl}^0 - X_{ijk}^0 X_{IKl}^0 - X_{jkl}^0 X_{iJL}^0 \right) |\mathcal{M}_m(p_1, \dots, \tilde{p}_I, \tilde{p}_L, \dots, p_{m+2})|^2 \\
&\quad J_m^{(m)}(p_1, \dots, \tilde{p}_I, \tilde{p}_L, \dots, p_{m+2}),
\end{aligned} \tag{2.36}$$

And then the real-virtual counterterm,

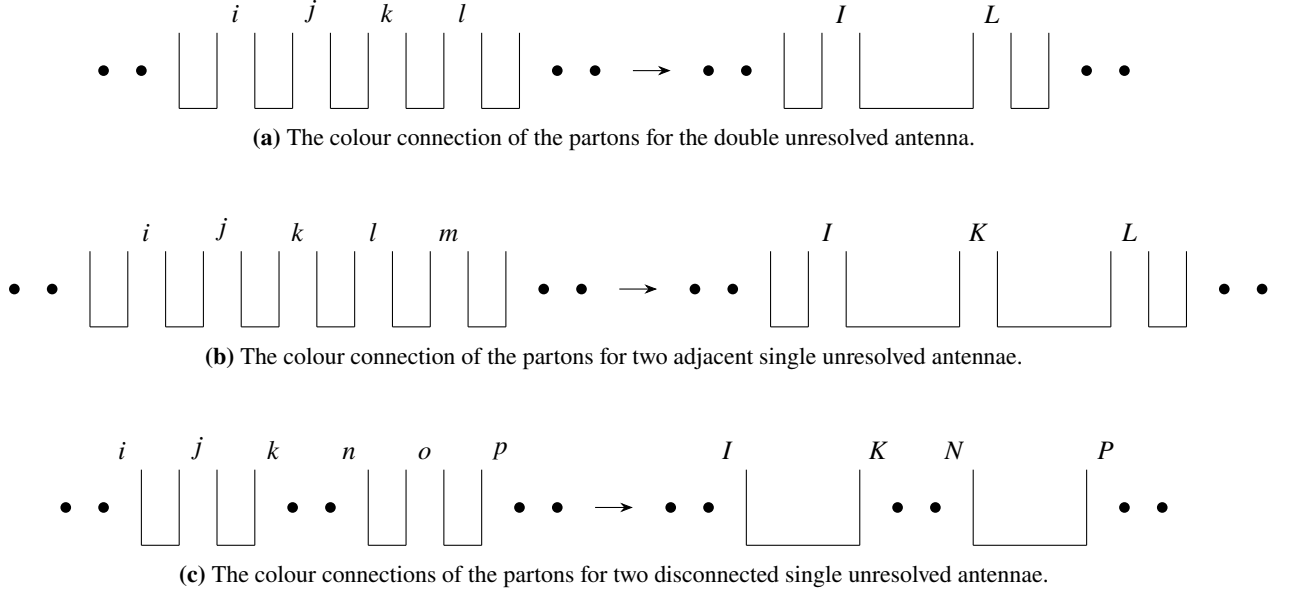
$$\begin{aligned}
d\sigma^{T,a} &= \sum_{m+1} d\Phi_{m+1}(p_1, \dots, p_{m+1}; q) \frac{1}{S_{m+1}} \\
&\quad \sum_{i,k} -X_{ijk}^0(s_{ik}) |\mathcal{M}_{m+1}(p_1, \dots, p_i, p_k, \dots, p_{m+1})|^2 J_m^{(m+1)}(p_1, \dots, p_i, p_k, \dots, p_{m+1}), \\
&\quad \text{with } X_{ijk}^0 = C(\epsilon, 1) \int d\Phi_{X_{ijk}} X_{ijk}^0, \\
d\sigma^{T,b} &= \sum_{m+1} d\Phi_{m+1}(p_1, \dots, p_{m+1}; q) \frac{1}{S_{m+1}} \\
&\quad \sum_j X_{ijk}^0 |\mathcal{M}_m^1(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1})|^2 J_m^{(m)}(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1}) \\
&\quad + X_{ijk}^1 |\mathcal{M}_m(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1})|^2 J_m^{(m)}(p_1, \dots, \tilde{p}_I, \tilde{p}_K, \dots, p_{m+1}), \\
&\quad \text{with } X_{ijk}^1 \rightarrow X_{ijk}^1 + \frac{\beta_0}{\epsilon} \frac{(4\pi)^\epsilon e^{-\epsilon\gamma_E}}{8\pi^2} X_{ijk}^0 \left( s_{ijk}^{-\epsilon} - (\mu^2)^{-\epsilon} \right),
\end{aligned} \tag{2.37}$$

$d\sigma^{S,c}$ ,  $d\sigma^{S,d}$ , and  $d\sigma^{T,c}$  are only required for processes with three or more hard partons, beyond those considered in the current text. Their complete form is written in [5].

The iterated mapping for the four-parton system  $(i, j, k, l)$  proceeds in two stages,

$$\begin{aligned} (i, j, k) &\rightarrow (\tilde{I}, \tilde{K}) \\ (\tilde{I}, \tilde{K}, l) &\rightarrow (\tilde{\tilde{I}}, \tilde{\tilde{L}}) \end{aligned} \quad (2.38)$$

and the three characteristic colour-connection configurations are shown in Fig. 2.3.



**Figure 2.3:** The three characteristic double-unresolved colour-connection configurations at NNLO: (a) two colour-connected unresolved partons, (b) two adjacent single-unresolved antennae, and (c) two disconnected single-unresolved antennae.

For the double-real subtraction terms, the corresponding phase-space factorisation generalises the NLO  $3 \rightarrow 2$  mapping to the four-parton antenna configuration,

$$d\Phi_{m+2}(p_1, \dots, p_{m+2}; q) = d\Phi_m(p_1, \dots, \tilde{p}_I, \tilde{p}_L, \dots, p_{m+2}; q) d\Phi_{X_{ijkl}}(p_i, p_j, p_k, p_l; q = \tilde{p}_I + \tilde{p}_L). \quad (2.39)$$

For a two-parton reduced configuration, this becomes

$$d\Phi_4 = \Phi_2 d\Phi_{X_{ijkl}}, \quad (2.40)$$

so that the four-parton antenna phase space  $d\Phi_{X_{ijkl}}$  provides the integration measure for the tree-level four-parton antennae  $X_{ijkl}^0$ .

## 2.4.2 Antenna Functions

In this thesis, the complete set of massless quark–antiquark final–final antenna functions required up to NNLO is considered. All antennae share the same underlying hard  $q\bar{q}$  radiator pair, with configurations containing up to one unresolved parton at NLO and up to two unresolved partons at NNLO. In addition, the corresponding two-parton virtual contributions, derived from the quark–antiquark form factor, are included. Although these two-parton objects do not describe unresolved real radiation, they enter the

antenna framework alongside the real-radiation antenna functions and will, for convenience, be referred to as antenna functions throughout this thesis.

Each antenna is generally determined by its external state and the hard parton pair it collapses to. In this thesis, the hard parton pair is always  $q\bar{q}$ . Up to NNLO, quark-antiquark final-final antennae are organised into three families, which differ in the type of particle the hard parton pair particles emit. *A*-type antennae are characterised by gluon emissions, leading to final states like  $qgg\bar{q}$  at NNLO or  $qg\bar{q}$  at NLO; *C*-type antennae are characterised by a final-state quark-antiquark pair with the original hard pair's flavour, as  $q\bar{q}q\bar{q}$ ; and *B*-type by a quark-antiquark pair of a different flavour, notated as  $q\bar{q}q'\bar{q}'$ . Following QCD vertex rules, described in Section B.1, *A*-type antennae may exist at any order, while *B*- and *C*-type antennae can only exist at NNLO.

Throughout the remainder of this thesis, a generic *X*-type antenna is denoted by  $X_n^l$ , where  $n$  is the number of final-state partons and  $l$  the number of loops. Where applicable, antennae are decomposed into leading- and subleading-colour components, denoted by  $X_n^l$  and  $\tilde{X}_n^l$ , respectively. For loop antennae, an additional closed-quark-loop contribution, denoted by  $\hat{X}_n^l$ , may be present. Here, “leading” and “subleading” refer to the corresponding colour structures in the expansion in the number of colours  $N$ .

For a given antenna type  $X_n^l$ , the collection of its different colour components is referred to as the  $X_n^l$  antenna set. Depending on the antenna, some of these components may vanish; in particular, not all antenna sets contain non-zero subleading-colour or, at loop level, closed-quark-loop contributions. The  $A_2^2$  set requires an additional contribution, denoted by  $\check{A}_2^2$ . This additional term does not represent a new colour component. Rather, it distinguishes the one-loop-squared contribution from the interference between the two-loop and tree-level amplitudes. Both contributions have the same two-parton final state and enter at NNLO, and are therefore grouped within the  $A_2^2$  set for convenience. The antenna functions (and  $q\bar{q}$  form factors) considered are organised by order, loop number, and final state in Table 2.3.

| Final state                               | Tree level             | One loop                            | Two loop                                           |
|-------------------------------------------|------------------------|-------------------------------------|----------------------------------------------------|
| $\gamma^* \rightarrow q\bar{q}$           | $A_2^0$                | $A_2^1$                             | $A_2^2, \tilde{A}_2^2, \hat{A}_2^2, \check{A}_2^2$ |
| $\gamma^* \rightarrow qg\bar{q}$          | $A_3^0$                | $A_3^1, \tilde{A}_3^1, \hat{A}_3^1$ | —                                                  |
| $\gamma^* \rightarrow qgg\bar{q}$         | $A_4^0, \tilde{A}_4^0$ | —                                   | —                                                  |
| $\gamma^* \rightarrow qq'\bar{q}'\bar{q}$ | $B_4^0$                | —                                   | —                                                  |
| $\gamma^* \rightarrow qq\bar{q}\bar{q}$   | $C_4^0$                | —                                   | —                                                  |

**Table 2.3:** Quark-antiquark antenna families used in this work, including the  $q\bar{q}$  form factors [5]. Final states are ordered with the two hard radiator listed first and last, following that paper's convention.

Antenna functions are constructed from physical, colour-ordered matrix elements (shown in Table 2.3) and are normalised to the corresponding two-parton matrix element of the underlying hard radiator pair. To define the notation used below, the matrix-element contributions at successive perturbative orders are written as,

$$\begin{aligned}
|\mathcal{M}_n^0|^2 &= \mathcal{M}_n^{0\dagger} \mathcal{M}_n^0 \\
|\mathcal{M}_n^1|^2 &= 2 \operatorname{Re} \left( \mathcal{M}_n^{0\dagger} \mathcal{M}_n^1 \right) \\
|\mathcal{M}_n^2|^2 &= 2 \operatorname{Re} \left( \mathcal{M}_n^{0\dagger} \mathcal{M}_n^2 \right) + \mathcal{M}_n^{1\dagger} \mathcal{M}_n^1.
\end{aligned} \tag{2.41}$$

Here,  $\mathcal{M}_n^l$  denotes the  $l$ -loop amplitude for an  $n$ -parton final state. Thus,  $|\mathcal{M}_n^1|^2$  denotes the tree–one-loop interference, while  $|\mathcal{M}_n^2|^2$  contains both the tree–two-loop interference and the one-loop-squared contribution.

At tree level ( $l = 0$ ), the three- and four-parton antenna functions are defined as [5],

$$\begin{aligned} X_{ijk}^0 &= S_{ijk,IK} \frac{|\mathcal{M}_{ijk}^0|^2}{|\mathcal{M}_{IK}^0|^2} \\ X_{ijkl}^0 &= S_{ijkl,IL} \frac{|\mathcal{M}_{ijkl}^0|^2}{|\mathcal{M}_{IL}^0|^2} \end{aligned} \quad (2.42)$$

Here,  $i, j, k$  and  $i, j, k, l$  label the partons in the three- and four-parton configurations, respectively, while  $I, K$  and  $I, L$  denote the corresponding hard radiators in the reduced two-parton configuration and  $S$  denotes the symmetry factor associated to the antenna, which accounts both for potential identical particle symmetries and for the presence of more than one antenna in the basic two-parton process. For all  $q\bar{q}$  antenna topologies considered in this work, the corresponding symmetry factors are equal to unity.

At one loop ( $l = 1$ ), the two- and three-parton antenna functions are defined as [5],

$$\begin{aligned} X_{ij}^1 &= S_{ij,IJ} \frac{|\mathcal{M}_{ij}^1|^2}{|\mathcal{M}_{IJ}^0|^2} \\ X_{ijk}^1 &= S_{ijk,IK} \frac{|\mathcal{M}_{ijk}^1|^2}{|\mathcal{M}_{IK}^0|^2} - X_{ijk}^0 \frac{|\mathcal{M}_{IK}^1|^2}{|\mathcal{M}_{IK}^0|^2}. \end{aligned} \quad (2.43)$$

where the second term removes the one-loop correction associated with the reduced two-parton process.

At two loops ( $l = 2$ ), the two-parton antenna considered in this work contributes to the  $A_2^2$  antenna set and is defined by

$$X_{ij}^2 = S_{ij,IJ} \frac{|\mathcal{M}_{ij}^2|^2}{|\mathcal{M}_{IJ}^0|^2}. \quad (2.44)$$

Loop antennae are additionally coupling-renormalised according to Eq. (2.22) whenever the corresponding reduced process carries powers of the strong coupling. This procedure is discussed further in Subsection 2.4.4. Finally, all antenna functions are normalised to the two-parton Born squared matrix element,  $|\mathcal{M}_2^0|^2$ , such that the Born-level two-parton antenna satisfies  $A_2^0 = 1$ .

In addition to the colour-ordered amplitudes used to define individual antenna functions, it is useful to introduce notation for the complete Born-normalised matrix-element contribution associated with a given physical final state. Such a contribution may contain several colour orderings, flavour assignments, or coupling configurations, together with the corresponding interference terms. This notation will be particularly useful below when decomposing the complete matrix elements into their individual antenna contributions. It is defined as,

$$|M_{p_1 p_2 \dots p_n}^l|^2 = \frac{1}{|\mathcal{M}_2^0|^2} \sum_{r=0}^l \mathcal{M}_n^{r\dagger}(p_1, \dots, p_n) \mathcal{M}_n^{l-r}(p_1, \dots, p_n), \quad (2.45)$$

where more than one colour ordering, flavour assignment, or coupling configuration contributes to the specified final state, the right-hand side is understood to include the complete sum over the corresponding amplitude products, including any interference terms. Thus,  $|M_{p_1 \dots p_n}^l|^2$  denotes the complete process-level contribution, whereas  $\mathcal{M}_n^l(p_1, \dots, p_n)$  denotes an amplitude with a specified ordering or assignment from which the individual antenna contributions are subsequently identified.

The complete matrix-element contributions in Eq. (2.45) also contain the colour factors associated with the underlying QCD process. After the relevant amplitude products are formed, the colour indices

must therefore be contracted and summed. The resulting colour algebra separates the matrix element into distinct colour structures, whose coefficients are used to identify the corresponding antenna colour components. The colour identities and decomposition required for this procedure are introduced in the following subsection.

### 2.4.3 Colour Algebra and the Antenna Colour Decomposition

The interactions of quarks and gluons are determined by the local  $SU(3)_c$  gauge symmetry of QCD. For the colour algebra relevant here, quarks transform in the fundamental representation and carry a colour index  $i = 1, \dots, N$ , while gluons transform in the adjoint representation and carry an index  $a = 1, \dots, N^2 - 1$ , with  $N = 3$  in QCD. The generators of the  $SU(N)$  Lie algebra in the fundamental representation are denoted by  $T^a$  and satisfy the normalisation

$$\text{Tr}(T^a T^b) = T_R \delta^{ab}, \quad T_R = \frac{1}{2}. \quad (2.46)$$

When an amplitude is squared, the colour factors of the amplitude and its complex conjugate must be contracted and summed over the corresponding colour indices. As a simple example, consider a colour structure containing a single generator  $T^a$ . Since the generators in the fundamental representation are Hermitian,  $(T^a)^\dagger = T^a$ , the corresponding colour sum can be written as

$$\sum_{i_1, i_2, a} (T^a)_{i_1 i_2} (T^a)_{i_2 i_1} = \sum_a \text{Tr}(T^a T^a) = \sum_a T_R \delta^{aa}. \quad (2.47)$$

Since the adjoint representation of  $SU(N)$  has dimension  $N^2 - 1$ , summing over the adjoint index gives

$$\sum_a T_R \delta^{aa} = \frac{1}{2} (N^2 - 1). \quad (2.48)$$

For processes containing two quark-antiquark colour lines, such as the four-quark final states relevant to the  $B_4^0$  and  $C_4^0$  antennae, the colour algebra involves four fundamental colour indices. A useful relation in this case is the  $SU(N)$  completeness relation, also known as the *colour Fierz identity* [2, 4],

$$\sum_a (T^a)_{i_1 i_2} (T^a)_{i_3 i_4} = \frac{1}{2} \left( \delta_{i_1 i_4} \delta_{i_2 i_3} - \frac{1}{N} \delta_{i_1 i_2} \delta_{i_3 i_4} \right). \quad (2.49)$$

We also introduce the quadratic Casimir invariants of the fundamental and adjoint representations, denoted by  $C_F$  and  $C_A$ , respectively. They are defined through

$$\sum_a T^a T^a = C_F \mathbf{1}, \quad f^{acd} f^{bcd} = C_A \delta^{ab}. \quad (2.50)$$

For  $SU(N)$ , with the normalisation  $T_R = 1/2$  adopted above, these relations give

$$C_F = \frac{N^2 - 1}{2N} \quad (2.51)$$

$$C_A = N.$$

The combination  $2C_F$  follows directly from the fundamental Casimir invariant above,

$$2C_F = N - \frac{1}{N}, \quad (2.52)$$

and will be used as a normalisation throughout the present work.

After performing the colour contractions and sums, the complete matrix-element contributions can be organised according to their distinct colour structures. For the  $q\bar{q}$  antennae considered in this work, these give rise to leading- and subleading-colour contributions and, at loop level, contributions associated with closed quark loops. The corresponding colour-stripped antenna components are conventionally distinguished by their notation: the leading-colour component carries no additional accent, the subleading-colour component is denoted by a tilde, and the closed-quark-loop component by a hat. The explicit identification of these components from the relevant matrix elements is presented in the following subsection.

#### 2.4.4 Antenna Construction

Each antenna set considered in Table 2.3 is now built explicitly in this subsection. As mentioned, every antenna is normalised by the Born-level squared amplitude given as,

$$|\mathcal{M}_2^0|^2 = 4(1 - \epsilon)q^2. \quad (2.53)$$

The LO antenna then becomes,

$$A_2^0(1_q, 2_{\bar{q}}) = \frac{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2}{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2} = 1. \quad (2.54)$$

Advancing to the NLO tree-level antenna, the relevant amplitude  $\mathcal{M}_3^0(1_q, 3_g, 2_{\bar{q}})$  is the sum of the two tree-level diagrams of Fig. 2.1, with the gluon emitted from either the quark or the antiquark line, as given in Eq. (2.10); both diagrams are therefore already contained within the same squared amplitude. The antenna becomes,

$$A_3^0(1_q, 3_g, 2_{\bar{q}}) = \frac{|\mathcal{M}_3^0(1_q, 3_g, 2_{\bar{q}})|^2}{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2}. \quad (2.55)$$

The NLO one-loop antenna is obtained from the interference between the tree-level and one-loop two-parton amplitudes, normalised to the corresponding Born-level squared amplitude,

$$A_2^1(1_q, 2_{\bar{q}}) = \frac{2 \operatorname{Re} \left( \mathcal{M}_2^{0\dagger}(1_q, 2_{\bar{q}}) \mathcal{M}_2^1(1_q, 2_{\bar{q}}) \right)}{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2}. \quad (2.56)$$

The NNLO tree-level  $A$ -type antenna set is the first case considered here to contain more than one colour component. For the  $qgg\bar{q}$  final state, the two gluons can appear in two different colour orderings along the  $q\bar{q}$  colour line. The complete squared matrix element therefore contains contributions from both colour orderings, together with their interference. After performing the colour algebra and extracting the common colour normalisation introduced above, these contributions can be organised into leading-

and subleading-colour structures. The resulting decomposition is

$$\begin{aligned}
M_{qg\bar{q}}^0 &= \frac{1}{|\mathcal{M}_2^0|^2} \left( N \left( |\mathcal{M}_4^0(1_q, 3_g, 4_g, 2_{\bar{q}})|^2 + |\mathcal{M}_4^0(1_q, 4_g, 3_g, 2_{\bar{q}})|^2 \right) \right. \\
&\quad \left. - \frac{1}{N} |\mathcal{M}_4^0(1_q, 3_g, 4_g, 2_{\bar{q}}) + \mathcal{M}_4^0(1_q, 4_g, 3_g, 2_{\bar{q}})|^2 \right) \\
&= N \left( A_4^0(1_q, 3_g, 4_g, 2_{\bar{q}}) + A_4^0(1_q, 4_g, 3_g, 2_{\bar{q}}) \right) - \frac{1}{N} \tilde{A}_4^0(1_q, 3_g, 4_g, 2_{\bar{q}})
\end{aligned} \tag{2.57}$$

This decomposition identifies the two components of the  $A_4^0$  antenna set: the leading-colour antenna  $A_4^0$  and the subleading-colour antenna  $\tilde{A}_4^0$ . For all antenna functions considered in this work, details of their explicit calculation using *AntCalc* are presented in Chapter 4, while the complete analytic results are collected in Appendix A.

The NNLO one-loop antenna set combines the features encountered separately in the NLO one-loop and NNLO tree-level antennae. It is obtained from the tree–one-loop interference of the three-parton matrix element and, before ultraviolet renormalisation, decomposes into three colour components,

$$\begin{aligned}
M_{qg\bar{q}}^1 &= \frac{2 \operatorname{Re} \left( \mathcal{M}_3^{0\dagger}(1_q, 3_g, 2_{\bar{q}}) \mathcal{M}_3^1(1_q, 3_g, 2_{\bar{q}}) \right)}{|\mathcal{M}_2^0|^2} \\
&= N A_{3,\text{bare}}^1(1_q, 3_g, 2_{\bar{q}}) - \frac{1}{N} \tilde{A}_{3,\text{bare}}^1(1_q, 3_g, 2_{\bar{q}}) + N_f \hat{A}_{3,\text{bare}}^1(1_q, 3_g, 2_{\bar{q}}).
\end{aligned} \tag{2.58}$$

The only diagram contributing to  $\hat{A}_{3,\text{bare}}^1$  is a massless-quark self-energy insertion on the external, on-shell gluon leg. Since this bubble carries the on-shell gluon’s momentum  $p_3$ , and  $p_3^2 = 0$ , it is a scaleless integral and therefore vanishes identically in dimensional regularisation, such that  $\hat{A}_{3,\text{bare}}^1 \equiv 0$ . The quark-loop ( $N_f$ ) term that appears below originates entirely from the coupling-renormalisation counterterm.

Ultraviolet renormalisation of the one-loop contribution is performed according to Eq. (2.22), yielding

$$M_{qg\bar{q}}^1 = N \left( A_{3,\text{ren}}^1 + \frac{11}{6\epsilon} A_3^0 \right) - \frac{1}{N} \tilde{A}_{3,\text{ren}}^1 + N_f \left( \hat{A}_{3,\text{ren}}^1 - \frac{1}{3\epsilon} A_3^0 \right). \tag{2.59}$$

After ultraviolet renormalisation, the three-parton contribution still contains the one-loop correction associated with the underlying  $q\bar{q}$  two-parton configuration. This contribution must therefore be subtracted in the definition of the one-loop three-parton antenna, yielding

$$M_{qg\bar{q}}^1 = N \left( A_3^1 + A_3^0 \left( \frac{11}{6\epsilon} + A_2^1 \right) \right) - \frac{1}{N} \left( \tilde{A}_3^1 + A_2^1 A_3^0 \right) + N_f \left( \hat{A}_3^1 - \frac{1}{3\epsilon} A_3^0 \right) \tag{2.60}$$

This procedure defines the final NNLO one-loop antenna set, consisting of  $A_3^1$ ,  $\tilde{A}_3^1$ , and  $\hat{A}_3^1$ .

The NNLO two-loop antenna-set contains two distinct terms: the interference between the tree-level and two-loop amplitudes, and the squared one-loop amplitude. These need to be UV renormalised but

no subtraction associated with a lower-multiplicity antenna is required in this case. We obtain,

$$\begin{aligned}
M_{q\bar{q}}^2 &= \frac{2 \operatorname{Re}(\mathcal{M}_2^{0\dagger}(1_q, 2_{\bar{q}})\mathcal{M}_2^2(1_q, 2_{\bar{q}})) + \mathcal{M}_2^{1\dagger}(1_q, 2_{\bar{q}})\mathcal{M}_2^1(1_q, 2_{\bar{q}})}{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2} = \frac{|\mathcal{M}_2^{2,[2\times 0]}|^2 + |\mathcal{M}_2^{2,[1\times 1]}|^2}{|\mathcal{M}_2^0|^2} \\
&= NA_{2,\text{bare}}^2 + \frac{1}{N}\tilde{A}_{2,\text{bare}}^2 + N_f\hat{A}_{2,\text{bare}}^2 + \left(N - \frac{1}{N}\right)\check{A}_{2,\text{bare}}^2 \\
&= N\left(A_2^2 + \frac{11}{6\epsilon}A_2^1\right) + \frac{1}{N}\tilde{A}_2^2 + N_f\left(\hat{A}_2^2 - \frac{1}{3\epsilon}A_2^1\right) + \left(N - \frac{1}{N}\right)\check{A}_2^2
\end{aligned} \tag{2.61}$$

defining the four components that make up the set  $\{A_2^2, \tilde{A}_2^2, \hat{A}_2^2, \check{A}_2^2\}$ . The  $\check{A}_2^2$  component arises from the self-interference of the one-loop amplitude and carries the single, undivided colour weight  $(N - \frac{1}{N})$ : unlike the tree–two-loop interference, it is not further decomposed into separate leading- and subleading-colour pieces.

The remaining NNLO tree-level antenna functions considered in this work arise from four-quark final states in the decay of a colour-singlet current. For non-identical quark flavours, the corresponding matrix element gives rise to the  $B_4^0$  antenna, whereas for identical flavours additional interference contributions are present, defining the  $C_4^0$  antenna. A further contribution, conventionally denoted by  $\hat{B}_4^0$ , arises from configurations in which the electroweak current couples to the secondary quark line. This contribution is not required for the antenna functions considered in this thesis and will therefore not be discussed further. The complete four-quark matrix-element contribution can be decomposed as

$$\begin{aligned}
M_{4q}^0 &= \frac{1}{|\mathcal{M}_2^0|^2} \left( \sum_{q'} |\mathcal{M}_4^0(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}})|^2 + |\mathcal{M}_4^0(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}})|^2 \right) \\
&= \left[ N_f B_4^0(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}}) + N_{f,\gamma} \hat{B}_4^0(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}}) \right. \\
&\quad \left. - \frac{1}{N} \left( C_4^0(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}}) + C_4^0(2_{\bar{q}}, 4_{\bar{q}}, 3_q, 1_q) \right) \right]
\end{aligned} \tag{2.62}$$

where  $N_{f,\gamma}$  is the charge weighted sum of the quark flavours, given as,

$$N_{f,\gamma} = \frac{(\sum_q e_q)^2}{\sum_q e_q^2}. \tag{2.63}$$

To isolate the  $B_4^0$  and  $C_4^0$  antennae from the complete four-quark matrix element, it is useful to first decompose the corresponding amplitude into its colour structures. Using the  $SU(N)$  Fierz identity, the tree-level amplitude for  $\gamma^* \rightarrow q\bar{q}q'\bar{q}'$  can be written as,

$$\begin{aligned}
\mathcal{M}_4^0(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}}) &= ie_q g^2 \delta_{q_1 q_2} \delta_{q_3 q_4} \left( \delta_{i_1 i_4} \delta_{i_3 i_2} - \frac{1}{N} \delta_{i_1 i_2} \delta_{i_3 i_4} \right) \mathcal{M}_{q\bar{q}-\gamma}(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}}) \\
&\quad + (1 \leftrightarrow 3, 2 \leftrightarrow 4).
\end{aligned} \tag{2.64}$$

Here,  $\mathcal{M}_{q\bar{q}-\gamma}$  denotes the colour-stripped contribution in which the  $q_1 \bar{q}_2$  pair couples to the electroweak current. The  $B_4^0$  antenna is obtained from the squared colour-stripped amplitude, whereas the  $C_4^0$  antenna

is defined by the interference between the two identical-quark assignments:

$$\begin{aligned} B_4^0(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}}) &= \frac{|\mathcal{M}_{q\bar{q}-\gamma}(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}})|^2}{|\mathcal{M}_2^0|^2} \\ C_4^0(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}}) &= -\frac{\text{Re}\left(\mathcal{M}_{q\bar{q}-\gamma}(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}})\mathcal{M}_{q\bar{q}-\gamma}^\dagger(1_q, 4_{\bar{q}}, 3_q, 2_{\bar{q}})\right)}{|\mathcal{M}_2^0|^2}. \end{aligned} \quad (2.65)$$

### 2.4.5 Phase-Space Factorisation

The previous subsections focused on the construction, colour decomposition, and normalisation of the unintegrated antenna functions. In the subtraction formalism, these functions enter local counterterms that must subsequently be integrated analytically over the corresponding antenna phase space. At NLO, as introduced in Eq. (2.13), this gives the integrated subtraction term

$$\int_1 d\sigma^S \quad (2.66)$$

where  $\int_1$  denotes the integration associated with one unresolved parton in the antenna phase space. This notation should not be interpreted as an integration over the momentum of the unresolved parton alone. For a three-parton antenna  $X_{ijk}^l$ , the antenna phase space contains the momenta of the unresolved parton  $j$  and of the two hard radiators  $i$  and  $k$ . The integration is made possible by the exact factorisation of the full  $(m+1)$ -particle phase space into the phase space of the reduced  $m$ -particle configuration and the antenna phase space, as already given in Eq. (2.31). The antenna contribution can therefore be integrated independently of the reduced  $m$ -parton hard process. Schematically, the antenna phase-space integration acts on the antenna function, while the reduced Born contribution remains factorised,

$$\int_1 d\sigma^S \sim \sum_{\text{antennae}} \left[ \int d\Phi_{X_{ijk}} X_{ijk}^0 \right] d\sigma^B. \quad (2.67)$$

At NNLO, the same construction applies to four-parton antennae  $X_{ijkl}^0$ , whose antenna phase space contains two unresolved partons.

For the final-final antennae considered here, the antenna phase-space integrals can be related directly to the standard three- and four-particle phase spaces, by setting  $m = 2, 3$  in Eq. (2.31). Specialising the phase-space factorisation to a two-particle reduced configuration gives

$$\begin{aligned} d\Phi_3 &= d\Phi_2 d\Phi_{X_{ijk}} \\ d\Phi_4 &= d\Phi_2 d\Phi_{X_{ijkl}}. \end{aligned} \quad (2.68)$$

Since the antenna functions  $X_n^l$  depend only on invariants constructed from the antenna momenta  $p_i, p_j, p_k, \dots$ , they are independent of the reduced two-particle phase space  $d\Phi_2$ . The latter can therefore be integrated separately, giving the constant

$$\Phi_2 = \int d\Phi_2. \quad (2.69)$$

The antenna phase-space integral can therefore be expressed as [5],

$$\int d\Phi_{X_{ijkl\dots}} X_n^l = \frac{1}{\Phi_2} \int d\Phi_n X_n^l. \quad (2.70)$$

Including the overall normalisation factor  $C(\epsilon, k)$  introduced in Section 2.3, the integrated antenna  $\mathcal{X}_n^l$  is then defined, generally<sup>2</sup>, as

$$\mathcal{X}_n^l = C(\epsilon, k) \int d\Phi_{X_{ijkl\dots}} \mathcal{X}_n^l. \quad (2.71)$$

For the two-parton antennae  $A_2^1$  and  $A_2^2$ , no unresolved real radiation is present and therefore no additional antenna phase space needs to be isolated. Their integrated counterparts are obtained directly from the corresponding two-particle phase-space integration, together with the loop integrations entering the virtual contributions [12]. Thus, while these antennae are also integrated over phase space, the phase-space factorisation of Eq. (2.70), relevant for antennae containing unresolved radiation, is not required.  $A_2^0$  also has no unresolved parton, but is excluded here for a different reason: it is the Born-level antenna itself, not an integrated subtraction ingredient.

## 2.5 Phase-Space Integration and IBP Reduction

In the previous sections, antenna functions were introduced together with their role in the local subtraction of infrared singularities. Eq. (2.71) defined their integration over the unresolved phase space, leading from an unintegrated antenna  $X_n^l$  to its integrated counterpart  $\mathcal{X}_n^l$ . For loop-level antennae ( $l \geq 1$ ), the loop momenta must additionally be integrated.

Both the unintegrated and integrated forms are required in a subtraction calculation, but they play different roles. The unintegrated antenna functions provide local counterterms for real-radiation matrix elements and are therefore required in numerical Monte Carlo (MC) phase-space integration. They must retain the dependence on the local kinematics of each phase-space point, which can be expressed in terms of Mandelstam invariants  $s_{ij}$ . This allows the subtraction term to reproduce the singular behaviour of the real-emission matrix element point by point in the unresolved limits. For loop-level antennae, the term *unintegrated* refers here to the unresolved phase space: the loop-momentum integration is first performed analytically, leaving an antenna function that depends on the external kinematic invariants and can be evaluated locally within the phase-space integration. The subsequent analytic integration over the unresolved phase space then produces the corresponding integrated antenna function, whose explicit infrared poles enter the cancellation against virtual contributions.

From Eq. (2.70), we have seen that through phase-space factorisation, the antennae are integrated over the antenna phase space  $\Phi_X$ . If required, they are also integrated over their loop momenta  $\ell_r$ ,  $r = 1, \dots, l$ . The propagator or Mandelstam-invariant denominators entering these integrals may generally be defined as,

$$D_j = \left( \sum_{r=1}^l c_{jr} \ell_r + r_j \right)^2 - m_j^2 + i0 \quad (2.72)$$

where  $r_j$  is built from momenta held fixed during the loop integration, including the mapped hard partons and unresolved antenna momenta. For the massless, on-shell external partons considered throughout this work,  $r_j$  reduces to a combination of Mandelstam invariants  $s_{ij}$ .

An algebraic expansion of the integrand of an antenna function can generate multiple integrals with the same kinematic structure but different numerator and denominator powers, since any numerator built from scalar products can itself be absorbed into the same propagator basis  $\{D_j\}$ . These integrals may then be organised into integral families, which fix the denominators and integration variables, and may

<sup>2</sup> This definition is general for antenna functions that are already cleanly and completely given in terms of Mandelstam invariants and  $\epsilon$ . This is only automatically true for antenna where  $l = 0$ . When  $l \geq 1$ , which happens for all loop antennae by definition, a prior loop-momentum integration step is necessary. This step is explored further in Section 2.5.

then be systematically reduced. These families are denoted as the complete collection [13],

$$\mathcal{F}[\{D_j\}; \{\ell_r\}, d\Phi_X] = \{I(a_1, \dots, a_N)\}, \quad a_j \in \mathbb{Z} \quad (2.73)$$

with,

$$I(\vec{a}) = \int \left[ \prod_{r=1}^l d^d \ell_r \right] d\Phi_X \frac{1}{D_1^{a_1} \dots D_N^{a_N}}, \quad \vec{a} = (a_1, \dots, a_N). \quad (2.74)$$

The integrals within each family are not independent. Through integral reduction, they can be expressed in terms of a finite and substantially smaller set of master integrals (MI),

$$I(\vec{a}) = \sum_{k=1}^{N_{\text{MI}}} c_k(\vec{a}, \epsilon, \{s_{ij}\}) M_k \quad (2.75)$$

where  $M_k$  are the master integrals and  $c_k$  are coefficients that depend on the integral indices, the dimensional regulator  $\epsilon$ , and the relevant kinematic invariants. The original integration problem is therefore separated into two tasks: determining the reduction coefficients  $c_k$  and evaluating the much smaller set of master integrals  $M_k$ . Once these are known, every required integral in the family can be reconstructed from the same master-integral basis. In  $\vec{a}$ , positive  $a_j$  indices denote propagator powers, while non-positive indices account for absent denominators or irreducible scalar products in the numerator.

As stated previously, MC integrators require the unintegrated antenna functions in terms of the Mandelstam invariants  $s_{ij}$ , since they must evaluate the subtractions at a sampled physical external phase-space point  $p$ . The loop-level antenna required by an MC is ultimately evaluated as a function of the external invariants. Its calculation, however, involves tensor loop integrals with loop-momentum dependence, which must be reduced before the antenna can be evaluated at a sampled phase-space point. This reduction is performed through the Passarino-Veltman (PaVe) reduction mechanism, described in detail in Section B.2, which reduces the tensor loop integrals to linear combinations of scalar one-loop functions, usually denoted by  $A_0, B_0, \dots$ , while keeping the phase-space variables and integration untouched, as [14],

$$\int d^d \ell \frac{\ell^{\mu_1} \dots \ell^{\mu_R}}{\prod_j D_j^{a_j}} \xrightarrow{\text{PaVe}} \sum_k c_k(\{s_{ij}\}) \{A_0, B_0, C_0, D_0, \dots\}_k. \quad (2.76)$$

To finally reach the integrated antenna functions  $\mathcal{X}_n^l$ , the remaining phase-space integration is performed through Integration By Parts (IBP) reduction, combined with the method of reverse unitarity to bring the on-shell phase-space constraints into the same algebraic framework as the loop integrals. Both methods, together with the resulting Laurent expansion of the master integrals, are described in full in Subsection 2.5.1.

## 2.5.1 IBP Identities and the Laporta Algorithm

### 2.5.1.1 IBP Identities

The integral family has already been defined in Eq. (2.74). This definition allows for the factorisation of the integral into a loop and a phase-space integral term. That factorisation results in,

$$I(\vec{a}) = \int d\Phi_X I(\vec{a}; \{D_1, \dots, D_N\}), \quad I(\vec{a}; \{D_1, \dots, D_N\}) = \int \prod_{r=1}^l d^d \ell_r \frac{1}{D_1^{a_1} \dots D_N^{a_N}}. \quad (2.77)$$

The overall normalisation factors are irrelevant to the identities and will, therefore, be suppressed throughout the following explanation. The denominators  $D_j$  were already defined generally in Eq. (2.72). Take one loop momentum  $\ell_s$ , for  $s = 1, \dots, l$ , and one vector  $v^\mu$  from the available loop and external momenta. Dimensional regularisation allows us to set a total derivative to zero inside  $\mathcal{I}$ , as [13, 15],

$$0 = \int \prod_{r=1}^l d^d \ell_r \frac{\partial}{\partial \ell_s^\mu} \left[ v^\mu \frac{1}{D_1^{a_1} \dots D_N^{a_N}} \right] \quad (2.78)$$

In dimensional regularisation, analytic continuation justifies the vanishing of the surface term at infinity, leaving an exact relation among regulated integrals<sup>3</sup>.

Expanding the derivative gives,

$$0 = \int \prod_{r=1}^l d^d \ell_r \left[ \frac{\partial \ell_s \cdot v}{\prod_j D_j^{a_j}} - \sum_{j=1}^N a_j \frac{v^\mu \partial \ell_s^\mu D_j}{D_1^{a_1} \dots D_j^{a_j+1} \dots D_N^{a_N}} \right]. \quad (2.79)$$

For these quadratic denominators  $D_j$ , differentiation with respect to  $\ell_s^\mu$  is linear in momenta,

$$\partial \ell_s^\mu D_j = 2c_{js} \left( \sum_r c_{jr} \ell_r + r_j \right)_\mu \quad (2.80)$$

and therefore,  $v^\mu \partial \ell_s^\mu D_j$  is a scalar product. Scalar products may be expressed in terms of chosen denominators and irreducible scalar products. The latter are represented by negative family indices  $a_j$ , while zero indices denote absent denominators. We are then able to write schematically,

$$\sum_{\vec{b}} \rho_{\vec{b}}(\vec{a}, \epsilon, \{s_{ij}\}) \mathcal{I}(\vec{b}) = 0 \quad (2.81)$$

where each vector  $\vec{b}$  labels a family member obtained from  $\vec{a}$  through index shifts in one or more denominator powers.  $\rho_{\vec{b}}$  is the corresponding coefficient in the IBP relation.

An example of the application of this method is written in Section B.3.

### 2.5.1.2 Reverse Unitarity and Cut Integrals

In the previous subsection, we have derived the IBP identities relative to  $\mathcal{I}$ . As described in Eq. (2.77), the entire integrated antenna is not merely  $\mathcal{I}$ , but rather,

$$I(\vec{a}) = \int d\Phi_X \mathcal{I}(\vec{a}). \quad (2.82)$$

The unresolved phase-space measure is not originally an ordinary loop integral, however. Instead, it has on-shell constraints, which have to be considered and transformed before  $I(\vec{a})$  is reducible via the present method.

As described in Subsection 2.4.5, phase-space factorisation isolates the antenna measure through  $d\Phi_{m+r} = d\Phi_m d\Phi_X$ , or in the two-hard-radiator final-final case we are considering throughout the present work,  $d\Phi_X = d\Phi_n / d\Phi_2$ , for  $n$  final-state particles. That  $n$ -particle phase space is given generally

<sup>3</sup>This method is explored further and completely in Subsubsection 4.1.1.4.

as,

$$d\Phi_n(q : p_1, \dots, p_n) = (2\pi)^d \delta^{(d)}\left(q - \sum_{i=1}^n p_i\right) \prod_{i=1}^n \frac{d^d p_i}{(2\pi)^{d-1}} \delta^+(p_i^2 - m_i^2) \quad (2.83)$$

where the  $\delta^{(d)}$ -function enforces overall momentum conservation, and each  $\delta^+$ -function imposes the mass-shell condition for the  $i$ -th final-state particle, with the superscript + selecting its positive-energy solution. This may be written as,

$$\delta^+(p_i^2 - m_i^2) = \theta(p_i^0) \delta(p_i^2 - m_i^2) \quad (2.84)$$

This  $\delta^+$ -function may be reexpressed as a discontinuity as follows [13, 16],

$$2\pi i \delta^+(p_i^2 - m_i^2) = \left[ \frac{1}{p_i^2 - m_i^2 - i0} - \frac{1}{p_i^2 - m_i^2 + i0} \right]_+ \quad (2.85)$$

where the  $\pm i0$  terms specify the propagator prescriptions. The subscript + represents the positive-energy cut selection.

These objects are called *cut propagators*, because they come about from cutting a propagator line, thus putting the represented particle on-shell, as a real final-state particle. Algebraically, it also allows phase-space constraints to be written in the same denominator language as virtual propagators. A cut propagator associated with a required final-state particle must be retained during the reduction and sectors in which such a cut is absent do not represent the physical phase-space integral and thus, vanish [13]. Using the discontinuity relation to refactor the  $\delta$ -functions, we are able to rewrite the integrated antenna expression schematically as,

$$\mathcal{X}_n^l \sim \int \left[ \prod_{r=1}^l d^d \ell_r \right] \left[ \prod_u d^d p_u \right] \frac{N_s(\ell_r, p_u; \tilde{p}_{\text{hard}})}{\prod_\alpha D_{v,\alpha}^{a_\alpha} \prod_\beta D_{c,\beta}^{a_\beta}} \quad (2.86)$$

where  $D_{v,\alpha}$  are ordinary virtual propagators,  $D_{c,\beta}$  are cut propagators,  $\tilde{p}_{\text{hard}}$  is the hard-radiator combined momenta and  $p_u$  are integrated unresolved momenta.

After rewriting the integral in this way, the total derivative introduced in the previous Subsubsection 2.5.1.1 may now be taken with respect to any loop momentum  $\ell_r^\mu$  or integrated unresolved momentum  $p_u^\mu$ . This process produces relations involving both virtual and cut denominators. This then allows us to reduce loop and phase-space integrals simultaneously using IBP reduction [13, 16].

An example of the method used to rewrite  $d\Phi_3$  in terms of cut propagators is available in Section B.4.

### 2.5.1.3 Laporta Algorithm

At NLO and NNLO these reduction systems become very complex and their solution ever more time-intensive. The Laporta algorithm is a systematic-elimination method that uses IBP relations to express more complicated integrals in terms of a smaller set of unreduced ones (master integrals) [17].

The algorithm first classifies the family members into sectors. These sectors are defined as,

$$\theta_j \equiv \theta(a_j) = \begin{cases} 1, & a_j > 0, \\ 0, & a_j \leq 0, \end{cases} \quad \vec{\theta} = (\theta_1, \dots, \theta_N). \quad (2.87)$$

After fixing an integral family and identifying its admissible sectors, the Laporta algorithm assigns an

ordering to the resulting integrals according to their complexity. This hierarchy is based on the number of propagators present  $t$ , the excess denominator power  $r$ , and the numerator complexity encoded through negative indices  $s^4$ . These quantities are defined using each integral's  $a$ -index, taken from the family's  $\vec{a}$  vector, as previously defined. They are, generally,

$$t = \sum_{j=1}^N \theta(a_j), \quad r = \sum_{j=1}^N \max(a_j - 1, 0), \quad s = \sum_{j=1}^N \max(-a_j, 0). \quad (2.88)$$

Here,  $\theta(a_j)$  records whether denominator  $D_j$  is present. The hierarchy is defined by comparing  $t$  first, then  $r$  and finally  $s$ . The algorithm uses this chosen hierarchy to select, within each IBP relation, the most complex integral to eliminate first.

An example of the application of the Laporta algorithm is written by hand in Section B.5.

After determining the integral families and sectors present, we apply the IBP identities to the seed integrals. These seed integrals are a part of a finite, bounded set of integrals chosen from each relevant sector, whose IBP identities must be sufficiently numerous to reduce every target integral and every non-zero subtopology reached along the way.

After generating IBP identities for the selected seed integrals, any relation containing these representatives is first solved for  $I(1, 1, 1)$ , the highest-ranked member, and the resulting expression is substituted recursively into the remaining system [17].

IBP reduction may produce lower sectors, but may also generate new integrals from the initially relevant sectors. Lastly, some sectors may be cancelled due to symmetry, if one denominator is shifted to another under loop-momentum shift or a relabelling of external momenta. If two sectors become equivalent, one is discarded from the independent reduction and the other is retained as its representative. In short, a sector is deemed relevant if it is an original target sector, or a lower sector reached from another via IBP reduction. The relevant sector must also be non-scaleless in a chosen kinematics, not redundant under a family symmetry and, for a cut family, it must retain all required cut denominators throughout.

It is important to note that cut families add a physical restriction: if an IBP reduction step cancels a cut propagator, the sector it produces is missing one of the required on-shell final-state particles and it no longer represents the intended phase-space. If that happens, the integral is set to zero [13].

The reduction is complete when the remaining non-zero family members cannot be eliminated by the available identities. These survivors form a master-integral basis.

## 2.5.2 Master Integrals

The IBP reduction method reduces the integral to a linear combination of master integrals as,

$$I(\vec{a}) = \sum_{k=1}^{N_{\text{MI}}} c_k(\vec{a}, \epsilon, \{s_{ij}\}) M_k \quad (2.89)$$

where  $c_k$  are the reduction coefficients and represent what comes from the reduction algebra and  $M_k$  are the MI. Our remaining task is evaluating the finite set  $M_k$  [17].

Master integrals are not uniquely defined: an integral is a master only relative to the chosen family, kinematics, IBP system, and basis. This means that a different basis with different masters may be used to reduce the same integrals and span the exact same integral space [17].

---

<sup>4</sup>The present  $t$ ,  $r$  and  $s$  metrics are not to be taken as unique Laporta orderings, as these vary by implementation. Instead, they are illustrative objects.

Master integrals are scalar integrals with a fixed denominator/cut structure. They are generically represented as,

$$M_k = \int \left[ \prod_r d^d \ell_r \right] \left[ \prod_u d^d p_u \right] \frac{1}{\prod_\alpha D_{v,\alpha}^{a_\alpha} \prod_\beta D_{c,\beta}^{a_\beta}}. \quad (2.90)$$

For integrated antennae, the cut denominators in these master integrals encode the real unresolved phase space. Hence, a master may be a pure phase-space volume, a phase-space integral with invariant denominators, or a loop-plus-phase-space cut integral. The MI are also not necessarily finite; in fact, in dimensional regularisation they commonly contain  $1/\epsilon$  poles [13].

For a massless fully inclusive antenna system,  $q^2$  is often the only independent kinematic scale once the dimensional-regularisation normalisation factors have been separated. Through dimensional analysis, the MI are strongly constrained to take the form,

$$M_k(q^2, \epsilon) = \left( q^2 \right)^{\lambda_k + \kappa_k \epsilon} f_k(\epsilon), \quad f_k(\epsilon) \sim \frac{\Gamma^m(1 - a\epsilon)}{\Gamma^n(1 - b\epsilon)} \quad (2.91)$$

where  $\lambda_k$  is fixed by mass dimension and  $f_k$  is dimensionless, depending only on  $\epsilon$  [13]. Here,  $\Gamma$  denotes the Euler Gamma function, the analytic continuation of the factorial to complex arguments, defined for  $\text{Re}(z) > 0$  by

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt, \quad (2.92)$$

and extended to the rest of the complex plane, away from its poles at  $z = 0, -1, -2, \dots$ , by analytic continuation.

When evaluating master integrals, different types often call for different approaches. Simple one-scale phase-space masters often integrate directly after the phase-space is parametrised. Angular and energy-fraction integrations typically reduce to Beta functions. When a master depends on a non-trivial kinematic variable, such as  $x = m^2/q^2$ , the most common approach is to differentiate the master with respect to that variable. The derivative produces integrals in the same family, which IBP is then able to reduce to the master basis, as [18],

$$\frac{d}{dx} \vec{M}(x, \epsilon) = A(x, \epsilon) \vec{M}(x, \epsilon). \quad (2.93)$$

where  $\vec{M} = (M_1, \dots, M_{N_{\text{MI}}})^T$  is the vector of masters and  $A$  is a matrix of coefficient functions. Boundary conditions can then be used to determine the solution. The MI used through this work and the methods used to compute them are discussed in further detail in Section 4.

Once the MI are evaluated, we expand them around  $\epsilon = 0$ , as a Laurent series [13],

$$M_k = \sum_{n=n_{\min}}^{\infty} m_{k,n} \epsilon^n \quad (2.94)$$

to expose their pole and finite structure. The lowest power  $n_{\min}$  can be, and often is, negative, resulting in an expansion similar to,

$$M_k = \frac{m_{k,-2}}{\epsilon^2} + \frac{m_{k,-1}}{\epsilon} + m_{k,0} + O(\epsilon). \quad (2.95)$$

The required expansion depth is determined by the reduction coefficient. If  $c_k \sim \epsilon^{-p}$ , the  $M_k$  must be expanded through to  $\epsilon^p$  to obtain the finite contribution to  $c_k M_k$ .

The reduction techniques introduced in this chapter provide the computational tools required for the analytic integration of the antenna functions considered in this work. In particular, the IBP identities are

systematically solved through the Laporta algorithm, reducing the phase-space and loop integrals to a finite set of master integrals. As described in Chapter 3, this reduction procedure is implemented within the *AntCalc* workflow and is subsequently applied in Chapter 4 to obtain the integrated antenna functions in dimensional regularisation. The resulting Laurent expansions in  $\epsilon$  provide the explicit infrared pole structure required for their use in higher-order subtraction at NLO and NNLO.

## Chapter 3

# Package Framework

A central contribution of this thesis is the development of *AntCalc*<sup>1</sup>, a *Wolfram Language* package created from scratch as part of this thesis work to automate the symbolic construction and integration of QCD antenna functions. The package provides direct access to both unintegrated and integrated antenna functions and implements the computational framework described in the previous chapter. The results presented in this thesis were produced with *AntCalc*, version 0.3.0 $\beta$ 2, corresponding to commit 4ceb991.

*AntCalc* makes use of *FeynCalc* [19, 20, 21, 22] for symbolic amplitude manipulation and interfaces with *LiteRed2* [23] for IBP reduction. Both are external dependencies that must be installed by the user prior to loading *AntCalc*; the package’s own installation procedure only installs the *AntCalc* paclet itself. Its internal architecture is organised as a profile-driven route system: route-specific information is centralised in profiles, which guide the selection and configuration of the appropriate build, reduction, and integration workflows. This establishes a common contract between the package components, reducing the amount of route-specific logic that must be duplicated when existing functionality is extended or new routes are introduced.

The package supports massless and massive antenna routes. All routes were validated against the established literature, with all massless routes being validated against [5] except for the two-loop antenna set, which was validated against [24]. The single massive route results were validated against [25]. All routes are public: none is held back as a partial or experimental implementation, and each produces stable and reproducible results notwithstanding the current beta version. The release-acceptance procedure that ensures this is described in Section 3.7.

*AntCalc*’s inner working structure can be described using the diagram in Fig. 3.1.

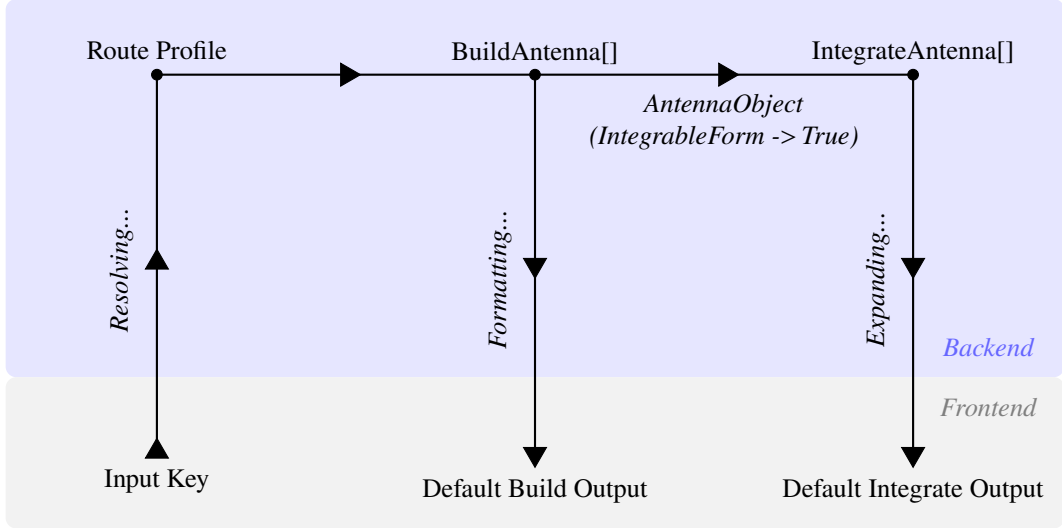
### 3.1 Design Philosophy and Scope

*AntCalc* was built to combine the symbolic amplitude capabilities of the *FeynCalc* environment with the IBP-reduction capabilities of *LiteRed2*, adapting and orchestrating the parts of those systems required for the construction and integration of QCD antenna functions.

The central design objective of the package is to return antenna functions in the conventional forms required for higher-order QCD calculations using the antenna subtraction method. Though it supports a wide variety of options to inspect and configure the backend operations, the default outputs of its two central functions are designed to reflect this objective. For example, the unintegrated one-loop

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<sup>1</sup>Source code available at <https://github.com/HenFar/AntCalc/tree/release/0.3.x>.



**Figure 3.1:** Profile-driven architecture of *AntCalc*. Route profiles configure the build and integration workflows, while the build-side route context carries the evolving calculation data. The *AntennaObject* provides the integration-ready handoff between the two stages.

three-parton antennae  $\{A_3^1, \tilde{A}_3^1, \hat{A}_3^1\}$  returned by `BuildAntenna[...]` are expressed after Passarino-Veltman reduction, in terms of scalar one-loop functions and the external kinematic invariants, following the conventional representation used in the literature [5]. By default, `IntegrateAntenna[...]` returns the integrated antennae as Laurent series in  $\epsilon$ , with the truncation order chosen to match the conventions used in the antenna literature: the NLO antennae  $\mathcal{A}_2^1$  and  $\mathcal{A}_3^0$  are expanded through  $O(\epsilon^2)$ , while the NNLO antennae  $\mathcal{A}_4^0$ ,  $\mathcal{A}_3^1$ ,  $\mathcal{A}_2^2$ ,  $\mathcal{B}_4^0$ , and  $\mathcal{C}_4^0$  are returned through  $O(\epsilon^0)$  [5, 24]. Furthermore, the massive NLO tree-level antenna  $\mathcal{A}_3^{0,(m_q)}$  is given up to  $O(\epsilon)$  [25].

Its structure was also designed to be modular and extensible. Each of the two central functions (`BuildAntenna` and `IntegrateAntenna`) is implemented as an encapsulated public module that communicates with the rest of the codebase using the profile system: the route profile selects and configures the appropriate workflows and the route context carries the evolving results and diagnostics between stages. Using route profiles and route contexts, as described in Section 3.2, facilitates the addition of further modules while reducing duplicated logic and direct dependencies. These new modules need not be compatible with previous implemented functions through function-specific couplings. They need only conform with the profile system, being selected and configured by route profiles, and accepting inputs from and delivering outputs to the route contexts.

Currently, *AntCalc* is organised by mass scale, into massless and massive routes. All massless quark-antiquark final-final antenna functions through NNLO have been made public. Alongside these, a single massive route,  $A_3^0$ , is finalised, validated, and likewise public. This is the first massive antenna of what is expected to be a complete expansion of *AntCalc*'s capabilities into the massive regime for quark-antiquark final-final antennae. The antenna functions currently implemented are summarised in Table 3.1.

## 3.2 Profile-Driven Architecture

As described in Section 3.1, *AntCalc* relies on a profile-driven routing system. This system maps the route key to its route-specific profile and workflow.

A public antenna route key is a three-element list which describes a canonical route whose execution

| Perturbative Order | Antenna                                                                 | Validated Against  |
|--------------------|-------------------------------------------------------------------------|--------------------|
| LO                 | $A_2^0$                                                                 | Born normalisation |
| NLO                | $A_2^1, A_3^0$                                                          | [5, 24]            |
| NNLO               | $A_3^1, \tilde{A}_3^1, \hat{A}_3^1, A_4^0, \tilde{A}_4^0, B_4^0, C_4^0$ | [5, 24]            |
| NNLO               | $A_2^2, \tilde{A}_2^2, \hat{A}_2^2, \check{A}_2^2$                      | [24]               |
| NLO, massive       | $A_3^{0,(m_q)}$                                                         | [25]               |

**Table 3.1:** Overview of the antenna functions currently implemented in *AntCalc*, by perturbative order and validation source.

may be refined by options. The route key takes the form,

$$\{\text{Antenna Type, Final-State Multiplicity, Loop Order}\}. \quad (3.1)$$

As an example, the route key for the one-loop three-parton antenna set associated with the full-process amplitude  $M_{qg\bar{q}}^1$  is  $\{A, 3, 1\}$ . This key represents the complete set of colour contributions, comprising the leading-colour antenna  $A_3^1$ , the subleading-colour antenna  $\tilde{A}_3^1$ , and the quark-loop contribution  $\hat{A}_3^1$ . The default public output is therefore the ordered list  $\{A_3^1, \tilde{A}_3^1, \hat{A}_3^1\}$ .

The package links this key to the profile it identifies and follows that profile's directions on how the route should be constructed and/or integrated. The data produced while the route executes is also stored in the route context association. The profile system consists of a main profile and two further profiles: one build-side, handling reduction, and one integration-side. The principal build-side profile is `AntennaProfile[key]`, which specifies information such as antenna type, multiplicity and loop order, colour normalisation, extraction rules, components, etc. `AntennaReductionProfile[key]` is separate, but still build-side. It specifies the Passarino-Veltman-reduction preferences for the appropriate routes. In this case, up to NNLO, these are the  $\{A, 2, 1\}$  and  $\{A, 3, 1\}$  loop routes.

`AntennaIntegrationProfile[key]` is the integration-side profile, which specifies its side's backend, IBP basis families, expansion depth, etc. The specifics of each stage's profiles and options will be discussed in their respective Sections 3.3 and 3.5.

While profiles select and configure computation paths, the workflows themselves perform the calculations. The route dispatcher and workflow code consult the profiles to select and configure each stage. As a representative example, which will be further explained and explored in the two upcoming sections, let us consider the call `BuildAntenna[A, 2, 1]`. Its full route profile is,

```
AntennaProfile[{A, 2, 1}] :=
  <|
    "Key" -> {A, 2, 1},
    "Name" -> "A21",
    "AntennaType" -> A,
    "NumFinalParticles" -> 2,
    "LoopOrder" -> 1,
    "TreeAmplitude" -> AntennaAmplitude[{A, 2, 0}],
    "BornInterference" -> BornInterference[],
    "Extraction" -> "LoopScalar",
    "ColourNorm" -> colourNorm,
```

```

"ExcludedParticles" ->
  {S[_], V[1], V[2], V[3], U[1], U[2], U[3], U[4]},
"DropSumOver" -> True,
"ReductionProfile" -> AntennaReductionProfile[{A, 2, 1}],
"ConventionProfile" -> BuildAntennaConventionProfile[{A, 2, 1}]
|>

```

In this profile, the key, name, antenna type, number of final particles (multiplicity), and loop order have been discussed before. The settings used in this route are explained in Table 3.2.

| Item              | Use                                                                       |
|-------------------|---------------------------------------------------------------------------|
| TreeAmplitude     | Selects the tree-level amplitude. In this case, it is $\mathcal{M}_2^0$ . |
| BornInterference  | Generates the general normalisation as $ \mathcal{M}_2^0 ^2$ .            |
| Extraction        | Selects how the components are extracted.                                 |
| ColourNorm        | Colour normalisation: $N - 1/N$ .                                         |
| ExcludedParticles | Restricts <i>FeynArts</i> internal particle insertions.                   |
| DropSumOver       | Replaces every <code>SumOver[]</code> statement with 1.                   |
| ReductionProfile  | Points to the reduction profile.                                          |
| ConventionProfile | Points to the normalisation-convention ruleset profile.                   |

**Table 3.2:** Overview of the settings for the  $A_2^1$  route used as a representative example of the route system used in *AntCalc*.

### 3.3 The Build Stage

The build stage constructs the unintegrated antenna functions from their corresponding matrix elements. Its main public interface is `BuildAntenna[type, multiplicity, loop order]`. The workflow begins with a route key specified through a public call, for example `BuildAntenna[A, 3, 1]`. From this key, *AntCalc* identifies the appropriate construction workflow and the ingredients required for the corresponding antenna set. The relevant amplitudes are then constructed and combined according to the route definition, after which the resulting expressions are processed into the required unintegrated antenna functions. Internally, the intermediate results are stored in an association containing the different components of the calculation. The final step reformats this internal representation into the antenna expression, or ordered set of antenna components, returned to the user. The build workflow can therefore be summarised as follows,

```

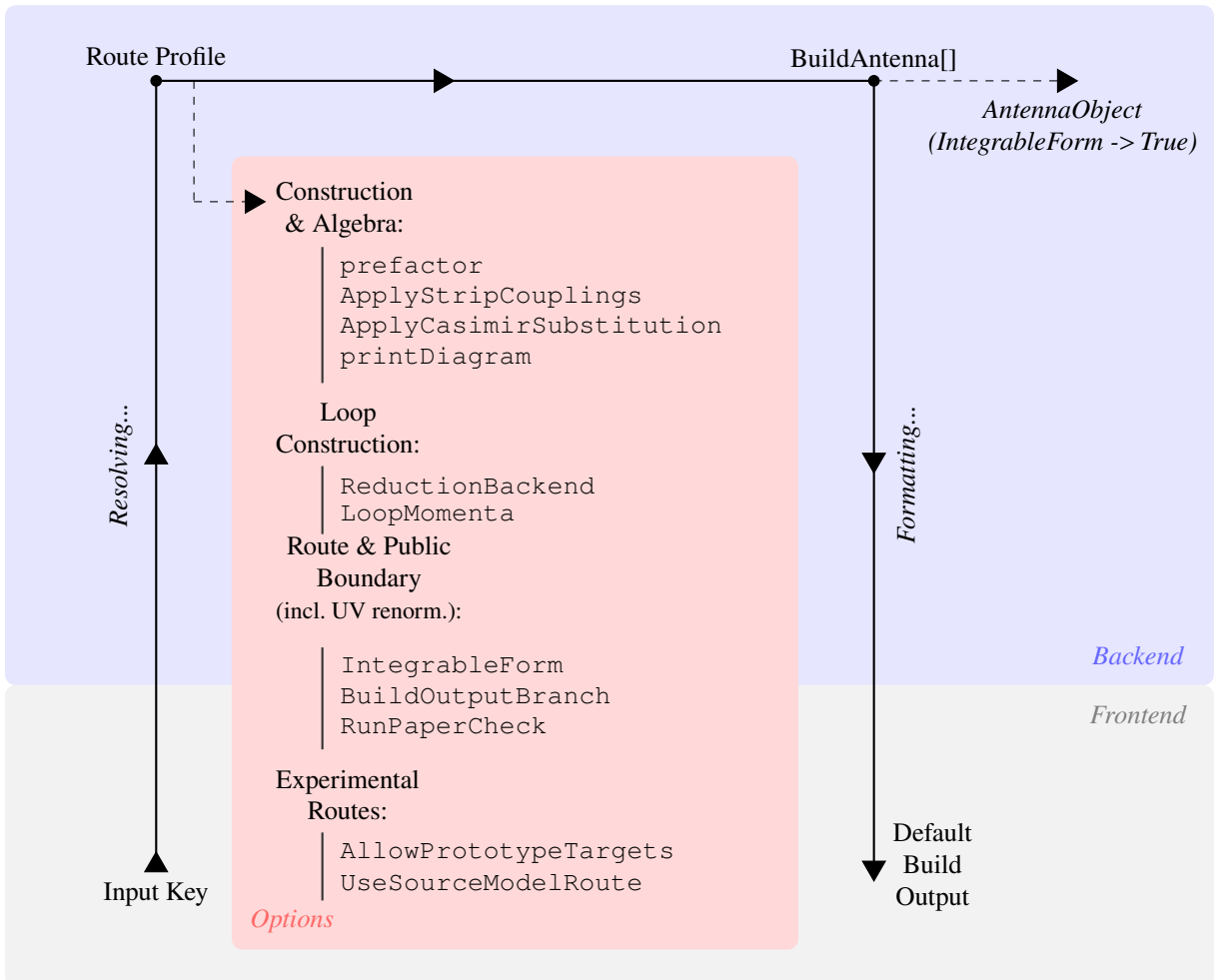
BuildAntenna[type, multiplicity, loop order]
  -> route key;
  -> loop-order dispatcher;
  -> route-specific build workflow;
  -> build-data association / route context;
  -> public-output boundary;
  -> output: expression, component list, AntennaObject, diagnostics,
      or run record.

```

The loop-order dispatcher is a central element of this workflow. Using the route key, it directs each antenna to the appropriate computational path, since different loop orders require different computational treatments, most notably the tensor reduction required for the loop-momentum dependence of one- and two-loop antennae.

Functionally, the route context for this stage holds the profile, the generated amplitude, the computed and normalised interference, the extracted components, and the diagnostics; some routes retain additional route-specific intermediates.

A diagram in the spirit of the one in Fig. 3.1, but only relative to the build stage is shown below, in Fig. 3.2.



**Figure 3.2:** Profile-driven architecture of *AntCalc*'s Build stage. The principal build-specific options to the `BuildAntenna` function are displayed on the red card.

For loop routes, this public-output boundary is also where ultraviolet renormalisation is applied. Renormalisation is not a discrete internal stage of the build workflow; instead, it is applied to the unintegrated one- and two-loop antennae precisely at the point where they cross from the internal (backend) representation into the public antenna expression returned to the user, using the coupling rescaling of Eq. (2.22). This is why no separate renormalisation step appears explicitly in Fig. 3.2: it coincides with the “Route & Public Boundary” crossing already shown there.

### 3.3.1 Tree-Level Routes

As mentioned, the workflow handles distinct loop-order routes differently, though the differences only appear after the dispatcher is called. Tree-level routes like those relative to the  $A_3^0$  antenna, start by retrieving the profile and computing the relevant process amplitude. It uses the profile’s `Production` field to select the physical construction path, and the `Extraction` and `ColourNorm` fields to select the extraction mode and the colour normalisation used for each antenna. The colour normalisation used by default has been described previously in Eq. (2.52), and the `Extraction` mode also introduces the Born normalisation. The routes selected in the `Production` and `Extraction` fields depend on the methodology required to compute the antenna functions and retrieve the components, if required. The public tree-level antenna functions are constructed using different production and extraction modes, as described in Table 3.3.

| Production Mode               | Extraction Mode            | Current Route          |
|-------------------------------|----------------------------|------------------------|
| SelfInterference              | BornScalar                 | $A_2^0$                |
| SelfInterference              | TreeColourCoefficients     | $A_3^0$                |
| ColourOrderedAntenna          | TreeColourCoefficients     | $A_4^0, \tilde{A}_4^0$ |
| SectorSelfInterference        | ColourNormScalar           | $B_4^0$                |
| SectorSymmetrisedInterference | MinusTwoColourNormOverSUNN | $C_4^0$                |
| SelfInterference              | TreeColourCoefficients     | $A_3^{0,(m_q)}$        |

**Table 3.3:** Overview of the tree-level production and extraction modes.

These production paths all lead to the extraction of the required antenna functions, but the way in which they do it changes with the antenna’s multiplicity. Each production and extraction mode is *AntCalc*’s operational implementation of the construction derived, in mathematical terms, for the corresponding *Current Route* entry in Subsection 2.4.4. On the extraction side,  $A_2^0$  is set to 1; both  $A_3^0$  and the  $A_4^0$ -set antennae are extracted from the usual  $NX_n^l + \frac{1}{N}\tilde{X}_n^l$  structure; and the  $B_4^0$  and  $C_4^0$  antennae are extracted from the more involved inner structure of the  $M_{4q}^0$  full-process amplitude, requiring a correspondingly more complex extraction process.

### 3.3.2 One-Loop Routes

One-loop routes are dispatched to `BuildLoopAntennaData` where the primary backend reduction method is selected through the nested `ReductionProfile`, which by default is Passarino-Veltman reduction, as described in Section B.2.

The production method consists of generating the tree-level and one-loop amplitudes, using `TreeAmplitude` and `MAmpOneLoop`, respectively, and then forming the tree-loop interference. The interference is colour- and one-loop-normalised, as described in Eqs. (2.20) and (2.52). This normalised

interference is then reduced and its components extracted. The production and extraction modes implemented for the supported one-loop routes are summarised in Table 3.4, below.

| Production Mode             | Extraction Mode        | Current Route                       |
|-----------------------------|------------------------|-------------------------------------|
| TreeAmplitude + MAmpOneLoop | LoopScalar             | $A_2^1$                             |
| TreeAmplitude + MAmpOneLoop | LoopColourCoefficients | $A_3^1, \tilde{A}_3^1, \hat{A}_3^1$ |

**Table 3.4:** Overview of the one-loop production and extraction modes.

The two one-loop routes make use of a hybrid production mode that combines the `TreeAmplitude` and `MAmpOneLoop` objects. As shown in Table 3.2, the former is  $\mathcal{M}_n^0$ , while the latter is its one-loop counterparts  $\mathcal{M}_n^1$ . These two amplitudes are used as described in Subsection 2.4.4. Their extraction modes are the one-loop analogous to the methods used for  $A_2^0$  and  $A_3^0$ , as discussed in Subsection 3.3.1.

### 3.3.3 Two-Loop Routes

Loop-order-two routes are dispatched to `BuildTwoLoopAntennaData`. Within the massless  $q\bar{q}$  final-final antennae considered up to NNLO, the only two-loop antenna set is  $A_2^2$ . Much like for all other loop routes, Passarino-Veltman reduction is used on the build side to remove the two loop momenta  $\ell_1$  and  $\ell_2$ , while IBP reduction is subsequently used to obtain the full result in terms of its  $\epsilon$ -poles. It is remarkably different from the rest, however. The four-component set is not generated like the  $A_4^0$  and  $A_3^1$  sets were, rather its four components come from two distinct source contributions, as discussed in Subsection 2.4.4. Their production and extraction modes are similar and handled as a bundle inside *AntCalc*, but, as shown in Table 3.5, their sources differ,

| Production Mode | Extraction Mode        | Current Route                       |
|-----------------|------------------------|-------------------------------------|
| TwoLoopInterf   | TwoLoopTTermComponents | $A_2^2, \tilde{A}_2^2, \hat{A}_2^2$ |
| OneLoopSelf     | TwoLoopTTermComponents | $\check{A}_2^2$                     |

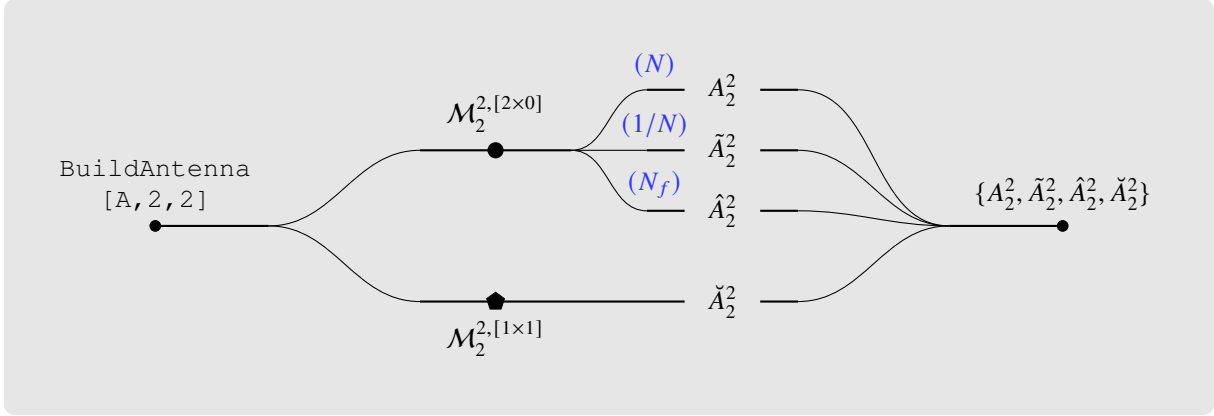
**Table 3.5:** Overview of the two-loop contributions and extraction modes.

One of the ways in which this route is particular inside *AntCalc*'s architecture is illustrated by their different production, but equal extraction modes. This route may be schematically represented as shown in Fig. 3.3, below.

The programme reads the profile and generates the two necessary amplitudes, for the two separate loops and for the one-loop self-interference case,  $\mathcal{M}_2^{2,[2\times 0]}$  and  $\mathcal{M}_2^{2,[1\times 1]}$ . The former amplitude generates three antenna-set components, while the latter produces only one, but all four are extracted using the same usual method used for tree-level and one-loop antenna functions, applied to two loops.

## 3.4 From the Build Stage to Integration

The build stage described above produces the unintegrated antenna functions in terms of the external kinematic invariants  $s_{ij}$ . This is the natural public representation of the antenna and is suitable for its use as a local subtraction term in a numerical phase-space integration, as discussed in Chapter 2. A standard `BuildAntenna[...]` call therefore returns either a single antenna expression or, where several antenna-set components are present, an ordered list made up of those components.



**Figure 3.3:** Diagram showing the build-side workflow leading to the generation of the  $A_2^2$  antenna set. The two subsets are separated by the interferences used in their generation.

For the analytic calculation of the integrated antennae, however, the final expression in terms of Mandelstam invariants is not by itself sufficient. The integration stage requires additional information from the construction of the antenna, including the route definition and the integral structures needed for the subsequent reduction and phase-space integration. *AntCalc* therefore provides the option `IntegrableForm -> True`. With this option, `BuildAntenna[...]` returns an `AntennaObject` that retains both the constructed antenna and the additional information required by `IntegrateAntenna[...]`. This object provides the interface between the build and integration stages.

At this point, *AntCalc* can therefore provide the unintegrated antennae in the kinematic form required for local subtraction. The next stage addresses the complementary part of the antenna subtraction procedure: their analytic integration over the unresolved phase space in dimensional regularisation. These integrated antennae are required to combine the subtraction terms with the virtual contributions and achieve the cancellation of explicit infrared poles. The implementation of this integration stage is described in the following section.

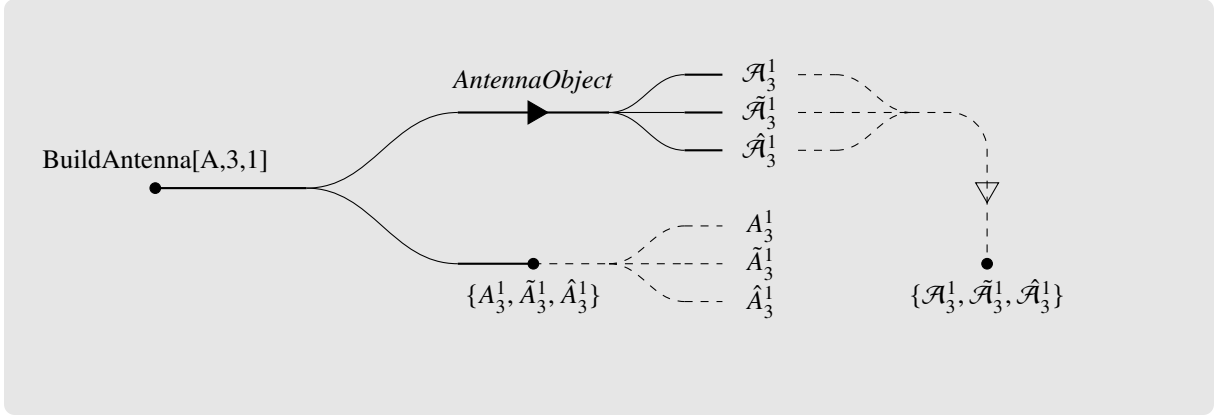
### 3.5 The Integration Stage

The integration stage performs the analytic integration of the antenna functions constructed in the build stage over the corresponding unresolved phase space in  $d = 4 - 2\epsilon$  dimensions, as in Eq. (2.71). The integrated antennae are denoted with calligraphic letters  $\mathcal{X}_n^l$ .

Within *AntCalc*, this stage is initiated by passing the `AntennaObject` produced by `BuildAntenna[...]` to `IntegrateAntenna[...]`. A diagram representing this architecture and how it splits from an unchanged `BuildAntenna[...]` call is shown in Figure 3.4.

The route key stored in this object determines the appropriate integration workflow, including the required reduction procedure, master integrals, and normalisation conventions. After these steps have been applied, `IntegrateAntenna[...]` returns the integrated antenna, or the corresponding set of antenna components, expanded as a Laurent series in  $\epsilon$ . The integration workflow is structured as follows,

```
IntegrateAntenna[AntennaObject]
-> recover route key and resolve profile;
```



**Figure 3.4:** Diagram of the build-only and build-and-integrate routes: the combined route is shown above, the build-only route below. The dashed lines mark the options to separate the result into single components (build-only route) or merge them into one ordered list (combined route). The  $A_3^1$  antenna set is used as an example.

```
-> choose integration backend;
-> complete IBP/Passarino-Veltman reduction;
-> substitute master integrals when applicable;
-> extract public integrated antennae;
-> output: selected component, list, diagnostics, run record, or
    master-combination view.
```

It is relevant to note that `IntegrateAntenna` requires the profile to complete its function like `BuildAntenna`, since it also encodes the relevant routing information for the reduction and integration process.

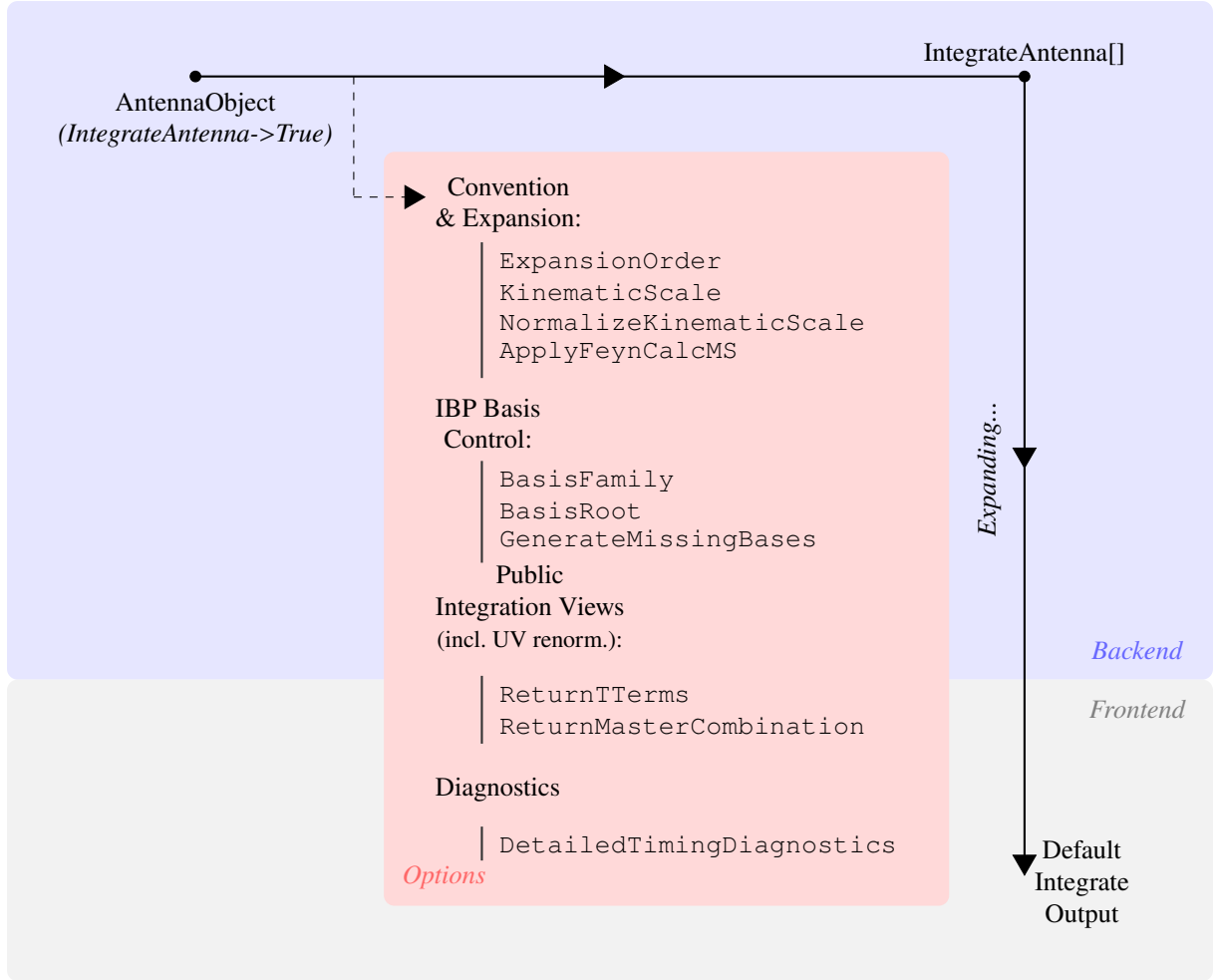
Much like for `BuildAntenna`, the route state is not held in a single context object but spread across several objects, accessed as required: the key and profile, the input (unintegrated) antenna and selected component, the IBP bases and master integrals, the series result, and the diagnostics.

An integration-stage-specific diagram in the spirit of the one in Fig. 3.1 can also be drawn in further detail. That diagram is shown in Fig. 3.5.

As for the build stage, ultraviolet renormalisation at the integration stage is applied independently, at the analogous public-output boundary: once the integrated one- and two-loop antennae are extracted, immediately before they are returned by `IntegrateAntenna[...]`. This mirrors, rather than reuses, the renormalisation applied during the build stage, since the two calls can be made independently of one another. No separate renormalisation step therefore appears in Fig. 3.5 either: it coincides with the “Public Integration Views” crossing shown there.

### 3.5.1 Integration Profiles and Selection

The integration-specific profile for each route selects the kind of backend process (and external package workflow used), and the basis family required for the process. The antennae are divided by topology, that is, number of final-state particles and number of loops, and all of those that share the same topology in the current implementation, share the same basis-family set. Since IBP reduction acts on the kinematic integral structure and is independent of the colour coefficients, the components of a set are reduced separately. *AntCalc* can therefore integrate the complete set or a single selected component, avoiding unnecessary reductions. The exception to this rule is the  $A_2^2$  set, in which different sets are used depending on configuration. A diagram of this route’s multi-configuration structure is shown in



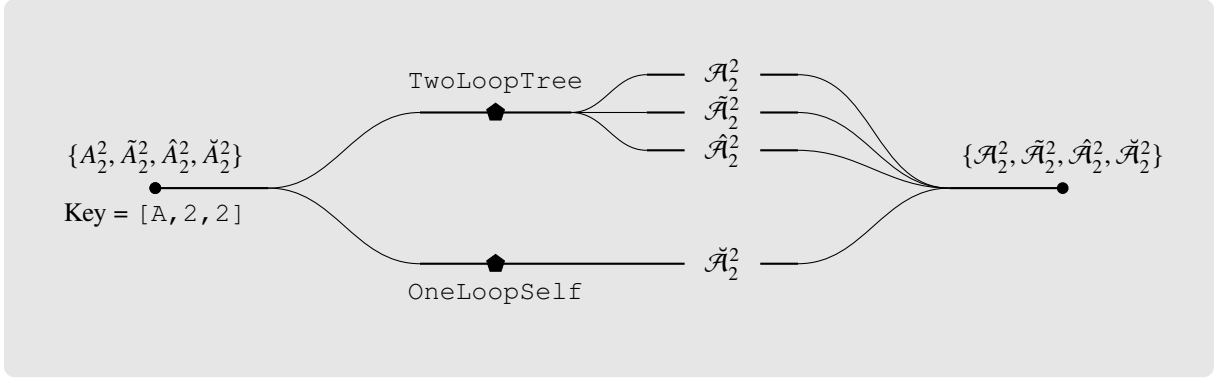
**Figure 3.5:** Profile-driven architecture of *AntCalc*'s Integration stage. The principal integration-only options are shown inside the red card.

Figure 3.6; comparing it with the upper path of Figure 3.4 shows how the  $A_2^2$  route differs from the single-family routes used for  $A_3^1$  and most others.

The  $A_2^1$  antenna is also an exception to the use of integral families: its loop integrals are reduced directly to scalar one-loop integrals using Passarino-Veltman reduction, whose known analytic expressions are then substituted and expanded as Laurent series in  $\epsilon$ . An overview of these is shown in the following Table 3.6.

| Default Backend                        | Basis-Family Set           | Current Route                                                                |
|----------------------------------------|----------------------------|------------------------------------------------------------------------------|
| Passarino-Veltman / <i>FeynHelpers</i> | massless two-parton vertex | $\mathcal{A}_2^1$                                                            |
| IBP / <i>LiteRed2</i>                  | X30                        | $\mathcal{A}_3^0$                                                            |
| IBP / <i>LiteRed2</i>                  | X40                        | $\mathcal{A}_4^0, \tilde{\mathcal{A}}_4^0, \mathcal{B}_4^0, \mathcal{C}_4^0$ |
| IBP / <i>LiteRed2</i>                  | A31                        | $\mathcal{A}_3^1, \tilde{\mathcal{A}}_3^1, \hat{\mathcal{A}}_3^1$            |
| IBP / <i>LiteRed2</i>                  | A22TwoLoopTree             | $\mathcal{A}_2^2, \tilde{\mathcal{A}}_2^2, \hat{\mathcal{A}}_2^2$            |
| IBP / <i>LiteRed2</i>                  | A22OneLoopSelf             | $\tilde{\mathcal{A}}_2^2$                                                    |
| IBP / <i>LiteRed2</i>                  | MX30                       | $\mathcal{A}_3^{0,(m_q)}$                                                    |

**Table 3.6:** Overview of the integration-side route-profiles.



**Figure 3.6:** Diagram showing the component extraction steps and integration-basis families used in the integration-side workflow of the  $A_2^2$  antenna set.

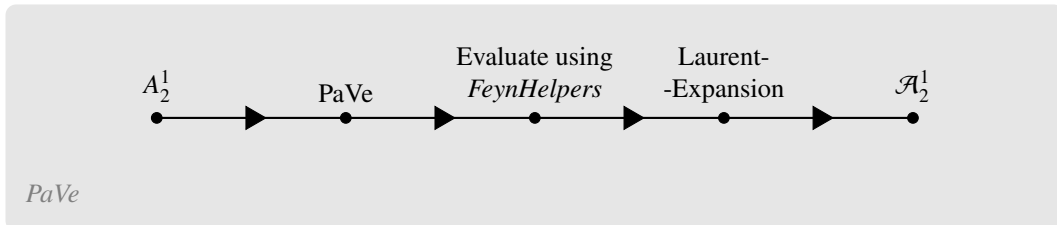
As on the build side, the integration profile does not itself perform any calculation; rather, it specifies the information required to execute the corresponding integration workflow. Consider, for example, the integration of the  $B_4^0$  antenna. From the route key stored in the `AntennaObject`, *AntCalc* identifies the appropriate integration profile and the integral family associated with the  $X_4^0$  phase-space topology. The unintegrated antenna is then mapped onto this integral family and reduced by IBP, using *LiteRed2*, to the corresponding set of master integrals. Their known analytic expressions are subsequently substituted to obtain the integrated antenna.

The master-integral basis used in these reductions is discussed in Chapter 4, where selected master integrals are evaluated explicitly and their role in the analytic integration of antenna functions is illustrated through representative examples.

### 3.5.2 Integration Routes

#### 3.5.2.1 Passarino-Veltman Route

The  $A_2^1$  antenna constitutes a special low-multiplicity virtual case and is the only integration route implemented without IBP reduction. As a two-parton one-loop form-factor contribution, its loop integrals can be treated directly using Passarino-Veltman reduction, which reduces the tensor loop integrals to scalar one-loop functions. *AntCalc* evaluates these scalar functions analytically using *Package-X* [26] through *FeynHelpers* [27]. The required branch and scale conventions are then applied, followed by the appropriate kinematic normalisation and Laurent expansion in  $\epsilon$ , yielding the integrated  $\mathcal{A}_2^1$  antenna. The workflow is summarised schematically in Fig. 3.7.

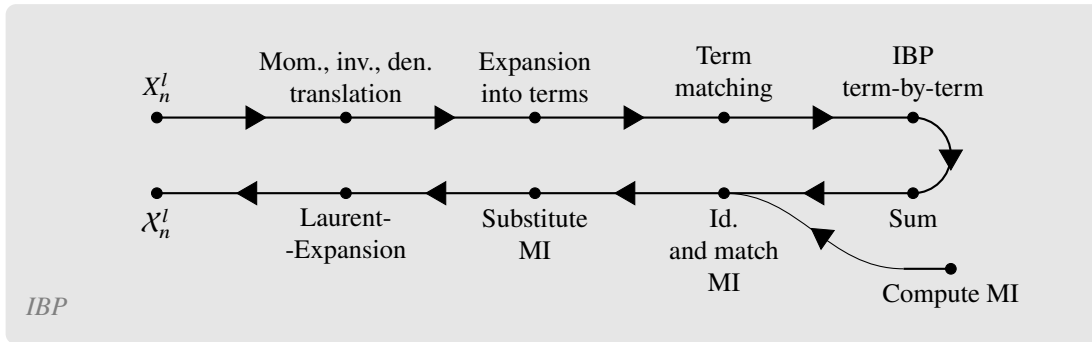


**Figure 3.7:** PaVe reduction for the  $A_2^1$  antenna integration.

### 3.5.2.2 IBP Route

All remaining antenna functions are integrated using IBP reduction, implemented through *LiteRed2* [23], following the formalism described in Section 2.5. The antenna integrand is first expressed in terms of the momenta, invariants, and denominator structures defining the relevant integral families. The appropriate phase-space cut propagators are then introduced, allowing the phase-space integrals to be treated within the same integral-family framework as loop integrals. After expanding the antenna into individual integral terms, each term is assigned to the corresponding integral family and reduced by *LiteRed2* to a linear combination of master integrals. The reduced terms are subsequently recombined to obtain the complete integrated antenna in terms of the corresponding master-integral basis.

For the public routes, the master integrals (MIs) identified by the IBP reduction are matched to their corresponding analytic expressions known from the literature and evaluated explicitly in this work. Their expressions are then substituted by *AntCalc*. Finally, the required normalisation and dimensional-regularisation conventions are applied, and the result is expanded as a Laurent series in  $\epsilon$ , yielding the integrated antenna function. Selected master integrals and explicit examples of their analytic evaluation are presented in Chapter 4, together with representative applications of this procedure to the integration of antenna functions. The complete IBP-based integration workflow is summarised schematically in Fig. 3.8.



**Figure 3.8:** IBP reduction workflow for the remaining antennae integrations. The MI computation is external to *AntCalc*’s internal operation, and hence is represented by an external incoming arrow.

## 3.6 Higher-Level Functions

Besides the two canonical functions, *BuildAntenna* and *IntegrateAntenna*, *AntCalc* also supports higher-level functions. This layer does not introduce new antenna-construction or integration methodology; instead it orchestrates the two canonical functions to compute observables such as the *R*-ratio, to assemble the topology-grouped combinations used in it, and to streamline repeated build or integrate evaluation. The five higher-level functions are:

- *BuildAndIntegrateAntenna* — chains *BuildAntenna*[..., *IntegrableForm* -> *True*] into *IntegrateAntenna* in a single call, forwarding the component-selection and Laurent-expansion-order controls to their respective stages.
- *BuildAllAntennae* — builds every public antenna function in one call.
- *BuildAndIntegrateAllAntennae* — builds and integrates every public antenna function in one call. Both bulk wrappers can be limited to a chosen perturbative order.

- `TObject` — assembles integrated antennae into the topology-grouped combinations used in the  $R$ -ratio, defined in Subsection 5.2.1 of Chapter 5.
- `BuildRRatio` — assembles the massless hadronic  $R$ -ratio from the integrated antenna ingredients required at the selected perturbative order.

The  $R$ -ratio produced by `BuildRRatio` is the observable used for *AntCalc*'s global validation. Called as `BuildRRatio[SMQCD, maxOrder -> ...]`<sup>2</sup>, it generates the  $R$ -ratio through NNLO and, by default, returns the finite coefficient after application of the observable convention bridge. This result has been validated against a literature reference target, as discussed in Chapter 5.

### 3.7 Validation and Development Diagnostics

As outlined in the introduction, the implementation is validated at two levels. Locally, each integrated antenna function is compared against its published Laurent expansion [5, 24] and, where applicable, against the expected eikonal factors and Altarelli–Parisi splitting functions in the relevant unresolved limits (Chapter 4). Globally, the integrated antennae are assembled into the inclusive  $R$ -ratio and compared with the known finite result (Chapter 5).

As discussed in Section 3.2, the route system can retain selected diagnostic information from an evaluation. That makes it easy to locate any inconsistency during a route calculation, which can then be inspected using the built-in diagnostics tool — for example, by retrieving the antenna ingredients of an  $R$ -ratio order and checking how their  $\epsilon$ -poles cancel.

The `Ingredients` association is only one part of the diagnostic information. For a complete retained history of a calculation, *AntCalc* can instead return an `AntennaRunRecord`, which records the relevant route stages and metadata. This information was used to trace the source of discrepancies during development; the resulting validation comparisons are presented in Chapter 5.

#### 3.7.1 Amplitude-Level and Release-Acceptance Validation

The checks described above validate an antenna function against an external reference: a published Laurent series, or a known finite observable. *AntCalc* also implements a complementary validation family that requires no such reference. The function `VerifyWardIdentity[type, numFinalParticles, loopOrder]` acts directly on the unsquared amplitude produced by the build stage, before spin summation, colour decomposition, or phase-space integration is performed. For a chosen external gluon leg, it replaces that gluon's polarisation vector by its own momentum,  $\varepsilon^\mu(k) \rightarrow k^\mu$ , while keeping every other external gluon polarisation explicitly transverse, and checks whether the resulting expression vanishes. This is the standard amplitude-level Ward identity, applied leg by leg to the raw generated amplitude rather than inferred from a property of the extracted or integrated antenna. It is currently applied to every massless public route with an external gluon leg, that is, to  $A_3^0$ ,  $A_4^0$  and the  $A_3^1$  set, and returns a structured pass, fail, or time-out report for each checked leg. Because it depends only on the internal gauge structure of the generated amplitude, and not on comparison with any external result, it certifies a property the literature-comparison checks above cannot: that the amplitude's diagrams were assembled with the correct relative signs and couplings before extraction even begins. In principle, this makes it applicable to a route for which no literature result exists at all.

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<sup>2</sup>The label `SMQCD` is not standard in the literature. It is used here to denote the massless QCD sector of the Standard Model.

Separately from this route-level testing, *AntCalc*'s development also relies on a release-acceptance practice. Before a version is prepared, every public route is re-evaluated in a fresh kernel, with any stored or cached intermediate result disabled, and its build, integration, and one-shot outputs are compared against each other and against the literature target discussed above. Each run is recorded, together with its timing and the specific checks it satisfied, as dated evidence. This practice has already caught, and allowed the correction of, release-preparation issues that were not visible from the final Laurent-series outputs alone.

### 3.7.2 Runtime and Memory Characteristics

To characterise the computational cost and performance of the implemented public routes, each antenna was evaluated in a fresh *Wolfram Mathematica* kernel. Package loading was excluded from the measurements. Both the build and integration stages were timed using the `AbsoluteTiming` function. Memory usage was measured using `MaxMemoryUsed[]`, relative to the initial kernel baseline. These measurements were taken using a MacBook Pro (Mac16.8), with an Apple M4 Pro 12 core (8 performance and 4 efficiency) CPU, and 24 GB unified RAM, running macOS 26.6.2 (build 25G83), *Wolfram Mathematica* 15.0.0 for Mac OS X ARM (64-bit), and *AntCalc* 0.3.0 $\beta$ 2.

The lower-order routes, up to NLO, complete both stages in approximately one second, with the cost increasing substantially for the NNLO routes, in particular  $A_2^2$ , which surpasses 31 minutes. The  $A_4^0$  route has the largest memory requirement, reaching 567 MB above the kernel benchmark. The complete results are shown in Table 3.7.

| Route           | Build time | Integration time | End-to-end time | Peak memory above baseline |
|-----------------|------------|------------------|-----------------|----------------------------|
| $A_2^0$         | 0.09 s     | 0.49 s           | 0.58 s          | 22.7 MiB                   |
| $A_3^0$         | 0.30 s     | 0.85 s           | 1.15 s          | 24.8 MiB                   |
| $A_2^1$         | 0.57 s     | 0.40 s           | 0.98 s          | 19.3 MiB                   |
| $A_3^1$         | 74.67 s    | 15 min 27 s      | 16 min 42 s     | 195.3 MiB                  |
| $A_2^2$         | 21.79 s    | 31 min 06 s      | 31 min 27 s     | 227.7 MiB                  |
| $A_4^0$         | 3 min 03 s | 13 min 51 s      | 16 min 53 s     | 567.3 MiB                  |
| $B_4^0$         | 8.65 s     | 17.27 s          | 25.92 s         | 143.2 MiB                  |
| $C_4^0$         | 5.56 s     | 5 min 26 s       | 5 min 32 s      | 174.5 MiB                  |
| $A_3^{0,(m_q)}$ | 2.81 s     | 0.25 s           | 3.09 s          | 12.07 MiB                  |

**Table 3.7:** Representative resource measurements for the implemented public antenna routes. The build time measures `BuildAntenna[... , IntegrableForm -> True]`, followed by `IntegrateAntenna[...]` for the integration time. Each route was evaluated in a fresh *Wolfram* kernel. Peak memory is the maximum kernel-memory increase relative to the initial baseline.

## Chapter 4

# Antenna Construction and Integration: Selected Case Studies

The methods *AntCalc* uses to build and then integrate the antenna functions described in Chapter 2 were discussed in detail in Chapter 3. In this chapter we will construct all antenna functions that *AntCalc* is currently able to handle, by first building their unintegrated form, then their master combination form, that is before the master integral substitutions, and finally the  $\epsilon$ -series integrated antenna form.

The chapter will begin with a thorough example of the complete building and integration of an antenna, the  $A_3^0$  antenna — first by hand, and then using *AntCalc*, comparing the results at relevant steps and ensuring their equality. Finally, the unintegrated antenna is locally validated by computing its soft and collinear limits to yield the eikonal factor and the Altarelli–Parisi splitting function. This local validation method can also be used with other antenna functions whenever the corresponding unresolved limits are present. This first section is intended to showcase the inner algebraic steps *AntCalc* pursues internally. After this longer section, several smaller sections follow, each giving a much abbreviated walkthrough of the building, master combination, and integration of the remaining implemented antenna functions. At the very end, the one massive antenna,  $A_3^0$ , is also explored in some detail, and the differences between its route and that of its massless analogue are compared.

The massless antenna results shown in the present chapter were validated against the literature and, in Chapter 5, were globally validated by computing the hadronic massless  $R$ -ratio up to NNLO. A complete look through all antennae considered and used in the  $R$ -ratio, plus the massive extension, alongside their respective processes, and other relevant information is provided at the very end of this chapter in Section 4.6.

*AntCalc* makes use of different master integrals for each antenna, but it is noteworthy that components of the same antenna set share the same overall pool of master integrals. Table 4.1, below, shows the build-side and integration-side master integrals used by each antenna function, and it is visible that sets like  $A_2^2$  or  $A_3^1$  share the same overall pool of masters.

As described in Chapter 3, *AntCalc* makes use of two distinct integral reduction methods: IBP and PaVe. The second one is well established and used only for reducing unintegrated one- and two-loop antennae to their canonical form. As such, its method and its master integrals are discussed only in Appendix B.2. In this chapter, after each family is introduced and discussed in its own section, a final subsection discussing the master integrals used within it is present. This is due to the way in which the package builds the complete set at once, and only separates them at the end, before integrating. A lengthier step-by-step derivation of one of these masters,  $R_{8,a}$ , in this case, is present in Subsubsection 4.4.5.3.

| Antenna type(s)                      | Build-Side MI                   | Integration-Side MI          |
|--------------------------------------|---------------------------------|------------------------------|
| $A_2^0$                              | —                               | —                            |
| $A_3^0$                              | —                               | $R_3$                        |
| $A_4^0, \tilde{A}_4^0, B_4^0, C_4^0$ | —                               | $R_4, R_6, R_{8,a}, R_{8,b}$ |
| $A_2^1$                              | —                               | $B_0, C_0$                   |
| $A_3^1, \tilde{A}_3^1, \hat{A}_3^1$  | $B_0, C_0, D_0$                 | $V_{5,a}, V_{5,b}, V_8$      |
| $A_2^2$                              | $A_{22,LO}, A_3, A_4, B_0, C_0$ | $A_{22,LO}, A_3, A_4$        |
| $\tilde{A}_2^2$                      | $A_{22,LO}, A_3, A_4, A_6$      | $A_{22,LO}, A_3, A_4, A_6$   |
| $\hat{A}_2^2$                        | $A_4, B_0, C_0$                 | $A_4$                        |
| $\check{A}_2^2$                      | $A_{22,LO}$                     | $A_{22,LO}$                  |

**Table 4.1:** Overview of the master integrals used in the build- and integration-stage by each antenna type [5, 13].

These IBP MI are computed over different phase-spaces, depending on the number of final-state particles. These phase-space integrals are defined over all  $n$ -particle Mandelstam invariants for each specific phase-space. For that reason, we introduce the shorthands,

$$\begin{aligned}
s_{12} &= s_{12}, & d\mathcal{S}_2 &= ds_{12} \\
s_{123} &= s_{12} + s_{13} + s_{23}, & d\mathcal{S}_3 &= ds_{12}ds_{13}ds_{23} \\
s_{1234} &= s_{12} + s_{13} + s_{14} + s_{23} + s_{24} + s_{34}, & d\mathcal{S}_4 &= ds_{12}ds_{13}ds_{14}ds_{23}ds_{24}ds_{34}.
\end{aligned} \tag{4.1}$$

Using the shorthands, the differential phase-spaces are defined in  $d$  dimensions as [13, 16],

$$\begin{aligned}
d\Phi_2 &= (2\pi)^{2-d} s_{12}^{\frac{d-4}{2}} 2^{1-d} d\Omega_{d-1} d\mathcal{S}_2 \delta(q^2 - s_{12}) \\
d\Phi_3 &= (2\pi)^{3-2d} 2^{-1-d} (q^2)^{2-\frac{d}{2}} (s_{12}s_{13}s_{23})^{\frac{d-4}{2}} d\Omega_{d-1} d\Omega_{d-2} d\mathcal{S}_3 \delta(q^2 - s_{123}) \\
d\Phi_4 &= (2\pi)^{4-3d} (q^2)^{3-\frac{d}{2}} 2^{1-2d} (-\Delta_4)^{\frac{d-5}{2}} \Theta(-\Delta_4) \\
&\quad d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} d\mathcal{S}_4 \delta(q^2 - s_{1234}).
\end{aligned} \tag{4.2}$$

Here  $\Delta_4$  is the Gram determinant given by the Källén function  $\lambda$  [13, 16],

$$\Delta_4 = \lambda(s_{12}s_{34}, s_{13}s_{24}, s_{14}s_{23}), \quad \lambda(x, y, z) = x^2 + y^2 + z^2 - 2(xy + xz + yz). \tag{4.3}$$

Also,  $\Theta$  is the Heaviside step function and  $d\Omega$  the differential solid hyperspherical angle element in  $d$  dimensions, given as,

$$\Omega_n = \int d\Omega_n = \frac{2\pi^{\frac{n}{2}}}{\Gamma(n/2)}. \tag{4.4}$$

Some MI appear normalised using the  $S_\Gamma$  term, to make the infrared singularity cancellations explicit and absorb common dimensional factors. The normalising term is given as [24, 13],

$$S_\Gamma = \left( \frac{(4\pi)^\epsilon}{16\pi^2 \Gamma(1-\epsilon)} \right)^2. \tag{4.5}$$

It is also common to normalise with

$$S_{\Gamma,2} = P_2 S_{\Gamma}, \quad P_2 = \int d\Phi_2 = \frac{2^{-3+2\epsilon} \pi^{-1+\epsilon} \Gamma(1-\epsilon)}{\Gamma(2-2\epsilon)} (q^2)^{-\epsilon}. \quad (4.6)$$

To provide a general result and prevent notation conflicts, *AntCalc* returns its master integral combinations as expressions including  $j[\dots]$  functions, the default notation of *LiteRed2*. In this chapter we shall, at the master combination stage for each antenna, include a substitution of those functions for their general notation forms ( $A_3$ ,  $A_4$ ,  $R_3$ , etc.). The substitution arrays used are shown in full in Appendix B.7.

## 4.1 NLO Antenna Results

The simplest non-trivial SMQCD antenna is  $A_3^0$ . As such, it may be used as an example of how *AntCalc* handles the physics involved in building and then integrating it over the phase space. In this chapter, we are going to go over how this antenna is mathematically computed by hand, and how *AntCalc* handles those same calculations on its end.

### 4.1.1 The Tree-Level Three-Parton Antenna $A_3^0$

#### 4.1.1.1 Analytic Construction of the Unintegrated $A_3^0$ Antenna

The  $A_3^0$  is the tree-level final-final quark-antiquark-gluon antenna. It represents the normalised  $\gamma^*(p) \rightarrow q(k_1)g(k_3)\bar{q}(k_2)$  process. This process, in the current model, is able to produce two distinct topologies which differ by what quark line the gluon is emitted from. Those diagrams are shown below in Fig. 4.1.



(a) Gluon emitted from the quark line.

(b) Gluon emitted from the antiquark line.

**Figure 4.1:** Feynman diagrams depicting the two topologies described under the  $A_3^0$  antenna, corresponding to the process  $\gamma^* \rightarrow q(k_1)g(k_3)\bar{q}(k_2)$ .

The amplitude is given as follows directly from the diagrams,

$$\mathcal{M}_3^0 = e_q g_s (T^{a_3})_{i_1 i_2} \left[ \bar{u}(k_1) \not{\epsilon}^*(k_3) \frac{\not{k}_1 + \not{k}_3}{s_{13}} \not{\epsilon}(p) v(k_2) - \bar{u}(k_1) \not{\epsilon}^*(p) \frac{\not{k}_2 + \not{k}_3}{s_{23}} \not{\epsilon}(k_3) v(k_2) \right] \quad (4.7)$$

where  $(T^{a_3})_{i_1 i_2}$  represents the colour matrix for the quark and antiquark in positions 1 and 2 and gluon in position 3.

The amplitude is now squared as follows,

$$\begin{aligned}
|\mathcal{M}_3^0|^2 &= e_q^2 g_s^2 (T^{a_3})_{i_1 i_2} (T^{a_3})_{i_1 i_2} \left( M_A M_A^\dagger + M_B M_B^\dagger - M_A M_B^\dagger - M_B M_A^\dagger \right) \\
&\text{with } M_A = \bar{u}(k_1) \not{\epsilon}^*(k_3) \frac{\not{k}_1 + \not{k}_3}{s_{13}} \not{\epsilon}(p) v(k_2) = \bar{u}(k_1) \Gamma_A v(k_2), \\
&M_B = \bar{u}(k_1) \not{\epsilon}^*(p) \frac{\not{k}_2 + \not{k}_3}{s_{23}} \not{\epsilon}(k_3) v(k_2) = \bar{u}(k_1) \Gamma_B v(k_2).
\end{aligned} \tag{4.8}$$

which in total becomes

$$\begin{aligned}
&= \sum_{a_3, i_1, i_2} (T^{a_3})_{i_1 i_2} (T^{a_3})_{i_1 i_2} \\
&\quad \sum_{s_1, s_2, \lambda_g, \lambda_\gamma} \bar{u}(k_1, s_1) u(k_1, s_1) \left[ \Gamma_A \bar{\Gamma}_A + \Gamma_B \bar{\Gamma}_B - \Gamma_A \bar{\Gamma}_B - \Gamma_B \bar{\Gamma}_A \right] v(k_2, s_2) \bar{v}(k_2, s_2)
\end{aligned} \tag{4.9}$$

Each of the  $\Gamma$  terms becomes, using Dirac Algebra,

$$\begin{aligned}
&\sum_{s_1, s_2, \lambda_g, \lambda_\gamma} \bar{u}(k_1, s_1) u(k_1, s_1) (\Gamma_A \bar{\Gamma}_A) v(k_2, s_2) \bar{v}(k_2, s_2) = \\
&= \frac{1}{s_{13}^2} \sum_{\lambda_g, \lambda_\gamma} \varepsilon_\mu^* \varepsilon_\nu \varepsilon_\rho \varepsilon_\sigma^* \text{Tr} \left( \not{k}_1 \gamma^\mu (\not{k}_1 + \not{k}_3) \gamma^\rho \not{k}_2 \gamma^\sigma (\not{k}_1 + \not{k}_3) \gamma^\nu \right) \\
&= \frac{1}{s_{13}^2} g_{\mu\nu} g_{\rho\sigma} \text{Tr} \left( \not{k}_1 \gamma^\mu (\not{k}_1 + \not{k}_3) \gamma^\rho \not{k}_2 \gamma^\sigma (\not{k}_1 + \not{k}_3) \gamma^\nu \right) \\
&= \frac{1}{s_{13}^2} \text{Tr} \left( \not{k}_1 \gamma^\mu (\not{k}_1 + \not{k}_3) \gamma^\rho \not{k}_2 \gamma_\rho (\not{k}_1 + \not{k}_3) \gamma_\mu \right) \\
&= \left( \frac{2-d}{s_{13}} \right)^2 \text{Tr}(\not{k}_1 \not{P} \not{k}_2 \not{P}), \quad \not{P} = (\not{k}_1 + \not{k}_3) \\
&= 4 \left( \frac{2-d}{s_{13}} \right)^2 \left( 2(k_1 \cdot P)(k_2 \cdot P) - (k_1 \cdot k_2) P^2 \right) \\
&= 2(d-2)^2 \frac{s_{23}}{s_{13}}
\end{aligned} \tag{4.10}$$

$$\begin{aligned}
&\sum_{s_1, s_2, \lambda_g, \lambda_\gamma} \bar{u}(k_1, s_1) u(k_1, s_1) (\Gamma_B \bar{\Gamma}_B) v(k_2, s_2) \bar{v}(k_2, s_2) = \\
&= 2(d-2)^2 \frac{s_{13}}{s_{23}}
\end{aligned} \tag{4.11}$$

$$\begin{aligned}
&\sum_{s_1, s_2, \lambda_g, \lambda_\gamma} \bar{u}(k_1, s_1) u(k_1, s_1) (\Gamma_A \bar{\Gamma}_B) v(k_2, s_2) \bar{v}(k_2, s_2) = \\
&= 2(d-2) \frac{(d-4)s_{13}s_{23} - 2s_{12}^2 - 2s_{12}(s_{13} + s_{23})}{s_{13}s_{23}}
\end{aligned} \tag{4.12}$$

$$\begin{aligned}
\sum_{s_1, s_2, \lambda_g, \lambda_\gamma} \bar{u}(k_1, s_1) u(k_1, s_1) (\Gamma_B \bar{\Gamma}_A) v(k_2, s_2) \bar{v}(k_2, s_2) = \\
= 2(d-2) \frac{(d-4)s_{13}s_{23} - 2s_{12}^2 - 2s_{12}(s_{13} + s_{23})}{s_{13}s_{23}}
\end{aligned} \tag{4.13}$$

and computing the colour matrix sum as

$$\sum_{a_3, i_1, i_2} (T^{a_3})_{i_1 i_2} (T^{a_3})_{i_1 i_2} = \frac{1}{2} (N^2 - 1) \tag{4.14}$$

we are finally able to reach the final result, as follows, using dimensional regularisation as  $d \rightarrow 4 - 2\epsilon$ ,

$$\begin{aligned}
|\mathcal{M}_3^0|^2 &= e_q^2 g_s^2 (N^2 - 1) \left( 2(d-2)^2 \left( \frac{s_{13}}{s_{23}} + \frac{s_{23}}{s_{13}} \right) - 4(d-2) \frac{(d-4)s_{13}s_{23} - 2s_{12}^2 - 2s_{12}(s_{13} + s_{23})}{s_{13}s_{23}} \right) \\
&= 2(d-2) e_q^2 g_s^2 (N^2 - 1) \left( (d-2) \left( \frac{s_{13}}{s_{23}} + \frac{s_{23}}{s_{13}} \right) - 2 \frac{(d-4)s_{13}s_{23} - 2s_{12}^2 - 2s_{12}(s_{13} + s_{23})}{s_{13}s_{23}} \right) \\
&= \frac{8}{s_{13}s_{23}} \left( 2q^2 s_{12} + s_{13}^2 + s_{23}^2 - 2\epsilon \left( q^2 s_{12} + s_{13}^2 + s_{23}^2 + s_{13}s_{23} \right) + \epsilon^2 (s_{13} + s_{23})^2 \right)
\end{aligned} \tag{4.15}$$

The  $A_3^0$  antenna also requires the  $\mathcal{M}_2^0$  amplitude. This amplitude corresponds to the  $\gamma^* \rightarrow q\bar{q}$  Born process. The amplitude is

$$\mathcal{M}_2^0 = e_q \delta_{i_1 i_2} \bar{u}(k_1) \not{\epsilon}(p) v(k_2). \tag{4.16}$$

Applying the same spin, polarisation and colour sums as above, and considering that, for a two final-state particles system  $s_{12} = q^2$ , we get

$$\begin{aligned}
|\mathcal{M}_2^0|^2 &= 4e_q^2 (d-2) N s_{12} = 4e_q^2 (d-2) N q^2 \\
&= 8e_q^2 (1 - \epsilon) N q^2
\end{aligned} \tag{4.17}$$

The antenna is now finally ready to be built. That is accomplished using the following formula,

$$A_3^0 = \frac{1}{2g_s^2 C_F} \frac{|\mathcal{M}_3^0|^2}{|\mathcal{M}_2^0|^2}. \tag{4.18}$$

By taking  $C_F = (N^2 - 1)/(2N)$ , the formula yields, using Eqs. (4.15) and (4.17),

$$\begin{aligned}
A_3^0 &= \frac{2q^2 s_{12} + s_{13}^2 + s_{23}^2}{q^2 s_{13} s_{23}} - \epsilon \frac{(s_{13} + s_{23})^2}{q^2 s_{13} s_{23}} \\
&= \frac{1}{q^2} \left[ \left( \frac{s_{13}}{s_{23}} + \frac{s_{23}}{s_{13}} + 2 \frac{s_{12} q^2}{s_{13} s_{23}} \right) - \epsilon \left( 2 + \frac{s_{13}}{s_{23}} + \frac{s_{23}}{s_{13}} \right) \right]
\end{aligned} \tag{4.19}$$

Note that this result is already fully normalised: the  $2g_s^2 C_F$  prefactor, together with the cancellation of the

electric charge  $e_q^2$  between  $|\mathcal{M}_3^0|^2$  and  $|\mathcal{M}_2^0|^2$ , strips away every coupling-, charge-, and colour-dependent factor carried by the raw amplitudes above, leaving the purely kinematic result in Eq. (4.19). This is the same normalisation already folded directly into the  $|\mathcal{M}_n^0|^2$  notation used throughout Chapter 2's general antenna-construction formalism (Subsection 2.4.4); the derivation here is kept fully explicit, coupling and charge included, to show each step by hand.

#### 4.1.1.2 Phase-Space Integration through $R_3$

We now integrate the  $A_3^0$  antenna over the appropriate (3 final-state particle, in this case) phase-space. This integral is computed in  $d$  dimensions and over the phase-space measure in terms of the Mandelstam invariants ( $s_{13}$ ,  $q^2$ , etc.). The integral takes the form,

$$\begin{aligned}\mathcal{A}_3^0 &= C(\epsilon, 1) \int d\Phi_{ijk} A_3^0 \\ &= \mathcal{N}(\epsilon, 1) (4\pi)^{3-2d} \left(q^2\right)^{1-\frac{d}{2}} \int ds_{12} ds_{13} ds_{23} d\Omega (s_{12}s_{13}s_{23})^{\frac{d-4}{2}} \delta\left(q^2 - s_{12} - s_{13} - s_{23}\right) A_3^0.\end{aligned}\tag{4.20}$$

with  $d\Phi_{ijk} = d\Phi_3/\Phi_2$ .  $C(\epsilon, 1)$  is the integral normalisation which is here given as

$$C(\epsilon, 1) = 8\pi^2 (4\pi)^{-\epsilon} e^{\epsilon\gamma_E}.\tag{4.21}$$

This integral is quite complex and spans across all  $A_3^0$  terms. Accordingly, we are not going to compute this integral directly: instead we are going to use IBP reduction to reduce the expression to a linear combination of master integrals, compute those master integrals and then use the result to compute the integrated antenna. This process is explained in Subsection 2.5.1.

After applying IBP identities and reducing the integral to a linear combination of this master integral, we reach the following expression,

$$\mathcal{A}_3^0 = \frac{C(\epsilon, 1)}{\Phi_2} \frac{2(2 - 6\epsilon + 5\epsilon^2 - 2\epsilon^3)}{q^2 \epsilon^2} R_3\tag{4.22}$$

This antenna has a single master,  $R_3$ , discussed in Section 4. This master integral is the integral over phase-space itself,

$$R_3 = \int d\Phi_3 = \frac{2^{-7+4\epsilon} \pi^{-3+2\epsilon} \Gamma^3(1-\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)} \left(q^2\right)^{1-2\epsilon}.\tag{4.23}$$

#### 4.1.1.3 The $R_3$ Integral

This MI is used only on the integration of the massless  $A_3^0$  antenna. It is the direct integral over the three-final-state-particle phase-space. It is written as [28],

$$\begin{aligned}R_3 &= \int d\Phi_3 \\ &= \frac{1}{4} \frac{\Omega_{d-1} \Omega_{d-2}}{(2\pi)^{2d-3}} \iiint dE_1 dE_2 d\theta_{12} (E_1 E_2 \sin \theta_{12})^{d-3} \delta(q^2 - s_{123}) \\ &= \frac{1}{128 q^2 \pi^3} \int d\mathcal{S}_3 \delta(q^2 - s_{123}).\end{aligned}\tag{4.24}$$

After analytical evaluation the integrated result, which is the one substituted in the final integrated antenna expression after IBP reduction becomes [13],

$$R_3 = 2^{-7+4\epsilon} \pi^{-3+2\epsilon} \frac{\Gamma^3(1-\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)} (q^2)^{1-2\epsilon}. \quad (4.25)$$

#### 4.1.1.4 IBP Reduction of an Example Term

As an example of the IBP reduction method, we will consider the eikonal term of the unintegrated antenna in Eq. (4.19),

$$\mathcal{S}_{12}^{(3)} = \frac{2s_{12}}{s_{13}s_{23}}. \quad (4.26)$$

We define,

$$\begin{aligned} y_{13} &= \frac{s_{13}}{q^2}, \quad y_{23} = \frac{s_{23}}{q^2}, \quad y_{12} = \frac{s_{12}}{q^2} = 1 - y_{13} - y_{23} \\ J(a, b, c) &= \int_{\Delta} dy_{13} dy_{23} y_{13}^{a-1} y_{23}^{b-1} y_{12}^{c-1}, \quad \Delta : y_{13}, y_{23}, y_{12} \geq 0, \quad y_{13} + y_{23} + y_{12} = 1 \end{aligned} \quad (4.27)$$

and with these,

$$\begin{aligned} \Phi_3 &\equiv \int d\Phi_3 = R_3 \\ &= \mathcal{K}(q^2, \epsilon) \int_{\Delta} (y_{13} y_{23} y_{12})^{-\epsilon} dy_{13} dy_{23} \\ &= \mathcal{K}(q^2, \epsilon) J(1-\epsilon, 1-\epsilon, 1-\epsilon), \quad \mathcal{K}(q^2, \epsilon) = \frac{2^{-7+4\epsilon} \pi^{-3+2\epsilon}}{\Gamma(2-2\epsilon)} (q^2)^{1-2\epsilon}. \end{aligned} \quad (4.28)$$

We are also able to express the integral over the eikonal term using a  $J$  function as,

$$\int d\Phi_3 \mathcal{S}_{12}^{(3)} = \frac{2\mathcal{K}}{q^2} J(-\epsilon, -\epsilon, 2-\epsilon). \quad (4.29)$$

The IBP identity we use to reduce the integral is implemented as,

$$\begin{aligned} 0 &= \int_{\Delta} dy_{13} dy_{23} \frac{\partial}{\partial y_{13}} (y_{13}^a y_{23}^{b-1} y_{12}^{c-1}) \\ &= aJ(a, b, c) - (c-1)J(a+1, b, c-1) \end{aligned} \quad (4.30)$$

such that, for  $y_{13}$ ,

$$J(-\epsilon, -\epsilon, 2-\epsilon) = -\frac{1-\epsilon}{\epsilon} J(1-\epsilon, -\epsilon, 1-\epsilon) \quad (4.31)$$

and for  $y_{23}$ ,

$$J(1-\epsilon, -\epsilon, 2-\epsilon) = -\frac{1-\epsilon}{\epsilon} J(1-\epsilon, 1-\epsilon, 1-\epsilon). \quad (4.32)$$

By the symmetry  $y_{13} \leftrightarrow y_{12}$ ,

$$J(2-\epsilon, -\epsilon, 1-\epsilon) = J(1-\epsilon, -\epsilon, 2-\epsilon). \quad (4.33)$$

And hence,

$$J(-\epsilon, -\epsilon, 2-\epsilon) = \frac{(1-\epsilon)(2-3\epsilon)}{\epsilon^2} J(1-\epsilon, 1-\epsilon, 1-\epsilon) \quad (4.34)$$

which finally makes,

$$\int d\Phi_3 \frac{2s_{12}}{s_{13}s_{23}} = \frac{2(1-\epsilon)(2-3\epsilon)}{\epsilon^2 q^2} R_3. \quad (4.35)$$

#### 4.1.1.5 Master Integral Substitution and Laurent Expansion

Substituting Eq. (4.23) into Eq. (4.22) gives

$$\begin{aligned} \mathcal{A}_3^0 &= \frac{C(\epsilon, 1)}{\Phi_2} (q^2)^{-2\epsilon} (4\pi)^{-3+2\epsilon} (2-6\epsilon+5\epsilon^2-2\epsilon^3) \frac{\Gamma^3(1-\epsilon)}{\epsilon^2 \Gamma(3-3\epsilon) \Gamma(2-2\epsilon)} \\ &= (q^2)^{-\epsilon} e^{\epsilon\gamma_E} (1-2\epsilon)(2-2\epsilon+\epsilon^2) \frac{\Gamma^2(-\epsilon)}{\Gamma(3-3\epsilon)} \end{aligned} \quad (4.36)$$

and, after expanding the integrated expression as a Laurent series in  $\epsilon$ , we finally arrive at the infrared pole structure,

$$\begin{aligned} \mathcal{A}_3^0 &= (q^2)^{-\epsilon} \left[ \frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{57-7\pi^2}{12} + \epsilon \left( \frac{84-21\pi^2-8\psi^{(2)}(1)+108\psi^{(2)}(3)}{24} \right) \right. \\ &\quad \left. + \epsilon^2 \left( \frac{35640-3990\pi^2-71\pi^4-720\psi^{(2)}(1)+9720\psi^{(2)}(3)}{1440} \right) \right] \\ &= (q^2)^{-\epsilon} \left[ \frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{19}{4} - \frac{7\pi^2}{12} + \epsilon \left( \frac{109}{8} - \frac{7\pi^2}{8} - \frac{25\zeta_3}{3} \right) + \epsilon^2 \left( \frac{639}{16} - \frac{133\pi^2}{48} - \frac{71\pi^4}{1440} - \frac{25\zeta_3}{2} \right) \right] \end{aligned} \quad (4.37)$$

where the usual polygamma identities,  $\psi^{(2)}(1) = -2\zeta_3$  and  $\psi^{(2)}(3) = -2\zeta_3 + \frac{9}{4}$ , were employed.

#### 4.1.1.6 Implementation in *AntCalc*

Using *AntCalc*, building the antenna is very simple, since the function `BuildAntenna[type, numFinalParticles, numLoops]` already encodes the entire logical chain and is able to return all intermediate steps using the appropriate options.

First, by running the function naively we get the final result in Eq. (4.19),

```
In[2] := BuildAntenna[A, 3, 0] // Collect[#, Epsilon] &
Out[2] :=
```

$$\frac{2s_{12}}{q^2 s_{13}} + \frac{2s_{12}}{q^2 s_{23}} + \frac{2s_{12}^2}{q^2 s_{13} s_{23}} + \frac{s_{13}}{q^2 s_{23}} + \frac{s_{23}}{q^2 s_{13}} + \epsilon \left( \frac{-2}{q^2} - \frac{s_{13}}{q^2 s_{23}} - \frac{s_{23}}{q^2 s_{13}} \right)$$

We can also use the `ReturnRecord` option to access all intermediate steps, and check that *AntCalc* goes through the same steps as the by-hand derivation in Subsection 4.1.1. We can see that with,

```
In[3] :=
BuildAntenna[A, 3, 0, ReturnRecord->True] ["IntermediateSteps"]
Out[3] :=
Amplitude:
```

$$T_{i_1 i_2}^{a_3} \left[ \frac{\bar{u}(k_1) \not{\epsilon}^*(k_3) (\not{k}_1 + \not{k}_3) \not{\epsilon}(p) v(k_2)}{(k_1 + k_3)^2} - \frac{\bar{u}(k_1) \not{\epsilon}(p) (\not{k}_2 + \not{k}_3) \not{\epsilon}^*(k_3) v(k_2)}{(k_2 + k_3)^2} \right]$$

Interference:

$$\frac{4(\epsilon - 1)(N^2 - 1) \left( -2s_{12}^2 - 2s_{12}(s_{13} + s_{23}) + (\epsilon - 1)s_{13}^2 + 2\epsilon s_{13}s_{23} + (\epsilon - 1)s_{23}^2 \right)}{s_{13}s_{23}}$$

Result:

$$-\frac{2\epsilon}{q^2} + \frac{2s_{12}^2}{q^2 s_{13}s_{23}} + \frac{2s_{12}}{q^2 s_{13}} + \frac{2s_{12}}{q^2 s_{23}} + \frac{s_{13}}{q^2 s_{23}} - \frac{\epsilon s_{13}}{q^2 s_{23}} + \frac{s_{23}}{q^2 s_{13}} - \frac{\epsilon s_{23}}{q^2 s_{13}}$$

where `Amplitude`, `Interference` and `Result` respectively refer to the expressions in Eqs. (4.7), (4.15) and (4.19).

Integrating antennae using *AntCalc* is also simple. Every step in Subsubsection 4.1.1.2 is handled by a single function: `IntegrateAntenna[antenna]`.

Before going over how this function is used, a small clarification needs to be made: the `IntegrateAntenna` function is not compatible with the default output of the `BuildAntenna` function described earlier. This is so that the built antenna is immediately useful and easily readable. As described in Chapter 3, however, the IBP reduction steps are handled by the `LiteRed2` package, which does not expect Mandelstam invariants, but rather, momenta. Accordingly, to use `BuildAntenna`'s output in `IntegrateAntenna` we must toggle the relevant flag: `BuildAntenna[A, 3, 0, IntegrableForm -> True]`. This will return a large association with all required information for integrating the antenna.

We may now proceed to the integration. The whole process goes as follows,

```
In[4]:= intA30 = BuildAntenna[A, 3, 0, IntegrableForm -> True];
In[5]:= IntegrateAntenna[intA30, ExpansionOrder->2]
Out[5]:=
```

$$\frac{19}{4} + \epsilon^{-2} + \frac{3}{2\epsilon} - \frac{7\pi^2}{12} + \epsilon^2 \left( \frac{639}{16} - \frac{133\pi^2}{48} - \frac{71\pi^4}{1440} - \frac{25\zeta_3}{2} \right) + \epsilon \left( \frac{109}{8} - \frac{7\pi^2}{8} - \frac{25\zeta_3}{3} \right)$$

Much like with `BuildAntenna`, `IntegrateAntenna` also allows us to inspect its intermediate steps. Here, however, the available options show the master integral linear combination expression both with  $R_3$  and with it substituted by the appropriate expression, that is, the equivalent to Eqs. (4.22) and (4.36) before normalisation. The code and results are,

```
In[6]:= resA30 = BuildAndIntegrateAntenna[A, 3, 0,
ReturnMasterCombination->True];
In[7]:= resA30//Simplify//Collect[#, R3]&
Out[7]:=
```

$$\frac{-4(d-3)(4+d(\epsilon-2)-4\epsilon)}{(d-4)^2 q^2} R_3$$

```
In[8]:= resA30["MasterSubstitutedExpression"]//Simplify
Out[8]:=
```

$$-\left(q^2\right)^{-2\epsilon} \frac{2^{-5+4\epsilon}(d-3)(4+d(\epsilon-2)-4\epsilon)\pi^{-3+2\epsilon}\Gamma^3(1-\epsilon)}{(d-4)^2\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)}$$

The results here are given as a combination of  $d$  and  $\epsilon$ . This is a result of the way the function processes each step. Since these are snapshots of what is happening inside the function itself, they show the expressions before dimension regularisation is applied.

Finally, *AntCalc* also enables us to forgo the work we did building and the integrating the antenna with the use of the `BuildAndIntegrateAntenna[type, numFinalParticles, numLoops]` function. It goes over the whole process, as we did, and outputs the  $\epsilon$  Laurent series,

```
In[9] := BuildAndIntegrateAntenna[A, 3, 0, ExpansionOrder -> 2]
Out[9] :=
```

$$\frac{19}{4} + \epsilon^{-2} + \frac{3}{2\epsilon} - \frac{7\pi^2}{12} + \epsilon^2 \left( \frac{639}{16} - \frac{133\pi^2}{48} - \frac{71\pi^4}{1440} - \frac{25\zeta_3}{2} \right) + \epsilon \left( \frac{109}{8} - \frac{7\pi^2}{8} - \frac{25\zeta_3}{3} \right)$$

The agreement between the analytic derivation and the corresponding *AntCalc* outputs validates both the construction and the integration workflow for  $A_3^0$ . Its local soft and collinear behaviour is validated separately in the following subsection.

### 4.1.2 Local Validation of $A_3^0$

To be usable as a subtraction term, an antenna function's unintegrated form must also reproduce the singular behaviour of the corresponding matrix element locally in phase space. In particular,  $A_3^0$  must reproduce the universal eikonal factor when the gluon becomes soft and the Altarelli-Parisi splitting function in the relevant collinear limits. These two quantities provide local validation of the results obtained analytically and using *AntCalc*.

#### 4.1.2.1 Soft Limit and the Eikonal Factor

When a gluon becomes soft, gauge theory amplitudes factorise universally to the amplitude without the soft gluon times a universal soft current as [8],

$$|\mathcal{M}_{n+1}(\dots, i, j, k, \dots)|^2 \xrightarrow{j \rightarrow 0} -g_s^2 \sum_{i,k} T_i \cdot T_k \mathcal{S}_{ik}^{(j)} |\mathcal{M}_n(\dots, i, k, \dots)|^2 \quad (4.38)$$

where  $T_i$  are colour-charge operators and,

$$\mathcal{S}_{ik}^{(j)} = \frac{2s_{ik}}{s_{ij}s_{jk}} \quad (4.39)$$

is the eikonal factor.

Since this property is universal, we can use it to validate our results. For the  $A_3^0$  antenna, the eikonal factor may be computed as,

$$\mathcal{S}_{12}^{(3)} = \lim_{k_3 \rightarrow 0} A_3^0(1_q, 3_g, 2_{\bar{q}}). \quad (4.40)$$

After building the antenna using *AntCalc*, the limit may be computed using *Wolfram Mathematica* as follows,

```
In[2] := A30 = BuildAntenna[A, 3, 0];

softSubs = {s13 -> lambda * s13, s23 -> lambda * s23};
A30Soft = A30 /. softSubs;

A30SoftSeries = Series[A30Soft, {lambda, 0, 0}];
```

```
eikonalFactor = SeriesCoefficient[A30SoftSeries, -2] /. q2 -> s12
// Simplify;
Out[2] :=
```

$$\frac{2s_{12}}{s_{13}s_{23}}$$

This result agrees with the expected eikonal factor.

#### 4.1.2.2 Collinear Limit and the Altarelli–Parisi Splitting Function

In the same manner, when two partons  $i$  and  $j$  become collinear, the amplitude factorises to [8, 29],

$$|\mathcal{M}_{n+1}(\dots, i, j, \dots)|^2 \xrightarrow{i||j} \frac{8\pi\alpha_s}{s_{ij}} P_{IJ \rightarrow K}(z) |\mathcal{M}_n(\dots, K, \dots)|^2 \quad (4.41)$$

where  $K$  is the parent parton in which  $i$  and  $j$  recombine,  $z$  is the momentum fraction carried by one of the daughter partons, and  $P_{IJ \rightarrow K}$  is the splitting function. This function is universal but channel-dependent and, since it carries the spin and helicity structure of the collinear splitting, it depends on the partons involved. For  $A_3^0$ , where a gluon may become collinear with a quark, the function appears as,

$$P_{qg \rightarrow q}(z) = C_F \frac{1+z^2}{1-z}. \quad (4.42)$$

Noting that the building process already normalises the fundamental Casimir  $C_F$ , using *AntCalc* in *Wolfram Mathematica*, the AP splitting function may be computed as,

```
In[3] := A30 = BuildAntenna[A, 3, 0];

collinearSubs = {s13 -> lambda, s12 -> z q2, s23 -> (1 - z) q2,
Epsilon -> 0};
A30Collinear = A30 /. collinearSubs;

A30CollinearSeries = Series[A30Collinear, {lambda, 0, 0}];

apSplittingFunc = SeriesCoefficient[A30CollinearSeries, -1]
// Simplify;
Out[3] :=
```

$$\frac{1+z^2}{1-z}$$

This result agrees with the expected colour-stripped  $qg \rightarrow q$  AP kernel.

#### 4.1.3 The Virtual $A_2^1$ Contribution

To compute  $A_2^1$ , we may follow the same steps. This is a loop-antenna, however, which means that its build result has been PaVe reduced. It is given in terms of the  $B_0$  and  $C_0$  scalar one-loop function. The unintegrated antenna is computed, using *AntCalc*, as,

```
In[2] := BuildAntenna[A, 2, 1]
```

```
Out[2] :=
```

$$-\frac{1}{2} \left[ (3 + 2\epsilon) B_0(q^2; 0, 0) \right] - q^2 C_0(0, 0, q^2; 0, 0, 0)$$

The integrated final result is, then [5],

```
In[3] := BuildAndIntegrateAntenna[A, 2, 1, ExpansionOrder->2]
```

```
Out[3] :=
```

$$-\frac{1}{\epsilon^2} - \frac{3}{2\epsilon} - 4 + \frac{7\pi^2}{12} + \epsilon \left( -8 + \frac{7\pi^2}{8} + \frac{7\zeta_3}{3} \right) + \epsilon^2 \left( -16 + \frac{7\pi^2}{3} - \frac{73\pi^4}{1440} + \frac{7\zeta_3}{2} \right)$$

Note that, exclusively for this antenna, the same PaVe scalar functions appear in both the build- and integration-side reductions. No phase-space IBP reduction is required.

## 4.2 Two-Parton Contributions: the $A_2^2$ Set

The  $A_2^2$  route is computed similarly to  $A_2^1$ . This route, however, yields four different antenna components and is the longest-running calculation supported by *AntCalc* in its current version. Each component is presented in turn below.

### 4.2.1 The $A_2^2$ Antenna

The leading unintegrated antenna is computed using,

```
In[2] := BuildAntenna[A, 2, 2, Component->Leading]
```

```
Out[2] :=
```

$$\frac{11 \left[ (3 + 2\epsilon) B_0(s_{12}; 0, 0) + 2s_{12} C_0(0, 0, s_{12}; 0, 0, 0) \right]}{12\epsilon} + \dots$$

The result in terms of the antenna's master integrals yields,

```
In[3] := A22Lead = BuildAndIntegrateAntenna[A, 2, 2, Component->Leading,
ReturnMasterCombination->True];
```

```
In[4] := (A22Lead/.A22MISubs)//Collect[#, A3, A4, A6, Simplify]&
```

```
Out[4] :=
```

$$-\frac{(-2\epsilon^2 + \epsilon - 2)^2}{8\pi^4 \epsilon^2} A_{22,LO} + \frac{24\epsilon^6 - 40\epsilon^5 + 76\epsilon^4 - 88\epsilon^3 + 68\epsilon^2 - 33\epsilon + 6}{4\pi^4 \epsilon^3 (2\epsilon - 1) q^2} A_3 \\ + \frac{12\epsilon^6 + 20\epsilon^5 - 101\epsilon^4 + 103\epsilon^3 - 88\epsilon^2 + 47\epsilon - 6}{4\pi^4 \epsilon^2 (6\epsilon^2 - 13\epsilon + 6)} A_4$$

Finally, the epsilon-series result is computed as,

```
In[5] := BuildAndIntegrateAntenna[A, 2, 2, Component->Leading,
ExpansionOrder->0]
```

Out [5] :=

$$\frac{1}{4\epsilon^4} + \frac{17}{8\epsilon^3} + \frac{433 - 72\pi^2}{144\epsilon^2} + \frac{121350 - 44820\pi^2 + 15120\zeta_3}{25920\epsilon} + \frac{-45415 - 64590\pi^2 + 4734\pi^4 + 37440\zeta_3}{25920}$$

#### 4.2.2 The $\tilde{A}_2^2$ Antenna

The subleading unintegrated antenna is computed using,

In[2] := BuildAntenna[A, 2, 2, Component->Subleading]

Out [2] :=

$$24\epsilon^{14} \left( 391\pi^{12} - 1149918\pi^8\zeta_3 - 62009136\pi^4\zeta_3^2 - 823011840\zeta_3^3 \right) + \dots$$

The result in terms of the antenna's master integrals yields,

In[3] := A22Sub = BuildAndIntegrateAntenna[A, 2, 2, Component->Subleading, ReturnMasterCombination->True];

In[4] := (A22Sub/.A22MISubs)//Collect[#, A22LO, A3, A4, A6, Simplify]&

Out [4] :=

$$\frac{(-2\epsilon^2 + \epsilon - 2)^2}{8\pi^4\epsilon^2} A_{22,LO} + \frac{108\epsilon^7 + 780\epsilon^6 - 2641\epsilon^5 + 3305\epsilon^4 - 2476\epsilon^3 + 1210\epsilon^2 - 330\epsilon + 36}{8\pi^4\epsilon^3(8\epsilon^2 - 6\epsilon + 1)q^2} A_3 + \frac{-42\epsilon^6 - 285\epsilon^5 + 659\epsilon^4 - 618\epsilon^3 + 372\epsilon^2 - 138\epsilon + 20}{8\pi^4\epsilon^2(3\epsilon - 2)(4\epsilon - 1)} A_4 - \frac{(\epsilon^3 + 4\epsilon^2 - 2\epsilon + 2)(q^2)^2}{8\pi^4(4\epsilon - 1)} A_6$$

Finally, the epsilon-series result is computed as,

In[5] := BuildAndIntegrateAntenna[A, 2, 2, Component->Subleading, ExpansionOrder->0]

Out [5] :=

$$-\frac{1}{4\epsilon^4} - \frac{3}{4\epsilon^3} + \frac{-1476 + 312\pi^2}{576\epsilon^2} + \frac{-3978 + 864\pi^2 + 1536\zeta_3}{576\epsilon} + \frac{-10359 + 2850\pi^2 - 118\pi^4 + 4176\zeta_3}{576}$$

#### 4.2.3 The $\hat{A}_2^2$ Antenna

The quark-loop unintegrated antenna is computed using,

In[2] := BuildAntenna[A, 2, 2, Component->Nf]

Out [2] :=

$$-\frac{1}{6} \frac{(3 + 2\epsilon)B_0(s_{12}; 0, 0) + 2s_{12}C_0(0, 0, s_{12}; 0, 0, 0)}{\epsilon} + \dots$$

The result in terms of the antenna's master integrals yields,

```
In[3]:= A22Nf = BuildAndIntegrateAntenna[A, 2, 2, Component->Nf,
ReturnMasterCombination->True];
In[4]:= (A22Nf/.A22MISubs)//Collect[#, A4, Simplify]&
Out[4]:=
```

$$-\frac{(\epsilon - 1)(6\epsilon^3 - 5\epsilon^2 + 3\epsilon - 2)}{\pi^4 \epsilon (2\epsilon - 3)(3\epsilon - 2)} A_4$$

Finally, the epsilon-series result is computed as,

```
In[5]:= BuildAndIntegrateAntenna[A, 2, 2, Component->Nf,
ExpansionOrder->0]
Out[5]:=
```

$$-\frac{1}{4\epsilon^3} - \frac{1}{9\epsilon^2} + \frac{65 + 9\pi^2}{216\epsilon} + \frac{4085 - 546\pi^2 + 72\zeta_3}{1296}$$

#### 4.2.4 The $\check{A}_2^2$ Antenna

The unintegrated antennae are computed using,

```
In[2]:= BuildAntenna[A, 2, 2, Component->Breve]
Out[2]:=
```

$$\frac{\Gamma^2(1 + \epsilon)\Gamma^6(1 - \epsilon)}{4\epsilon^4\Gamma^2(2 - 2\epsilon)} + \dots$$

The expressions have been truncated here. These four components represent, respectively, the leading, subleading, quark-loop and self-interference terms, or,  $A_2^2$ ,  $\tilde{A}_2^2$ ,  $\hat{A}_2^2$  and  $\check{A}_2^2$ , in that order. Their complete unintegrated forms are lengthy and are, therefore collected in Appendix A. We may then obtain the integrated expressions as,

```
In[3]:= A22Breve = BuildAndIntegrateAntenna[A, 2, 2, Component->Breve,
ReturnMasterCombination->True];
In[4]:= (A22Breve/.A22MISubs)//Collect[#, A22LO, Simplify]&
Out[4]:=
```

$$\frac{(-2\epsilon^2 + \epsilon - 2)^2}{16\pi^4 \epsilon^2} A_{22,LO}$$

```
In[5]:= BuildAndIntegrateAntenna[A, 2, 2, Component->Breve,
ExpansionOrder->0]
Out[5]:=
```

$$\begin{aligned} & \frac{1}{4\epsilon^4} + \frac{3}{4\epsilon^3} + \frac{1}{\epsilon^2} \left( \frac{41}{16} - \frac{\pi^2}{24} \right) + \frac{1}{\epsilon} \left( 7 - \frac{\pi^2}{8} - \frac{7\zeta_3}{6} \right) \\ & + 18 - \frac{41\pi^2}{96} - \frac{7\pi^4}{480} - \frac{7\zeta_3}{2} \end{aligned}$$

The integrated Laurent series agree with the established  $\mathcal{A}_2^2$  set results [24] apart from the exception discussed in Subsubsection 4.2.4.1, below.

#### 4.2.4.1 A Note on Sources

During the construction of the  $\check{A}_2^2$  antenna the exact formulation of the  $\epsilon^{-1}$  antenna infrared pole failed to match the expression in Eq. (4.10) of [24], our main source. Using the paper's expression also resulted in a failed cancellation on the massless hadronic  $R$ -ratio's  $\epsilon^{-1}$  infrared pole on the NNLO term. Tracing this discrepancy through the assembly of colour channels revealed an uncanceled  $28C_F^2\zeta_3/(3\epsilon)$  residue, which pointed at a sign error in the  $7\zeta_3/(6\epsilon)$  term of the antenna.

This discrepancy with the main source was independently confirmed in the literature by Jakubčík, Marcoli, and Stagnitto [30] (Eq. B.7) formally identified and published the typographical error in Eq. (4.10) of [24]. In *AntCalc*, the  $\check{A}_2^2$  antenna is integrated directly from the one-loop self-interference matrix element, natively reproducing the corrected Laurent series of [30] and ensuring exact pole cancellation in the NNLO  $R$ -ratio term.

### 4.2.5 Master Integrals

#### 4.2.5.1 The $A_{22,\text{LO}}$ Integral

$A_{22,\text{LO}}$  is the first of the four  $A$ -type MI, and represents a double bubble product. These are loop momenta integrals, not phase-space integrals. This first MI is given as [5],

$$\begin{aligned}
A_{22,\text{LO}} &= \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2(p-k_1-k_2)^2 \ell^2(\ell-k_1-k_2)^2} \\
&= -\frac{\mu^{-4\epsilon}}{(16\pi^2)^2} \left[ B_0(q^2; 0, 0) \right]^2 \\
&= S_\Gamma(-q^2)^{-2\epsilon} \frac{\Gamma^2(1+\epsilon)\Gamma^6(1-\epsilon)}{\Gamma^2(2-2\epsilon)} \left( -\frac{1}{\epsilon^2} \right) \\
&= S_\Gamma(-q^2)^{-2\epsilon} \left[ -\frac{1}{\epsilon^2} - \frac{4}{\epsilon} - 12 + \mathcal{O}(\epsilon) \right].
\end{aligned} \tag{4.43}$$

#### 4.2.5.2 The $A_3$ Integral

$A_3$  represents a three-propagator two-loop sunset vacuum bubble, with  $q^2 = s_{12}$ . It is given as [5],

$$\begin{aligned}
A_3 &= \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p-\ell-k_1-k_2)^2} \\
&= \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2} B_0((p-k_1-k_2)^2; 0, 0) \\
&= S_\Gamma(-q^2)^{1-2\epsilon} \frac{\Gamma(1+2\epsilon)\Gamma^5(1-\epsilon)}{\Gamma(3-3\epsilon)} \left( -\frac{1}{2(1-2\epsilon)\epsilon} \right) \\
&= S_\Gamma(-q^2)^{1-2\epsilon} \left[ -\frac{1}{4\epsilon} - \frac{13}{8} + \mathcal{O}(\epsilon) \right].
\end{aligned} \tag{4.44}$$

#### 4.2.5.3 The $A_4$ Integral

$A_4$  is a four-propagator two-loop vacuum planar double bubble. It is given as a triangle diagram containing a bubble insertion on one of its legs. That gives [5],

$$\begin{aligned}
A_4 &= \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p - k_1 - k_2)^2 (p - \ell - k_1)^2} \\
&= \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2 (p - k_1 - k_2)^2} B_0((p - k_1)^2; 0, 0) \\
&= S_\Gamma(-q^2)^{-2\epsilon} \frac{\Gamma(1 - 2\epsilon)\Gamma(1 + \epsilon)\Gamma^4(1 - \epsilon)\Gamma(1 + 2\epsilon)}{\Gamma(2 - 3\epsilon)} \left( -\frac{1}{2(1 - 2\epsilon)\epsilon^2} \right) \\
&= S_\Gamma(-q^2)^{-2\epsilon} \left[ -\frac{1}{2\epsilon^2} - \frac{5}{2\epsilon} - \left( \frac{19}{2} + \frac{\pi^2}{6} \right) + O(\epsilon) \right].
\end{aligned} \tag{4.45}$$

#### 4.2.5.4 The $A_6$ Integral

Lastly, the  $A_6$  MI represents the two-loop two-point vertex. It is given as [5],

$$A_6 = \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p - k_1 - k_2)^2 (p - \ell)^2 (p - \ell - k_2)^2 (\ell - k_1)^2}. \tag{4.46}$$

Unlike the other A-type antennae, this one cannot be given in terms of the scalar one-loop functions  $B_0$ ,  $C_0$  or  $D_0$ . To compute this integral we have to start by expressing it as a Mellin-Barnes contour integral, then evaluate the residues, and finally expand as a Laurent series around  $\epsilon$ . This process gives,

$$\begin{aligned}
A_6 &= \frac{S_\Gamma(-q^2)^{-2-2\epsilon}}{(2\pi i)^2} \int_{-i\infty}^{i\infty} \int_{-i\infty}^{i\infty} dz_1 dz_2 \frac{\Gamma(-z_1)\Gamma(-z_2)\Gamma(2 + 2\epsilon + z_1 + z_2)\Gamma^2(-\epsilon - z_1)\Gamma^2(-\epsilon - z_2)}{\Gamma(-2\epsilon - z_1 - z_2)\Gamma(2 + \epsilon + z_1 + z_2)} \\
&= S_\Gamma(-q^2)^{-2-2\epsilon} \left[ -\frac{1}{\epsilon^4} + \frac{5\pi^2}{6\epsilon^2} + \frac{27}{\epsilon}\zeta_3 + \frac{23\pi^4}{36} + O(\epsilon) \right].
\end{aligned} \tag{4.47}$$

### 4.3 Three-Parton Contributions: the $A_3^1$ Set

Computing the  $A_3^1$  antenna set is very similar to the work done for the  $A_2^2$  set, with the big differentiator being that this set does not include a self-interference term. Each component is presented in turn below.

#### 4.3.1 The $A_3^1$ Antenna

The leading unintegrated antenna is computed using,

```

In[2] := BuildAntenna[A, 3, 1, Component->Leading]
Out[2] :=
      
$$\epsilon(1 + 3\epsilon)s_{13}s_{23} + \dots$$


```

The result in terms of the antenna's master integrals yields,

```

In[3] := A31Lead = BuildAndIntegrateAntenna[A, 3, 1, Component->Leading,
ReturnMasterCombination->True];
In[4] := (A31Lead/.A31MISubs)//Collect[#, V5a, V5b, V8, Simplify]&

```

Out [4] :=

$$\frac{i(9\epsilon^5 + 25\epsilon^4 - 34\epsilon^3 + 41\epsilon^2 - 20\epsilon + 4)}{2\pi^2\epsilon^3q^2}V_{5,b} - \frac{i(24\epsilon^6 - 40\epsilon^5 + 76\epsilon^4 - 88\epsilon^3 + 68\epsilon^2 - 33\epsilon + 6)}{2\pi^2\epsilon^3(2\epsilon - 1)q^2}V_{5,a}$$

Finally, the epsilon-series result is computed as,

```
In[5] := BuildAndIntegrateAntenna[A, 3, 1, Component->Leading,
ExpansionOrder->0]
```

Out [5] :=

$$-\frac{1}{4\epsilon^4} - \frac{31}{12\epsilon^3} + \frac{-9540 + 660\pi^2}{1440\epsilon^2} + \frac{-38820 + 3520\pi^2 + 11040\zeta_3}{1440\epsilon} + \frac{-156930 + 12240\pi^2 - 123\pi^4 + 55120\zeta_3}{1440} + \mathcal{O}(\epsilon)$$

### 4.3.2 The $\tilde{A}_3^1$ Antenna

The subleading unintegrated antenna is computed using,

```
In[2] := BuildAntenna[A, 3, 1, Component->Subleading]
```

Out [2] :=

$$\frac{2s_{12}\left[(-1 + \epsilon)s_{13} + \epsilon s_{23}\right]}{2(-1 + \epsilon)q^2s_{13}}D_0(0, 0, 0; q^2, s_{12}, s_{23}; 0, 0, 0, 0) + \dots$$

The result in terms of the antenna's master integrals yields,

```
In[3] := A31Sub = BuildAndIntegrateAntenna[A, 3, 1, Component->Subleading,
ReturnMasterCombination->True];
```

```
In[4] := (A31Sub/.A31MISubs)//Collect[#, V5a, V5b, V8, Simplify]&
```

Out [4] :=

$$\frac{i(\epsilon^3 + 4\epsilon^2 - 2\epsilon + 2)(q^2)^2}{2\pi^2(4\epsilon - 1)}V_8 - \frac{i(108\epsilon^7 + 780\epsilon^6 - 2641\epsilon^5 + 3305\epsilon^4 - 2476\epsilon^3 + 1210\epsilon^2 - 330\epsilon + 36)}{4\pi^2\epsilon^3(8\epsilon^2 - 6\epsilon + 1)q^2}V_{5,a} + \frac{i(49\epsilon^5 + 125\epsilon^4 - 242\epsilon^3 + 185\epsilon^2 - 94\epsilon + 20)}{2\pi^2\epsilon^3q^2}V_{5,b}$$

Finally, the epsilon-series result is computed as,

```
In[5] := BuildAndIntegrateAntenna[A, 3, 1, Component->Subleading,
ExpansionOrder->0]
```

Out [5] :=

$$-\frac{15 + 4\pi^2}{24\epsilon^2} + \frac{-19 + \pi^2 + 28\zeta_3}{4\epsilon} + \frac{1215\pi^2 + 56\pi^4 + 540(-35 + 18\zeta_3)}{720}$$

### 4.3.3 The $\hat{A}_3^1$ Antenna

The quark-loop unintegrated antenna is computed using,

```
In[2] := BuildAntenna[A, 3, 1, Component->Nf]
Out[2] :=
```

$$\frac{2}{3} \frac{s_{12}^2}{\epsilon q^2 s_{13} s_{23}} + \dots$$

Unlike  $A_3^1$  and  $\tilde{A}_3^1$ , the result in terms of the antenna's master integrals vanishes for  $\hat{A}_3^1$ ,

```
In[3] := A31Nf = BuildAndIntegrateAntenna[A, 3, 1, Component->Nf,
ReturnMasterCombination->True];
```

```
In[4] := (A31Nf/.A31MISubs)//Collect[#, V5a, V5b, V8, Simplify] &
Out[4] :=
```

$$0$$

since this route has no bare one-loop IBP contribution left to reduce.  $\hat{A}_3^1$ 's non-zero content instead enters afterwards, as a UV coupling-renormalisation counterterm built from the already-integrated tree-level result of Section 4.1.1; its leading piece,

$$\frac{1}{3\epsilon} \mathcal{A}_3^0, \quad (4.48)$$

already reproduces the  $\epsilon^{-3}$  and  $\epsilon^{-2}$  poles of the integrated result below exactly. Finally, the epsilon-series result is computed as,

```
In[5] := BuildAndIntegrateAntenna[A, 3, 1, Component->Nf,
ExpansionOrder->0]
```

```
Out[5] :=
```

$$\frac{1}{3\epsilon^3} + \frac{1}{2\epsilon^2} + \frac{114 - 14\pi^2}{72\epsilon} + \frac{327 - 21\pi^2 - 200\zeta_3}{72} + \mathcal{O}(\epsilon)$$

### 4.3.4 Master Integrals

#### 4.3.4.1 The $V_{5,a}$ Integral

$V_{5,a}$  is a one-loop bubble integrated over a three-particle phase space, written as,

$$\begin{aligned} V_{5,a} &= \text{Re} \left[ -i \int d\Phi_3 \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2(p - k_1 - k_2 - k_3)^2} \right] \\ &= \text{Re} \left[ \int d\Phi_3 B_0(q^2; 0, 0) \right]. \end{aligned} \quad (4.49)$$

Since  $q^2 = (p_1 + p_2 + p_3)^2$ , this MI factors to the three-particle phase-space integral of the  $B_0(q^2; 0, 0)$  scalar one-loop function, discussed in Subsubsection B.2.2.1. After paramtrising the phase space and applying the methods introduced in Subsubsection 4.4.5.3 we get [5],

$$\begin{aligned} V_{5,a} &= S_{\Gamma,2}(q^2)^{1-2\epsilon} \cos(\pi\epsilon) \left( -\frac{1}{\epsilon} \right) \frac{\Gamma^6(1-\epsilon)\Gamma(1+\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)} \\ &= S_{\Gamma,2}(q^2)^{1-2\epsilon} \left[ -\frac{1}{2\epsilon} - \frac{13}{4} + \mathcal{O}(\epsilon) \right]. \end{aligned} \quad (4.50)$$

#### 4.3.4.2 The $V_{5,b}$ Integral

$V_{5,b}$  is very similar in structure to  $V_{5,a}$ , also being the three-particle phase-space integral of a  $B_0$  scalar one-loop function. The integral is also computed in the same manner and yields [5],

$$\begin{aligned} V_{5,b} &= \int d\Phi_3 B_0(s_{13}; 0, 0) \\ &= S_{\Gamma,2}(q^2)^{1-2\epsilon} \cos(\pi\epsilon) \left(-\frac{1}{\epsilon}\right) \frac{\Gamma^5(1-\epsilon)\Gamma(1-2\epsilon)\Gamma(1+\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-4\epsilon)} \\ &= S_{\Gamma,2}(q^2)^{1-2\epsilon} \left[-\frac{1}{2\epsilon} - 4 + O(\epsilon)\right]. \end{aligned} \quad (4.51)$$

#### 4.3.4.3 The $V_8$ Integral

The  $V_8$  MI is the most complex of the three  $V$ -type MI. It is given as a four-denominator propagator loop and three-particle phase-space integral. That makes it be restatable as a Box function which, itself, may be restated as a  $D_0$  function, as discussed in Subsubsection B.2.2.3. It is written as [5],

$$\begin{aligned} V_8 &= \text{Re} \left[ -i \int d\Phi_3 \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2(p-k_1)^2(p-k_1-k_3)^2(p-k_1-k_2-k_3)^2} \right] \\ &= \text{Re} \left[ -i \int d\Phi_3 \frac{1}{s_{12}} \text{Box}(s_{13}, s_{23}, s_{12}) \right] \\ &= \left( \frac{(4\pi)^\epsilon \mu^{-2\epsilon}}{16\pi^2} \right) \frac{\Gamma^2(1-\epsilon)\Gamma(1+\epsilon)}{\Gamma(1-2\epsilon)} \int d\Phi_3 \frac{1}{s_{12}} \text{Re} \left[ D_0(0, 0, 0, q^2; s_{13}, s_{23}; 0, 0, 0, 0) \right] \end{aligned} \quad (4.52)$$

which, after normalisation and evaluation gives,

$$V_8 = S_{\Gamma,2}(q^2)^{-2-2\epsilon} \left[ -\frac{5}{2\epsilon^4} + \frac{9\pi^2}{2\epsilon^2} + \frac{89\zeta_3}{\epsilon} + \frac{13\pi^4}{180} + O(\epsilon) \right]. \quad (4.53)$$

### 4.4 Four-Parton Contributions

The double real-radiation antenna set is  $A_4^0$ , and it has two components,  $A_4^0$  and  $\tilde{A}_4^0$ . It represents the  $\gamma^* \rightarrow qgg\bar{q}$  process. As with  $B_4^0$  and  $C_4^0$  below, the complete unintegrated expressions for all four routes are lengthy and are collected in Appendix A.

#### 4.4.1 The $A_4^0$ Antenna

The leading unintegrated antenna is computed using,

```
In[2] := BuildAntenna[A, 4, 0, Component->Leading]
Out[2] :=
```

$$-\frac{3s_{12}}{2(-1+\epsilon)(q^2)^3} + \dots$$

The result in terms of the antenna's master integrals yields,

```
In[3] := A40Lead = BuildAndIntegrateAntenna[A, 4, 0, Component->Leading,
ReturnMasterCombination->True];
In[4] := (A40Lead/.X40MISubs)//Collect[#, R4, R6, R8a, R8b, Simplify] &
```

Out [4] :=

$$\left[ \frac{1632\epsilon^{10} - 13528\epsilon^9 + 55268\epsilon^8 - 136894\epsilon^7 + 220017\epsilon^6}{2(1-2\epsilon)^2\epsilon^4(6\epsilon^2-13\epsilon+6)(q^2)^2} + \frac{-238829\epsilon^5 + 177763\epsilon^4 - 89380\epsilon^3 + 28844\epsilon^2 - 5352\epsilon + 432}{2(1-2\epsilon)^2\epsilon^4(6\epsilon^2-13\epsilon+6)(q^2)^2} \right] R_4 + \frac{-12\epsilon^6 - 20\epsilon^5 + 101\epsilon^4 - 103\epsilon^3 + 88\epsilon^2 - 47\epsilon + 6}{\epsilon^2(6\epsilon^2-13\epsilon+6)} R_6$$

Finally, the epsilon-series result is computed as,

```
In[5] := BuildAndIntegrateAntenna[A, 4, 0, Component->Leading,
ExpansionOrder->0]
```

Out [5] :=

$$\frac{3}{4\epsilon^4} + \frac{65}{24\epsilon^3} + \frac{1}{\epsilon^2} \left( \frac{217}{18} - \frac{13\pi^2}{12} \right) + \frac{1}{\epsilon} \left( \frac{43223}{864} - \frac{589\pi^2}{144} - \frac{71\zeta_3}{4} \right) + \frac{1076717}{5184} - \frac{7955\pi^2}{432} + \frac{373\pi^4}{1440} - \frac{1327\zeta_3}{18} + O(\epsilon)$$

#### 4.4.2 The $\tilde{A}_4^0$ Antenna

The subleading unintegrated antenna is computed using,

```
In[2] := BuildAntenna[A, 4, 0, Component->Subleading]
```

Out [2] :=

$$-\frac{\epsilon}{(-1+\epsilon)q^2s_{13}} + \dots$$

The result in terms of the antenna's master integrals yields,

```
In[3] := A40Sub = BuildAndIntegrateAntenna[A, 4, 0, Component->Subleading,
ReturnMasterCombination->True];
```

```
In[4] := (A40Sub/.X40MISubs)//Collect[#, R4, R6, R8a, R8b, Simplify]&
```

Out [4] :=

$$\left[ \frac{26016\epsilon^{10} - 21800\epsilon^9 - 124084\epsilon^8 + 326554\epsilon^7 - 392893\epsilon^6}{2(1-2\epsilon)^2\epsilon^4(3\epsilon-2)(4\epsilon-1)(q^2)^2} + \frac{297255\epsilon^5 - 155324\epsilon^4 + 56744\epsilon^3 - 13782\epsilon^2 + 1960\epsilon - 120}{2(1-2\epsilon)^2\epsilon^4(3\epsilon-2)(4\epsilon-1)(q^2)^2} \right] R_4 + \frac{(\epsilon^3 + 4\epsilon^2 - 2\epsilon + 2)(q^2)^2}{2-8\epsilon} R_{8,a} + \frac{2(\epsilon^3 + 4\epsilon^2 - 2\epsilon + 2)(q^2)^2}{2-8\epsilon} R_{8,b} + \frac{-138\epsilon^6 - 341\epsilon^5 + 687\epsilon^4 - 620\epsilon^3 + 420\epsilon^2 - 150\epsilon + 20}{24\epsilon^4 - 22\epsilon^3 + 4\epsilon^2} R_6$$

Finally, the epsilon-series result is computed as,

```
In[5] := BuildAndIntegrateAntenna[A, 4, 0, Component->Subleading,
ExpansionOrder->0]
Out[5] :=
```

$$\frac{1}{\epsilon^4} + \frac{3}{\epsilon^3} + \frac{1}{\epsilon^2} \left( 13 - \frac{3\pi^2}{2} \right) + \frac{1}{\epsilon} \left( \frac{845}{16} - \frac{9\pi^2}{2} - \frac{80\zeta_3}{3} \right) + \frac{6921}{32} - \frac{473\pi^2}{24} + \frac{17\pi^4}{72} - 80\zeta_3 + O(\epsilon)$$

Finally, the  $\gamma^* \rightarrow q\bar{q}q\bar{q}$  process is represented by the  $B_4^0$  and  $C_4^0$  antennae. The  $B_4^0$  antenna describes non-identical quark flavours, while  $C_4^0$  gives the identical-flavour contribution.

#### 4.4.3 The $B_4^0$ Antenna

The  $B_4^0$  antenna building process is completed as usual using *AntCalc*,

```
In[2] := BuildAntenna[B, 4, 0]
Out[2] :=
```

$$-\frac{\epsilon s_{12}}{(-1 + \epsilon)q^2 s_{134}^2} + \dots$$

The result in terms of the antenna's master integrals yields,

```
In[3] := B40 = BuildAndIntegrateAntenna[B, 4, 0,
ReturnMasterCombination->True];
In[4] := (B40/.X40MISubs)//Collect[#, R4, R6, R8a, R8b, Simplify]&
Out[4] :=
```

$$\frac{2(6\epsilon^4 - 11\epsilon^3 + 8\epsilon^2 - 5\epsilon + 2)}{\epsilon(6\epsilon^2 - 13\epsilon + 6)} R_6 - \frac{2(48\epsilon^6 - 184\epsilon^5 + 335\epsilon^4 - 372\epsilon^3 + 249\epsilon^2 - 88\epsilon + 12)}{\epsilon^3(6\epsilon^2 - 13\epsilon + 6)(q^2)^2} R_4$$

Finally, the epsilon-series result is computed as,

```
In[5] := BuildAndIntegrateAntenna[B, 4, 0, ExpansionOrder->0]
Out[5] :=
```

$$-\frac{1}{12\epsilon^3} - \frac{7}{18\epsilon^2} + \frac{1}{\epsilon} \left( -\frac{407}{216} + \frac{11\pi^2}{72} \right) - \frac{11753}{1296} + \frac{77\pi^2}{108} + \frac{67\zeta_3}{18} + O(\epsilon)$$

#### 4.4.4 The $C_4^0$ Antenna

The  $C_4^0$  antenna building process is completed as usual using *AntCalc*,

```
In[2] := BuildAntenna[C, 4, 0]
Out[2] :=
```

$$-\frac{1}{2} \frac{\epsilon s_{12}}{(-1 + \epsilon)q^2 s_{123} s_{134}} + \dots$$

The result in terms of the antenna's master integrals yields,

```

In[3] := C40 = BuildAndIntegrateAntenna[C, 4, 0,
ReturnMasterCombination->True];
In[4] := (C40/.X40MISubs)//Collect[#, R4, R6, R8a, R8b, Simplify] &
Out[4] :=

```

$$\begin{aligned}
& \left[ \frac{4128\epsilon^{10} + 88\epsilon^9 - 44412\epsilon^8 + 115994\epsilon^7 - 154689\epsilon^6}{8(1-2\epsilon)^2\epsilon^4(3\epsilon-2)(4\epsilon-1)(q^2)^2} \right. \\
& \left. + \frac{132651\epsilon^5 - 78114\epsilon^4 + 31324\epsilon^3 - 8058\epsilon^2 + 1176\epsilon - 72}{8(1-2\epsilon)^2\epsilon^4(3\epsilon-2)(4\epsilon-1)(q^2)^2} \right] R_4 \\
& + \frac{(\epsilon^3 + 4\epsilon^2 - 2\epsilon + 2)(q^2)^2}{4 - 16\epsilon} R_{8,b} + \frac{54\epsilon^6 - 229\epsilon^5 + 631\epsilon^4 - 616\epsilon^3 + 324\epsilon^2 - 126\epsilon + 20}{96\epsilon^4 - 88\epsilon^3 + 16\epsilon^2} R_6
\end{aligned}$$

Finally, the epsilon-series result is computed as,

```

In[5] := BuildAndIntegrateAntenna[C, 4, 0, ExpansionOrder->0]
Out[5] :=

```

$$\frac{1}{\epsilon} \left( -\frac{13}{32} + \frac{\pi^2}{16} - \frac{\zeta_3}{4} \right) - \frac{339}{64} + \frac{17\pi^2}{48} - \frac{\pi^4}{45} + \frac{21\zeta_3}{8} + O(\epsilon)$$

## 4.4.5 Master Integrals

### 4.4.5.1 The $R_4$ Integral

The first of the tree-level NNLO MI is  $R_4$ , which corresponds to the direct integral over the four-final-state-particle phase-space. It is written as,

$$\begin{aligned}
R_4 &= \int d\Phi_4 \\
&= (2\pi)^{4-3d} (q^2)^{3-\frac{d}{2}} 2^{1-2d} \int dS_4 d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} \\
&\quad (-\Delta_4)^{\frac{d-5}{2}} \Theta(-\Delta_4) \delta(q^2 - S_4).
\end{aligned} \tag{4.54}$$

To make the integral boundary limits integrable, we start by factorising the phase space into a two-particle phase space and a tripole phase space factor,  $d\Phi_4 = d\Phi_2 d\Phi_T$  and then we map the six physical invariants  $s_{ij}$  into five dimensionless variables that range strictly between 0 and 1, making a five-dimensional hypercube. We start by defining  $y_{ij} = s_{ij}/q^2$  and  $y_{ijk} = s_{ijk}/q^2$ . Our five dimensionless variables become [13]:

- The rescaled triple-invariant of the emitter jet,  $y_{134} = \frac{s_{134}}{q^2}$ ;
- The longitudinal momentum fraction of the unresolved parton 3,  $z_1 = \frac{p_3 \cdot \tilde{p}_2}{\tilde{p}_{134} \cdot \tilde{p}_2}$ ;
- The rescaled longitudinal momentum fraction of unresolved parton 4,  $z_2 = (1 - z_1)t = \frac{p_4 \cdot \tilde{p}_2}{\tilde{p}_{134} \cdot \tilde{p}_2}$ ;
- The rescaled invariant,  $y_{14} = \frac{s_{14}}{q^2} = y_{134}(1 - z_1)v$ ;
- The alignment variable  $\chi$ , that parametrises the physical boundaries of the Gram determinant for  $y_{13}$  as,  $y_{13} = (y_{13,b} - y_{13,a})\chi + y_{13,a} = \Delta y_{13}\chi + y_{13,a}$ .

After applying the mapping, the MI becomes

$$R_4 = P_2 \frac{16^{-2} \pi^{-5+2\epsilon}}{\Gamma(1-2\epsilon)} (q^2)^{2-2\epsilon} \int_{[0,1]^5} dy_{134} dz_1 dt dv d\chi \left[ y_{134}(1-y_{134})(1-z_1) \right]^{1-2\epsilon} \left[ z_1 t(1-t)v(1-v) \right]^{-\epsilon} \left[ \chi(1-\chi) \right]^{-\frac{1}{2}-\epsilon}. \quad (4.55)$$

We get [5, 13],

$$\begin{aligned} R_4 &= P_2 (q^2)^{2-2\epsilon} \frac{2^{-11+8\epsilon} \pi^{-\frac{7}{2}+2\epsilon} \Gamma^3(1-\epsilon)}{\Gamma(3-3\epsilon) \Gamma\left(\frac{5}{2}-2\epsilon\right)} \\ &= S_{\Gamma,2} (q^2)^{2-2\epsilon} \frac{\Gamma^5(1-\epsilon) \Gamma(2-2\epsilon)}{\Gamma(3-3\epsilon) \Gamma(4-4\epsilon)} \\ &= S_{\Gamma} (q^2)^{2-2\epsilon} \left[ \frac{1}{12} + \frac{59}{72} \epsilon + O(\epsilon^2) \right]. \end{aligned} \quad (4.56)$$

#### 4.4.5.2 The $R_6$ Integral

The  $R_6$  MI is written as [13],

$$\begin{aligned} R_6 &= \int d\Phi_4 \frac{1}{s_{134}s_{234}} \\ &= (2\pi)^{4-3d} (q^2)^{3-\frac{d}{2}} 2^{1-2d} \int dS_4 d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} \frac{(-\Delta_4)^{\frac{d-5}{2}}}{s_{134}s_{234}} \Theta(-\Delta_4) \delta(q^2 - S_4). \end{aligned} \quad (4.57)$$

Because the  $s_{234}$  is non-linear in the tripole parametrisation, we may not continue using direct integration [13]. We may, however, use the Optical Theorem as a cut of the three-loop vacuum polarisation diagram  $I_6$  as,

$$2 \operatorname{Im} I_6 = -2P_2 \operatorname{Re} A_4 + 2R_6 \quad (4.58)$$

with  $I_6$  and  $A_4$  being known from the literature and written as,

$$\begin{aligned} I_6 &= \frac{(4\pi)^{3\epsilon}}{(16\pi^2)^3} \frac{\Gamma^6(1-\epsilon) \Gamma^3(1+\epsilon)}{3\epsilon^3 \Gamma^3(2-2\epsilon)} (-q^2)^{-3\epsilon} \\ &\quad \left[ 1 + \epsilon + \epsilon^2 + \epsilon^3(14\zeta_3 - 7) + \epsilon^4 \left( -67 + 14\zeta_3 + \frac{21\pi^4}{90} \right) + O(\epsilon^5) \right] \\ A_4 &= \left( \frac{(4\pi)^\epsilon}{16\pi^2} \frac{\Gamma(1+\epsilon) \Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)} \right)^2 (-q^2)^{-2\epsilon} \left[ -\frac{1}{2\epsilon^2} - \frac{5}{2\epsilon} - \left( \frac{19}{2} + \frac{\pi^2}{6} \right) \right. \\ &\quad \left. - \epsilon \left( \frac{65}{2} + \frac{5\pi^2}{6} - 2\zeta_3 \right) - \epsilon^2 \left( \frac{211}{2} + \frac{19\pi^2}{6} - 10\zeta_3 \right) + O(\epsilon^3) \right]. \end{aligned} \quad (4.59)$$

Note that  $A_4$  is described in Subsubsection 4.2.5.3.

We then apply the Minkowski boundary prescription [13]  $(q^2)^{-\epsilon} = (q^2)^{-\epsilon} e^{i\pi\epsilon}$ , isolate the imaginary

and real parts, and get, by using  $\mathcal{A}_4$  and  $\mathcal{I}_6$

$$\begin{aligned} R_6 &= S_\Gamma(q^2)^{-2\epsilon} \frac{\Gamma^6(1-\epsilon)\Gamma^2(1+\epsilon)}{6} \left( \frac{6\cos(2\pi\epsilon)}{\Gamma^2(1-2\epsilon)} \mathcal{A}_4 + \frac{\Gamma(1-\epsilon)\Gamma(1+\epsilon)\sin(3\pi\epsilon)}{\epsilon^3\pi\Gamma^2(2-2\epsilon)} \mathcal{I}_6 \right) \\ &= S_{\Gamma,2}(q^2)^{-2\epsilon} \left[ -1 + \frac{\pi^2}{6} + \epsilon \left( -12 + \frac{5\pi^2}{6} + 9\zeta_3 \right) + O(\epsilon^2) \right]. \end{aligned} \quad (4.61)$$

#### 4.4.5.3 The $R_{8,a}$ Integral

The  $R_{8,a}$  is the first of two MI with four denominator invariants. Its complete integral form is written as [13],

$$\begin{aligned} R_{8,a} &= \int d\Phi_4 \frac{1}{s_{13}s_{23}s_{14}s_{24}} \\ &= (2\pi)^{4-3d} (q^2)^{3-\frac{d}{2}} 2^{1-2d} \int dS_4 d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} \frac{(-\Delta_4)^{\frac{d-5}{2}}}{s_{13}s_{23}s_{14}s_{24}} \Theta(-\Delta_4) \delta(q^2 - S_4) \\ &= P_2(q^2)^{-2-2\epsilon} \frac{16^{-2+\epsilon} \pi^{-5+2\epsilon}}{\Gamma(1-2\epsilon)} \int_{[0,1]^5} dy_{134} dz_1 dt dv d\chi \\ &\quad \left[ (1-y_{134})(1-z_1) \right]^{-1-2\epsilon} \left[ tvz_1^{-\epsilon} (\Delta y_{13}\chi + y_{13,a}) \right]^{-1} \left[ \chi(1-\chi) \right]^{-\frac{1}{2}-\epsilon}. \end{aligned} \quad (4.62)$$

This is the most complex MI integration of the set. As such, its computation will be written in detail here as most of the methods used are transferable to other MI. We start by defining the integration order. The computation will advance as,

$$\chi \rightarrow v \rightarrow y_{134} \rightarrow t \rightarrow z_1$$

with the inclusion of a  $u$  integral, also over the  $[0, 1]$  interval in between the  $t$  and  $z_1$  integrals, which will be introduced later.

The first integral is over  $\chi$ , and gives,

$$I_\chi = \int_0^1 d\chi \left[ \chi(1-\chi) \right]^{-\frac{1}{2}-\epsilon} (\Delta y_{13}\chi + y_{13,a})^{-1} = \frac{\Delta y_{13}^{-2\epsilon} \Gamma^2\left(\frac{1}{2}-\epsilon\right)}{y_{13,a} \Gamma(1-2\epsilon)^2} {}_2F_1\left(1, \frac{1}{2}-\epsilon; 1-2\epsilon; -\frac{\Delta y_{13}}{y_{13,a}}\right) \quad (4.63)$$

where  ${}_2F_1(\dots)$  is a hypergeometric function. We now consider the definition  $y_{13,(a,b)} = y_{134}(A \mp B)$ ,  $A = \sqrt{(1-t)(1-v)}$ ,  $B = \sqrt{z_1 tv}$ , and  $Z = -B/A$ , and get,

$$-\frac{\Delta y_{13}}{y_{13,a}} = \frac{4Z}{(1+Z)^2}, \quad (1+Z)^2 = \frac{y_{13,a}}{y_{134}A^2}. \quad (4.64)$$

Finally, by taking,

$${}_2F_1\left(1, b; 2b; \frac{4Z}{(1+Z)^2}\right) = (1+Z)^2 {}_2F_1\left(1, \frac{3}{2}-b; b+\frac{1}{2}; Z^2\right) \quad (4.65)$$

and

$$16^\epsilon \Delta y_{13}^{-2\epsilon} = \left[ t(1-t)v(1-v)y_{134}^2 z_1 \right]^{-\epsilon} \quad (4.66)$$

we get,

$$I_\chi = \frac{\pi\Gamma(1-2\epsilon)}{y_{134}\Gamma^2(1-\epsilon)} \left( tv y_{134}^2 z_1 \right)^{-\epsilon} \left[ (1-t)(1-v) \right]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 tv}{(1-t)(1-v)}\right). \quad (4.67)$$

The  $R_{8,a}$  expression becomes,

$$R_{8,a} = P_2(q^2)^{-2-2\epsilon} \frac{16^{-2+\epsilon} \pi^{-4+2\epsilon}}{\Gamma^2(1-\epsilon)} \int_{[0,1]^4} dy_{134} dz_1 dt dv \left[ y_{134}(1-y_{134}) z_1(1-z_1) \right]^{-1-2\epsilon} \left[ t(1-t)v(1-v) \right]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 t v}{(1-t)(1-v)}\right). \quad (4.68)$$

We now continue to the  $v$  integral. To start, we restate the hypergeometric function as a series, which enables us to write,

$$\begin{aligned} I_v &= \int_0^1 dv \left[ v(1-v) \right]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 t v}{(1-t)(1-v)}\right) \\ &= \sum_{n \geq 0} \frac{(1+\epsilon)_n}{(1-\epsilon)_n} \left( \frac{z_1 t}{1-t} \right)^n \frac{\Gamma(-\epsilon-n)\Gamma(-\epsilon+n)}{\Gamma(-2\epsilon)} \\ &= -\frac{2\Gamma^2(1-\epsilon)}{\epsilon\Gamma(1-2\epsilon)} {}_2F_1\left(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\right). \end{aligned} \quad (4.69)$$

Plugging this expression back in and computing the trivial  $y_{134}$  integral we get,

$$\begin{aligned} R_{8,a} &= -P_2(q^2)^{-2-2\epsilon} \frac{2^{-7+8\epsilon} \pi^{-\frac{7}{2}+2\epsilon}}{\epsilon^2 \Gamma\left(\frac{1}{2}-2\epsilon\right)} \int_{[0,1]^2} dz_1 dt (1-z_1)^{-1-2\epsilon} \\ &\quad \left[ z_1 t(1-t) \right]^{-1-\epsilon} {}_2F_1\left(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\right). \end{aligned} \quad (4.70)$$

We now restate the hypergeometric function as an Euler integral as,

$${}_2F_1\left(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\right) = -\epsilon \int_0^1 du u^{-1-\epsilon} \left[ 1 + \frac{u z_1 t}{1-t} \right]^{-1} \quad (4.71)$$

and after computing the direct  $t$ ,  $u$  and  $z_1$  integrals, in that order, we get to the final expression,

$$\begin{aligned} R_{8,a} &= P_2(q^2)^{-2-2\epsilon} 2^{-7+4\epsilon} \pi^{-3+2\epsilon} \csc(\pi\epsilon) \Gamma(-2\epsilon) \left[ \frac{2^{8\epsilon} \pi}{\epsilon^2 \Gamma^2\left(\frac{1}{2}-2\epsilon\right)} {}_3F_2\left( \begin{matrix} -2\epsilon, -2\epsilon, -2\epsilon \\ 1-2\epsilon, -4\epsilon \end{matrix} ; 1 \right) - \right. \\ &\quad \left. - \frac{2\Gamma^3(-\epsilon)}{\Gamma(1-4\epsilon)\Gamma(1-\epsilon)\Gamma(1+\epsilon)\Gamma(-3\epsilon)} {}_4F_3\left( \begin{matrix} 1, -\epsilon, -\epsilon, -\epsilon \\ 1-\epsilon, 1+\epsilon, -3\epsilon \end{matrix} ; 1 \right) \right] \end{aligned} \quad (4.72)$$

which after a final normalisation becomes [13, 28],

$$\begin{aligned} R_{8,a} &= S_\Gamma(q^2)^{-2-2\epsilon} \left[ \frac{6}{\epsilon^2} \frac{\Gamma^5(1-\epsilon)\Gamma(1-2\epsilon)}{\Gamma(1-3\epsilon)\Gamma(1-4\epsilon)} {}_4F_3\left( \begin{matrix} 1, -\epsilon, -\epsilon, -\epsilon \\ 1-\epsilon, 1+\epsilon, -3\epsilon \end{matrix} ; 1 \right) - \right. \\ &\quad \left. - \frac{1}{\epsilon^4} \frac{\Gamma^3(1-\epsilon)\Gamma(1+\epsilon)\Gamma^3(1-2\epsilon)}{\Gamma^2(1-4\epsilon)} {}_3F_2\left( \begin{matrix} -2\epsilon, -2\epsilon, -2\epsilon \\ -4\epsilon, 1-2\epsilon \end{matrix} ; 1 \right) \right] \\ &= S_{\Gamma,2}(q^2)^{-2-2\epsilon} \left[ \frac{5}{\epsilon^4} - \frac{20\pi^2}{3\epsilon^2} - \frac{126\zeta_3}{\epsilon} + \frac{7\pi^4}{18} + O(\epsilon) \right]. \end{aligned} \quad (4.73)$$

#### 4.4.5.4 The $R_{8,b}$ Integral

The second of the two four-denominator-invariant MI is  $R_{8,b}$ . It is written as [13],

$$R_{8,b} = \int d\Phi_4 \frac{1}{s_{13}s_{134}s_{23}s_{234}}. \quad (4.74)$$

Due to the presence of  $s_{234}$ , this MI must be computed using the Optical Theorem, much like  $R_6$  was. For this case, we are able to map  $R_{8,b}$  as one of the cuts of the three-loop vacuum polarisation diagram  $I_8$  as,

$$2 \operatorname{Im} I_8 = -2P_2 \operatorname{Re} A_6 + 4 \operatorname{Re} V_8 + R_{8,a} + 4R_{8,b}. \quad (4.75)$$

with,

$$I_8 = S + \mathcal{O}(\epsilon^0), \quad \operatorname{Im} S = -\frac{13\pi^4}{90}; \quad (4.76)$$

$$V_8 = -i \int d\Phi_3 \frac{1}{s_{12}} \operatorname{Box}(s_{13}, s_{13}, s_{12}); \quad (4.77)$$

$$A_6 = \left( \frac{(4\pi)^\epsilon}{16\pi^2} \frac{\Gamma(1+\epsilon)\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)} \right)^2 (-q^2)^{-2-2\epsilon} \left[ -\frac{1}{\epsilon^4} + \frac{5\pi^2}{6\epsilon^2} + \frac{23\zeta_3}{\epsilon} + \frac{103\pi^4}{180} + \mathcal{O}(\epsilon) \right]. \quad (4.78)$$

Note that the  $A_6$  and  $V_8$  integrals are described in Subsubsections 4.2.5.4 and 4.3.4.3. In those Subsubsections, the integrals are given normalised using the usual  $S_\Gamma$  prefactor.

Using  $\mathcal{V}_8$ ,  $\mathcal{A}_6$  and  $\mathcal{R}_{8,a}$  to denote the series parts of  $\operatorname{Re} V_8$ ,  $P_2 \operatorname{Re} A_6$  and  $R_{8,a}$ , respectively, we get [13],

$$\begin{aligned} R_{8,b} &= \frac{1}{2} \operatorname{Im} S + S_\Gamma (q^2)^{-2-2\epsilon} \left( \frac{1}{2} \frac{\cos(2\pi\epsilon)\Gamma^6(1-\epsilon)\Gamma^2(1-\epsilon)}{\Gamma^2(1-2\epsilon)} \mathcal{A}_6 - \mathcal{V}_8 - \frac{1}{4} \mathcal{R}_{8,a} \right) \\ &= S_\Gamma (q^2)^{-2-2\epsilon} \left[ \frac{3}{4\epsilon^4} - \frac{17\pi^2}{12\epsilon^2} - \frac{44\zeta_3}{\epsilon} - \frac{61\pi^4}{60} + \mathcal{O}(\epsilon) \right]. \end{aligned} \quad (4.79)$$

### 4.5 Exploratory Massive Extension: $A_3^0$

The massive  $A_3^0$  route is currently the only massive route supported in *AntCalc*. Its inner workflow is similar to that of the massless  $A_3^0$  route with the added complexity of the quarks being massive.

Kinematically, instead of considering only the centre of mass energy as a scale, the quark mass must also be considered. Hence,  $q^2$  and  $m_q^2$  are scales, and they combine to create the relevant threshold variable  $r$ ,

$$r = 1 - \frac{4m_q^2}{q^2}. \quad (4.80)$$

The phase space has  $k_1^2 = k_2^2 = m_q^2$  and  $k_3^2 = 0$ . The cut families also shift the massless cuts by a mass term as,

$$\begin{aligned} k_{1,2}^2 &\rightarrow m_q^2 - k_{1,2}^2, \\ k_3^2 &\rightarrow k_3^2. \end{aligned} \quad (4.81)$$

This also implies using massive Mandelstam invariants which, in this case, with  $k_3$  presenting the massless

gluon's momentum, become,

$$s_{ij} = (q - k_k)^2 - m_i^2 - m_j^2, \quad \{i, j, k\} = \{1, 2, 3\}. \quad (4.82)$$

*LiteRed2*, then, has to be updated with these new families before recognising the topology. It will then generate two master integrals. In the current *AntCalc* implementation, the IBP reduction step produces two MI which do not exactly match the  $I_1^{(m_q, 0, m_q)}$  and  $I_2^{(m_q, 0, m_q)}$  integrals described in Subsection 4.5.1. Instead, it produces  $J_1$  and  $J_2$ , shown below using *LiteRed2*'s  $j[\dots]$  notation, where the complete basis, named MX30Basis123, uses the denominators  $\{m_q^2 - k_1^2, m_q^2 - k_2^2, k_3^2, -s_{13}, -s_{23}\}$  in that order,

$$\begin{aligned} J_1^{(m_q, 0, m_q)} &= j[\text{MX30Basis123}, 1, 1, 1, 0, 0] = \mathcal{N}_{\text{LiteRed2}} \int \text{Cut} \left[ \frac{1}{D_1 D_2 D_3} \right], \\ J_2^{(m_q, 0, m_q)} &= j[\text{MX30Basis123}, 2, 1, 1, 0, 0] = \mathcal{N}_{\text{LiteRed2}} \int \text{Cut} \left[ \frac{1}{D_1^2 D_2 D_3} \right]. \end{aligned} \quad (4.83)$$

Here,  $\mathcal{N}_{\text{LiteRed2}}$  represents the normalisation implicit in the cut-integral convention used for the family in *LiteRed2*. These two MI do not directly equal the two literature masters, but they can be mapped to them. We need here to introduce a conversion factor of  $-1/4$  between the literature and *AntCalc*'s conventions. Using it, we can complete the mapping as,

$$\begin{aligned} I_1^{(m_q, 0, m_q)} &= -\frac{1}{4} J_1^{(m_q, 0, m_q)}, \\ I_2^{(m_q, 0, m_q)} &= -\frac{1}{4} \left( a J_1^{(m_q, 0, m_q)} + b J_2^{(m_q, 0, m_q)} \right). \end{aligned} \quad (4.84)$$

with,

$$\begin{aligned} a &= \frac{(1 - 2\epsilon)m_q^2 + (1 - \epsilon)q^2}{3(1 - \epsilon)}, \\ b &= \frac{m_q^2(4m_q^2 - q^2)}{3(1 - \epsilon)}. \end{aligned} \quad (4.85)$$

#### 4.5.1 Exploratory Massive Master Integrals

*AntCalc* implements the complete massive  $A_3^0$  build-and-integrate pipeline; the route remains exploratory in the sense that, unlike the massless set, it has not yet been checked against a global observable (see Table 4.3). To integrate it, two new MI are required. These two massive MI are,

$$\begin{aligned} I_1^{(m_q, 0, m_q)} &= \int d\Phi_X^{(m_q, 0, m_q)} \\ I_2^{(m_q, 0, m_q)} &= \int d\Phi_X^{(m_q, 0, m_q)} s_{13} \end{aligned} \quad (4.86)$$

where  $\Phi_X^{(m_q, 0, m_q)}$  is the massive tripole phase space and is defined as,

$$d\Phi_X^{(m_q, 0, m_q)} = \frac{d\Phi_3^{(m_q, 0, m_q)}(q; k_1, k_3, k_2)}{d\Phi_2^{(m_q, m_q)}(q; \tilde{k}_I, \tilde{k}_K)}, \quad p \rightarrow q(k_1, m_q) + g(k_3, 0) + \bar{q}(k_2, m_q) \quad (4.87)$$

and  $m_q$  is the quark mass. Note that the two- and three-particle massive phase spaces are described as in Eq. (4.2) with  $k_1^2 = k_2^2 = m_q^2$ ,  $k_3^2 = 0$ , and  $q^2 > 4m_q^2$ .

These two integrals evaluate to [25],

$$\begin{aligned} I_1^{(m_q, 0, m_q)} &= \left(q^2\right)^{1-\epsilon} r^{2-2\epsilon} 2^{-2\epsilon} \pi^{-2+\epsilon} \frac{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)}{\Gamma(6-6\epsilon)} {}_2F_1\left(\frac{1}{2}, 2-2\epsilon; \frac{7}{2}-3\epsilon; r\right) \\ I_2^{(m_q, 0, m_q)} &= \left(q^2\right)^{2-\epsilon} r^{3-2\epsilon} 2^{1-2\epsilon} \pi^{-2+\epsilon} \frac{\Gamma(3-2\epsilon)\Gamma(4-3\epsilon)}{\Gamma(8-6\epsilon)} {}_2F_1\left(\frac{1}{2}, 3-2\epsilon; \frac{9}{2}-3\epsilon; r\right) \end{aligned} \quad (4.88)$$

where,

$$r = 1 - \frac{4m_q^2}{q^2}. \quad (4.89)$$

## 4.5.2 Build and Integrate Results

To build and integrate the massive  $A_3^0$  in *AntCalc*, we use the same commands as for the massless  $A_3^0$  route, as discussed in Subsubsection 4.1.1.1, with the added option `quarkMass -> mQ`, where  $mQ$  is the quark mass, so far written here as  $m_q$ . The building process is completed as,

```
In[2] := BuildAntenna[A, 3, 0, quarkMass->mQ]
Out[2] :=
```

$$\left[ -4(-2 + \epsilon)m_q^4 \left( s_{13}^2 + s_{23}^2 \right) \right] + \dots$$

The integration process follows in the same usual way, noting that much like in all other integrated cases, the default output is given with  $q^2 \rightarrow 1$ , making here  $r \rightarrow 1 - 4m_q^2$ ,

```
In[3] := BuildAndIntegrateAntenna[A, 3, 0, quarkMass->mQ,
ExpansionOrder->0]
Out[3] :=
```

$$\begin{aligned} &\frac{1}{2100m_q^2(1+2m_q^2)^2\pi^2} \left[ \frac{1}{\epsilon} \left( 70m_q^2(-1+2m_q^2+8m_q^4) {}_2F_1\left(\frac{1}{2}, 2; \frac{7}{2}; 1-4m_q^2\right) + \dots \right) \right. \\ &\quad \left. + \left( 10(-1+2m_q^2+8m_q^4) {}_2F_1\left(\frac{1}{2}, 2; \frac{7}{2}; 1-4m_q^2\right) + \dots \right) \right] + O(\epsilon) \end{aligned}$$

These expressions agree with the established literature results [25]. The full unintegrated and integrated antenna expressions were truncated in this section. They are shown completely in Appendix A.

## 4.6 Overview of Implemented Routes

Having covered both the massless antenna set and the exploratory massive extension, this section collects every implemented route in one place. Table 4.2 summarises how each route is constructed and integrated; Table 4.3 summarises the corresponding validation evidence and status. Together, they replace the individual validation statements given throughout this chapter with a single point of reference.

| Route                                                              | Process / Scope                                          | API Call <sup>1</sup>    | Construction Check                                                                    | Integration Route                                                                                                                              |
|--------------------------------------------------------------------|----------------------------------------------------------|--------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| $A_2^0$                                                            | $\gamma^* \rightarrow q\bar{q}$ , LO                     | [A, 2, 0]                | Self-interference, Born-normalised reference                                          | Trivial (reference, $A_2^0 \equiv 1$ )                                                                                                         |
| $A_3^0$                                                            | $\gamma^* \rightarrow qg\bar{q}$ , NLO real              | [A, 3, 0]                | Self-interference; colour-coefficient extraction; matched to by-hand derivation       | IBP / <i>LiteRed2</i> , $\times 30$ basis; MI: $R_3$                                                                                           |
| $A_2^1$                                                            | $\gamma^* \rightarrow q\bar{q}$ , NLO virtual            | [A, 2, 1]                | Tree-loop interference; PaVe-reduced scalar extraction                                | PaVe / <i>FeynHelpers</i> ; MI: $B_0, C_0$                                                                                                     |
| $A_3^1$ set ( $A_3^1, \tilde{A}_3^1, \hat{A}_3^1$ )                | $\gamma^* \rightarrow qg\bar{q}$ , NNLO, one loop        | [A, 3, 1]                | Tree-loop interference; colour-coefficient extraction (leading/subleading/quark-loop) | IBP / <i>LiteRed2</i> , $A_{31}$ basis; build-side MI are PaVe scalars ( $B_0, C_0, D_0$ , Table 4.1); integration MI: $V_{5,a}, V_{5,b}, V_8$ |
| $A_2^2$ set ( $A_2^2, \tilde{A}_2^2, \hat{A}_2^2, \check{A}_2^2$ ) | $\gamma^* \rightarrow q\bar{q}$ , NNLO, two loops        | [A, 2, 2]                | Two-loop tree interference and one-loop self-interference (two source contributions)  | IBP / <i>LiteRed2</i> , <i>A22TwoLoopTree/A22OneLoopSelf</i> bases; MI: $A_{22,LO}, A_3, A_4$ (also $A_6$ for $\tilde{A}_2^2$ )                |
| $A_4^0, \tilde{A}_4^0$                                             | $\gamma^* \rightarrow qgg\bar{q}$ , NNLO, double-real    | [A, 4, 0]                | Colour-ordered amplitude; colour-coefficient extraction                               | IBP / <i>LiteRed2</i> , $\times 40$ basis; MI: $R_4, R_6, R_{8,a}, R_{8,b}$                                                                    |
| $B_4^0$                                                            | $\gamma^* \rightarrow q\bar{q}q'\bar{q}'$ , NNLO         | [B, 4, 0]                | Sector self-interference (direct amplitude only)                                      | IBP / <i>LiteRed2</i> , $\times 40$ basis; MI: $R_4, R_6, R_{8,a}, R_{8,b}$                                                                    |
| $C_4^0$                                                            | $\gamma^* \rightarrow q\bar{q}q\bar{q}$ , NNLO           | [C, 4, 0]                | Sector symmetrised interference (direct and exchange)                                 | IBP / <i>LiteRed2</i> , $\times 40$ basis; MI: $R_4, R_6, R_{8,a}, R_{8,b}$                                                                    |
| Massive $A_3^0$                                                    | $\gamma^* \rightarrow q(m_q)g\bar{q}(m_q)$ , exploratory | [A, 3, 0, quarkMass->mQ] | Same route as massless $A_3^0$ , with the quarkMass option                            | IBP / <i>LiteRed2</i> , massive family; MI mapped from <i>LiteRed2</i> 's $J_1, J_2$ to the literature $I_1^{(m_q,0,m_q)}, I_2^{(m_q,0,m_q)}$  |

**Table 4.2:** Construction and integration overview of every antenna route implemented in *AntCalc*.

| Route                               | Literature Reference (Order Checked) | Local Validation                                    | Global Validation                  | Status          |
|-------------------------------------|--------------------------------------|-----------------------------------------------------|------------------------------------|-----------------|
| $A_2^0$                             | — (reference normalisation)          | Not applicable                                      | LO normalisation of the $R$ -ratio | By construction |
| $A_3^0$                             | [5] (NLO)                            | Eikonal factor and AP splitting function reproduced | Feeds $r_{\text{NLO}}$             | Pass            |
| $A_2^1$                             | [5] (NLO)                            | Not applicable                                      | Feeds $r_{\text{NLO}}$             | Pass            |
| $A_3^1$ set                         | [5, 24] (NNLO)                       | Not performed                                       | Feeds $r_{\text{NNLO}}$            | Pass            |
| $A_2^2, \tilde{A}_2^2, \hat{A}_2^2$ | [24] (NNLO)                          | Not applicable                                      | Feeds $r_{\text{NNLO}}$            | Pass            |
| $\check{A}_2^2$                     | [30] (NNLO)                          | Not applicable                                      | Feeds $r_{\text{NNLO}}$            | Pass            |
| $A_4^0, \tilde{A}_4^0$              | [5] (NNLO)                           | Not performed                                       | Feeds $r_{\text{NNLO}}$            | Pass            |
| $B_4^0$                             | [5] (NNLO)                           | Not performed                                       | Feeds $r_{\text{NNLO}}$            | Pass            |

Table 4.3 continued.

| Route           | Literature Reference<br>(Order Checked) | Local Validation | Global Validation                                            | Status                                                            |
|-----------------|-----------------------------------------|------------------|--------------------------------------------------------------|-------------------------------------------------------------------|
| $C_4^0$         | [5] (NNLO)                              | Not performed    | Feeds $r_{\text{NNLO}}$                                      | Pass                                                              |
| Massive $A_3^0$ | [25]                                    | Not performed    | Not applicable<br>(does not feed the<br>massless $R$ -ratio) | Literature agreement; no<br>global observable valida-<br>tion yet |

**Table 4.3:** Validation evidence and status of every antenna route implemented in *AntCalc*. “Not applicable” marks routes with no real unresolved parton to take a soft/collinear limit of  $(A_2^0, A_2^1, A_2^2)$ ; “Not performed” marks routes where such a limit exists but was not checked in this work — see the chapter introduction for why  $A_3^0$  was chosen to demonstrate the method rather than repeating it on every route. The massive  $A_3^0$  route is deliberately marked as less mature than the massless set: it agrees with the literature locally, but has not yet been checked against any global observable.

## Chapter 5

# Global Validation: The Massless Hadronic R-Ratio

As mentioned in Chapter 1, the antenna functions that were computed using *AntCalc* in the previous Chapter 4 now need to be globally validated. This validation is undertaken by the computation of the massless hadronic  $R$ -ratio up to NNLO. This observable is particularly useful for validating these antenna functions due to the fact that it is fully inclusive: it requires the entire antenna suite, whose poles cancel only in the inclusive sum, as discussed in Chapter 2. Its result at NLO and NNLO is also known and has been computed and validated through methods other than Antenna Subtraction [31].

### 5.1 R-Ratio Definition

The hadronic  $R$ -ratio is usually defined as the ratio of the total cross-section for  $e^+e^-$  annihilation into hadrons to the point-like cross-section for  $e^+e^- \rightarrow \mu^+\mu^-$ . It is mathematically described as follows,

$$\begin{aligned} R_{\text{hadr}} &= \frac{\sigma(e^+e^- \rightarrow \text{hadrons})}{\sigma(e^+e^- \rightarrow \mu^+\mu^-)} \\ &= \frac{3q^2}{4\pi\alpha_{\text{em}}^2} \sigma(e^+e^- \rightarrow \text{hadrons}). \end{aligned} \quad (5.1)$$

Looking closely, the  $\sigma(e^+e^- \rightarrow \text{hadrons})$  term is what QCD predicts for  $\gamma^* \rightarrow \text{hadrons}$ , computed via the optical theorem [13, 32]. What makes the  $R$ -ratio a clean QCD observable, and a good benchmark for *AntCalc* is that the ratio features terms for higher perturbative orders as a perturbative series in  $\alpha_s$  [31],

$$R_{\text{hadr}} = \left( c_{\text{LO}} + \frac{\alpha_s}{\pi} c_{\text{NLO}} + \left( \frac{\alpha_s}{\pi} \right)^2 c_{\text{NNLO}} + \mathcal{O}(\alpha_s^3) \right) \quad (5.2)$$

here the perturbative coefficients,  $c_{\text{LO}}$ ,  $c_{\text{NLO}}$  and  $c_{\text{NNLO}}$ , respectively receive contributions from the multiparticle final states at the relative orders via their respective antennae, as described in Chapter 2. For example,  $A_3^0$  as described in Chapter 4 contributes to  $c_{\text{NLO}}$  [5]. Moreover, the leading-order term  $R^{(0)}$  simply counts the number of quark colours and flavours weighted by their electric charges [2],

$$R^{(0)} = N_c \sum_q e_q^2 \quad (5.3)$$

since  $R_{\text{hadr}} = R^{(0)}(1 + \mathcal{O}(\alpha_s)) = R^{(0)}R(\alpha_s)$ .

### 5.2 Global Validation with *AntCalc*

As discussed, the hadronic  $R$ -ratio is a very useful quantity to compute using the integrated antenna functions obtained from *AntCalc*. As such, the programme provides an  $R$ -ratio calculator. It is the longest-running function in the whole package. Its result gives,

```
In[2] := BuildRRatio[SMQCD]
```

```
Out[2] :=
```

$$1 + \left(\frac{\alpha_s}{2\pi}\right) \frac{3}{2} C_F + \left(\frac{\alpha_s}{2\pi}\right)^2 C_F \left[ N \left( \frac{243}{16} - 11\zeta_3 \right) + \frac{3}{16N} + N_f \left( -\frac{11}{4} + 2\zeta_3 \right) \right]$$

As expected, the results for each perturbative order are finite. This validates the mutual normalisation and assembly of the integrated antennae: at each perturbative order, their infrared poles cancel in the inclusive sum, leaving a finite coefficient. Moreover, the result matches the literature's [24, 31].

### 5.2.1 $T$ -objects

The antenna functions computed in Chapter 4 are colour-stripped purely kinematic objects. For subtraction, as discussed in Subsection 2.4.1, we need to assemble differential cross-section subtraction terms, like  $d\sigma_{\text{NNLO}}^S$ , from contributions across several distinct final-state topologies. For each topology, every relative antenna-set component must be included, correctly colour-ordered and normalised using the common  $2C_F T_{q\bar{q}}^2$  factor, where  $T_{q\bar{q}}^2$  is the  $T$ -object relative to the LO  $\gamma^* \rightarrow q\bar{q}$  process. We define these  $T$ -objects as this per-topology, colour-restored assembly.

Each antenna-set is organised in its own  $T$ -object, which is notated accordingly, with a subscript relaying the final-state particles, e.g.  $q\bar{q}g$ , and a superscript that relays the perturbation order as 2 for LO, 4 for NLO, and 6 for NNLO. As stated, all  $T$ -objects at NLO and NNLO are normalised by the LO  $T$ -object relative to the  $A_2^0$  antenna. This  $T$ -object equals,

$$T_{q\bar{q}}^2 = N |\mathcal{M}_2^0|^2 = 4N(1 - \epsilon)q^2. \quad (5.4)$$

The remaining relevant  $T$ -objects, for the coming  $R$ -ratio assembly, follow directly from the integrated antennae computed in Chapter 4. Writing

$$P = 2C_F T_{q\bar{q}}^2, \quad (5.5)$$

the NLO  $T$ -objects are,

$$T_{q\bar{q}}^4 = P A_2^1, \quad T_{q\bar{q}g}^4 = P A_3^0, \quad (5.6)$$

and the NNLO  $T$ -objects are,

$$T_{q\bar{q}}^6 = P \left( N A_2^2 + \frac{1}{N} \tilde{A}_2^2 + N_f \hat{A}_2^2 + \left( N - \frac{1}{N} \right) \tilde{\hat{A}}_2^2 \right), \quad (5.7)$$

$$T_{q\bar{q}g}^6 = P \left[ N \left( A_3^1 + A_2^1 A_3^0 \right) - \frac{1}{N} \left( \tilde{A}_3^1 + A_2^1 A_3^0 \right) + N_f \hat{A}_3^1 \right], \quad (5.8)$$

$$T_{q\bar{q}q'\bar{q}'}^6 = P (N_f - 1) B_4^0, \quad (5.9)$$

$$T_{q\bar{q}q\bar{q}}^6 = P \left( B_4^0 - \frac{1}{N} C_4^0 \right), \quad (5.10)$$

$$T_{q\bar{q}gg}^6 = P \left( N A_4^0 - \frac{1}{N} \tilde{A}_4^0 \right). \quad (5.11)$$

We can now integrate the antennae and get the integrated  $T$ -objects, notated as  $\mathcal{T}$ . These  $\mathcal{T}$ -objects are the ones used when assembling the  $R$ -ratio in the next section. Let us now consider the two NLO  $\mathcal{T}$ -objects,  $\mathcal{T}_{q\bar{q}}^4$  and  $\mathcal{T}_{q\bar{q}g}^6$ , each relative to the generating processes of the  $A_2^1$  and  $A_3^0$  antennae. As discussed previously, via the KLN theorem, the virtual correction poles of the first antenna must cancel against the real emission poles of the second. That can be seen as follows,

$$\begin{aligned} r_{\text{NLO}} &= \frac{\mathcal{T}_{q\bar{q}}^4 + \mathcal{T}_{q\bar{q}g}^6}{T_{q\bar{q}}^2} = 2C_F \left( \mathcal{A}_2^1 + \mathcal{A}_3^0 \right) \\ &= 2C_F \left[ \left( -\frac{1}{\epsilon^2} - \frac{3}{2\epsilon} - 4 + \frac{7\pi^2}{12} + \dots \right) + \left( \frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{19}{4} - \frac{7\pi^2}{12} + \dots \right) \right] = \frac{3}{2} C_F. \end{aligned} \quad (5.12)$$

### 5.2.2 R-Ratio Assembly

The R-ratio is assembled using these  $\mathcal{T}$ -objects. We simply sum the normalised objects order-by-order as,

$$\begin{aligned}
 R &= \frac{R_{\text{had}}}{R^{(0)}} = R_{\text{LO}} + R_{\text{NLO}} + R_{\text{NNLO}} \\
 &= r_{\text{LO}} + \frac{\alpha_s}{2\pi} r_{\text{NLO}} + \left(\frac{\alpha_s}{2\pi}\right)^2 r_{\text{NNLO}} \\
 &= 1 + \frac{\alpha_s}{2\pi} \frac{\mathcal{T}_{q\bar{q}}^4 + \mathcal{T}_{q\bar{q}g}^4}{T_{q\bar{q}}^2} + \left(\frac{\alpha_s}{2\pi}\right)^2 \frac{\mathcal{T}_{q\bar{q}}^6 + \mathcal{T}_{q\bar{q}g}^6 + \mathcal{T}_{q\bar{q}q\bar{q}}^6 + \mathcal{T}_{q\bar{q}q'\bar{q}'}^6 + \mathcal{T}_{q\bar{q}gg}^6}{T_{q\bar{q}}^2} + \mathcal{O}(\alpha_s^3).
 \end{aligned} \tag{5.13}$$

Expanding the  $\mathcal{T}$ -objects, we are able to describe the R-ratio formula using the integrated antenna functions as,

$$\begin{aligned}
 r_{\text{LO}} &= 1 \\
 r_{\text{NLO}} &= 2C_F (\mathcal{A}_2^1 + \mathcal{A}_3^0) \\
 r_{\text{NNLO}} &= 2C_F \left[ N(\mathcal{A}_2^1 \mathcal{A}_3^0 + \mathcal{A}_3^1 + \mathcal{A}_4^0 + \mathcal{A}_2^2) - \frac{1}{N}(\mathcal{A}_2^1 \mathcal{A}_3^0 + \tilde{\mathcal{A}}_3^1 + \tilde{\mathcal{A}}_4^0 + C_4^0 - \tilde{\mathcal{A}}_2^2) \right. \\
 &\quad \left. + N_f(\mathcal{B}_4^0 + \hat{\mathcal{A}}_3^1 + \hat{\mathcal{A}}_2^2) + 2C_F \tilde{\mathcal{A}}_2^2 \right].
 \end{aligned} \tag{5.14}$$

Assembling these pieces together, using the integrated antenna results from *AntCalc*, we finally get,

$$R = 1 + \left(\frac{\alpha_s}{2\pi}\right) \frac{3}{2} C_F + \left(\frac{\alpha_s}{2\pi}\right)^2 C_F \left[ N \left( \frac{243}{16} - 11\zeta_3 \right) + \frac{3}{16N} + N_f \left( -\frac{11}{4} + 2\zeta_3 \right) \right] \tag{5.15}$$

which is the exact massless hadronic R-ratio coefficient through NNLO [24] and is also exactly the expression we get by running `BuildRRatio[SMQCD]`.

### 5.3 Comparison with Experimental Data

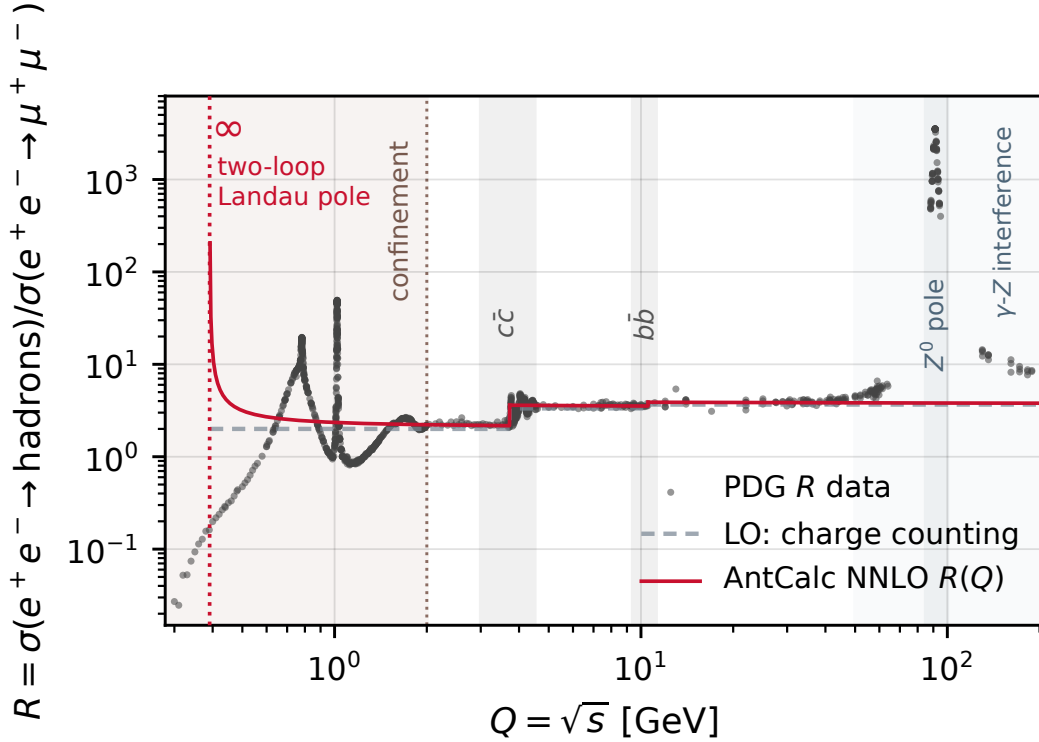
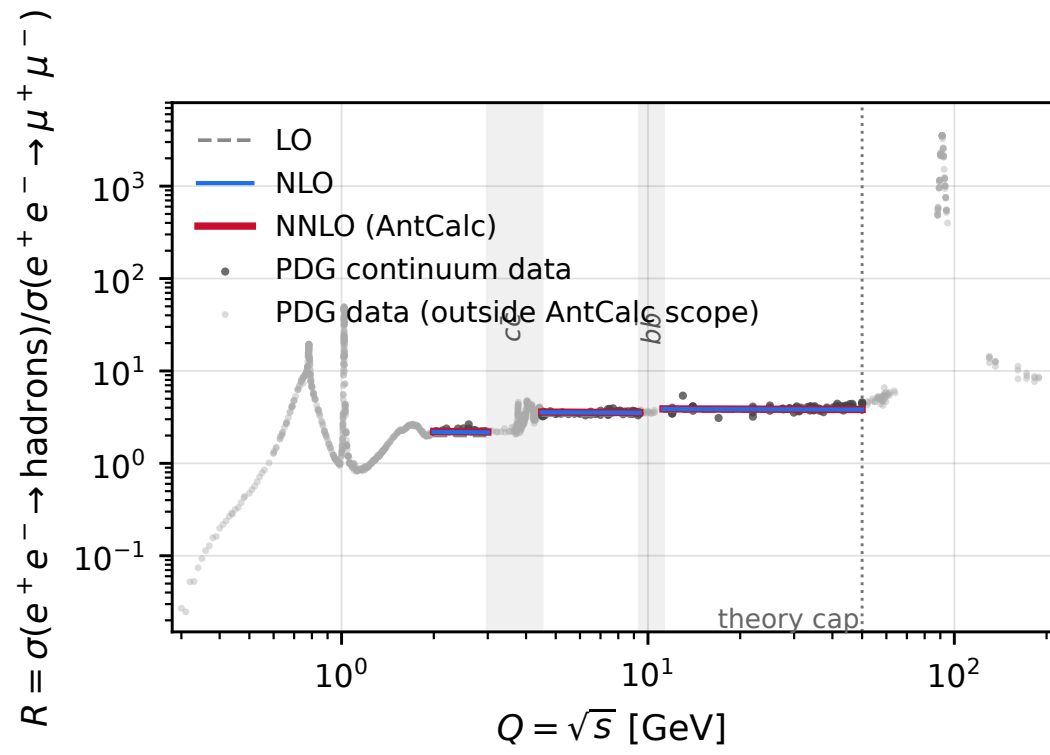


Figure 5.1

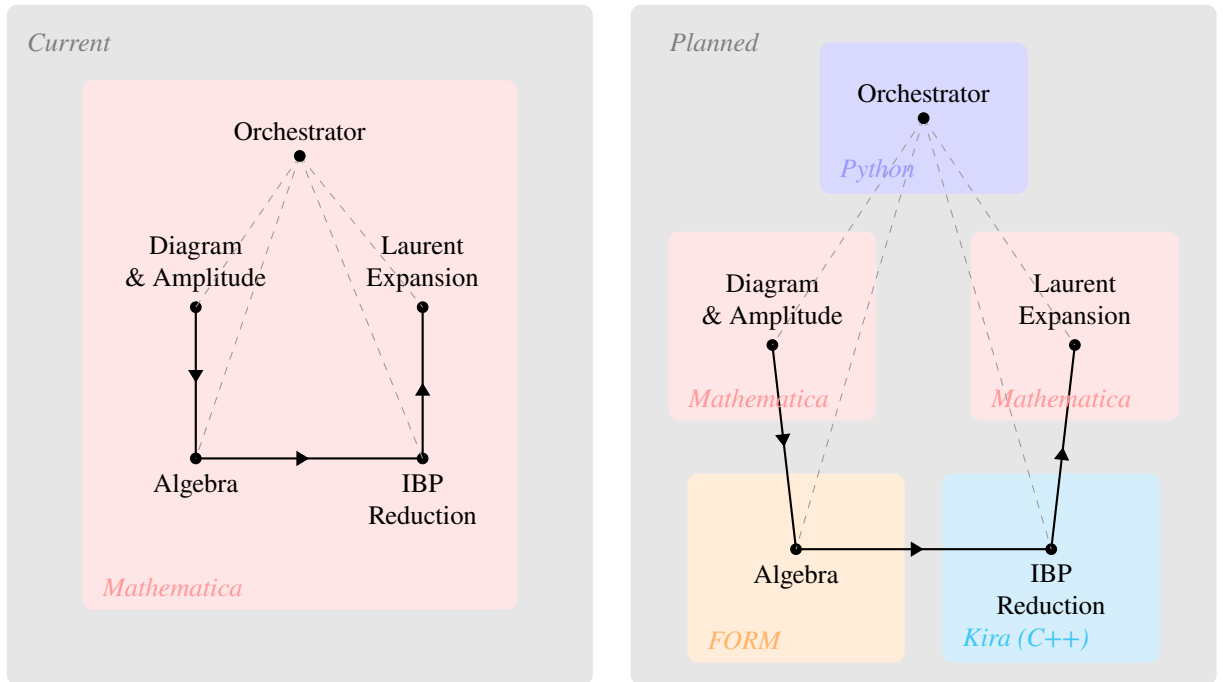


**Figure 5.2**

## Chapter 6

# Future Work

In its current version, *AntCalc* is implemented entirely in *Wolfram Mathematica*. This is the right language to build a trial-stage programme of its nature due to its forgiving mathematical environment and its general adoption throughout the field. But this choice of language comes at a cost: efficiency. Though convenient, *Mathematica* is slow and memory-hungry. In its current state, running the full `BuildRRatio` call takes one and a half to two hours and, on machines with less RAM, may cause the calculation to crash halfway through. It is then apparent that the next step in *AntCalc*'s development is taking it mostly out of the *Mathematica* environment and making it a truly modular, multi-language programme that exploits only the best characteristics of each tool, including *Mathematica*, keeping the current high-level workflow. The current plan for this step is to make the orchestrator a *Python* programme and to store the profiles in plain, machine-readable files, such as those in the JSON format. Then, retain *FeynArts* and *FeynCalc* for diagram generation and amplitude manipulation, respectively, but hand the algebraic manipulations required to obtain the antenna, or its components, over to *FORM*. Then, lastly, make IBP reduction faster and more stable by replacing *LiteRed2* with *Kira*, written in C++, keeping *Mathematica* as an algebraic simplifier.



**Figure 6.1:** Diagram showing the current and planned workflow steps relative to their programming languages.

This step will then allow us to keep expanding *AntCalc*'s profile library without fear of hitting a computational bottleneck. The first point of expansion would be to the rest of the massive quark-antiquark final-final antenna functions, especially  $A_4^0$ ,  $B_4^0$  and  $C_4^0$  at first, and then  $A_3^1$  and  $A_2^2$ . After these are completed, expanding the scope to other relevant antenna types, particularly the quark-gluon ( $D$  and  $E$ ) and the gluon-gluon ( $F$ ,  $G$  and  $H$ ) antennae, both in the massless and in the massive regimes becomes the logical next step. Lastly, extensions to kinematics other than final-final, namely initial-initial and initial-final configurations, and higher perturbative orders, particularly  $N^3\text{LO}$ , antennae would also be relevant.

A further, more structural, direction is to generalise *AntCalc* from a validated library of known antenna functions into a construction tool for antenna functions that have not yet been computed at all. The profile-driven architecture described in Chapter 3 already separates the process-specific description of a route, its profile, from the shared build and integration workflows; letting a user supply their own process description, rather than only selecting an existing route key, is a natural continuation of that design rather than a new one. Such user-defined routes would not, by definition, have a published result to validate against. This is precisely the gap that the amplitude-level Ward-identity check described in Subsection 3.7.1 is suited to: since it certifies gauge invariance directly from the generated amplitude, independently of any external reference, it, and checks like it, could be run automatically on a newly constructed antenna as a first, literature-independent certificate of correctness, before any attempt is made to compare it with an established result.

Connections to other established tools would also be pertinent. A connection to a Feynman diagram generation tool such as *MadGraph* [33] and one to a parton-level event generator such as *NNLOJet* [34] could allow us to generate more complex antennae, from more complex topologies and also enable the numerical validation of the results.

# Chapter 7

## Conclusion

This thesis developed the *Mathematica* package *AntCalc* to remove a friction that recurs at the start of most new antenna-subtraction calculations: before the intended physics can even be pursued, a group must first reconstruct and integrate the antenna functions it needs from scratch, much as Feynman-diagram algebra once had to be carried out by hand before tools such as *FeynCalc* became standard. *AntCalc* removes this initial barrier, providing a validated, ready-to-use library of antenna functions that can be loaded directly into a *Mathematica* notebook, making their construction and everyday use easier, more standardised, and streamlined.

The resulting framework was implemented completely within *Mathematica*, using the *FeynCalc* environment (including *FeynArts* and *FeynHelpers*) to handle the building stage and *LiteRed2* for the IBP reduction in the integration stage. That framework was shown to produce all massless quark-antiquark final-final antennae through NNLO. It was also shown to be able to be extended to the massive regime by implementing the massive  $A_3^0$  route in its entirety. It begins by using *FeynArts* to produce one or more Feynman diagrams relevant for the process under study and generates their amplitudes. It then develops the amplitude into one antenna or multiple antenna components using *FeynCalc*. The unintegrated antenna is then handed to *LiteRed2* for IBP reduction and, finally, after the pre-computed MI are identified and substituted, the integrated antenna is Laurent-expanded around the dimensional regularisation parameter  $\epsilon$ .

These unintegrated and integrated antennae were shown to be in agreement with established results in the literature. Furthermore, the massless  $A_3^0$  antenna, whose workflow was shown in great detail, was locally validated by computing the eikonal factor and the Altarelli–Parisi splitting function, using a method that may be used to validate other appropriate antennae. All massless integrated antenna results were also validated when we computed the massless hadronic R-ratio.

The package is beginning to reach the computational limits of its current workflow, running completely within *Mathematica*. As such, the most pressing piece of future work is the overhaul to a new, truly modular and multi-language, workflow, that is able to harness the best characteristics of different tools, like Kira for IBP reduction and FORM for algebraic manipulation, without compromising efficiency. Once this overhaul is done, *AntCalc* could be expanded to handle much more complex cases, particularly those in the massive regime, other antenna types, such as quark-gluon, other kinematic configurations, such as initial-final antennae, and those in higher perturbative orders, such as N<sup>3</sup>LO.

# Appendix A

## Complete Antenna Expressions

This appendix includes the complete expressions of the antennae computed using *AntCalc*. It shows the unintegrated and integrated antennae, along with the commands used to compute them.

A note on conventions: the unintegrated antenna functions below are given exactly as *AntCalc* returns them, in terms of  $s_{ij}$  and  $q^2$ . The integrated antennae, however, are quoted at  $q^2 = 1$ , matching *IntegrateAntenna*'s default convention. This does not break generality as established in Subsection 2.5.2 and Eq. (2.91), a quantity of this kind scales with  $q^2$  as a single power fixed by its mass dimension and  $\epsilon$ -scaling, so the complete  $q^2$ -dependent result is recovered by multiplying the expression given here by  $(q^2)^{\lambda+\kappa\epsilon}$  for the appropriate  $\lambda$  and  $\kappa$ . The explicit case of  $\mathcal{A}_3^0$ , which scales as  $(q^2)^{-\epsilon}$ , is worked out in full in Eq. (4.36). We must also here introduce the shorthand,

$$C_\epsilon \equiv \frac{(4\pi)^\epsilon e^{-\gamma_E \epsilon}}{8\pi^2} = \frac{1}{C(\epsilon, 1)} \quad (\text{A.1})$$

as the one-loop normalisation factor.

### A.1 LO Antenna

#### A.1.1 $A_2^0$

Running `BuildAntenna[A, 2, 0]`, we get the built result,

$$A_2^0 = 1. \quad (\text{A.2})$$

By running `BuildAndIntegrateAntenna[A, 2, 0]` we get the integrated result,

$$\mathcal{A}_2^0 = \frac{1}{4} + \epsilon \frac{9}{8} - \epsilon^2 \left( \frac{9}{16} - \frac{\pi^2}{48} \right). \quad (\text{A.3})$$

### A.2 NLO Antennae

#### A.2.1 $A_2^1$

Running `BuildAntenna[A, 2, 0]`, we get the built result,

$$A_2^1 = -\frac{1}{2}(3 + 2\epsilon)B_0(q^2; 0, 0) - q^2 C_0(0, 0, q^2; 0, 0, 0). \quad (\text{A.4})$$

By running `BuildAndIntegrateAntenna[A, 2, 1]` we get the integrated result,

$$\mathcal{A}_2^1 = -\frac{1}{\epsilon^2} - \frac{3}{2\epsilon} - 4 + \frac{7\pi^2}{12} + \epsilon \left( \frac{7\zeta_3}{3} - 8 + \frac{7\pi^2}{8} \right) + \epsilon^2 \left( \frac{7\zeta_3}{2} - 16 + \frac{7\pi^2}{3} - \frac{73\pi^4}{1440} \right) + \mathcal{O}(\epsilon^3). \quad (\text{A.5})$$

### A.2.2 $A_3^0$

Running `BuildAntenna[A, 3, 0]`, we get the built result,

$$A_3^0 = \frac{2s_{12}^2 + 2s_{12}(s_{13} + s_{23})}{q^2 s_{13} s_{23}} - \epsilon \frac{2}{q^2} + (1 - \epsilon) \frac{s_{13}^2 + s_{23}^2}{q^2 s_{13} s_{23}}. \quad (\text{A.6})$$

By running `BuildAndIntegrateAntenna[A, 3, 0]` we get the integrated result,

$$\begin{aligned} \mathcal{A}_3^0 = & \frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{19}{4} - \frac{7\pi^2}{12} + \epsilon \left( -\frac{25\zeta_3}{3} + \frac{109}{8} - \frac{7\pi^2}{8} \right) \\ & + \epsilon^2 \left( -\frac{25\zeta_3}{2} + \frac{639}{16} - \frac{133\pi^2}{48} - \frac{71\pi^4}{1440} \right) + \mathcal{O}(\epsilon^3). \end{aligned} \quad (\text{A.7})$$

## A.3 NNLO Antennae

### A.3.1 $A_2^2$

Running `BuildAntenna[A, 2, 2, Component->Leading]`, we get the built result,

$$A_2^2 = A_2^{2,(0)} + (-s_{12})^{-\epsilon} C_\epsilon A_2^{2,(-\epsilon)} + s_{12}^{-2\epsilon} A_2^{2,(-2\epsilon)}. \quad (\text{A.8})$$

where,

$$A_2^{2,(0)} = \frac{1}{16\epsilon^4} - \frac{1}{\epsilon^2} \left( \frac{9}{32} - \frac{11\pi^2}{96} \right) - \frac{1}{\epsilon} \left( \frac{377}{64} - \frac{55\pi^2}{96} \right) + \frac{863}{384} - \frac{319\pi^2}{576} - \frac{13\zeta_3}{36} \quad (\text{A.9})$$

$$A_2^{2,(-\epsilon)} = -\frac{11}{6\epsilon^3} + \frac{11}{4\epsilon^2} + \frac{22}{3\epsilon} + \frac{44}{3} + \mathcal{O}(\epsilon) \quad (\text{A.10})$$

$$\begin{aligned} A_2^{2,(-2\epsilon)} = & \frac{3}{16\epsilon^4} + \frac{7}{24\epsilon^3} + \frac{1}{\epsilon^2} \left( \frac{155}{288} - \frac{37\pi^2}{96} \right) + \frac{1}{\epsilon} \left( \frac{5597}{1728} - \frac{355\pi^2}{288} + \frac{7\zeta_3}{12} \right) \\ & - \frac{193531}{10368} - \frac{577\pi^2}{1728} + \frac{263\pi^4}{1440} + \frac{73\zeta_3}{12}. \end{aligned} \quad (\text{A.11})$$

By running `BuildAndIntegrateAntenna[A, 2, 2, Component->Leading]` we get the integrated result,

$$\begin{aligned} \mathcal{A}_2^2 = & \frac{1}{4\epsilon^4} + \frac{17}{8\epsilon^3} + \frac{1}{\epsilon^2} \left( \frac{433}{144} - \frac{\pi^2}{2} \right) + \frac{1}{\epsilon} \left( \frac{4045}{864} - \frac{83\pi^2}{48} + \frac{7\zeta_3}{12} \right) \\ & - \frac{9083}{5184} - \frac{2153\pi^2}{864} + \frac{263\pi^4}{1440} + \frac{13\zeta_3}{9} + \mathcal{O}(\epsilon). \end{aligned} \quad (\text{A.12})$$

### A.3.2 $\tilde{A}_2^2$

Running `BuildAntenna[A, 2, 2, Component->Subleading]`, we get the built result,

$$\tilde{A}_2^2 = \tilde{A}_2^{2,(0)} + s_{12}^{-2\epsilon} \tilde{A}_2^{2,(-2\epsilon)}. \quad (\text{A.13})$$

where,

$$\begin{aligned} \tilde{A}_2^{2,(0)} = & -\frac{3}{16\epsilon^4} - \frac{9}{16\epsilon^3} - \frac{1}{\epsilon^2} \left( -\frac{89}{32} + \frac{11\pi^2}{32} \right) + \frac{1}{\epsilon} \left( \frac{533}{64} - \frac{11\pi^2}{6} - \frac{13\zeta_3}{12} \right) \\ & - \frac{1803}{128} - \frac{341\pi^2}{192} + \frac{79\pi^4}{2880} - \frac{39\zeta_3}{8}. \end{aligned} \quad (\text{A.14})$$

$$\begin{aligned} \tilde{A}_2^{2,(-2\epsilon)} = & -\frac{1}{16\epsilon^4} - \frac{3}{16\epsilon^3} + \frac{1}{\epsilon^2} \left( \frac{7}{32} + \frac{19\pi^2}{96} \right) + \frac{1}{\epsilon} \left( -\frac{975}{64} + \frac{10\pi^2}{3} + \frac{15\zeta_3}{4} \right) \\ & - \frac{499}{128} + \frac{1291\pi^2}{192} - \frac{233\pi^4}{960} + \frac{97\zeta_3}{8}. \end{aligned} \quad (\text{A.15})$$

By running `BuildAndIntegrateAntenna[A, 2, 2, Component->Subleading]` we get the integrated result,

$$\begin{aligned}\mathcal{A}_2^2 = & -\frac{1}{4\epsilon^4} - \frac{3}{4\epsilon^3} - \frac{1}{\epsilon^2} \left( \frac{41}{16} - \frac{13\pi^2}{24} \right) + \frac{1}{\epsilon} \left( -\frac{221}{32} + \frac{3\pi^2}{2} + \frac{8\zeta_3}{3} \right) \\ & - \frac{1151}{64} + \frac{475\pi^2}{96} - \frac{59\pi^4}{288} + \frac{29\zeta_3}{4} + \mathcal{O}(\epsilon).\end{aligned}\tag{A.16}$$

### A.3.3 $\hat{A}_2^2$

Running `BuildAntenna[A, 2, 2, Component->Nf]`, we get the built result,

$$\hat{A}_2^2 = (-s_{12})^{-\epsilon} C_\epsilon \hat{A}_2^{2,(-\epsilon)} + s_{12}^{-2\epsilon} \hat{A}_2^{2,(-2\epsilon)}.\tag{A.17}$$

where,

$$\hat{A}_2^{2,(-\epsilon)} = \frac{1}{3\epsilon^3} - \frac{1}{2\epsilon^2} - \frac{4}{3\epsilon} - \frac{8}{3}\tag{A.18}$$

$$\hat{A}_2^{2,(-2\epsilon)} = \frac{1}{12\epsilon^3} + \frac{7}{18\epsilon^2} + \frac{1}{\epsilon} \left( \frac{353}{216} - \frac{11\pi^2}{72} \right) + \frac{7541}{1296} - \frac{77\pi^2}{108} - \frac{13\zeta_3}{18}.\tag{A.19}$$

By running `BuildAndIntegrateAntenna[A, 2, 2, Component->Nf]` we get the integrated result,

$$\hat{\mathcal{A}}_2^2 = -\frac{1}{4\epsilon^3} - \frac{1}{9\epsilon^2} + \frac{1}{\epsilon} \left( \frac{65}{216} + \frac{\pi^2}{24} \right) + \frac{4085}{1296} - \frac{91\pi^2}{216} + \frac{\zeta_3}{18} + \mathcal{O}(\epsilon).\tag{A.20}$$

### A.3.4 $\check{A}_2^2$

Running `BuildAntenna[A, 2, 2, Component->Breve]`, we get the built result,

$$\check{A}_2^2 = \frac{\Gamma^6(1-\epsilon)\Gamma^2(1+\epsilon)}{\Gamma^2(2-2\epsilon)} \frac{(-2+\epsilon-2\epsilon^2)^2 (\pi^4\epsilon^4 - 80\zeta_3\epsilon^3 - 20\pi^2\epsilon^2 + 120)}{1920\epsilon^4}.\tag{A.21}$$

By running `BuildAndIntegrateAntenna[A, 2, 2, Component->Breve]` we get the integrated result,

$$\begin{aligned}\check{\mathcal{A}}_2^2 = & \frac{1}{4\epsilon^4} + \frac{3}{4\epsilon^3} + \frac{1}{\epsilon^2} \left( \frac{41}{16} - \frac{\pi^2}{24} \right) + \frac{1}{\epsilon} \left( 7 - \frac{\pi^2}{8} - \frac{7\zeta_3}{6} \right) \\ & + 18 - \frac{41\pi^2}{96} - \frac{7\pi^4}{480} - \frac{7\zeta_3}{2} + \mathcal{O}(\epsilon).\end{aligned}\tag{A.22}$$

### A.3.5 $A_3^1$

Running `BuildAntenna[A, 3, 1, Component->Leading]`, we get the built result,

$$\begin{aligned}A_3^1 = & \frac{1}{2(\epsilon-1)q^2} \left( (-s_{13})^{-\epsilon} C_\epsilon A_3^{1,(s_{13})} + (-s_{23})^{-\epsilon} C_\epsilon A_3^{1,(s_{23})} + (-q^2)^{-\epsilon} C_\epsilon A_3^{1,(q^2)} \right. \\ & \left. + \left[ \frac{\pi^2 q^2}{4} + \frac{1}{4} \log^2 \left( \frac{s_{13}}{s_{23}} \right) + \text{Li}_2 \left( 1 - \frac{q^2}{s_{13}} \right) + \text{Li}_2 \left( 1 - \frac{q^2}{s_{23}} \right) \right] A_3^{1,(\text{Trans.})} \right).\end{aligned}\tag{A.23}$$

where,

$$A_3^{1,(s_{13})} = \mathcal{F}_1(s_{12}; s_{13}, s_{23}), \quad A_3^{1,(s_{23})} = \mathcal{F}_1(s_{12}; s_{23}, s_{13})\tag{A.24}$$

$$\begin{aligned}\mathcal{F}_1(s_{12}; x, y) = & \frac{1}{4(q^2)^2 (s_{12} + y)^2} \left[ \frac{1}{\epsilon} \left( 10s_{12}^2 + 4s_{12}(x + 4y) + 3y(x + 2y) \right) \right. \\ & \left. - \frac{1}{x} \left( s_{12}^3 - s_{12}^2(23x - 3y) - s_{12} \left( 13x^2 + 36xy - 3y^2 \right) - 10x^2y - 13xy^2 + y^3 \right) \right]\end{aligned}\tag{A.25}$$

$$A_3^{1,(q^2)} = \frac{\mathcal{P} \left( 20s_{12}^2(q^2)^2 + 28s_{12}q^2s_{13}s_{23} + 11s_{13}^2s_{23}^2 \right)}{12\epsilon(q^2)^2 s_{13}s_{23} (s_{12}q^2 + 13s_{23})^2} + \mathcal{O}(\epsilon), \quad \mathcal{P} = (q^2)^2 + s_{12}^2 - 2s_{13}s_{23} \quad (\text{A.26})$$

$$A_3^{1,(\text{Trans.})} = -\frac{1}{2q^2} A_3^0. \quad (\text{A.27})$$

By running `BuildAndIntegrateAntenna[A, 3, 1, Component->Leading]` we get the integrated result,

$$\begin{aligned} \mathcal{A}_3^1 = & -\frac{1}{4\epsilon^4} - \frac{31}{12\epsilon^3} - \frac{1}{\epsilon^2} \left( \frac{53}{8} - \frac{11\pi^2}{24} \right) - \frac{1}{\epsilon} \left( \frac{647}{24} - \frac{22\pi^2}{9} - \frac{23\zeta_3}{3} \right) \\ & - \frac{5231}{48} + \frac{17\pi^2}{2} - \frac{41\pi^4}{480} + \frac{689\zeta_3}{18} + \mathcal{O}(\epsilon). \end{aligned} \quad (\text{A.28})$$

### A.3.6 $\tilde{A}_3^1$

Running `BuildAntenna[A, 3, 1, Component->Subleading]`, we get the built result,

$$\begin{aligned} \tilde{A}_3^1 = & \frac{\mathcal{R}(\epsilon, s_{13}, s_{23})}{2(\epsilon-1)q^2} \left[ (-s_{12}^{-\epsilon}) \tilde{A}_3^{1,(s_{12})} + (-s_{13})^{-\epsilon} \tilde{A}_3^{1,(s_{13})} + (-s_{23})^{-\epsilon} \tilde{A}_3^{1,(s_{23})} + \right. \\ & + \frac{(-q^2)^{-\epsilon}}{(s_{12}+s_{13})^3(s_{12}+s_{23})^3} \tilde{A}_3^{1,(q^2)} + \epsilon^2(2\epsilon-1) \left( \frac{\pi^2}{4} + \text{Li}_2 \left( 1 - \frac{q^2}{s_{12}} \right) \right) \tilde{A}_3^{1,(\text{Trans. } s_{12})} \\ & \left. + \left( \text{Li}_2 \left( 1 - \frac{q^2}{s_{13}} \right) + \frac{1}{4} \log^2 \left( \frac{s_{12}}{s_{13}} \right) \right) \tilde{A}_3^{1,(\text{Trans. } s_{13})} + \left( \text{Li}_2 \left( 1 - \frac{q^2}{s_{23}} \right) + \frac{1}{4} \log^2 \left( \frac{s_{12}}{s_{23}} \right) \right) \tilde{A}_3^{1,(\text{Trans. } s_{23})} \right] \quad (\text{A.29}) \\ \mathcal{R}(\epsilon, s_{13}, s_{23}) = & \frac{C\epsilon}{\epsilon^2(2\epsilon-1)s_{13}s_{23}} \end{aligned}$$

where, using  $u = s_{13} + s_{23}$  and  $v = s_{13}s_{23}$ ,

$$\begin{aligned} \tilde{A}_3^{1,(s_{12})} = & -4s_{12}^2u^2 + 4\epsilon \left( 3s_{12}^2u^2 - s_{12}^2v + s_{12}u^3 - 2s_{12}uv + u^4 - u^2v \right) \\ & - 2\epsilon^2u \left( 4s_{12}^2u + 4s_{12}u^2 - 4s_{12}v + 7u^3 - 6uv \right) + 2\epsilon^3u^2 \left( 5u^2 - 4v \right) + 4\epsilon^4u^4 \end{aligned} \quad (\text{A.30})$$

$$\tilde{A}_3^{1,(s_{13})} = \mathcal{G}(s_{12}, s_{13}, s_{23}), \quad \tilde{A}_3^{1,(s_{23})} = \mathcal{G}(s_{12}, s_{23}, s_{13}) \quad (\text{A.31})$$

$$\begin{aligned} \mathcal{G}(s_{12}, x, y) = & -2(s_{12}+y)^2 [2(s_{12}+y)^2 - 2y(s_{12}+y) + y^2] + \epsilon \left( 12(s_{12}+y)^4 \right. \\ & + 4(s_{12}+y)^3(x-3y) + 4(s_{12}+y)^2(x^2-xy+2y^2) - 2xy(s_{12}+y)(x-2y) + x^2y^2 \Big) \\ & - \epsilon^2 \left( 8(s_{12}+y)^4 + (s_{12}+y)^3(8x-7y) + (s_{12}+y)^2(14x^2+xy+10y^2) + xy(s_{12}+y)(x+6y) \right. \\ & + x^2y^2 \Big) + \epsilon^3(s_{12}+y) \left( y(s_{12}+y)^2 + (s_{12}+y)^2 + (s_{12}+y)(10x^2+14xy+5y^2) \right. \\ & \left. + 5x^2y \right) + \epsilon^4(s_{12}+y)(x+y) \left( (s_{12}+y)(4x-y) + 2xy \right) \end{aligned} \quad (\text{A.32})$$

$$\begin{aligned} \tilde{A}_3^{1,(q^2)} = & 2s_{12}^5(3u^2-2v) + 12s_{12}^4u(u^2-v) + s_{12}^3(9u^4-10u^3v-8v^2) + s_{12}^2u(3u^4-16v^2) \\ & + 4s_{12}v(u^4-2u^2v-v^2) + 2uv^2(u^2-2v^2) - \epsilon u \left( 2s_{12}^5u + 4s_{12}^4(u^2-v) + 2s_{12}^3u(3u^2-5v) \right. \\ & \left. + 4s_{12}^2(u^4-v^2) + s_{12}uv(9u^2-8v) + 2v^2(3u^2-2v) \right) \end{aligned} \quad (\text{A.33})$$

$$\tilde{A}_3^{1,(\text{Trans. } s_{12})} = 4s_{12}^2 + 2s_{12}(x+y) + x^2 + y^2 - 4\epsilon \left( s_{12}^2 + s_{12}(x+y) + x^2 + xy + y^2 \right)$$

$$+\epsilon^2\left(4(x+y)^2-2xy\right)+\epsilon^3(x+y)^2 \quad (\text{A.34})$$

$$\tilde{A}_3^{1,(\text{Trans. } s_{13})} = \mathcal{H}(s_{12}, s_{13}, s_{23}), \quad \tilde{A}_3^{1,(\text{Trans. } s_{23})} = \mathcal{H}(s_{12}, s_{23}, s_{13}) \quad (\text{A.35})$$

$$\begin{aligned} \mathcal{H}(s_{12}, x, y) &= s_{12}^2 + (s_{12} + y)^2 - 2\epsilon\left(s_{12}^2 + s_{12}(x+y) + x^2 + xy + y^2\right) \\ &\quad + \epsilon^2\left(3x(x+y) + y^2\right) + \epsilon^3x(x+y). \end{aligned} \quad (\text{A.36})$$

By running `BuildAndIntegrateAntenna[A, 3, 1, Component->Subleading]` we get the integrated result,

$$\tilde{\mathcal{A}}_3^1 = -\frac{1}{\epsilon^2}\left(\frac{5}{8} - \frac{\pi^2}{6}\right) - \frac{1}{\epsilon}\left(\frac{19}{4} - \frac{\pi^2}{4} - 7\zeta_3\right) - \frac{105}{4} + \frac{27\pi^2}{16} + \frac{7\pi^4}{90} + \frac{27\zeta_3}{2} + \mathcal{O}(\epsilon). \quad (\text{A.37})$$

### A.3.7 $\hat{A}_3^1$

Running `BuildAntenna[A, 3, 1, Component->Nf]`, we get the built result,

$$\begin{aligned} \hat{A}_3^1 &= \frac{4s_{12}^2}{3s_{13}s_{23}} + \frac{4s_{12}}{3s_{13}} + \frac{4s_{12}}{3s_{23}} + \frac{4s_{23}}{3s_{13}} + \frac{4s_{13}}{3s_{23}} + \frac{4}{3} \\ &\quad - \frac{1}{\epsilon}\left(\frac{4s_{12}^2}{3s_{13}s_{23}} + \frac{4s_{12}}{3s_{23}} + \frac{2s_{23}}{3s_{13}} + \frac{2s_{13}}{3s_{23}}\right) \\ &\quad - \epsilon\left(\frac{2s_{13}}{3s_{23}} + \frac{2s_{23}}{3s_{13}} + \frac{4}{3}\right). \end{aligned} \quad (\text{A.38})$$

By running `BuildAndIntegrateAntenna[A, 3, 1, Component->Nf]` we get the integrated result,

$$\hat{\mathcal{A}}_3^1 = \frac{1}{3\epsilon^3} + \frac{1}{2\epsilon^2} + \frac{1}{\epsilon}\left(\frac{19}{12} - \frac{7\pi^2}{36}\right) + \frac{109}{24} - \frac{7\pi^2}{24} - \frac{25\zeta_3}{9} + \mathcal{O}(\epsilon). \quad (\text{A.39})$$

### A.3.8 $A_4^0$

Running `BuildAntenna[A, 4, 0, Component->Leading]`, we get the built result,

$$A_4^0 = \frac{1}{2(\epsilon-1)q^2}\left(A_4^{0,(0)} + \epsilon A_4^{0,(1)} + \epsilon^2 A_4^{0,(2)} + \epsilon^3 A_4^{0,(3)}\right) \quad (\text{A.40})$$

where,

$$\begin{aligned} A_4^{0,(0)} &= \frac{2s_{12}s_{14}}{s_{13}s_{134}^2} + \frac{s_{14}s_{23}}{s_{13}s_{134}^2} - \frac{4s_{23}}{s_{13}s_{134}} - \frac{4s_{14}s_{24}}{s_{13}s_{134}^2} - \frac{s_{12}s_{24}}{s_{13}s_{134}s_{234}} - \frac{s_{24}}{s_{134}s_{234}} - \frac{5s_{24}}{s_{134}^2} \\ &\quad - \frac{2s_{24}s_{34}}{s_{13}s_{134}^2} - \frac{s_{12}s_{34}}{s_{13}s_{134}s_{234}} - \frac{2s_{34}}{s_{13}s_{24}} - \frac{2s_{12}s_{34}}{s_{13}s_{134}s_{24}} - \frac{2s_{12}s_{34}}{s_{13}s_{234}s_{24}} - \frac{s_{12}s_{34}}{s_{134}s_{234}s_{24}} - \frac{4s_{12}^2s_{34}}{s_{13}s_{134}s_{234}s_{24}} \\ &\quad - \frac{s_{12}s_{34}}{s_{234}^2s_{24}} - \frac{2s_{13}s_{34}}{s_{234}^2s_{24}} - \frac{s_{12}s_{34}}{s_{13}s_{134}^2} - \frac{2s_{12}}{s_{13}s_{134}} + \frac{2}{s_{134}} - \frac{s_{12}s_{14}}{s_{13}s_{134}s_{234}} + \frac{s_{14}s_{23}}{s_{13}s_{134}s_{234}} \\ &\quad - \frac{s_{12}s_{23}}{s_{13}s_{134}s_{234}} + \frac{2s_{12}}{s_{134}s_{234}} - \frac{s_{13}}{s_{134}s_{234}} - \frac{2s_{12}^2}{s_{13}s_{134}s_{234}} + \frac{2}{s_{234}} - \frac{4s_{14}}{s_{13}s_{24}} - \frac{5s_{12}s_{14}}{s_{13}s_{134}s_{24}} \\ &\quad + \frac{2s_{14}s_{23}}{s_{13}s_{134}s_{24}} - \frac{4s_{23}}{s_{13}s_{24}} - \frac{3s_{12}s_{23}}{s_{13}s_{134}s_{24}} - \frac{2s_{12}}{s_{13}s_{24}} - \frac{4s_{12}^2}{s_{13}s_{134}s_{24}} - \frac{2s_{12}}{s_{234}s_{24}} - \frac{3s_{12}s_{14}}{s_{13}s_{234}s_{24}} \\ &\quad - \frac{s_{12}s_{14}}{s_{134}s_{234}s_{24}} - \frac{2s_{12}^2s_{14}}{s_{13}s_{134}s_{234}s_{24}} - \frac{4s_{14}}{s_{234}s_{24}} + \frac{2s_{14}s_{23}}{s_{13}s_{234}s_{24}} + \frac{6s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{24}} + \frac{s_{14}s_{23}}{s_{134}s_{234}s_{24}} \\ &\quad - \frac{5s_{12}s_{23}}{s_{13}s_{234}s_{24}} - \frac{s_{12}s_{23}}{s_{134}s_{234}s_{24}} - \frac{2s_{12}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}} - \frac{4s_{12}^2}{s_{13}s_{234}s_{24}} - \frac{2s_{12}^2}{s_{134}s_{234}s_{24}} - \frac{s_{12}s_{13}}{s_{134}s_{234}s_{24}} \\ &\quad - \frac{4s_{12}^3}{s_{13}s_{134}s_{234}s_{24}} - \frac{2s_{12}s_{23}}{s_{234}^2s_{24}} - \frac{4s_{13}s_{23}}{s_{234}^2s_{24}} + \frac{s_{14}s_{23}}{s_{234}^2s_{24}} - \frac{3s_{12}s_{14}^2}{s_{13}s_{134}^2s_{34}} - \frac{s_{24}^2}{s_{134}s_{234}s_{34}} \\ &\quad - \frac{4s_{12}s_{14}}{s_{13}s_{134}s_{34}} - \frac{s_{14}}{s_{134}s_{34}} - \frac{6s_{12}s_{14}}{s_{134}^2s_{34}} + \frac{2s_{14}^2s_{23}}{s_{13}s_{134}^2s_{34}} - \frac{6s_{14}s_{23}}{s_{13}s_{134}s_{34}} - \frac{3s_{23}}{s_{134}s_{34}} \end{aligned}$$

$$\begin{aligned}
& -\frac{4s_{14}^2s_{24}}{s_{13}s_{134}^2s_{34}} - \frac{6s_{14}s_{24}}{s_{134}^2s_{34}} + \frac{2s_{24}}{s_{134}s_{34}} - \frac{s_{12}s_{14}s_{24}}{s_{13}s_{134}s_{234}s_{34}} - \frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{14}s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}} \\
& - \frac{s_{23}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{12}s_{24}}{s_{134}s_{234}s_{34}} - \frac{4s_{13}s_{24}}{s_{134}s_{234}s_{34}} - \frac{4s_{13}s_{24}}{s_{134}^2s_{34}} - \frac{3s_{12}s_{24}}{s_{234}^2s_{34}} - \frac{4s_{13}s_{24}}{s_{234}^2s_{34}} \\
& + \frac{4s_{12}}{s_{134}s_{34}} + \frac{s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{34}} + \frac{4s_{12}}{s_{234}s_{34}} + \frac{2s_{13}}{s_{234}s_{34}} - \frac{s_{12}s_{14}}{s_{13}s_{234}s_{34}} - \frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}} \\
& - \frac{3s_{14}}{s_{234}s_{34}} + \frac{s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{6s_{14}s_{23}}{s_{134}s_{234}s_{34}} - \frac{s_{12}s_{23}}{s_{13}s_{234}s_{34}} - \frac{s_{12}s_{23}}{s_{134}s_{234}s_{34}} \\
& - \frac{s_{13}s_{23}}{s_{134}s_{234}s_{34}} - \frac{s_{23}}{s_{234}s_{34}} - \frac{2s_{12}^2}{s_{13}s_{234}s_{34}} + \frac{8s_{12}^2}{s_{134}s_{234}s_{34}} - \frac{s_{13}^2}{s_{134}s_{234}s_{34}} + \frac{s_{12}s_{13}}{s_{134}s_{234}s_{34}} \\
& - \frac{4s_{14}^2}{s_{13}s_{24}s_{34}} - \frac{3s_{12}s_{14}^2}{s_{13}s_{134}s_{24}s_{34}} + \frac{3s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}} - \frac{4s_{23}^2}{s_{13}s_{24}s_{34}} - \frac{4s_{12}s_{14}}{s_{13}s_{24}s_{34}} - \frac{s_{12}s_{14}}{s_{134}s_{24}s_{34}} \\
& - \frac{6s_{12}^2s_{14}}{s_{13}s_{134}s_{24}s_{34}} + \frac{5s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}} - \frac{6s_{14}s_{23}}{s_{13}s_{24}s_{34}} + \frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{24}s_{34}} - \frac{4s_{12}s_{23}}{s_{13}s_{24}s_{34}} - \frac{s_{12}s_{23}}{s_{134}s_{24}s_{34}} \\
& - \frac{2s_{12}^2}{s_{134}s_{24}s_{34}} - \frac{2s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{5s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} + \frac{4s_{12}s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}} \\
& - \frac{3s_{12}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} + \frac{3s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}} + \frac{4s_{12}s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{s_{14}^2s_{23}}{s_{134}s_{234}s_{24}s_{34}} - \frac{4s_{12}s_{23}}{s_{234}s_{24}s_{34}} \\
& + \frac{s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{24}s_{34}} + \frac{s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}} + \frac{s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}} + \frac{8s_{12}^2s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}} - \frac{6s_{14}s_{23}}{s_{234}s_{24}s_{34}} \\
& - \frac{6s_{12}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}} - \frac{s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{24}s_{34}} - \frac{3s_{12}s_{23}^2}{s_{234}^2s_{24}s_{34}} - \frac{4s_{13}s_{23}^2}{s_{234}^2s_{24}s_{34}} + \frac{2s_{14}s_{23}^2}{s_{234}^2s_{24}s_{34}} - \frac{3s_{12}s_{13}}{s_{134}^2s_{34}} \\
& - \frac{6s_{12}s_{23}}{s_{234}^2s_{34}} - \frac{6s_{13}s_{23}}{s_{234}^2s_{34}} + \frac{2}{s_{34}} - \frac{3s_{12}}{s_{134}^2} - \frac{3s_{12}}{s_{234}^2} - \frac{5s_{13}}{s_{234}^2} - \frac{3s_{14}^2}{s_{134}s_{34}^2} \\
& - \frac{5s_{12}s_{14}^2}{s_{134}^2s_{34}^2} - \frac{s_{14}s_{24}^2}{s_{134}s_{234}s_{34}^2} - \frac{3s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} - \frac{s_{12}s_{24}^2}{s_{234}^2s_{34}^2} - \frac{s_{13}s_{24}^2}{s_{234}^2s_{34}^2} + \frac{6s_{12}s_{14}}{s_{134}s_{34}^2} \\
& - \frac{s_{13}s_{14}}{s_{134}s_{34}^2} - \frac{2s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2} + \frac{4s_{14}}{s_{34}^2} + \frac{3s_{14}^3s_{23}}{s_{13}s_{134}s_{34}^2} - \frac{4s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2} - \frac{3s_{14}s_{23}}{s_{134}s_{34}^2} + \frac{s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2} \\
& - \frac{s_{13}s_{23}}{s_{134}s_{34}^2} + \frac{4s_{23}}{s_{34}^2} - \frac{3s_{14}^2s_{24}}{s_{134}s_{34}^2} + \frac{4s_{14}s_{24}}{s_{134}s_{34}^2} + \frac{2s_{12}s_{24}}{s_{234}s_{34}^2} - \frac{2s_{12}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} \\
& - \frac{s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{s_{14}s_{24}}{s_{234}s_{34}^2} + \frac{s_{14}^2s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{s_{23}s_{24}}{s_{234}s_{34}^2} \\
& - \frac{3s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{6s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{s_{13}^2s_{24}}{s_{134}^2s_{34}^2} - \frac{2s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2} + \frac{s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2} + \frac{2s_{12}s_{13}}{s_{134}s_{34}^2} \\
& + \frac{s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{34}^2} + \frac{s_{14}s_{23}^2}{s_{13}s_{234}s_{34}^2} + \frac{3s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} - \frac{3s_{23}^2}{s_{234}s_{34}^2} + \frac{s_{14}^2s_{23}}{s_{13}s_{234}s_{34}^2} + \frac{3s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2} \\
& + \frac{6s_{12}s_{23}}{s_{234}s_{34}^2} + \frac{4s_{13}s_{23}}{s_{234}s_{34}^2} + \frac{2s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{34}^2} - \frac{2s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{3s_{14}s_{23}}{s_{234}s_{34}^2} \\
& - \frac{s_{13}^2s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{3s_{14}^2s_{23}^2}{s_{13}s_{134}s_{24}s_{34}^2} - \frac{4s_{14}s_{23}^2}{s_{13}s_{24}s_{34}^2} + \frac{s_{14}s_{23}^2}{s_{134}s_{24}s_{34}^2} + \frac{3s_{14}^3s_{23}}{s_{13}s_{134}s_{24}s_{34}^2} \\
& - \frac{4s_{14}^2s_{23}}{s_{13}s_{24}s_{34}^2} + \frac{6s_{12}s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}^2} + \frac{s_{14}^2s_{23}}{s_{134}s_{24}s_{34}^2} - \frac{8s_{12}s_{14}s_{23}}{s_{13}s_{24}s_{34}^2} + \frac{2s_{12}s_{14}s_{23}}{s_{134}s_{24}s_{34}^2} \\
& - \frac{2s_{14}^3s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} + \frac{3s_{14}s_{23}^3}{s_{13}s_{234}s_{24}s_{34}^2} - \frac{2s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} + \frac{3s_{14}^2s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2} - \frac{4s_{12}s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} \\
& + \frac{s_{14}^2s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2} + \frac{6s_{12}s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2} + \frac{s_{13}s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2} - \frac{4s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2} + \frac{3s_{14}s_{23}^3}{s_{234}^2s_{24}s_{34}^2} \\
& - \frac{s_{12}s_{13}}{s_{134}^2s_{34}^2} - \frac{5s_{12}s_{23}^2}{s_{234}^2s_{34}^2} - \frac{3s_{13}s_{23}^2}{s_{234}^2s_{34}^2}
\end{aligned}$$

(A.41)

$$\begin{aligned}
A_4^{0,(1)} = & \frac{4s_{12}^3}{s_{13}s_{134}s_{234}s_{24}} + \frac{2s_{12}^2}{s_{13}s_{134}s_{234}} + \frac{4s_{12}^2}{s_{13}s_{134}s_{24}} + \frac{2s_{12}^2s_{14}}{s_{13}s_{134}s_{234}s_{24}} + \frac{2s_{12}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}} + \frac{4s_{12}^2}{s_{13}s_{234}s_{24}} + \frac{2s_{12}^2}{s_{134}s_{234}s_{24}} \\
& + \frac{2s_{12}^2}{s_{13}s_{234}s_{34}} - \frac{8s_{12}^2}{s_{134}s_{234}s_{34}} + \frac{6s_{12}^2s_{14}}{s_{13}s_{134}s_{24}s_{34}} + \frac{2s_{12}^2}{s_{134}s_{24}s_{34}} - \frac{8s_{12}^2s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{6s_{12}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}} - \frac{s_{12}s_{14}}{s_{13}s_{134}^2} \\
& + \frac{2s_{12}s_{24}}{s_{13}s_{134}s_{234}} + \frac{2s_{12}s_{34}}{s_{13}s_{134}s_{24}} - \frac{2s_{12}s_{14}s_{34}}{s_{13}s_{134}s_{234}s_{24}} - \frac{2s_{12}s_{23}s_{34}}{s_{13}s_{134}s_{234}s_{24}} + \frac{2s_{12}s_{34}}{s_{13}s_{234}s_{24}} + \frac{2s_{12}s_{34}}{s_{234}^2s_{24}} + \frac{2s_{12}s_{34}}{s_{13}s_{134}^2} \\
& + \frac{2s_{12}}{s_{13}s_{134}} + \frac{s_{12}s_{14}}{s_{13}s_{134}s_{234}} + \frac{6s_{12}}{s_{134}s_{234}} + \frac{5s_{12}s_{14}}{s_{13}s_{134}s_{24}} + \frac{5s_{12}s_{23}}{s_{13}s_{134}s_{24}} + \frac{2s_{12}}{s_{13}s_{24}} + \frac{5s_{12}s_{14}}{s_{13}s_{234}s_{24}} \\
& + \frac{5s_{12}s_{23}}{s_{13}s_{234}s_{24}} + \frac{s_{12}s_{23}}{s_{134}s_{234}s_{24}} + \frac{2s_{12}s_{13}}{s_{134}s_{234}s_{24}} + \frac{2s_{12}}{s_{234}s_{24}} - \frac{s_{12}s_{23}}{s_{234}^2s_{24}} + \frac{3s_{12}s_{14}^2}{s_{13}s_{134}^2s_{34}} + \frac{4s_{12}s_{14}}{s_{13}s_{134}s_{34}} \\
& + \frac{4s_{12}s_{14}}{s_{134}^2s_{34}} + \frac{s_{12}s_{14}s_{24}}{s_{13}s_{134}s_{234}s_{34}} + \frac{s_{12}s_{24}}{s_{134}s_{234}s_{34}} + \frac{5s_{12}s_{24}}{s_{234}^2s_{34}} - \frac{4s_{12}}{s_{134}s_{34}} + \frac{s_{12}s_{14}}{s_{13}s_{234}s_{34}} \\
& - \frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{s_{12}s_{23}}{s_{13}s_{234}s_{34}} - \frac{s_{12}s_{23}}{s_{134}s_{234}s_{34}} + \frac{s_{12}s_{13}}{s_{134}s_{234}s_{34}} - \frac{4s_{12}}{s_{234}s_{34}} \\
& + \frac{3s_{12}s_{14}^2}{s_{13}s_{134}s_{24}s_{34}} + \frac{4s_{12}s_{14}}{s_{13}s_{24}s_{34}} + \frac{s_{12}s_{14}}{s_{134}s_{24}s_{34}} - \frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{24}s_{34}} + \frac{4s_{12}s_{23}}{s_{13}s_{24}s_{34}} + \frac{s_{12}s_{23}}{s_{134}s_{24}s_{34}} \\
& - \frac{4s_{12}s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{3s_{12}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} - \frac{4s_{12}s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}} - \frac{s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{24}s_{34}} - \frac{s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}} \\
& + \frac{s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{24}s_{34}} + \frac{4s_{12}s_{23}}{s_{234}s_{24}s_{34}} + \frac{3s_{12}s_{23}^2}{s_{234}^2s_{24}s_{34}} + \frac{5s_{12}s_{13}}{s_{134}^2s_{34}} + \frac{4s_{12}s_{23}}{s_{234}^2s_{34}} + \frac{5s_{12}}{s_{134}^2} \\
& + \frac{5s_{12}}{s_{234}^2} + \frac{6s_{12}s_{14}^2}{s_{134}^2s_{34}^2} + \frac{2s_{12}s_{24}^2}{s_{234}^2s_{34}^2} - \frac{6s_{12}s_{14}}{s_{134}s_{34}^2} + \frac{8s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{12}s_{24}}{s_{234}s_{34}^2} \\
& - \frac{2s_{12}s_{13}}{s_{134}s_{34}^2} - \frac{2s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{34}^2} + \frac{4s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{6s_{12}s_{23}}{s_{234}s_{34}^2} - \frac{6s_{12}s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}^2} + \frac{8s_{12}s_{14}s_{23}}{s_{13}s_{24}s_{34}^2} \\
& - \frac{2s_{12}s_{14}s_{23}}{s_{134}s_{24}s_{34}^2} + \frac{4s_{12}s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} - \frac{6s_{12}s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2} + \frac{2s_{12}s_{13}^2}{s_{134}^2s_{34}^2} + \frac{6s_{12}s_{23}^2}{s_{234}^2s_{34}^2} \\
& + \frac{8s_{14}}{s_{13}s_{134}} - \frac{2s_{14}s_{23}}{s_{13}s_{134}^2} + \frac{8s_{23}}{s_{13}s_{134}} + \frac{8s_{14}s_{24}}{s_{13}s_{134}^2} + \frac{4s_{24}}{s_{134}s_{234}} + \frac{14s_{24}}{s_{134}^2} \\
& + \frac{6s_{24}s_{34}}{s_{13}s_{134}^2} + \frac{4s_{34}}{s_{13}s_{134}} + \frac{6s_{34}}{s_{13}s_{24}} + \frac{4s_{34}}{s_{234}s_{24}} + \frac{6s_{13}s_{34}}{s_{234}^2s_{24}} + \frac{4}{s_{134}} + \frac{2s_{14}}{s_{134}s_{234}} \\
& + \frac{2s_{23}}{s_{134}s_{234}} + \frac{4s_{13}}{s_{134}s_{234}} + \frac{4}{s_{234}} + \frac{8s_{14}}{s_{13}s_{24}} - \frac{2s_{14}s_{23}}{s_{13}s_{134}s_{24}} + \frac{8s_{23}}{s_{13}s_{24}} + \frac{2s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}} \\
& + \frac{8s_{14}}{s_{234}s_{24}} + \frac{2s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}} - \frac{2s_{14}s_{23}}{s_{13}s_{234}s_{24}} + \frac{8s_{23}}{s_{234}s_{24}} + \frac{8s_{13}s_{23}}{s_{234}^2s_{24}} - \frac{2s_{14}s_{23}}{s_{234}^2s_{24}} \\
& + \frac{8s_{14}^2}{s_{13}s_{134}s_{34}} + \frac{2s_{24}^2}{s_{134}s_{234}s_{34}} + \frac{7s_{14}}{s_{134}s_{34}} + \frac{s_{14}^2s_{23}}{s_{13}s_{134}^2s_{34}} + \frac{6s_{14}s_{23}}{s_{13}s_{134}s_{34}} + \frac{s_{14}s_{23}}{s_{134}^2s_{34}} \\
& + \frac{5s_{23}}{s_{134}s_{34}} + \frac{8s_{14}^2s_{24}}{s_{13}s_{134}^2s_{34}} + \frac{11s_{14}s_{24}}{s_{134}^2s_{34}} - \frac{2s_{24}}{s_{134}s_{34}} + \frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}} - \frac{2s_{14}s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}} \\
& + \frac{6s_{13}s_{24}}{s_{134}s_{234}s_{34}} + \frac{2s_{24}}{s_{234}s_{34}} + \frac{11s_{13}s_{24}}{s_{134}^2s_{34}} + \frac{11s_{13}s_{24}}{s_{234}^2s_{34}} + \frac{2s_{13}}{s_{134}s_{34}} - \frac{2s_{13}}{s_{234}s_{34}} \\
& + \frac{5s_{14}}{s_{234}s_{34}} - \frac{s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{2s_{14}s_{23}}{s_{134}s_{234}s_{34}} + \frac{s_{13}s_{23}}{s_{134}s_{234}s_{34}} + \frac{7s_{23}}{s_{234}s_{34}} + \frac{2s_{13}^2}{s_{134}s_{234}s_{34}} \\
& + \frac{8s_{14}^2}{s_{13}s_{24}s_{34}} - \frac{5s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}} + \frac{8s_{23}^2}{s_{13}s_{24}s_{34}} - \frac{5s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}} + \frac{6s_{14}s_{23}}{s_{13}s_{24}s_{34}} \\
& - \frac{5s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} - \frac{s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}} + \frac{8s_{23}^2}{s_{234}s_{24}s_{34}} - \frac{5s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}} - \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}} \\
& + \frac{6s_{14}s_{23}}{s_{234}s_{24}s_{34}} + \frac{8s_{13}s_{23}^2}{s_{234}^2s_{24}s_{34}} + \frac{s_{14}s_{23}^2}{s_{234}^2s_{24}s_{34}} + \frac{11s_{13}s_{23}}{s_{234}^2s_{34}} + \frac{s_{14}s_{23}}{s_{234}^2s_{34}} - \frac{2}{s_{34}} \\
& + \frac{14s_{13}}{s_{234}^2} + \frac{3s_{14}^2}{s_{134}s_{34}^2} + \frac{s_{14}s_{24}^2}{s_{134}s_{234}s_{34}^2} + \frac{5s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} + \frac{3s_{13}s_{24}^2}{s_{234}^2s_{34}^2} + \frac{s_{13}s_{14}}{s_{134}s_{34}^2}
\end{aligned}$$

$$\begin{aligned}
& -\frac{4s_{14}}{s_{34}^2} - \frac{3s_{14}^3s_{23}}{s_{13}s_{134}s_{34}^2} + \frac{4s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2} + \frac{2s_{14}^2s_{23}}{s_{134}^2s_{34}^2} + \frac{3s_{14}s_{23}}{s_{134}s_{34}^2} - \frac{3s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2} \\
& + \frac{s_{13}s_{23}}{s_{134}s_{34}^2} - \frac{4s_{23}}{s_{34}^2} + \frac{3s_{14}^2s_{24}}{s_{134}^2s_{34}^2} - \frac{4s_{14}s_{24}}{s_{134}s_{34}^2} - \frac{2s_{13}s_{14}s_{24}}{s_{134}^2s_{34}^2} - \frac{s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} \\
& + \frac{s_{14}s_{24}}{s_{234}s_{34}^2} - \frac{s_{14}^2s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}^2} - \frac{4s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{s_{23}s_{24}}{s_{234}s_{34}^2} + \frac{5s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2} \\
& + \frac{3s_{13}^2s_{24}}{s_{134}^2s_{34}^2} - \frac{2s_{13}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{3s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{34}^2} - \frac{s_{14}s_{23}^2}{s_{13}s_{234}s_{34}^2} - \frac{s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} \\
& + \frac{3s_{23}^2}{s_{234}s_{34}^2} - \frac{s_{14}^2s_{23}}{s_{13}s_{234}s_{34}^2} - \frac{s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{4s_{13}s_{23}}{s_{234}s_{34}^2} - \frac{4s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{3s_{14}s_{23}}{s_{234}s_{34}^2} \\
& + \frac{s_{13}^2s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{3s_{14}^2s_{23}^2}{s_{13}s_{134}s_{24}s_{34}^2} + \frac{4s_{14}s_{23}^2}{s_{13}s_{24}s_{34}^2} - \frac{s_{14}s_{23}^2}{s_{134}s_{24}s_{34}^2} - \frac{3s_{14}^3s_{23}}{s_{13}s_{134}s_{24}s_{34}^2} + \frac{4s_{14}^2s_{23}}{s_{13}s_{24}s_{34}^2} \\
& - \frac{s_{14}^2s_{23}}{s_{134}s_{24}s_{34}^2} + \frac{2s_{14}^2s_{23}^3}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} - \frac{3s_{14}s_{23}^3}{s_{13}s_{234}s_{24}s_{34}^2} + \frac{2s_{14}^3s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} - \frac{3s_{14}^2s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2} \\
& - \frac{s_{14}^2s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2} - \frac{s_{13}s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2} + \frac{4s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2} - \frac{3s_{14}s_{23}^3}{s_{234}^2s_{24}s_{34}^2} + \frac{3s_{13}s_{23}^2}{s_{234}^2s_{34}^2} + \frac{2s_{14}s_{23}^2}{s_{234}^2s_{34}^2}
\end{aligned} \tag{A.42}$$

$$\begin{aligned}
A_4^{0,(2)} = & -\frac{s_{13}^2}{s_{134}s_{234}s_{34}} - \frac{2s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}^2s_{24}}{s_{134}^2s_{34}^2} - \frac{s_{12}s_{13}^2}{s_{134}^2s_{34}^2} - \frac{6s_{13}s_{34}}{s_{234}^2s_{24}} - \frac{7s_{13}}{s_{134}s_{234}} - \frac{s_{12}s_{13}}{s_{134}s_{234}s_{34}} \\
& - \frac{4s_{13}s_{23}}{s_{234}^2s_{24}} - \frac{12s_{13}s_{24}}{s_{134}s_{234}s_{34}} - \frac{7s_{13}s_{24}}{s_{134}^2s_{34}} - \frac{7s_{13}s_{24}}{s_{234}^2s_{34}} - \frac{s_{13}}{s_{134}s_{34}} + \frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}} - \frac{2s_{13}s_{23}}{s_{134}s_{234}s_{34}} \\
& - \frac{2s_{12}s_{13}}{s_{134}s_{234}s_{34}} + \frac{s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}} - \frac{4s_{13}s_{23}^2}{s_{234}^2s_{24}s_{34}} - \frac{2s_{12}s_{13}}{s_{134}^2s_{34}} - \frac{5s_{13}s_{23}}{s_{234}^2s_{34}} - \frac{11s_{13}}{s_{234}^2} \\
& - \frac{2s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}s_{24}^2}{s_{234}^2s_{34}^2} + \frac{2s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{24}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} \\
& + \frac{2s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}s_{23}s_{24}}{s_{234}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{7s_{24}}{s_{134}s_{234}} \\
& - \frac{11s_{24}}{s_{134}^2} - \frac{2s_{34}}{s_{134}s_{234}} + \frac{s_{12}s_{34}}{s_{134}s_{234}s_{24}} - \frac{8s_{34}}{s_{234}s_{24}} - \frac{3s_{12}s_{34}}{s_{234}^2s_{24}} - \frac{11}{s_{134}} - \frac{4s_{14}}{s_{134}s_{234}} \\
& - \frac{4s_{23}}{s_{134}s_{234}} + \frac{2s_{12}}{s_{134}s_{234}} - \frac{11}{s_{234}} - \frac{s_{12}}{s_{134}s_{24}} + \frac{s_{12}s_{14}}{s_{134}s_{234}s_{24}} - \frac{4s_{14}}{s_{234}s_{24}} - \frac{s_{14}s_{23}}{s_{134}s_{234}s_{24}} \\
& - \frac{2s_{12}s_{23}}{s_{134}s_{234}s_{24}} - \frac{8s_{23}}{s_{234}s_{24}} + \frac{3s_{12}s_{23}}{s_{234}^2s_{24}} + \frac{3s_{14}s_{23}}{s_{234}^2s_{24}} - \frac{s_{24}^2}{s_{134}s_{234}s_{34}} - \frac{5s_{14}}{s_{134}s_{34}} + \frac{2s_{12}s_{14}}{s_{134}^2s_{34}} \\
& - \frac{s_{14}s_{23}}{s_{134}^2s_{34}} - \frac{2s_{23}}{s_{134}s_{34}} - \frac{5s_{14}s_{24}}{s_{134}^2s_{34}} - \frac{2s_{14}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{23}s_{24}}{s_{134}s_{234}s_{34}} - \frac{2s_{12}s_{24}}{s_{134}s_{234}s_{34}} \\
& - \frac{s_{24}}{s_{234}s_{34}} - \frac{2s_{12}s_{24}}{s_{234}^2s_{34}} + \frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{2s_{14}}{s_{234}s_{34}} + \frac{4s_{14}s_{23}}{s_{134}s_{234}s_{34}} + \frac{2s_{12}s_{23}}{s_{134}s_{234}s_{34}} \\
& - \frac{5s_{23}}{s_{234}s_{34}} + \frac{s_{14}s_{23}}{s_{134}s_{24}s_{34}} + \frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}} - \frac{8s_{23}^2}{s_{234}s_{24}s_{34}} - \frac{s_{14}^2s_{23}}{s_{134}s_{234}s_{24}s_{34}} - \frac{3s_{14}s_{23}^2}{s_{234}^2s_{24}s_{34}} \\
& + \frac{2s_{12}s_{23}}{s_{234}^2s_{34}} - \frac{s_{14}s_{23}}{s_{234}^2s_{34}} - \frac{4s_{12}}{s_{134}} - \frac{4s_{12}}{s_{234}} - \frac{s_{12}s_{14}^2}{s_{134}^2s_{34}^2} - \frac{s_{12}s_{24}^2}{s_{234}^2s_{34}^2} \\
& - \frac{2s_{14}^2s_{23}}{s_{134}^2s_{34}^2} + \frac{2s_{12}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} \\
& - \frac{2s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{s_{12}s_{23}^2}{s_{234}^2s_{34}^2} - \frac{2s_{14}s_{23}^2}{s_{234}^2s_{34}^2} - \frac{2s_{12}s_{24}^2}{s_{13}s_{134}s_{234}s_{24}} - \frac{8s_{14}}{s_{13}s_{134}} \\
& + \frac{3s_{12}s_{14}}{s_{13}s_{134}} + \frac{3s_{14}s_{23}}{s_{13}s_{134}} - \frac{4s_{23}}{s_{13}s_{134}} - \frac{4s_{14}s_{24}}{s_{13}s_{134}} - \frac{s_{12}s_{24}}{s_{13}s_{134}s_{234}} - \frac{6s_{24}s_{34}}{s_{13}s_{134}} \\
& - \frac{8s_{34}}{s_{13}s_{134}} + \frac{s_{12}s_{34}}{s_{13}s_{134}s_{234}} + \frac{s_{12}s_{34}}{s_{13}s_{134}s_{24}} + \frac{s_{12}s_{34}}{s_{13}s_{234}s_{24}} + \frac{2s_{14}s_{23}s_{34}}{s_{13}s_{134}s_{234}s_{24}} - \frac{6s_{34}}{s_{13}s_{24}}
\end{aligned}$$

$$\begin{aligned}
& -\frac{3s_{12}s_{34}}{s_{13}s_{134}^2} - \frac{s_{12}}{s_{13}s_{234}} - \frac{2s_{12}s_{14}}{s_{13}s_{134}s_{234}} - \frac{s_{14}s_{23}}{s_{13}s_{134}s_{234}} + \frac{s_{12}s_{23}}{s_{13}s_{134}s_{234}} + \frac{3s_{12}s_{14}}{s_{13}s_{134}s_{24}} \\
& -\frac{4s_{14}}{s_{13}s_{24}} - \frac{s_{14}s_{23}}{s_{13}s_{134}s_{24}} - \frac{2s_{12}s_{23}}{s_{13}s_{134}s_{24}} - \frac{4s_{23}}{s_{13}s_{24}} - \frac{2s_{12}s_{14}}{s_{13}s_{234}s_{24}} + \frac{3s_{12}s_{23}}{s_{13}s_{234}s_{24}} \\
& -\frac{s_{14}s_{23}}{s_{13}s_{234}s_{24}} - \frac{8s_{14}^2}{s_{13}s_{134}s_{34}} - \frac{3s_{14}^2s_{23}}{s_{13}s_{134}^2s_{34}} - \frac{4s_{14}^2s_{24}}{s_{13}s_{134}^2s_{34}} + \frac{s_{14}s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}} - \frac{s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{34}} \\
& + \frac{2s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{s_{14}s_{23}}{s_{13}s_{234}s_{34}} - \frac{4s_{14}^2}{s_{13}s_{24}s_{34}} + \frac{2s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}} - \frac{4s_{23}^2}{s_{13}s_{24}s_{34}} - \frac{3s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}} \\
& - \frac{3s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} + \frac{2s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}}
\end{aligned} \tag{A.43}$$

$$\begin{aligned}
A_4^{0,(3)} &= \frac{2s_{12}s_{34}}{s_{13}s_{134}s_{234}} + \frac{2s_{12}s_{34}}{s_{13}s_{134}s_{24}} + \frac{2s_{12}s_{34}}{s_{13}s_{234}s_{24}} + \frac{2s_{12}s_{34}}{s_{134}s_{234}s_{24}} + \frac{2s_{12}s_{34}}{s_{234}^2s_{24}} + \frac{2s_{12}s_{34}}{s_{13}s_{134}^2} + \frac{4s_{12}}{s_{134}s_{234}} \\
&+ \frac{2s_{12}}{s_{134}^2} + \frac{2s_{12}}{s_{234}^2} - \frac{2s_{14}s_{23}}{s_{13}s_{134}^2} + \frac{2s_{24}}{s_{134}s_{234}} + \frac{2s_{24}}{s_{134}^2} + \frac{2s_{24}s_{34}}{s_{13}s_{134}^2} + \frac{4s_{34}}{s_{13}s_{134}} \\
&+ \frac{4s_{34}}{s_{134}s_{234}} + \frac{2s_{34}}{s_{13}s_{24}} + \frac{4s_{34}}{s_{234}s_{24}} + \frac{2s_{13}s_{34}}{s_{234}^2s_{24}} + \frac{2}{s_{134}} - \frac{2s_{14}s_{23}}{s_{13}s_{134}s_{234}} + \frac{2s_{13}}{s_{134}s_{234}} \\
&+ \frac{2}{s_{234}} - \frac{2s_{14}s_{23}}{s_{13}s_{134}s_{24}} - \frac{2s_{14}s_{23}}{s_{13}s_{234}s_{24}} - \frac{2s_{14}s_{23}}{s_{134}s_{234}s_{24}} - \frac{2s_{14}s_{23}}{s_{234}^2s_{24}} + \frac{2s_{13}}{s_{234}^2}.
\end{aligned} \tag{A.44}$$

By running `BuildAndIntegrateAntenna[A, 4, 0, Component->Leading]` we get the integrated result,

$$\begin{aligned}
\mathcal{A}_4^0 &= \frac{3}{4\epsilon^4} + \frac{65}{24\epsilon^3} + \frac{434 - 39\pi^2}{36\epsilon^2} - \frac{1}{\epsilon} \left( -\frac{43223}{864} + \frac{589\pi^2}{144} + \frac{71\zeta_3}{4} \right) \\
&+ \frac{1076717}{5184} - \frac{7955\pi^2}{432} + \frac{373\pi^4}{1440} - \frac{1327\zeta_3}{18} + O(\epsilon).
\end{aligned} \tag{A.45}$$

### A.3.9 $\tilde{A}_4^0$

Running `BuildAntenna[A, 4, 0, Component->Subleading]`, we get the built result,

$$\tilde{A}_4^0 = \frac{1}{2(\epsilon - 1)q^2} \left( \tilde{A}_4^{0,(0)} + \epsilon \tilde{A}_4^{0,(1)} + \epsilon^2 \tilde{A}_4^{0,(2)} + \epsilon^3 \tilde{A}_4^{0,(3)} \right) \tag{A.46}$$

where,

$$\begin{aligned}
\tilde{A}_4^{0,(0)} &= -\frac{2}{s_{14}s_{23}\mathcal{D}_{4,gg}} \left[ -\frac{2s_{12}s_{34}^2}{s_{13}s_{134}^2s_{14}} + \frac{4s_{14}^3s_{23}}{s_{13}s_{134}^2s_{34}^2} \right. \\
&- \frac{2s_{14}^2(2s_{12} - 3s_{23} + 2s_{24})}{s_{13}s_{134}^2s_{34}} - \frac{2(3s_{12}s_{14} + 3s_{12}s_{34} - 3s_{14}s_{23} + 2s_{14}s_{24} - s_{23}s_{34} + s_{24}s_{34})}{s_{13}s_{134}^2} \\
&- \frac{2s_{12}s_{34}(s_{12} + s_{24})}{s_{13}s_{134}s_{14}s_{23}} - \frac{2s_{12}s_{34}(s_{12} + s_{23})}{s_{13}s_{134}s_{14}s_{24}} \\
&- \frac{2s_{12}^2(2s_{12} + s_{14} + s_{24} + 2s_{34})}{s_{13}s_{134}s_{23}s_{234}} - \frac{2s_{12}(2s_{12} + 2s_{24} + s_{34})}{s_{13}s_{134}s_{23}} \\
&- \frac{2s_{14}^2s_{23}^2(2s_{12} + s_{14} + s_{23})}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} + \frac{2s_{14}s_{23}(4s_{12}^2 + 2s_{12}s_{14} + 2s_{12}s_{23} - s_{14}s_{23})}{s_{13}s_{134}s_{234}s_{24}s_{34}} \\
&- \frac{2s_{12}(2s_{12}^2 + s_{12}s_{14} + s_{12}s_{23} + 2s_{12}s_{34} - 3s_{14}s_{23})}{s_{13}s_{134}s_{234}s_{24}} - \frac{2s_{14}^2s_{23}(2s_{12} + s_{14} - s_{24})}{s_{13}s_{134}s_{234}s_{34}^2} \\
&+ \frac{2s_{14}(4s_{12}^2 + 2s_{12}s_{14} + 2s_{12}s_{23} + s_{23}^2 + s_{23}s_{24})}{s_{13}s_{134}s_{234}s_{34}} - \frac{2(2s_{12}^2 - 2s_{12}s_{14} + s_{12}s_{23} + s_{12}s_{24} + s_{12}s_{34} - s_{14}s_{23})}{s_{13}s_{134}s_{234}} \\
&+ \frac{4s_{14}^2s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{134}s_{24}s_{34}^2} - \frac{2s_{14}(4s_{12}^2 + 2s_{12}s_{14} - 2s_{12}s_{23} - 3s_{14}s_{23} - 3s_{23}^2)}{s_{13}s_{134}s_{24}s_{34}} \\
&- \frac{2(4s_{12}^2 + 3s_{12}s_{14} + 2s_{12}s_{23} + s_{12}s_{34} - s_{14}s_{23} - s_{23}^2)}{s_{13}s_{134}s_{24}} - \frac{8s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{4s_{14}(s_{12} - 2s_{23} + s_{24})}{s_{13}s_{134}s_{34}} + \frac{2(s_{12} + s_{14} - 2s_{23} + s_{24})}{s_{13}s_{134}} \\
& - \frac{4s_{12}^3}{s_{13}s_{14}s_{23}s_{24}} - \frac{2s_{12}^2}{s_{13}s_{14}s_{23}} - \frac{2s_{12}^2}{s_{13}s_{14}s_{24}} - \frac{2s_{12}}{s_{13}s_{14}} \\
& - \frac{2s_{12}s_{34}(s_{12} + s_{14})}{s_{13}s_{23}s_{234}s_{24}} - \frac{2s_{12}(2s_{12} + 2s_{14} + s_{34})}{s_{13}s_{23}s_{234}} - \frac{2s_{12}^2}{s_{13}s_{23}s_{24}} \\
& + \frac{4s_{14}s_{23}^2(2s_{12} + s_{14} + s_{23})}{s_{13}s_{234}s_{24}s_{34}^2} - \frac{2s_{23}(4s_{12}^2 - 2s_{12}s_{14} + 2s_{12}s_{23} - 3s_{14}^2 - 3s_{14}s_{23})}{s_{13}s_{234}s_{24}s_{34}} \\
& - \frac{2(4s_{12}^2 + 2s_{12}s_{14} + 3s_{12}s_{23} + s_{12}s_{34} - s_{14}^2 - s_{14}s_{23})}{s_{13}s_{234}s_{24}} + \frac{4s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{234}s_{34}^2} \\
& - \frac{4(2s_{12}^2 + s_{12}s_{23} - s_{14}^2 - s_{14}s_{23})}{s_{13}s_{234}s_{34}} - \frac{2(2s_{12} - s_{14})}{s_{13}s_{234}} \\
& - \frac{6s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{24}s_{34}^2} + \frac{2(4s_{12}^2 - 2s_{14}^2 - 3s_{14}s_{23} - 2s_{23}^2)}{s_{13}s_{24}s_{34}} \\
& - \frac{2(s_{12} + 2s_{14} + 2s_{23} + s_{34})}{s_{13}s_{24}} + \frac{2s_{14}s_{23}}{s_{13}s_{34}^2} + \frac{2s_{23}}{s_{13}s_{34}} \\
& + \frac{4s_{13}^3s_{24}}{s_{134}s_{14}s_{34}^2} - \frac{2s_{13}^2(2s_{12} + 2s_{23} - 3s_{24})}{s_{134}^2s_{14}s_{34}} \\
& - \frac{2(3s_{12}s_{13} + 3s_{12}s_{34} + 2s_{13}s_{23} - 3s_{13}s_{24} + s_{23}s_{34} - s_{24}s_{34})}{s_{134}^2s_{14}} \\
& - \frac{4(2s_{12}s_{13}^2 + 4s_{12}s_{13}s_{14} + 2s_{12}s_{14}^2 + s_{13}^2s_{23} - s_{13}^2s_{24} + s_{13}s_{14}s_{23} + s_{13}s_{14}s_{24} - s_{14}^2s_{23} + s_{14}^2s_{24})}{s_{134}^2s_{34}^2} \\
& - \frac{10(2s_{12}s_{13} + 2s_{12}s_{14} + s_{13}s_{23} + s_{14}s_{24})}{s_{134}^2s_{34}} - \frac{2(8s_{12} + s_{23} + s_{24})}{s_{134}^2} \\
& - \frac{2s_{13}^2s_{24}^2(2s_{12} + s_{13} + s_{24})}{s_{134}s_{14}s_{23}s_{234}s_{34}^2} + \frac{2s_{13}s_{24}(4s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{24} - s_{13}s_{24})}{s_{134}s_{14}s_{23}s_{234}s_{34}} \\
& - \frac{2s_{12}(2s_{12}^2 + s_{12}s_{13} + s_{12}s_{24} + 2s_{12}s_{34} - 3s_{13}s_{24})}{s_{134}s_{14}s_{23}s_{234}} + \frac{4s_{13}^2s_{24}(2s_{12} + s_{13} + s_{24})}{s_{134}s_{14}s_{23}s_{34}^2} \\
& - \frac{2s_{13}(4s_{12}^2 + 2s_{12}s_{13} - 2s_{12}s_{24} - 3s_{13}s_{24} - 3s_{24}^2)}{s_{134}s_{14}s_{23}s_{34}} \\
& - \frac{2(4s_{12}^2 + 3s_{12}s_{13} + 2s_{12}s_{24} + s_{12}s_{34} - s_{13}s_{24} - s_{24}^2)}{s_{134}s_{14}s_{23}} \\
& - \frac{2s_{12}^2(2s_{12} + s_{13} + s_{23} + 2s_{34})}{s_{134}s_{14}s_{234}s_{24}} - \frac{2s_{13}^2s_{24}(2s_{12} + s_{13} - s_{23})}{s_{134}s_{14}s_{234}s_{34}^2} \\
& + \frac{2s_{13}(4s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{24} + s_{23}s_{24} + s_{24}^2)}{s_{134}s_{14}s_{234}s_{34}} \\
& - \frac{2(2s_{12}^2 - 2s_{12}s_{13} + s_{12}s_{23} + s_{12}s_{24} + s_{12}s_{34} - s_{13}s_{24})}{s_{134}s_{14}s_{234}} \\
& - \frac{2s_{12}(2s_{12} + 2s_{23} + s_{34})}{s_{134}s_{14}s_{24}} - \frac{8s_{13}^2s_{24}}{s_{134}s_{14}s_{34}^2} \\
& + \frac{4s_{13}(s_{12} + s_{23} - 2s_{24})}{s_{134}s_{14}s_{34}} + \frac{2(s_{12} + s_{13} + s_{23} - 2s_{24})}{s_{134}s_{14}} \\
& - \frac{2s_{13}s_{24}^2(2s_{12} - s_{14} + s_{24})}{s_{134}s_{23}s_{234}s_{34}^2} + \frac{2s_{24}(4s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{24} + s_{13}^2 + s_{13}s_{14})}{s_{134}s_{23}s_{234}s_{34}} \\
& - \frac{2(2s_{12}^2 + s_{12}s_{13} + s_{12}s_{14} - 2s_{12}s_{24} + s_{12}s_{34} - s_{13}s_{24})}{s_{134}s_{23}s_{234}} \\
& + \frac{4s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{134}s_{23}s_{34}^2} - \frac{4(2s_{12}^2 + s_{12}s_{13} - s_{13}s_{24} - s_{24}^2)}{s_{134}s_{23}s_{34}} - \frac{2(2s_{12} - s_{24})}{s_{134}s_{23}}
\end{aligned}$$

$$\begin{aligned}
& - \frac{2s_{14}s_{23}^2(2s_{12} - s_{13} + s_{23})}{s_{134}s_{234}s_{24}s_{34}^2} + \frac{2s_{23}(4s_{12}^2 + 2s_{12}s_{14} + 2s_{12}s_{23} + s_{13}s_{14} + s_{14}^2)}{s_{134}s_{234}s_{24}s_{34}} \\
& - \frac{2(2s_{12}^2 + s_{12}s_{13} + s_{12}s_{14} - 2s_{12}s_{23} + s_{12}s_{34} - s_{14}s_{23})}{s_{134}s_{234}s_{24}} \\
& - \frac{2(4s_{12}s_{13}s_{23} + 6s_{12}s_{13}s_{24} + 6s_{12}s_{14}s_{23} + 4s_{12}s_{14}s_{24} - s_{13}s_{14}s_{23} - s_{13}s_{14}s_{24} - s_{13}s_{23}s_{24} - s_{14}s_{23}s_{24})}{s_{134}s_{234}s_{34}^2} \\
& + \frac{2}{s_{134}s_{234}s_{34}} \left( 8s_{12}^2 - 2s_{12}s_{13} - 2s_{12}s_{14} - 2s_{12}s_{23} - 2s_{12}s_{24} - s_{13}^2 - 2s_{13}s_{14} + s_{13}s_{24} \right. \\
& \quad \left. - s_{14}^2 + s_{14}s_{23} - s_{23}^2 - 2s_{23}s_{24} - s_{24}^2 \right) \\
& - \frac{2(2s_{12} + s_{13} + s_{14} + s_{23} + s_{24})}{s_{134}s_{234}} \\
& + \frac{4s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{134}s_{24}s_{34}^2} - \frac{4(2s_{12}^2 + s_{12}s_{14} - s_{14}s_{23} - s_{23}^2)}{s_{134}s_{24}s_{34}} - \frac{2(2s_{12} - s_{23})}{s_{134}s_{24}} \\
& + \frac{4(4s_{12}s_{13} + 4s_{12}s_{14} - s_{13}^2 - 2s_{13}s_{14} + s_{13}s_{23} - s_{13}s_{24} - s_{14}^2 - s_{14}s_{23} + s_{14}s_{24})}{s_{134}s_{34}^2} \\
& + \frac{2(8s_{12} - s_{13} - s_{14} + s_{23} + s_{24})}{s_{134}s_{34}} + \frac{4}{s_{134}} \\
& - \frac{2s_{12}s_{34}(s_{12} + s_{13})}{s_{14}s_{23}s_{234}s_{24}} + \frac{4s_{13}s_{24}^2(2s_{12} + s_{13} + s_{24})}{s_{14}s_{23}s_{234}s_{34}^2} \\
& - \frac{2s_{24}(4s_{12}^2 - 2s_{12}s_{13} + 2s_{12}s_{24} - 3s_{13}^2 - 3s_{13}s_{24})}{s_{14}s_{23}s_{234}s_{34}} \\
& - \frac{2(4s_{12}^2 + 2s_{12}s_{13} + 3s_{12}s_{24} + s_{12}s_{34} - s_{13}^2 - s_{13}s_{24})}{s_{14}s_{23}s_{234}} - \frac{2s_{12}^2}{s_{14}s_{23}s_{24}} \\
& - \frac{6s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{14}s_{23}s_{34}^2} + \frac{2(4s_{12}^2 - 2s_{13}^2 - 3s_{13}s_{24} - 2s_{24}^2)}{s_{14}s_{23}s_{34}} \\
& - \frac{2(s_{12} + 2s_{13} + 2s_{24} + s_{34})}{s_{14}s_{23}} - \frac{2s_{12}(2s_{12} + 2s_{13} + s_{34})}{s_{14}s_{234}s_{24}} \\
& + \frac{4s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{14}s_{234}s_{34}^2} - \frac{4(2s_{12}^2 + s_{12}s_{24} - s_{13}^2 - s_{13}s_{24})}{s_{14}s_{234}s_{34}} \\
& - \frac{2(2s_{12} - s_{13})}{s_{14}s_{234}} + \frac{2s_{13}s_{24}}{s_{14}s_{34}^2} + \frac{2s_{24}}{s_{14}s_{34}} \\
& - \frac{2s_{12}s_{34}^2}{s_{23}s_{234}s_{24}} + \frac{4s_{13}s_{24}^3}{s_{23}s_{234}s_{34}^2} \\
& - \frac{2s_{24}^2(2s_{12} - 3s_{13} + 2s_{14})}{s_{23}s_{234}s_{34}} - \frac{2(3s_{12}s_{24} + 3s_{12}s_{34} - 3s_{13}s_{24} - s_{13}s_{34} + 2s_{14}s_{24} + s_{14}s_{34})}{s_{23}s_{234}^2} \\
& - \frac{8s_{13}s_{24}^2}{s_{23}s_{234}s_{34}^2} + \frac{4s_{24}(s_{12} - 2s_{13} + s_{14})}{s_{23}s_{234}s_{34}} \\
& + \frac{2(s_{12} - 2s_{13} + s_{14} + s_{24})}{s_{23}s_{234}} - \frac{2s_{12}}{s_{23}s_{24}} + \frac{2s_{13}s_{24}}{s_{23}s_{34}^2} + \frac{2s_{13}}{s_{23}s_{34}} \\
& + \frac{4s_{14}s_{23}^3}{s_{234}s_{24}s_{34}^2} - \frac{2s_{23}^2(2s_{12} + 2s_{13} - 3s_{14})}{s_{234}s_{24}s_{34}} \\
& - \frac{2(3s_{12}s_{23} + 3s_{12}s_{34} + 2s_{13}s_{23} + s_{13}s_{34} - 3s_{14}s_{23} - s_{14}s_{34})}{s_{234}^2s_{24}} \\
& - \frac{4(2s_{12}s_{23}^2 + 4s_{12}s_{23}s_{24} + 2s_{12}s_{24}^2 + s_{13}s_{23}^2 + s_{13}s_{23}s_{24} - s_{13}s_{24}^2 - s_{14}s_{23}^2 + s_{14}s_{23}s_{24} + s_{14}s_{24}^2)}{s_{234}^2s_{34}^2} \\
& - \frac{10(2s_{12}s_{23} + 2s_{12}s_{24} + s_{13}s_{23} + s_{14}s_{24})}{s_{234}^2s_{34}} - \frac{2(8s_{12} + s_{13} + s_{14})}{s_{234}^2}
\end{aligned}$$

$$\begin{aligned}
& -\frac{8s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2} + \frac{4s_{23}(s_{12} + s_{13} - 2s_{14})}{s_{234}s_{24}s_{34}} \\
& + \frac{2(s_{12} + s_{13} - 2s_{14} + s_{23})}{s_{234}s_{24}} \\
& + \frac{4\left(4s_{12}s_{23} + 4s_{12}s_{24} + s_{13}s_{23} - s_{13}s_{24} - s_{14}s_{23} + s_{14}s_{24} - s_{23}^2 - 2s_{23}s_{24} - s_{24}^2\right)}{s_{234}s_{34}^2} \\
& + \frac{2(8s_{12} + s_{13} + s_{14} - s_{23} - s_{24})}{s_{234}s_{34}} + \frac{4}{s_{234}} \\
& + \frac{2s_{14}s_{23}}{s_{24}s_{34}^2} + \frac{2s_{14}}{s_{24}s_{34}} - \frac{4(2s_{12} - s_{13} - s_{14} - s_{23} - s_{24})}{s_{34}^2} - \frac{4}{s_{34}} \Big] \tag{A.47}
\end{aligned}$$

$$\begin{aligned}
\tilde{A}_4^{0,(1)} = & \frac{2}{s_{14}s_{23}\mathcal{D}_{4,gg}} \left[ \frac{4s_{12}s_{34}^2}{s_{13}s_{134}^2s_{14}} - \frac{4s_{14}^3s_{23}}{s_{13}s_{134}^2s_{34}^2} + \frac{2s_{14}^2(2s_{12} - 3s_{23} + 4s_{24})}{s_{13}s_{134}^2s_{34}} \right. \\
& + \frac{2(3s_{12}s_{14} + 6s_{12}s_{34} - 6s_{14}s_{23} + 4s_{14}s_{24} - 2s_{23}s_{34} + 3s_{24}s_{34})}{s_{13}s_{134}^2} \\
& + \frac{2s_{12}s_{34}(s_{12} + 2s_{24} + s_{34})}{s_{13}s_{134}s_{14}s_{23}} + \frac{2s_{12}s_{34}(s_{12} + 2s_{23} + s_{34})}{s_{13}s_{134}s_{14}s_{24}} \\
& + \frac{2s_{12}(2s_{12}^2 + s_{12}s_{14} + s_{12}s_{24} + 2s_{14}s_{24} + s_{14}s_{34} + s_{24}s_{34} + 2s_{34}^2)}{s_{13}s_{134}s_{23}s_{234}} \\
& + \frac{4s_{12}(s_{12} + 2s_{24} + s_{34})}{s_{13}s_{134}s_{23}} + \frac{2s_{14}^2s_{23}^2(2s_{12} + s_{14} + s_{23})}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} \\
& - \frac{4s_{12}s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{134}s_{234}s_{24}s_{34}} \\
& + \frac{2(2s_{12}^3 + s_{12}^2s_{14} + s_{12}^2s_{23} - s_{12}s_{14}s_{34} - s_{12}s_{23}s_{34} + s_{14}^2s_{23} + s_{14}s_{23}^2)}{s_{13}s_{134}s_{234}s_{24}} \\
& + \frac{2s_{14}^2s_{23}(2s_{12} + s_{14} - s_{24})}{s_{13}s_{134}s_{234}s_{34}^2} \\
& - \frac{2s_{14}(4s_{12}^2 + 2s_{12}s_{14} + 2s_{12}s_{23} + 3s_{14}s_{23} + 2s_{14}s_{24} + s_{23}^2 + s_{23}s_{24})}{s_{13}s_{134}s_{234}s_{34}} \\
& + \frac{2(2s_{12}^2 + 3s_{12}s_{14} + s_{12}s_{23} + s_{12}s_{24} + 2s_{12}s_{34} - s_{14}^2 - 2s_{14}s_{23} - s_{14}s_{24} - 2s_{14}s_{34})}{s_{13}s_{134}s_{234}} \\
& - \frac{4s_{14}^2s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{134}s_{24}s_{34}^2} \\
& + \frac{2s_{14}(4s_{12}^2 + 2s_{12}s_{14} - 2s_{12}s_{23} - 3s_{14}s_{23} - 5s_{23}^2)}{s_{13}s_{134}s_{24}s_{34}} \\
& + \frac{2(4s_{12}^2 + 3s_{12}s_{14} + 4s_{12}s_{23} + 2s_{12}s_{34} - 2s_{14}s_{23} - 2s_{23}^2 - s_{23}s_{34})}{s_{13}s_{134}s_{24}} \\
& + \frac{8s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2} + \frac{4s_{14}(-s_{12} + 2s_{14} + 2s_{23} - 2s_{24})}{s_{13}s_{134}s_{34}} \\
& + \frac{2(-s_{12} + 2s_{14} + 4s_{23} - 2s_{24} + s_{34})}{s_{13}s_{134}} + \frac{4s_{12}(s_{12}^2 + s_{12}s_{34} + s_{34}^2)}{s_{13}s_{14}s_{23}s_{24}} \\
& + \frac{2s_{12}(s_{12} + s_{34})}{s_{13}s_{14}s_{23}} + \frac{2s_{12}(s_{12} + s_{34})}{s_{13}s_{14}s_{24}} + \frac{2s_{12}}{s_{13}s_{14}} \\
& + \frac{2s_{12}s_{34}(s_{12} + 2s_{14} + s_{34})}{s_{13}s_{23}s_{234}s_{24}} + \frac{4s_{12}(s_{12} + 2s_{14} + s_{34})}{s_{13}s_{23}s_{234}} \\
& + \frac{2s_{12}(s_{12} + s_{34})}{s_{13}s_{23}s_{24}} + \frac{4s_{12}}{s_{13}s_{23}} \\
& - \frac{4s_{14}s_{23}^2(2s_{12} + s_{14} + s_{23})}{s_{13}s_{234}s_{24}s_{34}^2} \\
& + \frac{2s_{23}(4s_{12}^2 - 2s_{12}s_{14} + 2s_{12}s_{23} - 5s_{14}^2 - 3s_{14}s_{23})}{s_{13}s_{234}s_{24}s_{34}}
\end{aligned}$$

$$\begin{aligned}
& + \frac{2(4s_{12}^2 + 4s_{12}s_{14} + 3s_{12}s_{23} + 2s_{12}s_{34} - 2s_{14}^2 - 2s_{14}s_{23} - s_{14}s_{34})}{s_{13}s_{234}s_{24}} \\
& - \frac{4s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{234}s_{34}^2} + \frac{4(2s_{12}^2 + s_{12}s_{23} - 2s_{14}^2 - s_{14}s_{23})}{s_{13}s_{234}s_{34}} \\
& + \frac{4(s_{12} - s_{14})}{s_{13}s_{234}} + \frac{6s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{13}s_{24}s_{34}^2} \\
& + \frac{8(-s_{12}^2 + s_{14}^2 + s_{14}s_{23} + s_{23}^2)}{s_{13}s_{24}s_{34}} + \frac{2(-2s_{12} + 3s_{14} + 3s_{23} + s_{34})}{s_{13}s_{24}} \\
& - \frac{2s_{14}s_{23}}{s_{13}s_{34}^2} - \frac{2(2s_{14} + s_{23})}{s_{13}s_{34}} - \frac{2}{s_{13}} \\
& - \frac{4s_{13}^3s_{24}}{s_{134}^2s_{14}s_{34}^2} + \frac{2s_{13}^2(2s_{12} + 4s_{23} - 3s_{24})}{s_{134}^2s_{14}s_{34}} \\
& + \frac{2(3s_{12}s_{13} + 6s_{12}s_{34} + 4s_{13}s_{23} - 6s_{13}s_{24} + 3s_{23}s_{34} - 2s_{24}s_{34})}{s_{134}^2s_{14}} \\
& + \frac{4(2s_{12}s_{13}^2 + 4s_{12}s_{13}s_{14} + 2s_{12}s_{14}^2 + s_{13}^2s_{23} - s_{13}^2s_{24} + s_{13}s_{14}s_{23} + s_{13}s_{14}s_{24} - s_{14}^2s_{23} + s_{14}^2s_{24})}{s_{134}^2s_{34}^2} \\
& + \frac{2(10s_{12}s_{13} + 10s_{12}s_{14} + 9s_{13}s_{23} + 2s_{13}s_{24} + 2s_{14}s_{23} + 9s_{14}s_{24})}{s_{134}^2s_{34}} \\
& + \frac{4(5s_{12} + s_{23} + s_{24})}{s_{134}^2} \\
& + \frac{2s_{13}^2s_{24}^2(2s_{12} + s_{13} + s_{24})}{s_{134}s_{14}s_{23}s_{234}s_{34}^2} \\
& - \frac{4s_{12}s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{134}s_{14}s_{23}s_{234}s_{34}} \\
& + \frac{2(2s_{12}^3 + s_{12}^2s_{13} + s_{12}^2s_{24} - s_{12}s_{13}s_{34} - s_{12}s_{24}s_{34} + s_{13}^2s_{24} + s_{13}s_{24}^2)}{s_{134}s_{14}s_{23}s_{234}} \\
& - \frac{4s_{13}^2s_{24}(2s_{12} + s_{13} + s_{24})}{s_{134}s_{14}s_{23}s_{34}^2} \\
& + \frac{2s_{13}(4s_{12}^2 + 2s_{12}s_{13} - 2s_{12}s_{24} - 3s_{13}s_{24} - 5s_{24}^2)}{s_{134}s_{14}s_{23}s_{34}} \\
& + \frac{2(4s_{12}^2 + 3s_{12}s_{13} + 4s_{12}s_{24} + 2s_{12}s_{34} - 2s_{13}s_{24} - 2s_{24}^2 - s_{24}s_{34})}{s_{134}s_{14}s_{23}} \\
& + \frac{2s_{12}(2s_{12}^2 + s_{12}s_{13} + s_{12}s_{23} + 2s_{13}s_{23} + s_{13}s_{34} + s_{23}s_{34} + 2s_{34}^2)}{s_{134}s_{14}s_{234}s_{24}} \\
& + \frac{2s_{13}^2s_{24}(2s_{12} + s_{13} - s_{23})}{s_{134}s_{14}s_{234}s_{34}^2} \\
& - \frac{2s_{13}(4s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{24} + 2s_{13}s_{23} + 3s_{13}s_{24} + s_{23}s_{24} + s_{24}^2)}{s_{134}s_{14}s_{234}s_{34}} \\
& + \frac{2(2s_{12}^2 + 3s_{12}s_{13} + s_{12}s_{23} + s_{12}s_{24} + 2s_{12}s_{34} - s_{13}^2 - s_{13}s_{23} - 2s_{13}s_{24} - 2s_{13}s_{34})}{s_{134}s_{14}s_{234}} \\
& + \frac{4s_{12}(s_{12} + 2s_{23} + s_{34})}{s_{134}s_{14}s_{24}} + \frac{8s_{13}^2s_{24}}{s_{134}s_{14}s_{34}^2} \\
& + \frac{4s_{13}(-s_{12} + 2s_{13} - 2s_{23} + 2s_{24})}{s_{134}s_{14}s_{34}} + \frac{2(-s_{12} + 2s_{13} - 2s_{23} + 4s_{24} + s_{34})}{s_{134}s_{14}} \\
& + \frac{2s_{13}s_{24}^2(2s_{12} - s_{14} + s_{24})}{s_{134}s_{23}s_{234}s_{34}^2} \\
& - \frac{2s_{24}(4s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{24} + s_{13}^2 + s_{13}s_{14} + 3s_{13}s_{24} + 2s_{14}s_{24})}{s_{134}s_{23}s_{234}s_{34}} \\
& + \frac{2(2s_{12}^2 + s_{12}s_{13} + s_{12}s_{14} + 3s_{12}s_{24} + 2s_{12}s_{34} - 2s_{13}s_{24} - s_{14}s_{24} - s_{24}^2 - 2s_{24}s_{34})}{s_{134}s_{23}s_{234}}
\end{aligned}$$

$$\begin{aligned}
& - \frac{4s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{134}s_{23}s_{34}^2} + \frac{4(2s_{12}^2 + s_{12}s_{13} - s_{13}s_{24} - 2s_{24}^2)}{s_{134}s_{23}s_{34}} \\
& + \frac{4(s_{12} - s_{24})}{s_{134}s_{23}} + \frac{2s_{14}s_{23}^2(2s_{12} - s_{13} + s_{23})}{s_{134}s_{234}s_{24}s_{34}^2} \\
& - \frac{2s_{23}(4s_{12}^2 + 2s_{12}s_{14} + 2s_{12}s_{23} + s_{13}s_{14} + 2s_{13}s_{23} + s_{14}^2 + 3s_{14}s_{23})}{s_{134}s_{234}s_{24}s_{34}} \\
& + \frac{2(2s_{12}^2 + s_{12}s_{13} + s_{12}s_{14} + 3s_{12}s_{23} + 2s_{12}s_{34} - s_{13}s_{23} - 2s_{14}s_{23} - s_{23}^2 - 2s_{23}s_{34})}{s_{134}s_{234}s_{24}} \\
& + \frac{2(4s_{12}s_{13}s_{23} + 6s_{12}s_{13}s_{24} + 6s_{12}s_{14}s_{23} + 4s_{12}s_{14}s_{24} - s_{13}s_{14}s_{23} - s_{13}s_{14}s_{24} - s_{13}s_{23}s_{24} - s_{14}s_{23}s_{24})}{s_{134}s_{234}s_{34}^2} \\
& + \frac{2}{s_{134}s_{234}s_{34}} \left( (-8s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{14} + 2s_{12}s_{23} + 2s_{12}s_{24} + s_{13}^2 + 2s_{13}s_{14} + 2s_{13}s_{23} + s_{14}^2 \right. \\
& \quad \left. + 2s_{14}s_{24} + s_{23}^2 + 2s_{23}s_{24} + s_{24}^2) \right) \\
& + \frac{2(16s_{12} + 3s_{13} + 3s_{14} + 3s_{23} + 3s_{24})}{s_{134}s_{234}} \\
& - \frac{4s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{134}s_{24}s_{34}^2} + \frac{4(2s_{12}^2 + s_{12}s_{14} - s_{14}s_{23} - 2s_{23}^2)}{s_{134}s_{24}s_{34}} \\
& + \frac{4(s_{12} - s_{23})}{s_{134}s_{24}} \\
& + \frac{4(-4s_{12}s_{13} - 4s_{12}s_{14} + s_{13}^2 + 2s_{13}s_{14} - s_{13}s_{23} + s_{13}s_{24} + s_{14}^2 + s_{14}s_{23} - s_{14}s_{24})}{s_{134}s_{34}^2} \\
& + \frac{2(-8s_{12} + 5s_{13} + 5s_{14} - 3s_{23} - 3s_{24})}{s_{134}s_{34}} \\
& + \frac{2s_{12}s_{34}(s_{12} + 2s_{13} + s_{34})}{s_{14}s_{23}s_{234}s_{24}} \\
& - \frac{4s_{13}s_{24}^2(2s_{12} + s_{13} + s_{24})}{s_{14}s_{23}s_{234}s_{34}^2} \\
& + \frac{2s_{24}(4s_{12}^2 - 2s_{12}s_{13} + 2s_{12}s_{24} - 5s_{13}^2 - 3s_{13}s_{24})}{s_{14}s_{23}s_{234}s_{34}} \\
& + \frac{2(4s_{12}^2 + 4s_{12}s_{13} + 3s_{12}s_{24} + 2s_{12}s_{34} - 2s_{13}^2 - 2s_{13}s_{24} - s_{13}s_{34})}{s_{14}s_{23}s_{234}} \\
& + \frac{2s_{12}(s_{12} + s_{34})}{s_{14}s_{23}s_{24}} + \frac{6s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{14}s_{23}s_{34}^2} \\
& + \frac{8(-s_{12}^2 + s_{13}^2 + s_{13}s_{24} + s_{24}^2)}{s_{14}s_{23}s_{34}} + \frac{2(-2s_{12} + 3s_{13} + 3s_{24} + s_{34})}{s_{14}s_{23}} \\
& + \frac{4s_{12}(s_{12} + 2s_{13} + s_{34})}{s_{14}s_{234}s_{24}} - \frac{4s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{14}s_{234}s_{34}^2} \\
& + \frac{4(2s_{12}^2 + s_{12}s_{24} - 2s_{13}^2 - s_{13}s_{24})}{s_{14}s_{234}s_{34}} + \frac{4(s_{12} - s_{13})}{s_{14}s_{234}} \\
& + \frac{4s_{12}}{s_{14}s_{24}} - \frac{2s_{13}s_{24}}{s_{14}s_{34}^2} - \frac{2(2s_{13} + s_{24})}{s_{14}s_{34}} - \frac{2}{s_{14}} \\
& + \frac{4s_{12}s_{34}^2}{s_{23}s_{234}^2s_{24}} - \frac{4s_{13}s_{24}^3}{s_{23}s_{234}^2s_{34}^2} \\
& + \frac{2s_{24}^2(2s_{12} - 3s_{13} + 4s_{14})}{s_{23}s_{234}^2s_{34}} \\
& + \frac{2(3s_{12}s_{24} + 6s_{12}s_{34} - 6s_{13}s_{24} - 2s_{13}s_{34} + 4s_{14}s_{24} + 3s_{14}s_{34})}{s_{23}s_{234}^2} \\
& + \frac{8s_{13}s_{24}^2}{s_{23}s_{234}s_{34}^2} + \frac{4s_{24}(-s_{12} + 2s_{13} - 2s_{14} + 2s_{24})}{s_{23}s_{234}s_{34}} \\
& + \frac{2(-s_{12} + 4s_{13} - 2s_{14} + 2s_{24} + s_{34})}{s_{23}s_{234}} + \frac{2s_{12}}{s_{23}s_{24}}
\end{aligned}$$

$$\begin{aligned}
& -\frac{2s_{13}s_{24}}{s_{23}s_{34}^2} - \frac{2(s_{13} + 2s_{24})}{s_{23}s_{34}} - \frac{2}{s_{23}} \\
& -\frac{4s_{14}s_{23}^3}{s_{23}^2s_{24}s_{34}^2} + \frac{2s_{23}^2(2s_{12} + 4s_{13} - 3s_{14})}{s_{23}^2s_{24}s_{34}} \\
& + \frac{2(3s_{12}s_{23} + 6s_{12}s_{34} + 4s_{13}s_{23} + 3s_{13}s_{34} - 6s_{14}s_{23} - 2s_{14}s_{34})}{s_{23}^2s_{24}} \\
& + \frac{4(2s_{12}s_{23}^2 + 4s_{12}s_{23}s_{24} + 2s_{12}s_{24}^2 + s_{13}s_{23}^2 + s_{13}s_{23}s_{24} - s_{13}s_{24}^2 - s_{14}s_{23}^2 + s_{14}s_{23}s_{24} + s_{14}s_{24}^2)}{s_{23}^2s_{34}^2} \\
& + \frac{2(10s_{12}s_{23} + 10s_{12}s_{24} + 9s_{13}s_{23} + 2s_{13}s_{24} + 2s_{14}s_{23} + 9s_{14}s_{24})}{s_{23}^2s_{34}} \\
& + \frac{4(5s_{12} + s_{13} + s_{14})}{s_{23}^2} + \frac{8s_{14}s_{23}^2}{s_{23}s_{24}s_{34}^2} \\
& + \frac{4s_{23}(-s_{12} - 2s_{13} + 2s_{14} + 2s_{23})}{s_{23}s_{24}s_{34}} \\
& + \frac{2(-s_{12} - 2s_{13} + 4s_{14} + 2s_{23} + s_{34})}{s_{23}s_{24}} \\
& + \frac{4(-4s_{12}s_{23} - 4s_{12}s_{24} - s_{13}s_{23} + s_{13}s_{24} + s_{14}s_{23} - s_{14}s_{24} + s_{23}^2 + 2s_{23}s_{24} + s_{24}^2)}{s_{23}s_{34}^2} \\
& + \frac{2(-8s_{12} - 3s_{13} - 3s_{14} + 5s_{23} + 5s_{24})}{s_{23}s_{34}} \\
& - \frac{2s_{14}s_{23}}{s_{24}s_{34}^2} - \frac{2(s_{14} + 2s_{23})}{s_{24}s_{34}} - \frac{2}{s_{24}} \\
& + \left[ \frac{4(2s_{12} - s_{13} - s_{14} - s_{23} - s_{24})}{s_{34}^2} - \frac{8}{s_{34}} \right] \tag{A.48}
\end{aligned}$$

$$\begin{aligned}
\tilde{A}_4^{0,(2)} = & \frac{1}{s_{14}s_{23}\mathcal{D}_{4,gg}} \left[ -\frac{4s_{14}^2s_{24}}{s_{13}s_{134}s_{34}} - \frac{2(3s_{12}s_{34} - 3s_{14}s_{23} + 2s_{14}s_{24} + 3s_{24}s_{34})}{s_{13}s_{134}^2} - \frac{2s_{12}s_{34}(s_{24} + 2s_{34})}{s_{13}s_{134}s_{14}s_{23}} \right. \\
& - \frac{2s_{12}s_{34}(s_{23} + 2s_{34})}{s_{13}s_{134}s_{14}s_{24}} \\
& - \frac{2s_{12}(2s_{14}s_{24} + s_{14}s_{34} + s_{24}s_{34} + 3s_{34}^2)}{s_{13}s_{134}s_{23}s_{234}} - \frac{2s_{12}(2s_{24} + s_{34})}{s_{13}s_{134}s_{23}} \\
& - \frac{2s_{34}(s_{12}s_{34} - s_{14}s_{23})}{s_{13}s_{134}s_{234}s_{24}} + \frac{4s_{14}^2(s_{23} + s_{24})}{s_{13}s_{134}s_{234}s_{34}} \\
& - \frac{2(2s_{12}s_{14} + 2s_{12}s_{34} - s_{14}^2 - 2s_{14}s_{23} - s_{14}s_{24} - 3s_{14}s_{34})}{s_{13}s_{134}s_{234}} \\
& + \frac{4s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}} - \frac{2(2s_{12}s_{23} + 3s_{12}s_{34} - 3s_{14}s_{23} - s_{23}^2 - 2s_{23}s_{34})}{s_{13}s_{134}s_{24}} \\
& - \frac{4s_{14}(2s_{14} - s_{24})}{s_{13}s_{134}s_{34}} - \frac{2(3s_{14} + 2s_{23} - s_{24} + 2s_{34})}{s_{13}s_{134}} \\
& - \frac{6s_{12}s_{34}^2}{s_{13}s_{14}s_{23}s_{24}} - \frac{2s_{12}s_{34}}{s_{13}s_{14}s_{23}} - \frac{2s_{12}s_{34}}{s_{13}s_{14}s_{24}} \\
& - \frac{2s_{12}s_{34}(s_{14} + 2s_{34})}{s_{13}s_{23}s_{234}s_{24}} - \frac{2s_{12}(2s_{14} + s_{34})}{s_{13}s_{23}s_{234}} \\
& - \frac{2s_{12}s_{34}}{s_{13}s_{23}s_{24}} - \frac{4s_{12}}{s_{13}s_{23}} + \frac{4s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}} \\
& - \frac{2(2s_{12}s_{14} + 3s_{12}s_{34} - s_{14}^2 - 3s_{14}s_{23} - 2s_{14}s_{34})}{s_{13}s_{234}s_{24}} \\
& + \frac{4s_{14}^2}{s_{13}s_{234}s_{34}} + \frac{2s_{14}}{s_{13}s_{234}} - \frac{4(s_{14}^2 + s_{23}^2)}{s_{13}s_{24}s_{34}} \\
& - \frac{2(s_{14} + s_{23})}{s_{13}s_{24}} + \frac{4s_{14}}{s_{13}s_{34}} + \frac{2}{s_{13}} \tag{A.48}
\end{aligned}$$

$$\begin{aligned}
& - \frac{4s_{13}^2 s_{23}}{s_{134}^2 s_{14} s_{34}} - \frac{2(3s_{12} s_{34} + 2s_{13} s_{23} - 3s_{13} s_{24} + 3s_{23} s_{34})}{s_{134}^2 s_{14}} \\
& - \frac{4(2s_{13} s_{23} + s_{13} s_{24} + s_{14} s_{23} + 2s_{14} s_{24})}{s_{134}^2 s_{34}} - \frac{2(2s_{12} + s_{23} + s_{24})}{s_{134}^2} \\
& - \frac{2s_{34}(s_{12} s_{34} - s_{13} s_{24})}{s_{134} s_{14} s_{23} s_{24}} + \frac{4s_{13} s_{24}^2}{s_{134} s_{14} s_{23} s_{34}} \\
& - \frac{2(2s_{12} s_{24} + 3s_{12} s_{34} - 3s_{13} s_{24} - s_{24}^2 - 2s_{24} s_{34})}{s_{134} s_{14} s_{23}} \\
& - \frac{2s_{12}(2s_{13} s_{23} + s_{13} s_{34} + s_{23} s_{34} + 3s_{34}^2)}{s_{134} s_{14} s_{23} s_{24}} \\
& + \frac{4s_{13}^2 (s_{23} + s_{24})}{s_{134} s_{14} s_{23} s_{34}} - \frac{2(2s_{12} s_{13} + 2s_{12} s_{34} - s_{13}^2 - s_{13} s_{23} - 2s_{13} s_{24} - 3s_{13} s_{34})}{s_{134} s_{14} s_{23} s_{34}} \\
& - \frac{2s_{12}(2s_{23} + s_{34})}{s_{134} s_{14} s_{24}} - \frac{4s_{13}(2s_{13} - s_{23})}{s_{134} s_{14} s_{34}} \\
& - \frac{2(3s_{13} - s_{23} + 2s_{24} + 2s_{34})}{s_{134} s_{14}} + \frac{4s_{24}^2 (s_{13} + s_{14})}{s_{134} s_{23} s_{23} s_{34}} \\
& - \frac{2(2s_{12} s_{24} + 2s_{12} s_{34} - 2s_{13} s_{24} - s_{14} s_{24} - s_{24}^2 - 3s_{24} s_{34})}{s_{134} s_{23} s_{23} s_{34}} \\
& + \frac{4s_{24}^2}{s_{134} s_{23} s_{34}} + \frac{2s_{24}}{s_{134} s_{23}} + \frac{4s_{23}^2 (s_{13} + s_{14})}{s_{134} s_{23} s_{24} s_{34}} \\
& - \frac{2(2s_{12} s_{23} + 2s_{12} s_{34} - s_{13} s_{23} - 2s_{14} s_{23} - s_{23}^2 - 3s_{23} s_{34})}{s_{134} s_{23} s_{24}} \\
& + \frac{8(s_{13} + s_{14})(s_{23} + s_{24})}{s_{134} s_{23} s_{34}} - \frac{4(4s_{12} + 2s_{13} + 2s_{14} + 2s_{23} + 2s_{24} + s_{34})}{s_{134} s_{23} s_{34}} \\
& + \frac{4s_{23}^2}{s_{134} s_{24} s_{34}} + \frac{2s_{23}}{s_{134} s_{24}} - \frac{4(2s_{13} + 2s_{14} - s_{23} - s_{24})}{s_{134} s_{34}} - \frac{4}{s_{134}} \\
& - \frac{2s_{12} s_{34}(s_{13} + 2s_{34})}{s_{14} s_{23} s_{23} s_{24}} + \frac{4s_{13}^2 s_{24}}{s_{14} s_{23} s_{23} s_{34}} \\
& - \frac{2(2s_{12} s_{13} + 3s_{12} s_{34} - s_{13}^2 - 3s_{13} s_{24} - 2s_{13} s_{34})}{s_{14} s_{23} s_{23} s_{34}} \\
& - \frac{2s_{12} s_{34}}{s_{14} s_{23} s_{24}} - \frac{4(s_{13}^2 + s_{24}^2)}{s_{14} s_{23} s_{34}} - \frac{2(s_{13} + s_{24})}{s_{14} s_{23}} \\
& - \frac{2s_{12}(2s_{13} + s_{34})}{s_{14} s_{23} s_{24}} + \frac{4s_{13}^2}{s_{14} s_{23} s_{34}} + \frac{2s_{13}}{s_{14} s_{23} s_{34}} \\
& - \frac{4s_{12}}{s_{14} s_{24}} + \frac{4s_{13}}{s_{14} s_{34}} + \frac{2}{s_{14}} \\
& - \frac{4s_{14} s_{24}^2}{s_{23} s_{23}^2 s_{34}} - \frac{2(3s_{12} s_{34} - 3s_{13} s_{24} + 2s_{14} s_{24} + 3s_{14} s_{34})}{s_{23} s_{23}^2} \\
& + \frac{4s_{24}(s_{14} - 2s_{24})}{s_{23} s_{23} s_{34}} - \frac{2(2s_{13} - s_{14} + 3s_{24} + 2s_{34})}{s_{23} s_{23} s_{34}} \\
& + \frac{4s_{24}}{s_{23} s_{34}} + \frac{2}{s_{23}} - \frac{4s_{13} s_{23}^2}{s_{23}^2 s_{24} s_{34}} \\
& - \frac{2(3s_{12} s_{34} + 2s_{13} s_{23} + 3s_{13} s_{34} - 3s_{14} s_{23})}{s_{23}^2 s_{24}} \\
& - \frac{4(2s_{13} s_{23} + s_{13} s_{24} + s_{14} s_{23} + 2s_{14} s_{24})}{s_{23}^2 s_{34}} - \frac{2(2s_{12} + s_{13} + s_{14})}{s_{23}^2} \\
& + \frac{4s_{23}(s_{13} - 2s_{23})}{s_{23} s_{24} s_{34}} + \frac{2(s_{13} - 2s_{14} - 3s_{23} - 2s_{34})}{s_{23} s_{24}} \\
& + \frac{4(s_{13} + s_{14} - 2s_{23} - 2s_{24})}{s_{23} s_{34}} - \frac{4}{s_{23} s_{34}} + \frac{4s_{23}}{s_{24} s_{34}} + \frac{2}{s_{24}} \Big]
\end{aligned}
\tag{A.49}$$

$$\begin{aligned}
\bar{A}_4^{0,(3)} = \frac{1}{s_{14}s_{23}\mathcal{D}_{4,gg}} & \left[ -\frac{2s_{12}s_{34}^2}{s_{13}s_{134}^2s_{14}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{14}s_{23}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{14}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{23}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{14}s_{23}s_{24}} \right. \\
& - \frac{2s_{12}s_{34}^2}{s_{13}s_{23}s_{234}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{134}s_{14}s_{234}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{14}s_{23}s_{234}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{23}s_{234}s_{24}} \\
& + \frac{2s_{34}(s_{23} + s_{24})}{s_{13}s_{134}^2} + \frac{2s_{14}s_{34}}{s_{13}s_{134}s_{234}} + \frac{2s_{23}s_{34}}{s_{13}s_{134}s_{24}} \\
& + \frac{2s_{14}s_{34}}{s_{13}s_{234}s_{24}} + \frac{2s_{34}(s_{23} + s_{24})}{s_{134}^2s_{14}} + \frac{2s_{24}s_{34}}{s_{134}s_{14}s_{23}} \\
& + \frac{2s_{13}s_{34}}{s_{134}s_{14}s_{234}} + \frac{2s_{24}s_{34}}{s_{134}s_{23}s_{234}} + \frac{2s_{23}s_{34}}{s_{134}s_{234}s_{24}} \\
& + \frac{2s_{13}s_{34}}{s_{14}s_{23}s_{234}} + \frac{2s_{34}(s_{13} + s_{14})}{s_{23}s_{234}^2} + \frac{2s_{34}(s_{13} + s_{14})}{s_{234}^2s_{24}} \\
& \left. + \frac{6s_{34}}{s_{13}s_{134}} + \frac{4s_{34}}{s_{13}s_{24}} + \frac{6s_{34}}{s_{134}s_{14}} + \frac{8s_{34}}{s_{134}s_{234}} + \frac{4s_{34}}{s_{14}s_{23}} + \frac{6s_{34}}{s_{23}s_{234}} + \frac{6s_{34}}{s_{234}s_{24}} \right] \quad (A.50)
\end{aligned}$$

using,

$$\mathcal{D}_{4,gg} = s_{13}s_{24}s_{34}^2s_{134}^2s_{234}^2$$

By running `BuildAndIntegrateAntenna[A, 4, 0, Component->Subleading]` we get the integrated result,

$$\begin{aligned}
\bar{\mathcal{A}}_4^0 = \frac{1}{\epsilon^4} + \frac{3}{\epsilon^3} + \frac{1}{\epsilon^2} \left( 13 - \frac{3\pi^2}{2} \right) + \frac{1}{\epsilon} \left( \frac{845}{16} - \frac{9\pi^2}{2} - \frac{80\zeta_3}{3} \right) \\
+ \frac{6921}{32} - \frac{473\pi^2}{24} + \frac{17\pi^4}{72} - 80\zeta_3 + \mathcal{O}(\epsilon). \quad (A.51)
\end{aligned}$$

### A.3.10 $B_4^0$

Running `BuildAntenna[B, 4, 0]`, we get the built result,

$$\begin{aligned}
B_4^0 = \epsilon^2 & \left( \frac{4s_{12}}{s_{134}s_{234}} + \frac{2s_{12}}{s_{134}^2} + \frac{2s_{12}}{s_{234}^2} - \frac{2s_{14}s_{23}}{s_{134}^2s_{34}} - \frac{2s_{13}s_{23}}{s_{134}^2s_{34}} - \frac{2s_{14}s_{24}}{s_{134}^2s_{34}} \right. \\
& - \frac{4s_{14}s_{24}}{s_{134}s_{234}s_{34}} - \frac{4s_{13}s_{24}}{s_{134}s_{234}s_{34}} - \frac{2s_{13}s_{24}}{s_{134}^2s_{34}} - \frac{2s_{13}s_{24}}{s_{234}^2s_{34}} - \frac{2s_{14}s_{24}}{s_{234}^2s_{34}} \\
& - \frac{4s_{14}s_{23}}{s_{134}s_{234}s_{34}} - \frac{4s_{13}s_{23}}{s_{134}s_{234}s_{34}} - \frac{2s_{13}s_{23}}{s_{234}^2s_{34}} - \left. \frac{2s_{14}s_{23}}{s_{234}^2s_{34}} \right) \\
& + \epsilon \left( \frac{4s_{12}^2}{s_{134}s_{234}s_{34}} + \frac{4s_{12}}{s_{134}s_{234}} + \frac{2s_{12}s_{14}}{s_{134}^2s_{34}} + \frac{2s_{12}s_{24}}{s_{134}s_{234}s_{34}} + \frac{2s_{12}s_{24}}{s_{234}^2s_{34}} + \frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}} \right. \\
& + \frac{2s_{12}s_{23}}{s_{134}s_{234}s_{34}} + \frac{2s_{12}s_{13}}{s_{134}s_{234}s_{34}} + \frac{2s_{12}s_{13}}{s_{134}^2s_{34}} + \frac{2s_{12}s_{23}}{s_{234}^2s_{34}} - \frac{2s_{12}}{s_{134}^2} - \frac{2s_{12}}{s_{234}^2} \\
& + \frac{4s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2} - \frac{4s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{4s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{4s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}}{s_{134}^2s_{34}} + \frac{4s_{13}s_{23}}{s_{134}^2s_{34}} \\
& + \frac{4s_{14}s_{24}}{s_{134}^2s_{34}} + \frac{4s_{14}s_{24}}{s_{134}s_{234}s_{34}} + \frac{2s_{13}s_{24}}{s_{134}^2s_{34}} + \frac{2s_{13}s_{24}}{s_{234}^2s_{34}} + \frac{4s_{14}s_{24}}{s_{234}^2s_{34}} + \frac{4s_{13}s_{23}}{s_{134}s_{234}s_{34}} \\
& + \frac{4s_{13}s_{23}}{s_{234}^2s_{34}} + \frac{2s_{14}s_{23}}{s_{234}^2s_{34}} - \frac{2s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}s_{24}^2}{s_{234}^2s_{34}^2} - \frac{2s_{14}^2s_{23}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2} \\
& + \frac{2s_{13}s_{14}s_{24}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2} \\
& - \frac{2s_{13}^2s_{24}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{23}s_{24}}{s_{234}^2s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} - \frac{2s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2} \\
& + \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{14}s_{23}^2}{s_{234}^2s_{34}^2} \left. \right) \\
& - \frac{4s_{12}}{s_{134}s_{234}} - \frac{2s_{12}s_{14}}{s_{134}^2s_{34}} - \frac{2s_{13}s_{23}}{s_{134}^2s_{34}} - \frac{2s_{14}s_{24}}{s_{134}^2s_{34}} - \frac{2s_{12}s_{24}}{s_{134}s_{234}s_{34}} - \frac{2s_{12}s_{24}}{s_{234}^2s_{34}}
\end{aligned}$$

$$\begin{aligned}
& -\frac{2s_{14}s_{24}}{s_{234}^2s_{34}} - \frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{2s_{12}s_{23}}{s_{134}s_{234}s_{34}} - \frac{4s_{12}^2}{s_{134}s_{234}s_{34}} - \frac{2s_{12}s_{13}}{s_{134}s_{234}s_{34}} - \frac{2s_{12}s_{13}}{s_{134}^2s_{34}} \\
& - \frac{2s_{12}s_{23}}{s_{234}^2s_{34}} - \frac{2s_{13}s_{23}}{s_{234}^2s_{34}} + \frac{2s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}s_{24}^2}{s_{234}^2s_{34}^2} - \frac{4s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2} + \frac{2s_{14}^2s_{23}}{s_{134}^2s_{34}^2} \\
& - \frac{2s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2} - \frac{2s_{13}s_{14}s_{24}}{s_{134}^2s_{34}^2} - \frac{2s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} \\
& + \frac{2s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{4s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}^2s_{24}}{s_{134}^2s_{34}^2} - \frac{4s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{2s_{13}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{2s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2} \\
& + \frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{4s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}^2}{s_{234}^2s_{34}^2}.
\end{aligned} \tag{A.52}$$

By running `BuildAndIntegrateAntenna[B, 4, 0]` we get the integrated result,

$$\mathcal{B}_4^0 = -\frac{1}{12\epsilon^3} - \frac{7}{18\epsilon^2} - \frac{1}{\epsilon} \frac{11(37 - 3\pi^2)}{216} - \frac{11753}{1296} + \frac{77\pi^2}{108} + \frac{67\zeta_3}{18} + \mathcal{O}(\epsilon). \tag{A.53}$$

### A.3.11 $C_4^0$

Running `BuildAntenna[C, 4, 0]`, we get the built result,

$$\begin{aligned}
C_4^0 = & \epsilon^3 \left( \frac{s_{13}s_{24}^2}{s_{123}s_{134}s_{23}s_{34}} + \frac{s_{13}s_{24}^2}{s_{123}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{24}^2}{s_{134}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{24}^2}{s_{23}s_{234}s_{34}} + \frac{s_{13}s_{24}}{s_{123}s_{134}s_{23}} \right. \\
& - \frac{s_{12}s_{24}}{s_{123}s_{134}s_{23}} + \frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}} - \frac{s_{12}s_{24}}{s_{123}s_{23}s_{234}} - \frac{s_{12}s_{24}}{s_{134}s_{23}s_{234}} + \frac{s_{13}s_{24}}{s_{134}s_{23}s_{234}} \\
& - \frac{s_{14}s_{24}}{s_{14}s_{24}} + \frac{s_{13}s_{24}}{s_{13}s_{24}} + \frac{s_{13}s_{24}}{s_{13}s_{24}} - \frac{s_{14}s_{24}}{s_{14}s_{24}} - \frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}} \\
& + \frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{13}s_{24}}{s_{234}^2s_{34}} - \frac{s_{14}s_{24}}{s_{234}^2s_{34}} - \frac{s_{12}s_{24}}{s_{23}s_{234}^2} + \frac{s_{13}s_{24}}{s_{23}s_{234}^2} \\
& + \frac{s_{14}}{s_{123}s_{134}} - \frac{s_{12}s_{34}}{s_{123}s_{134}s_{23}} - \frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}} - \frac{s_{12}s_{34}}{s_{134}s_{23}s_{234}} - \frac{s_{12}s_{34}}{s_{23}s_{234}^2} \\
& + \frac{2s_{13}}{s_{123}s_{134}} + \frac{s_{12}}{s_{123}s_{134}} + \frac{2s_{13}}{s_{123}s_{234}} + \frac{s_{14}}{s_{123}s_{234}} + \frac{s_{14}}{s_{134}s_{234}} \\
& + \frac{s_{12}}{s_{123}s_{234}} + \frac{s_{12}}{s_{134}s_{234}} + \frac{2s_{13}}{s_{134}s_{234}} - \frac{s_{14}s_{23}}{s_{123}s_{134}s_{34}} - \frac{s_{14}s_{23}}{s_{123}s_{234}s_{34}} \\
& - \frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}} - \frac{s_{14}s_{23}}{s_{234}^2s_{34}} + \frac{s_{12}}{s_{234}^2} + \frac{2s_{13}}{s_{234}^2} + \frac{s_{14}}{s_{234}^2} \Big) \\
& + \epsilon^2 \left( -\frac{s_{12}^2}{s_{123}s_{134}s_{23}} - \frac{s_{12}^2}{s_{134}s_{23}s_{234}} - \frac{3s_{12}s_{24}}{s_{123}s_{134}s_{23}} - \frac{2s_{12}s_{24}}{s_{123}s_{23}s_{234}} - \frac{2s_{12}s_{24}}{s_{134}s_{23}s_{234}} \right. \\
& - \frac{2s_{12}s_{34}}{s_{123}s_{134}s_{23}} - \frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}} + \frac{2s_{12}s_{34}}{s_{23}s_{234}^2} + \frac{2s_{12}}{s_{123}s_{134}} + \frac{s_{12}s_{14}}{s_{123}s_{134}s_{23}} \\
& - \frac{2s_{12}s_{13}}{s_{123}s_{134}s_{23}} - \frac{2s_{12}}{s_{123}s_{234}} + \frac{s_{12}}{s_{134}s_{234}} - \frac{s_{12}s_{13}}{s_{123}s_{23}s_{234}} + \frac{s_{12}s_{14}}{s_{123}s_{23}s_{234}} \\
& - \frac{s_{12}s_{13}}{s_{134}s_{23}s_{234}} + \frac{s_{12}s_{14}}{s_{123}s_{134}s_{34}} + \frac{s_{12}s_{13}s_{24}}{s_{123}s_{134}s_{23}s_{34}} - \frac{2s_{12}s_{24}}{s_{123}s_{234}s_{34}} + \frac{s_{12}s_{13}s_{24}}{s_{134}s_{23}s_{234}s_{34}} \\
& - \frac{2s_{12}s_{24}}{s_{234}^2s_{34}} + \frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{2s_{12}}{s_{234}^2} + \frac{2s_{14}}{s_{123}s_{134}} + \frac{2s_{13}s_{24}}{s_{123}s_{134}s_{23}} \\
& + \frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}} - \frac{2s_{14}s_{24}}{s_{134}s_{23}s_{234}} - \frac{2s_{13}s_{24}}{s_{23}s_{234}^2} - \frac{2s_{14}s_{24}}{s_{23}s_{234}^2} - \frac{2s_{13}}{s_{123}s_{234}} \\
& + \frac{s_{14}}{s_{123}s_{234}} - \frac{2s_{14}}{s_{134}s_{234}} - \frac{2s_{13}}{s_{134}s_{234}} - \frac{s_{14}^2}{s_{123}s_{134}s_{34}} + \frac{3s_{13}s_{24}^2}{s_{123}s_{134}s_{23}s_{34}} \\
& + \frac{2s_{13}s_{24}^2}{s_{123}s_{23}s_{234}s_{34}} + \frac{2s_{13}s_{24}^2}{s_{134}s_{23}s_{234}s_{34}} - \frac{2s_{13}s_{14}}{s_{123}s_{134}s_{34}} - \frac{2s_{14}s_{23}}{s_{123}s_{134}s_{34}} - \frac{3s_{14}s_{24}}{s_{123}s_{134}s_{34}} \\
& + \frac{2s_{13}s_{24}}{s_{123}s_{134}s_{34}} + \frac{s_{13}s_{14}s_{24}}{s_{123}s_{134}s_{23}s_{34}} + \frac{2s_{13}^2s_{24}}{s_{123}s_{134}s_{23}s_{34}} - \frac{2s_{14}s_{24}}{s_{123}s_{234}s_{34}} - \frac{2s_{14}s_{24}}{s_{134}s_{234}s_{34}} \Big)
\end{aligned}$$

$$\begin{aligned}
& + \frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{13}^2s_{24}}{s_{123}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{14}s_{24}}{s_{123}s_{23}s_{234}s_{34}} + \frac{s_{13}^2s_{24}}{s_{134}s_{23}s_{234}s_{34}} - \frac{2s_{13}s_{24}}{s_{234}^2s_{34}} \\
& - \frac{s_{14}^2}{s_{123}s_{234}s_{34}} - \frac{s_{13}s_{14}}{s_{123}s_{234}s_{34}} - \frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}} - \frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}} + \frac{2s_{14}s_{23}}{s_{234}^2s_{34}} \\
& - \frac{4s_{13}}{s_{234}^2} - \frac{2s_{14}}{s_{234}^2} \Big) \\
& + \epsilon \left( \frac{s_{12}^2}{s_{123}s_{134}s_{23}} - \frac{s_{12}^2}{s_{123}s_{23}s_{234}} + \frac{2s_{12}^2}{s_{134}s_{23}s_{234}} + \frac{2s_{12}s_{24}}{s_{123}s_{134}s_{23}} + \frac{s_{12}s_{24}}{s_{123}s_{23}s_{234}} \right. \\
& + \frac{s_{12}s_{24}}{s_{134}s_{23}s_{234}} + \frac{3s_{12}s_{24}}{s_{23}s_{234}^2} + \frac{s_{12}s_{34}}{s_{123}s_{134}s_{23}} - \frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}} + \frac{2s_{12}s_{34}}{s_{134}s_{23}s_{234}} \\
& - \frac{s_{12}s_{34}}{s_{23}s_{234}^2} - \frac{s_{12}}{s_{123}s_{134}} - \frac{s_{12}s_{14}}{s_{123}s_{134}s_{23}} - \frac{2s_{12}s_{13}}{s_{123}s_{134}s_{23}} + \frac{s_{12}}{s_{134}s_{234}} \\
& + \frac{s_{12}s_{13}}{s_{123}s_{23}s_{234}} - \frac{2s_{12}s_{14}}{s_{123}s_{23}s_{234}} + \frac{s_{12}s_{14}}{s_{134}s_{23}s_{234}} + \frac{s_{12}s_{13}}{s_{134}s_{23}s_{234}} - \frac{s_{12}s_{14}}{s_{123}s_{134}s_{24}} \\
& - \frac{s_{12}s_{13}s_{24}}{s_{123}s_{134}s_{23}s_{34}} + \frac{2s_{12}s_{24}}{s_{123}s_{234}s_{34}} + \frac{s_{12}s_{13}s_{24}}{s_{123}s_{23}s_{234}s_{34}} - \frac{2s_{12}s_{13}s_{24}}{s_{134}s_{23}s_{234}s_{34}} + \frac{4s_{12}s_{24}}{s_{234}^2s_{34}} \\
& - \frac{2s_{12}s_{13}s_{14}}{s_{123}s_{134}s_{23}s_{34}} + \frac{s_{12}s_{14}}{s_{123}s_{234}s_{34}} - \frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}} + \frac{s_{12}}{s_{234}^2} - \frac{s_{14}}{s_{123}s_{134}} \\
& - \frac{s_{13}s_{24}}{s_{123}s_{134}s_{23}} + \frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}} + \frac{2s_{14}s_{24}}{s_{134}s_{23}s_{234}} + \frac{s_{13}s_{24}}{s_{23}s_{234}^2} + \frac{4s_{14}s_{24}}{s_{23}s_{234}^2} \\
& - \frac{4s_{13}}{s_{123}s_{134}} + \frac{s_{14}}{s_{123}s_{234}} + \frac{s_{14}^2}{s_{123}s_{134}s_{34}} - \frac{2s_{13}s_{24}^2}{s_{123}s_{134}s_{23}s_{34}} - \frac{s_{13}s_{24}^2}{s_{123}s_{23}s_{234}s_{34}} \\
& - \frac{s_{13}s_{24}^2}{s_{134}s_{23}s_{234}s_{34}} - \frac{3s_{13}s_{24}^2}{s_{23}s_{234}^2s_{34}} - \frac{2s_{13}s_{14}}{s_{123}s_{134}s_{34}} + \frac{s_{14}s_{23}}{s_{123}s_{134}s_{34}} + \frac{2s_{14}s_{24}}{s_{123}s_{134}s_{34}} \\
& - \frac{s_{13}s_{24}}{s_{123}s_{134}s_{34}} - \frac{s_{13}s_{14}s_{24}}{s_{123}s_{134}s_{23}s_{34}} + \frac{s_{14}s_{24}}{s_{123}s_{234}s_{34}} + \frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}} \\
& - \frac{s_{13}^2s_{24}}{s_{123}s_{23}s_{234}s_{34}} - \frac{2s_{13}s_{14}s_{24}}{s_{123}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{14}s_{24}}{s_{134}s_{23}s_{234}s_{34}} - \frac{s_{13}^2s_{24}}{s_{134}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{24}}{s_{234}^2s_{34}} \\
& + \frac{3s_{14}s_{24}}{s_{234}^2s_{34}} + \frac{2s_{14}^2}{s_{123}s_{234}s_{34}} - \frac{s_{14}^2}{s_{134}s_{234}s_{34}} + \frac{s_{13}s_{14}}{s_{123}s_{234}s_{34}} + \frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}} \\
& + \frac{2s_{14}s_{23}}{s_{123}s_{234}s_{34}} - \frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}} - \frac{s_{14}s_{23}}{s_{234}^2s_{34}} + \frac{2s_{13}}{s_{234}^2} + \frac{s_{14}}{s_{234}^2} \Big) \\
& - \frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}} - \frac{s_{13}s_{24}}{s_{134}s_{23}s_{234}} - \frac{2s_{12}s_{24}}{s_{23}s_{234}^2} - \frac{2s_{14}s_{24}}{s_{23}s_{234}^2} + \frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}} \\
& - \frac{s_{12}s_{34}}{s_{134}s_{23}s_{234}} + \frac{2s_{13}}{s_{123}s_{134}} + \frac{2s_{12}s_{13}}{s_{123}s_{134}s_{23}} - \frac{s_{14}}{s_{123}s_{234}} + \frac{s_{14}}{s_{134}s_{234}} \\
& + \frac{s_{12}}{s_{123}s_{234}} - \frac{s_{12}}{s_{134}s_{234}} + \frac{s_{12}s_{14}}{s_{123}s_{23}s_{234}} - \frac{s_{12}s_{14}}{s_{134}s_{23}s_{234}} + \frac{s_{12}^2}{s_{123}s_{23}s_{234}} \\
& - \frac{s_{12}^2}{s_{134}s_{23}s_{234}} + \frac{2s_{13}s_{24}^2}{s_{23}s_{234}^2s_{34}} + \frac{2s_{13}s_{14}}{s_{123}s_{134}s_{34}} - \frac{s_{13}s_{24}}{s_{123}s_{234}s_{34}} - \frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}} \\
& - \frac{s_{12}s_{13}s_{24}}{s_{123}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{14}s_{24}}{s_{123}s_{23}s_{234}s_{34}} - \frac{s_{13}s_{14}s_{24}}{s_{134}s_{23}s_{234}s_{34}} + \frac{s_{12}s_{13}s_{24}}{s_{134}s_{23}s_{234}s_{34}} - \frac{2s_{12}s_{24}}{s_{234}^2s_{34}} \\
& - \frac{2s_{14}s_{24}}{s_{234}^2s_{34}} + \frac{2s_{12}s_{13}s_{14}}{s_{123}s_{134}s_{23}s_{34}} - \frac{s_{14}^2}{s_{123}s_{234}s_{34}} + \frac{s_{14}^2}{s_{134}s_{234}s_{34}} - \frac{s_{12}s_{14}}{s_{123}s_{234}s_{34}} \\
& + \frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{s_{14}s_{23}}{s_{123}s_{234}s_{34}} + \frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}}.
\end{aligned} \tag{A.54}$$

By running `BuildAndIntegrateAntenna[C, 4, 0]` we get the integrated result,

$$C_4^0 = \frac{1}{\epsilon} \frac{(-13 + 2\pi^2 - 8\zeta_3)}{32} - \frac{339}{64} + \frac{17\pi^2}{48} - \frac{\pi^4}{45} + \frac{21\zeta_3}{8} + O(\epsilon). \tag{A.55}$$

## A.4 Exploratory Massive $A_3^0$

Running `BuildAntenna[A, 3, 0, quarkMass->mQ]`, we get the built result,

$$\begin{aligned}
A_3^{0,(m_q)} = & \frac{1}{s_{13}s_{23}(2m_q^2 + q^2)} \left[ \frac{1}{\epsilon^2} \frac{m_q^2}{(2m_q^2 + q^2)^2} \left( m_q^2 (s_{13}^2 + 4s_{13}s_{23} + s_{23}^2) + s_{13}s_{23}s_{123} \right) \right. \\
& - \frac{1}{\epsilon} \frac{1}{4(2m_q^2 + q^2)} \left( 2m_q^2 (3s_{13}^2 + 4s_{13}s_{23} + 3s_{23}^2) + s_{12} (s_{13} + s_{23})^2 + (s_{13} + s_{23})^3 \right) \\
& - \frac{1}{4s_{13}s_{23}} \left( -8m_q^4 (s_{13}^2 + s_{23}^2) - 2m_q^2 (s_{12} (s_{13}^2 - s_{13}s_{23} + s_{23}^2) + (s_{13} + s_{23})^3) \right. \\
& \left. \left. + 2s_{12}s_{13}s_{23}s_{123} + s_{13}s_{23} (s_{13}^2 + s_{23}^2) \right) \right] + \mathcal{O}(\epsilon^3). \tag{A.56}
\end{aligned}$$

By running `BuildAndIntegrateAntenna[A, 3, 0, quarkMass->mQ]` we get the integrated result,

$$\begin{aligned}
\mathcal{A}_3^{0,(m_q)} = & \frac{2^{-2(1+\epsilon)} (1-\epsilon) \pi^{\epsilon-2} (1-4m_q^2)^{1-2\epsilon}}{\epsilon(2\epsilon-1)m_q^2(\epsilon-2m_q^2-1)} \left[ -\frac{24\Gamma(4-3\epsilon)\Gamma(3-2\epsilon)}{\Gamma(8-6\epsilon)} (\epsilon-1) (4m_q^2-1) \right. \\
& \cdot \left( \epsilon^2 (8m_q^2-2) + \epsilon (3-4m_q^2) - 4m_q^2-2 \right) {}_2F_1\left(\frac{1}{2}, 3-2\epsilon; \frac{9}{2}-3\epsilon; 1-4m_q^2\right) \\
& - \frac{4\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)}{\Gamma(6-6\epsilon)} \left( 4m_q^4 + \epsilon^3 (8m_q^2-2) + 6m_q^2 + \epsilon^2 (16m_q^4-12m_q^2+5) \right. \\
& \left. \left. - \epsilon (16m_q^4+4m_q^2+5) + 2 \right) {}_2F_1\left(\frac{1}{2}, 2-2\epsilon; \frac{7}{2}-3\epsilon; 1-4m_q^2\right) \right]. \tag{A.57}
\end{aligned}$$

# Appendix B

## Supporting Derivations

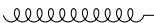
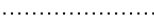
### B.1 Feynman Rules

For the massless final-final quark-antiquark antenna functions considered throughout this work, in CDR, with  $d = 4 - 2\epsilon$ , and using the Feynman gauge  $\lambda = 1$ , the required Feynman rules go as follows.

An application of these rules is shown in full in Subsection 4.1.1.1.

#### B.1.1 Propagators

Each internal line in a Feynman diagram represents a propagator: the way a quantum field carries an excitation from one spacetime point to another. These propagators differ particle-by-particle both in mathematical structure and in the way they are graphically represented in the diagrams. A description of propagators encountered throughout this work is present in Table B.1 [2, 9].

| Particle-type | Mathematical Representation                                                                                  | Graphical Representation                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Quark ( $q$ ) | $\frac{i\delta_{ij}\not{p}}{p^2 + i0}$ $\left(m_q \neq 0 : \frac{i(\not{p} + m_q)}{p^2 - m_q^2 + i0}\right)$ |  |
| Gluon ( $g$ ) | $\frac{-i\delta^{ab}g_{\mu\nu}}{p^2 + i0}$                                                                   |  |
| Ghost ( $c$ ) | $\frac{i\delta^{ab}}{p^2 + i0}$                                                                              |  |

**Table B.1:** Overview of the propagators found throughout the thesis, alongside their mathematical and graphical representations.

For an external massless quark,

$$\sum_s u_s(p)\bar{u}_s(p) = \not{p}, \quad \sum_s v_s(p)\bar{v}_s(p) = \not{p}. \quad (\text{B.1})$$

For external gluons,

$$\sum_\lambda \varepsilon_\mu^{(\lambda)}(p)\varepsilon_\nu^{(\lambda)*}(p) = -g_{\mu\nu} + t_g \quad (\text{B.2})$$

where  $t_g$  are gauge terms that vanish after summing a gauge-invariant amplitude or colour-ordered gauge-invariant partial amplitude.

## B.1.2 Vertices

These lines meet in vertices, which represent the way in which quantum fields interact. The four vertices encountered throughout this work are shown in Table B.2 [2, 9].

| Interaction         | Mathematical Representation                                                                                                                                                                                                                                                               | Graphical Representation |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| $q\bar{q}g$         | $ig_s\gamma^\mu (T^a)_{ij}$                                                                                                                                                                                                                                                               |                          |
| $ggg$               | $g_s f^{abc} (g^{\mu\nu}(p-q)^\rho + g^{\nu\rho}(q-r)^\mu + g^{\rho\mu}(r-p)^\nu)$<br>$p+q+r=0$                                                                                                                                                                                           |                          |
| $gggg$              | $-ig_s^2 \left( f^{abe} f^{cde} (g^{\mu\rho} g^{\nu\sigma} - g^{\mu\sigma} g^{\nu\rho}) \right.$<br>$\quad + f^{ace} f^{bde} (g^{\mu\nu} g^{\rho\sigma} - g^{\mu\sigma} g^{\nu\rho})$<br>$\quad \left. + f^{ade} f^{bce} (g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma}) \right)$ |                          |
| $c\bar{c}g$         | $-g_s f^{abc} p^\mu$<br>$p$ is the outgoing ghost momentum                                                                                                                                                                                                                                |                          |
| $\gamma^* q\bar{q}$ | $ieQ_q\gamma^\mu\delta_{ij}$                                                                                                                                                                                                                                                              |                          |

**Table B.2:** Overview of the vertices encountered throughout this thesis, alongside their mathematical and graphical representations.

## B.2 PaVe Reduction

At one-loop, amplitudes produce tensor integrals similar to those in Eq. (2.76) [14]. With the general massive<sup>1</sup> denominators defined as,

$$D_0 = \ell^2 - m_0^2 + i0, \quad D_i = (\ell + r_i)^2 - m_i^2 + i0 \quad (\text{B.3})$$

the tensor integral takes the form,

$$I_N^{\mu_1 \dots \mu_R} = \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{\ell^{\mu_1} \dots \ell^{\mu_R}}{D_0 D_1 \dots D_{N-1}} \quad (\text{B.4})$$

where  $R$  is the tensor rank,  $r_i$  are the combinations of external momenta and  $m_i$  is the mass associated with the  $i$ -th propagator. The reduction is carried out in  $d = 4 - 2\epsilon$  dimensions, so that,

$$g_{\mu\nu} g^{\mu\nu} = d. \quad (\text{B.5})$$

PaVe reduces the Lorentz indices in the numerator  $\ell^\mu \ell^\nu \ell^\rho \dots$  terms and expresses them in terms of external momenta  $p_i^\mu$ , the metric tensor  $g_{\mu\nu}$  and scalar one-loop integrals, through Lorentz covariance [14].

<sup>1</sup>Throughout this section, the general and massive case will be considered for completeness. The rest of the present work, however, heavily leans towards the massless regime. In that regime the internal masses vanish and the physical external partons satisfy  $p_a^2 = 0$ . The shifted momenta  $r_i$ , introduced in Eq. (B.3), entering propagators are combinations of external momenta and need not be lightlike.

For a rank-one  $N$ -point integral, the only available vectors are the independent external momenta, such that,

$$I_N^\mu = \sum_{i=1}^{N-1} p_i^\mu C_i \quad (\text{B.6})$$

where  $C_i$  are scalar coefficient functions. For rank-two, the relation becomes,

$$I_N^{\mu\nu} = \sum_{i,j=1}^{N-1} p_i^\mu p_j^\nu C_{ij} + g^{\mu\nu} C_{00} \quad (\text{B.7})$$

and for rank-three,

$$I_N^{\mu\nu\rho} = \sum_{i,j,k=1}^{N-1} p_i^\mu p_j^\nu p_k^\rho C_{ijk} + \sum_{i=1}^{N-1} (g^{\mu\nu} p_i^\rho + g^{\mu\rho} p_i^\nu + g^{\nu\rho} p_i^\mu) C_{00i} \quad (\text{B.8})$$

and so on for higher ranks.

The reduction mechanism works because scalar products involving the loop momentum can be rewritten using the propagator denominators as,

$$2\ell \cdot r_i = D_i - D_0 - r_i^2 + m_i^2 - m_0^2 \quad (\text{B.9})$$

which lets us cancel denominator terms, through a method like the following example set at  $i = 1$ ,

$$\frac{\ell \cdot r_1}{D_0 D_1} = \frac{1}{2} \left( \frac{1}{D_0} - \frac{1}{D_1} + \frac{1}{D_0 D_1} (-m_0^2 + m_1^2 - r_1^2) \right) \quad (\text{B.10})$$

thereby rewriting loop-momentum scalar products in terms of denominators and scalar integrals.

While using this reduction scheme, we might also encounter Gram matrices, defined as,

$$G_{ji} = 2p_j \cdot p_i. \quad (\text{B.11})$$

Contracting a general rank-one integral, as described in Eq. (B.6), with  $p_j$  gives,

$$p_j \cdot I_N = \sum_i (p_j \cdot p_i) C_i = \frac{1}{2} \sum_i G_{ji} C_i. \quad (\text{B.12})$$

If we now define  $R_j \equiv p_j \cdot I_N$ , we are then able to compute the  $C_i$  coefficients in the Lorentz decomposition by taking,

$$G_{ji} C_i = 2R_j \Rightarrow C_i = 2 \left( G^{-1} \right)_{ij} R_j. \quad (\text{B.13})$$

### B.2.1 Rank-One Bubble Example

To illustrate the method, let us consider the rank-one bubble,

$$B^\mu(p; m_0^2, m_1^2) = \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{\ell^\mu}{D_0 D_1} \quad (\text{B.14})$$

where the denominators follow Eq. (B.3). Lorentz covariance gives,

$$B^\mu = p^\mu B_1 \quad (\text{B.15})$$

Contracting with  $p_\mu$  yields,

$$p^2 B_1 = p_\mu B^\mu = \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{\ell \cdot p}{D_0 D_1}. \quad (\text{B.16})$$

Using the relation in Eq. (B.9), setting  $r_1 = p$ , we get,

$$p^2 B_1 = \frac{1}{2} \left[ A_0(m_0^2) - A_0(m_1^2) + (m_1^2 - m_0^2 - p^2) B_0(p^2; m_0^2, m_1^2) \right] \quad (\text{B.17})$$

and therefore<sup>2</sup>,

$$B^\mu = p^\mu \frac{A_0(m_0^2) - A_0(m_1^2) + (m_1^2 - m_0^2 - p^2) B_0}{2p^2}. \quad (\text{B.18})$$

---

<sup>2</sup>The result shown in Eq. (B.18) assumes generic kinematics with  $p^2 \neq 0$ . Exceptional cases are obtained through an appropriate limit or alternative reduction.

Here,  $A_0$  and  $B_0$  are the scalar one-loop integrals mentioned throughout. These two types can be defined as,

$$\begin{aligned} A_0(m^2) &= \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{1}{\ell^2 - m^2 + i0} \\ B_0(p^2; m_0^2, m_1^2) &= \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{1}{D_0 D_1}. \end{aligned} \quad (\text{B.19})$$

## B.2.2 PaVe Scalar Functions

As mentioned, only three of the MI considered are PaVe scalar functions. Those are  $B_0$ ,  $C_0$  and  $D_0$  and are the PaVe two-, three- and four-point scalar integrals, respectively. The PaVe method also features the  $A_0$  (one-point) scalar function, that does not feature in *AntCalc*. It also accounts for scalar integrals with more than four external legs, as those can be further reduced in four dimensions to combinations of  $D_0$  functions [10, 14].

### B.2.2.1 The $B_0$ Function

The  $B_0$  function represents a scalar one-loop diagram with two external lines and two internal propagators. It is represented, in  $d$  dimensions as,

$$B_0(k^2, m_0, m_1) = \frac{(2\pi\mu)^{4-d}}{i\pi^2} \int d^d p \frac{1}{(p^2 - m_0^2 + i\varepsilon) ((p+k)^2 - m_1^2 + i\varepsilon)} \quad (\text{B.20})$$

where  $k$  is the external momentum, and  $m_0$  and  $m_1$  are the internal masses.  $\mu$  is the t'Hooft normalisation scale.

It is also noteworthy that this integral, unlike higher-point ones, is UV divergent and generates the pole term  $\Delta_B$ ,

$$\Delta_B = \frac{2}{4-d} - \gamma_E + \log(4\pi) \quad (\text{B.21})$$

where  $\gamma_E$  is Euler's gamma constant. This function may also be written as an integral over the Feynman parameter  $x$ . That yields the explicit analytical form [14],

$$B_0(k^2, m_0, m_1) = \Delta_B - \int_0^1 dx \log \left[ \frac{k^2 x^2 - x(k^2 - m_0^2 + m_1^2) + m_1^2 - i\varepsilon}{\mu^2} \right] + \mathcal{O}(d-4). \quad (\text{B.22})$$

### B.2.2.2 The $C_0$ Function

The  $C_0$  function represents a scalar loop with three external lines and three internal propagators. It is a common feature in the vertex corrections of electroweak physics. It is written in  $d$  dimensions as,

$$C_0(k_1, k_2, m_0, m_1, m_2) = \frac{(2\pi\mu)^{4-d}}{i\pi^2} \int d^d p \frac{1}{(p^2 - m_0^2 + i\varepsilon) ((p+k_1)^2 - m_1^2 + i\varepsilon) ((p+k_1+k_2)^2 - m_2^2 + i\varepsilon)}. \quad (\text{B.23})$$

This integral is UV finite and may be rewritten as a double integral over the Feynman parameters  $x$  and  $y$  as,

$$C_0 \propto \int_0^1 dx \int_0^{1-x} dy [ax^2 + by^2 + cxy + dx + ey + f]^{-1} \quad (\text{B.24})$$

where the coefficients  $a, b, c, d, e$  and  $f$  are linear combinations of the external momenta and internal mass parameters [10].

### B.2.2.3 The $D_0$ Function

The  $D_0$  function describes a scalar loop with four external lines and four internal propagators. It is often called the scalar box integral. It is written as,

$$D_0(k_1, k_2, k_3, m_0, m_1, m_2, m_3) = \frac{(2\pi\mu)^{4-d}}{i\pi^2} \int d^d p \frac{1}{D_0 D_1 D_2 D_3} \quad (\text{B.25})$$

with denominator factors,

$$\begin{aligned} D_0 &= p^2 - m_0^2 + i\varepsilon \\ D_i &= \left( p + \sum_{a=1}^i k_a \right)^2 - m_i^2 + i\varepsilon. \end{aligned} \quad (\text{B.26})$$

Evaluating  $D_0$  is highly non-trivial as it features numerous physical cuts and branch points. The resulting analytical expression is expressed in terms of Spence functions (dilogarithms) [10].

### B.3 IBP Reduction Example

The method behind IBP reduction, as discussed in Subsection 2.5.1 may be made easier to visualise with an example. Consider a one-loop bubble, akin to that studied via PaVe reduction in Section B.2. Take,

$$\mathcal{I}(a_0, a_1) = \int d^d \ell \frac{1}{D_0^{a_0} D_1^{a_1}} \quad (\text{B.27})$$

where,

$$D_0 = \ell^2 - m_0^2, \quad D_1 = (\ell + p)^2 - m_1^2. \quad (\text{B.28})$$

We now choose  $v^\mu = \ell^\mu$ , and build and expand the relation from Eq. (2.78) as,

$$\begin{aligned} 0 &= \int d^d \ell \frac{\partial}{\partial \ell^\mu} \left( \frac{\ell^\mu}{D_0^{a_0} D_1^{a_1}} \right) \\ &= d\mathcal{I}(a_0, a_1) - 2a_0 \int d^d \ell \frac{\ell^2}{D_0^{a_0+1} D_1^{a_1}} - 2a_1 \int d^d \ell \frac{\ell \cdot (\ell + p)}{D_0^{a_0} D_1^{a_1+1}}. \end{aligned} \quad (\text{B.29})$$

We can now use,

$$\ell^2 = D_0 + m_0^2 \quad (\text{B.30})$$

$$2\ell \cdot (\ell + p) = D_0 + D_1 + m_0^2 + m_1^2 - p^2 \quad (\text{B.31})$$

and finally get the explicit shifted-index identity,

$$\begin{aligned} 0 &= (d - 2a_0 - a_1)\mathcal{I}(a_0, a_1) \\ &\quad - 2a_0 m_0^2 \mathcal{I}(a_0 + 1, a_1) \\ &\quad - a_1 \mathcal{I}(a_0 - 1, a_1 + 1) \\ &\quad - a_1 (m_0^2 + m_1^2 - p^2) \mathcal{I}(a_0, a_1 + 1). \end{aligned} \quad (\text{B.32})$$

### B.4 Three-Particle Phase-Space to Cut Propagators

Let us now consider the massless three-particle phase-space  $\Phi_3$ . It may be described as,

$$d\Phi_3(q; p_1, p_2, p_3) \propto d^d p_1 d^d p_2 d^d p_3 \delta^+(p_1^2) \delta^+(p_2^2) \delta^+(p_3^2) \delta^{(d)}(q - p_1 - p_2 - p_3). \quad (\text{B.33})$$

We start by using momentum conservation to eliminate  $p_3$ ,

$$p_3 = q - p_1 - p_2. \quad (\text{B.34})$$

This first step eliminates the rightmost  $\delta$ -function. Then, using reverse unitarity, each of the remaining  $\delta$ -functions may be rewritten as cut propagators. Schematically, it gives,

$$d\Phi_3 \sim d^d p_1 d^d p_2 \frac{1}{[p_1^2]_c [p_2^2]_c [(q - p_1 - p_2)^2]_c} \quad (\text{B.35})$$

with

$$\left[ \frac{1}{p^2} \right]_c \equiv \frac{1}{2\pi i} \left[ \frac{1}{p^2 - i0} - \frac{1}{p^2 + i0} \right]_+. \quad (\text{B.36})$$

## B.5 Laporta Algorithm Implementation Example

Consider an arbitrary family defined as  $\mathcal{F}[\{D_1, D_2, D_3\}; d\Phi]$ , with the denominators chosen as  $D_1 = s_{12}$ ,  $D_2 = s_{13}$  and  $D_3 = s_{23}$ . We may take the integrals below, all members of this family, and organise them using their respective sector vectors as follows,

$$I_1 = \int d\Phi \frac{s_{13}}{s_{12}} \Rightarrow I_1 = I(1, -1, 0), \quad (\text{B.37})$$

$$I_2 = \int d\Phi \frac{2}{s_{12}s_{13}s_{23}} \Rightarrow I_2 = 2I(1, 1, 1), \quad (\text{B.38})$$

$$I_3 = \int d\Phi \frac{1}{s_{12}^3} \Rightarrow I_3 = I(3, 0, 0) \quad (\text{B.39})$$

where  $d\Phi$  is the common phase-space measure. Importantly, two integrals may share a sector if they contain the same denominators, independently of those denominators' powers. Also, only positive indices denote denominator presence, meaning that integrals  $I_1$  and  $I_3$  in the example above are members of the same sector,  $I_1, I_3 \in (1, 0, 0)$ . In our example, only two sectors appear:  $(1, 0, 0)$  and  $(1, 1, 1)$ . These are our two target sectors. The three integrals are organised in terms of the  $t$ ,  $r$  and  $s$  as follows,

|               | $t$ | $r$ | $s$ |
|---------------|-----|-----|-----|
| $I(1, -1, 0)$ | 1   | 0   | 1   |
| $I(1, 1, 1)$  | 3   | 0   | 0   |
| $I(3, 0, 0)$  | 1   | 2   | 0   |

(B.40)

then, by order of complexity, the algorithm orders the integrals as,

$$I(1, 1, 1) \succ I(3, 0, 0) \succ I(1, -1, 0). \quad (\text{B.41})$$

## B.6 The Massive Extension

The main body of this thesis focused primarily on the massless regime, since the vast majority of the present work's chapters focus on that regime. This appendix collects the supporting extension of the discussed methods into the massive regime.

The fundamental change is the massive on-shell condition; the remainder of this section traces its consequences for the antenna construction and integration. Since final-state particles now have mass, their squared momenta are not set to zero, they are set to their squared mass.

$$p_q^2 = p_{\bar{q}}^2 = 0 \xrightarrow{\text{Massive Regime}} p_q^2 = p_{\bar{q}}^2 = m_q^2. \quad (\text{B.42})$$

where  $p_i$  represents the momentum and  $m_i$  represents the mass of particle  $i$ .

The antenna subtraction formalism and the reduction architecture that was developed and described throughout Chapter 2 are retained, however. What changes is the intermediate steps and kinematics: the massive on-shell condition changes the kinematics themselves and the unresolved limits, with soft radiation remaining singular but heavy-quark collinear behaviour becoming quasi-collinear. Massive integration is also different, as the characteristic phase-space scale in the massive regime is not equal to its massless counterpart. A new scale of  $m_q^2/q^2$  is also introduced and leads to distinct cut-integral families and master integrals [25].

### B.6.1 Massive Kinematics and Infrared Structure

In Subsection 2.2.1, the Mandelstam invariants were introduced for the massless regime. The general definition is,

$$\begin{aligned} s_{ij} &= (p_i + p_j)^2 = p_i^2 + 2p_i \cdot p_j + p_j^2 \\ &= m_i^2 + m_j^2 + 2p_i \cdot p_j. \end{aligned} \quad (\text{B.43})$$

Through momentum conservation, which states that  $q = \sum_i p_i$ , we also have to reconsider the  $q^2$  expression, as,

$$q^2 = \sum_i m_i^2 + 2 \sum_{i < j} p_i \cdot p_j. \quad (\text{B.44})$$

Soft radiation remains singular in the massive regime and therefore continues to generate explicit infrared poles after integration. A non-zero mass regulates collinear limits involving a massive parton, whereas collinear limits between two massless partons remain singular. If the two final-state collinear partons are massless, then the limit follows the massless

collinear limit from the massless regime. If the two partons are massive, no strict collinear singularity exists. But if one of the collinear partons is massive, consider that particle  $i$  of mass  $m_i$  and energy  $E_i$ , and the other is massless, consider that particle  $j$ , such that  $m_j = 0$ , then,

$$p_i \cdot p_j = E_i E_j (1 - \beta_i \cos \theta), \quad \beta_i = \sqrt{1 - \frac{m_i^2}{E_i^2}} < 1. \quad (\text{B.45})$$

As the angle between the partons  $\theta$  tends towards zero, the mass prevents the invariant from reaching the collinear singular surface,

$$\lim_{\theta \rightarrow 0} p_i \cdot p_j = E_i E_j (1 - \beta_i) \neq 0. \quad (\text{B.46})$$

This regulated collinear region still carries physically important information, particularly in the joint high-energy and small-angle region,  $E_i \gg m_i$  and  $\theta \ll 1$ ,

$$p_i \cdot p_j \sim \frac{E_i E_j}{2} \left( \theta^2 + \frac{m_i^2}{E_i^2} \right) \quad (\text{B.47})$$

and consequently since,

$$\int_0^{\theta_{\max}^2} d\theta^2 \frac{1}{p_i \cdot p_j} \sim \int_0^{\theta_{\max}^2} d\theta^2 \left( \theta^2 + \frac{m_i^2}{E_i^2} \right)^{-1} \sim \ln \left( \frac{E_i^2}{m_i^2} \right) \quad (\text{B.48})$$

the massive-massless quasi-collinear region replaces the strict collinear pole by a mass logarithm. The quasi-collinear limit, then, is the limit in which both the mass  $m_i$  and the opening angle  $\theta$  become small together. It is through this limit that the massive antennae recover the massless splitting behaviour, as  $m_i \rightarrow 0$  [25].

Mass does not regulate UV divergences, as  $\ell^2 \gg m_i^2, p_i^2$ . It is noteworthy, however, that in a theory with massive quarks, the quark mass and quark field must be renormalised as,

$$g_{s,0} = \mu^\epsilon Z_g g_s, \quad m_{q,0} = Z_m m_q, \quad \psi_{q,0} = \sqrt{Z_2^q} \psi_q. \quad (\text{B.49})$$

## B.6.2 Massive Phase-Space Integration and Reduction Methods

For final-state particles of momenta  $p_i$  and masses  $m_i$ , the phase-space becomes,

$$d\Phi_n(q; p_1, \dots, p_n) = (2\pi)^d \delta^{(d)} \left( q - \sum_{i=1}^n p_i \right) \prod_{i=1}^n \frac{d^d p_i}{(2\pi)^{d-1}} \delta^+(p_i^2 - m_i^2). \quad (\text{B.50})$$

At the level of the phase-space measure, the formal change from the massless case, as described in Subsubsection 2.5.1.2, is,

$$\delta^+(p_i^2) \rightarrow \delta^+(p_i^2 - m_i^2) \quad (\text{B.51})$$

which physically imposes that  $p_i^2 = m_i^2$  and  $p_i^0 > 0$ .

Phase-space factorisation still holds for the massive regime, but the momentum mapping must preserve the relevant mass-shell conditions. So,

$$d\Phi_{m+r}(\{p\}; q) = d\Phi_m(\{\tilde{p}\}; q) d\Phi_X. \quad (\text{B.52})$$

For a final-final antenna with hard radiator  $i, k$  and unresolved particles between them, this is schematically,

$$d\Phi_{m+r} = d\Phi_m(\dots, \tilde{p}_I, \tilde{p}_K, \dots; q) d\Phi_{X_{ijk}} \quad (\text{B.53})$$

while  $\tilde{p}_I^2 = m_i^2$ ,  $\tilde{p}_K^2 = m_k^2$  and momentum conservation is satisfied,

$$\tilde{p}_I + \tilde{p}_K = p_i + \left( \sum_{\text{unresolved } j} p_j \right) + p_k. \quad (\text{B.54})$$

Antenna integration is computed in the same manner as in the massless regime,

$$\mathcal{X}_n^l \sim \int d\Phi_X \left( \prod_{r=1}^l d^d \ell_r \right) X_n^l(\{p\}, \{\ell\}) \sim (q^2)^{\lambda+\kappa\epsilon} f\left(\left\{\frac{m^2}{q^2}\right\}, \epsilon\right). \quad (\text{B.55})$$

PaVe and IBP reduction remain applicable in the massive regime, following the same principles introduced for the massless case. For IBP reduction, however, the massive case is significantly harder analytically: the phase-space is non-trivial and has mass-dependent thresholds; the master integrals are generally multi-scale, and can involve logarithms, polylogarithms and more complicated transcendental functions; and lastly, boundary conditions are less direct, although common choices are the massless

limit at  $m \rightarrow 0$  and the threshold limit at  $\beta \rightarrow 0$  [25].

## B.7 Master Integral Substitution Arrays

As discussed in Section B.5, *AntCalc*'s IBP reduction, handled internally by *LiteRed2*, organises each antenna family's master integrals as members of that family's own basis, addressed through *LiteRed2*'s  $j[\text{basis}, \dots]$  notation rather than the named masters used throughout Chapter 4. At the master combination stage for each antenna, the raw  $j[\dots]$ -form result is substituted through a rule list specific to that family's basis, translating it into the named masters. The rule lists used are collected below, one per antenna family.

### B.7.1 The $A_2^2$ Set

```
In[1]:= A22MISubs = {
j[A22TwoLoopTreeBasis1, 0, 0, 0, 1, 1, 0, 1] -> A3,
j[A22TwoLoopTreeBasis1, 1, 0, 1, 1, 1, 0, 0] -> A4,
j[A22TwoLoopTreeBasis4, 1, 0, 1, 0, 0, 1, 0] -> A3,
j[A22TwoLoopTreeBasis4, 1, 0, 0, 1, 0, 1, 1] -> A4,
j[A22TwoLoopTreeBasis6, 0, 0, 1, 1, 0, 0, 1] -> A3,
j[A22TwoLoopTreeBasis6, 1, 0, 1, 1, 0, 1, 0] -> A4,
j[A22TwoLoopTreeBasis8, 0, 1, 0, 0, 0, 1, 1] -> A3,
j[A22TwoLoopTreeBasis8, 1, 0, 1, 1, 1, 0, 0] -> A4,
j[A22TwoLoopTreeBasis8, 1, 1, 1, 1, 1, 1, 0] -> A6,
j[A22OneLoopSelfBasis, 1, 0, 1, 1, 0, 1, 0] -> A22LO
};
```

### B.7.2 The $A_3^1$ Set

```
In[1]:= A31MISubs = {
j[A31Basis10, 1, 1, 1, 0, 0, 1, 0, 1, 0] -> V5b,
j[A31Basis12, 1, 1, 1, 0, 0, 1, 0, 1, 0] -> V5b,
j[A31Basis2, 1, 1, 1, 0, 0, 1, 0, 1, 0] -> V5b,
j[A31Basis6, 1, 1, 1, 0, 1, 0, 0, 1, 0] -> V5a,
j[A31Basis6, 1, 1, 1, 0, 0, 1, 0, 1, 0] -> V5b,
j[A31Basis9, 1, 1, 1, 1, 0, 0, 0, 1, 0] -> V5a,
j[A31Basis9, 1, 1, 1, 0, 1, 0, 0, 1, 0] -> V5b,
j[A31Basis3, 1, 1, 1, 0, 1, 0, 0, 1, 0] -> V5a,
j[A31Basis3, 1, 1, 1, 0, 0, 0, 1, 1, 0] -> V5b,
j[A31Basis4, 1, 1, 1, 0, 1, 0, 0, 1, 0] -> V5a,
j[A31Basis4, 1, 1, 1, 0, 0, 1, 0, 1, 0] -> V5b,
j[A31Basis4, 1, 1, 1, 0, 1, 1, 1, 1, 1] -> 2 V8,
j[A31Basis7, 1, 1, 1, 1, 0, 0, 0, 1, 0] -> V5a,
j[A31Basis7, 1, 1, 1, 0, 1, 0, 0, 1, 0] -> V5b,
j[A31Basis7, 1, 1, 1, 1, 1, 1, 1, 1, 0] -> V8,
j[A31Basis8, 1, 1, 1, 1, 0, 0, 0, 1, 0] -> V5a,
j[A31Basis8, 1, 1, 1, 0, 1, 0, 0, 1, 0] -> V5b
};
```

### B.7.3 The Four-Parton (X40) Set

```
In[1]:= X40MISubs = {
j[chainBasis1342, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[chainBasis1342, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[chainBasis1432, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
```

```

j[chainBasis1432, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[boxBasis1324, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[boxBasis1324, 1, 1, 1, 1, 1, 1, 1, 1, 0] -> R8a,
j[chainBasis1324, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[chainBasis1423, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[hybridBasis1234, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[hybridBasis1234, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[hybridBasis1234, 1, 1, 1, 1, 1, 0, 1, 1, 1] -> R8b,
j[hybridBasis1243, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[hybridBasis1243, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[hybridBasis1243, 1, 1, 1, 1, 1, 0, 1, 1, 1] -> R8b,
j[chainBasis1234, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[chainBasis1234, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[chainBasis2134, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[chainBasis2134, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[chainBasis2314, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[chainBasis2314, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[hybridBasis2431, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[hybridBasis2431, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[hybridBasis2431, 1, 1, 1, 1, 1, 0, 1, 1, 1] -> R8b
};

```

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